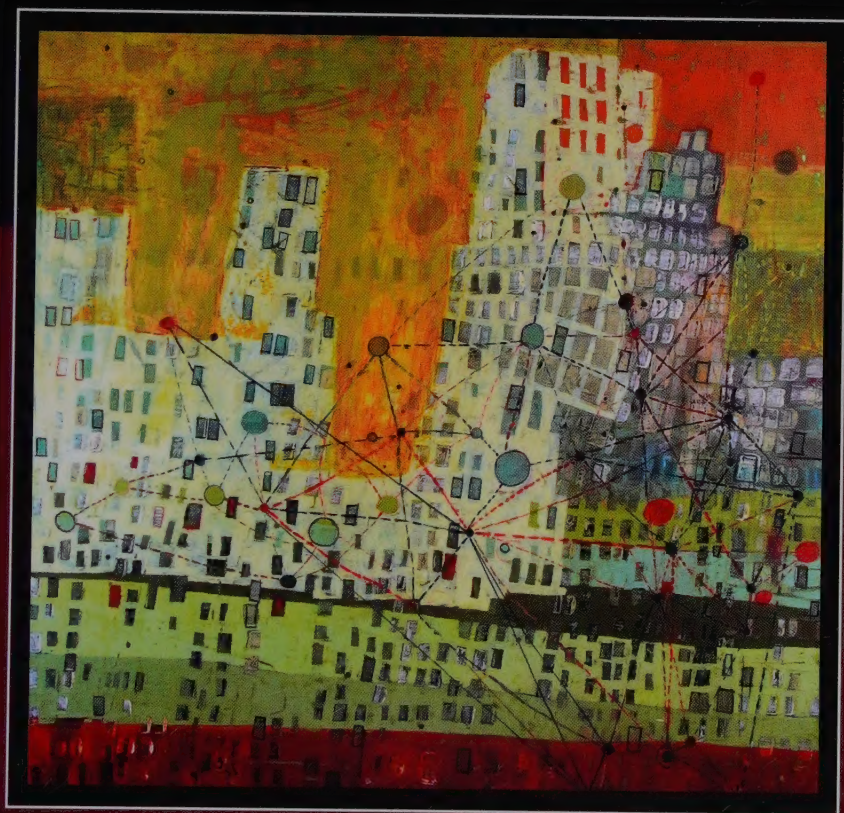


Egocentric Network Analysis

FOUNDATIONS, METHODS,
AND MODELS



Brea L. Perry, Bernice A. Pescosolido,
and Stephen P. Borgatti

Egocentric Network Analysis

Foundations, Methods, and Models

Egocentric network analysis is used widely across the social sciences, especially in anthropology, political science, economics, and sociology, and is increasingly being employed in communications, informatics, and business and marketing studies. Egocentric network analysis requires a unique set of data collection and analysis skills that overlap only minimally with other network methodologies. However, until now there has been no single reference for conceptualizing, collecting, and analyzing egocentric social network data. This comprehensive guide to study design, data collection, and analysis brings together the state of knowledge with the most effective research tools to guide newcomers to this field. It is illustrated with many engaging examples and graphics and assumes no prior knowledge. Covering the entire research process in a logical sequence, from conceptualizing research questions to interpreting findings, this volume provides a solid foundation for researchers at any stage of their career to learn and apply ego network methods.

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Egocentric Network Analysis

Foundations, Methods, and Models

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To Rus Perry
and
Joe Pescosolido
and
Joe Borgatti

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Preface

Networks and network terminology are everywhere. The past several decades have seen a significant increase in the use of network theory and methodology to address complexity and interdependence in relationships between actors or organisms at multiple levels of analysis. A broad literature spanning multiple disciplines suggests that networks critically influence a variety of human, societal, and technical processes and outcomes. In the social and behavioral sciences, the social network perspective is concerned with the structure of linkages between individuals or other social actors such as households, organizations, or communities; the resources shared by these network members; and the ideas and information flowing through those networks. A social network can be defined as a “set of individuals who are either directly or indirectly connected” (Lin & Peek 1999: 243). Although the individual units in a social network can be any of a wide array of entities (e.g., insects, animals, family groups, corporations, nations), we focus here on people.

We live in a “networked society” where the spread of digital information and communications technologies have fundamentally changed the social, political, economic, and cultural nature of human life. Castells (2004) and Wellman (1999) see this as a new social form – one in which advances in technological communications, from computers to cell phones, have reorganized societies and the way individuals are connected to one another. Though it is clear that connectedness matters, perhaps more so by the day, that may be where the clarity ends. Given the ubiquity of “networks,” it is hardly surprising that the term can evoke confusion for the public and researchers alike. Consider, for example, the business world’s use (or misuse) of the term “networking,” the popularity of the

notion of “six degrees of separation,” the millennials’ idea that social networks equal social media, as well as social scientists’ concerns about “social support” and “social capital.” Taken together, the result is a sense of mystification around what networks are, what they do, and what, possibly, “network science” might be.

That said, networks offer enough of a shared perspective and language to deconstruct terminological and scale differences that plague efforts at serious integration of insights across scientific disciplines (Pescosolido 2006). Perhaps most importantly, research on social networks is united by a common theoretical focus on ties between actors, but diverges methodologically. There are at least two distinct research designs: (1) whole (i.e., sociocentric) network analysis in which ties between nearly every member of a group or social context are represented, and (2) ego (i.e., egocentric or personal) network analysis in which individuals’ connections to their own personal community network are measured. In whole network research, every node in the network is also a research subject. In ego network research, we collect data from a sample of individuals who provide information about their immediate “network neighborhoods.” That is, ego network analysis diverges in its focus on individually bounded personal communities rather than a common boundary applicable to all respondents.

This book constitutes a comprehensive guide to ego network analysis, focusing on fundamental research design, methodological, and analytic issues. Ego network research has a long and rich tradition. It includes some of the most influential contributions to social network research, such as Bott’s (1955) seminal study in the 1950s of husbands’ and wives’ personal networks and their relationship to the enactment of marital roles. Compared to whole network designs, ego network research allows for greater flexibility in data collection and analysis as well as a broader scope of inference. As a result, ego network analysis requires a unique set of data collection and analysis skills and considerations that overlap only minimally with other network methods.

Despite the importance of ego network research, there is no comprehensive manual for the design and analysis of ego network data and substantive applications of this approach (but see Crossley et al. 2015 for an introductory volume). In contrast, there have been at least a dozen books that systematically cover whole network methods and/or analyses (e.g., Wasserman & Faust 1994; Scott 2012; Borgatti, Everett, & Johnson 2013). The absence of even one counterpart resource for researchers seeking guidance on the best practices in ego network analysis is

disconcerting. This book – long overdue – fills this gap, synthesizing a diverse and diffuse body of knowledge on ego network methodology and its applications. Our aim is to provide a broad survey, covering everything from the conception of research questions to the analysis and interpretation of data. This strategy prohibits us from addressing all relevant topics, and sometimes limits the depth of the material we can provide. Consequently, we direct readers to additional references, where appropriate.

Many people in our own professional and personal networks helped make this book possible. We are grateful to Ann McCranie, Mark Granovetter, Isidro Maya Jariego, J. Scott Long, Ajay Mehra, Kate Eddens, Raffaele Vacca, Erin Pullen, and anonymous reviewers for their comments and contributions. We would like to thank Mary Hannah and Alex Capshaw for invaluable editing and administrative assistance, and the Indiana University Network Science Institute for financial resources. In addition, we are appreciative of strong ongoing support for network science research and infrastructure from the Offices of the President, Provost, and Vice Provost for Research, the College of Arts and Sciences, and the Department of Sociology at Indiana University, as well as the LINKS Center for Social Network Analysis at the University of Kentucky. We offer special thanks to our immediate and extended families, and especially Brian, Eirelyn, Harper, Joe, and Roberta.

PART I

Theoretical and Methodological Foundations

The Nature of Networks

Theory and research in social networks are far from monolithic. Instead, there are a variety of network theories, a number of basic concepts, and also a set of shared principles. Together, these constitute what many have termed the network perspective. The purpose of this chapter is to lay out and illustrate some fundamentals of a network perspective. To this end, we will both highlight the features of this perspective that are distinctive and also explore the commonalities and overlap with similar approaches. Particularly important is how network research strives to move beyond the reduction of information to an individual-level trait. Instead, the network approach focuses more on the ties themselves, examining the composition of networks (e.g., are they made up of family members, friends, coworkers?), the nature of relationships and exchange between members of a network (e.g., do members of a network share information, resources?), and the structural positions of various actors (e.g., does a person have influence by virtue of direct or indirect access to others?).

I.1 WHAT ARE NETWORKS?

Simply stated, a network is a “set of actors and the ties among them” (Wasserman & Faust 1994) or “nothing more than a collection of objects connected to each other in some fashion” (Watts 2003a: 28). Social networks have also been termed the “structures of relationships linking social actors” (Marsden 2000: 2727). These relationships or ties are the most basic components of social experience, mapping the connections that individuals have to one another (Pescosolido 1991). The networks perspective is, if not unique, distinctive in that it embeds individuals and their

decisions, outcomes, and life chances in the larger social context of relationships, group membership, and community.

Doing network research requires a different way of thinking about how any phenomenon works. All network theories directed toward human action are based on a fundamental premise: social ties and interactions, rather than individual actors, represent a major “engine of action” underlying behavior. And while actors can be humans, network theories are not restricted to them, and can include larger social units like families, organizations, and nations, or smaller social units like insects, neurons, or proteins. So, the structure of ties between and among countries are critical in determining the global balance of economic and political power. Even at the molecular level, proteins interacting with other proteins in biological organisms, or even brain loci sending signals to other loci, may be just as important for understanding disease as traditional structures. For the sake of simplicity, we concentrate primarily on the nature of human actions, including the institutions and places that they build.

Thus, despite all differences in training, interests, or level of analysis, what network scientists share is the fundamental proposition that connectedness represents a basic vector of action. This means networks cannot simply be added as one more variable in a traditional model: use of the network paradigm changes the very nature of the model itself. Moreover, it is not one more utility in an economic cost-benefit analysis; it is not one more motivational factor that matters for action. Rather, it is a differential point of departure for theorizing, for thinking about data collection, and for doing analyses.

1.2 NETWORKS AS METHOD AND THEORY

In the field of network science, there has historically been a tension between the primacy of method and theory. On the one hand, network science is often seen as an analytic method, a tool for analyzing data. Network science *does* provide solutions to the analytic challenges posed by relational data. Typical linear regression models are often not appropriate for network data because these violate the assumption of independent observations. Networks imply the opposite – cases are, by their very nature, interdependent or connected. Thus, even from the earliest days when quantitative network data were collected, questions have arisen about how such data should be analyzed.

On the other hand, many researchers consider social networks to be a theoretical perspective that lends itself to many different kinds of

analysis (Borgatti & Halgin 2011). Networks as theory are based on the fundamental premise that interconnectedness represents the mechanism of action. In other words, social interaction, rather than individual motivation, calculation, or beliefs, are the principal bases of behavior. Individuals shape their everyday lives through consultation, suggestion, support, and nagging from others, not to mention conflict and competition. Furthermore, this perspective suggests that social networks set a context in formal organizations and institutions that influences what people do, how they feel, and what happens to them. Finally, network theorists observe that the patterns of relationships among actors – who is connected to whom and in what ways – has important consequences, determining what and how much is shared or flows from person to person in a network, and how much power or advantage individual actors possess. More generally, the position of individuals in a social network is thought to determine, in part, the opportunities and constraints they will encounter.

In reality, networks as method or as theory are mutually reinforcing propositions. Network science requires a new toolbox precisely because it brings to the table a different explanatory perspective. At its core, network science is about connectedness and its implications not only for individual actors, but also for the networked system as a whole. It shifts the focus from an object-oriented to an interaction-oriented approach. When considered broadly, a network science perspective can be applied to virtually any substantive area in which two or more actors or objects interact, from biological brain networks to global transport routes, making it extremely powerful. The most pressing challenges of our time, including technological, demographic, and environmental changes, are in complex systems where underlying interconnections and interdependencies are the key to scientific understanding and real-world solutions.

1.3 THE BUILDING BLOCKS OF NETWORKS

To provide a foundation for the remainder of the book, we briefly present some of the more frequently referenced terms here. This is neither an exhaustive nor a technical lexicon of network terminology; rather, the goal is to provide an orientation to network language and its basic variants (also see Monge & Contractor 2003). As required in the chapters that follow, these terms will be reintroduced, expanded, or illustrated. Others will be newly introduced in more complex or sophisticated nominal and operational definitions.

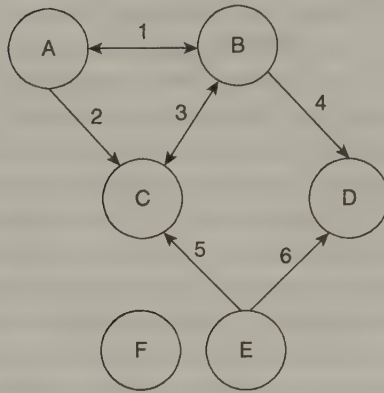


FIGURE 1.1 Representation of network ties in a sociogram

- **Network diagram:** A network diagram, also called a sociogram, is a picture of the relationships among members in a social network. Figure 1.1 is a network diagram.
- **Node:** The term “node” (alternatively “actor” or “agent”) refers to the units that are connected through networks. In the social sciences, nodes typically refer to individuals, but may be families (Padgett & Ansell 1993), organizations (Galaskiewicz 1985), nations (Alderson & Beckfield 2004), or any other entity that can form or maintain formal (e.g., legal, economic) or informal (e.g., friendship, gossip) ties. In Figure 1.1, the circles labeled A through F represent “nodes.”
- **Ego and alter:** We use “ego” throughout the book to refer to any node that is the current focus of attention. In ego network research, ego is the respondent. We use the term “alter” to refer to any actor with whom ego has a tie.
- **Ego network:** An ego network (also called a personal network or egocentric network) consists of the alters connected to ego, along with the ties between ego and alters, and ties among alters. When we refer to a person’s network in this book, we mean their ego network.
- **Ties:** The network connections between and among actors are referred to as “ties,” “links,” or “edges.” Ties can be directed (i.e., sent or received) or not directed (e.g., joint organizational memberships). In Figure 1.1, a tie is sent from B to D; D receives a tie from E; and A and B send and receive ties to each other. Double-headed arrows indicate “bidirectional,” “symmetrical,” or “reciprocal” ties. Ties may be binary (present/absent) or

	A	B	C	D	E	F
A	0	1	0	0	1	1
B	1	0	1	0	0	1
C	0	1	0	1	1	0
D	0	0	1	0	0	1
E	1	0	1	0	0	0
F	1	1	0	1	0	0

FIGURE 1.2 Matrix representation of a network

measured on a scale, as in duration or strength of tie. In Figure 1.1, ties are depicted using lines 1 through 6.

- **Adjacency matrix:** Network ties are often recorded as a square actor-by-actor matrix. A value in a given cell of the matrix indicates that there is a tie from the row actor to column actor (ties received; see Figure 1.2).

1.4 PRINCIPLES OF A NETWORK PERSPECTIVE

Our shared position is that network science is a perspective or framework that can be tailored to generate theories applicable to specific phenomena. The following basic principles establish key considerations in applying the network perspective to a particular substantive problem (adapted and revised from Pescosolido 2006).¹

Proposition 1: Connections are a key mechanism of social action. Social actors, whether individuals, organizations, or nations, shape and are shaped by consultation, formal agreements, information and resource sharing, support, and regulation (White, Boorman, & Brieger 1976). Individuals are neither puppets of the social structure nor purely rational, calculating individuals. Rather, they are “sociosyncratic,” both shaping and reacting to the networks in their environment (Pescosolido 1992). They are always seen as interdependent rather than independent (Wasserman & Faust 1994).

That said, differences in theorizing, while sharing this point, reflect varying degrees of structure and agency. For example, while Coleman (1986) saw all network interactions as “purposive” (under a more

¹ While these propositions build a foundation, many issues remain to be addressed in carrying them forward in the research process.

individual rational choice frame), others see “habitus” (under a cultural habit frame which sets predispositions) as playing a major role in day-to-day interactions (Bourdieu 1990). Perhaps most interesting, when considered simultaneously, these different theoretical perspectives give rise to a question: When might an actor change from habitus-driven interaction to purposive-driven interaction? For example, family history and tradition may cause a voter to generally pull a single party lever in the US presidential elections time after time. What factors would cause a shift toward a cost-benefit decision-making process? One answer postulated by network scientists would focus on change in the economic, social, and political networks that surround the individual (Beck et al. 2002).

Proposition 2: Human networks are fundamental building blocks of non-human entities such as institutions, cultures, communities, and social systems. Tilly (1984) argued that institutions and organizations represent the crystallization of network interactions that are so fundamental or routine that they shift from informal to formal structures. That is, networks set the culture within groups, formal organizations, and institutions for those who work in or are served by them. These network-based cultures, in turn, affect what people do, how they feel, and what happens to them (Pescosolido 1992). Put differently, networks can be thought of as a bridge between the macro and micro levels of society (Coleman 1990; Lin & Peek 1999). An individual’s dyadic relationships provide a basis for network structures and cultures, upon which communities are built and sustained.

For example, as Glisson and Hemmelgarn (1998) demonstrated for the operation of child mental health programs, it was not the number or training of the team therapists that mattered for whether children with mental health issues improved. It was whether the network of providers created a cultural climate that allowed for flexibility, consultation, and maximizing the skills of each member. Further, as Gould (1991) showed, the preexisting network ties among soldiers, not their bravery, training, or age, determined whether they stayed at their posts or deserted them during the insurgency in the Paris Commune of 1871. Units composed of soldiers from the same neighborhoods stood their ground because of their informal bonds to each other. Other units, who were a mix of previously unconnected individuals, scattered in the face of certain defeat.

Proposition 3: Four dimensions of social networks are distinct – structure, function, strength, and content. We find it useful to think about networks

in terms of these four dimensions. Network *structure* reflects the architectural aspect of networks, including the presence and patterns of linkages between actors. This is the most typical focus of network research, and common concerns are the overall size of the network, and also how tightly knit it is (e.g., density or how well alters know one another).

However, the structure of the network reflects only one aspect of what is important about networks. Network researchers are also concerned with *function* – or the types of exchanges, services, or supports accessible through ties to alters. That is, there are specific functions that ego networks serve, including providing emotional support (e.g., care, concern), instrumental aid (e.g., lending money, providing transportation, baby-sitting), appraisal (e.g., evaluating a problem or solution), and monitoring (e.g., making sure a person with diabetes watches his or her diet and takes insulin shots) (Pearlin & Aneshensel 1986). As Umberson (1987) found for married men, it is not only the support that comes with marriage that is essential to men's health, but also the kind of regulatory behavior that wives exert in influencing their husbands' health habits. Although providing emotional support and caring are important, so too are nagging and monitoring.

In network research, *strength* is often conflated with function because they are closely related. After all, your strongest ties are often the people who are most willing to do things for you. Network strength captures the intensity and duration of bonds between an ego and alters within the network (Marsden & Campbell 1984). Others define tie strength using the concept of durability, arguing that strong ties are characterized by commitment to maintaining a relationship (Shi et al. 2009). An individual's strongest ties are also often those with whom contact is most regular, though distant ties can sometimes be affectively close, such as old college friends.

Finally, network *content* matters because it taps into the substance of social networks. Attitudes, opinions, and beliefs, as well as more tangible experiences and collective memory, are held within networks (Emirbayer & Goodwin 1994). Network content also provides information about actors' active or potential access to economic or cultural capital and other resources. Network content is tied to function and strength, since these properties of networks often determine whether network content like knowledge, money, and skills is actually leveraged as support provision or toward instrumental ends. Likewise, if we conceptualize the structure of the network as a set of connections between people, then content represents the types of things that flow from or to

each person. In recent social media studies, for example, examining the emotional valence of tweets led to interesting and important understandings of how political information flows across Twitter networks (Conover et al. 2011).

Proposition 4: Network effects are a function of interactions among these four dimensions. To use a biological metaphor, structural aspects of networks make up the skeleton of social relationships. In contrast, the more substantive properties of networks (i.e., the function, strength, and content) are the muscle, blood, and skin of social networks. The four dimensions interact, often in complex ways, to influence individual outcomes and behaviors. For instance, structural elements (e.g., size) and network strength (e.g., intimacy) may tap the amount of potential influence that can be exerted by the network (i.e., the “push”). However, the network’s content (e.g., attitudes) and function (e.g., regulation) may indicate the direction of that influence (i.e., the “trajectory”).

For example, individuals with mental health problems in Puerto Rico are less likely to access the formal health care system if they have a large social network on whom they can depend for advice and care in the community – an interaction between structural and functional components of networks (Pescosolido, Brooks-Gardner, & Lubell 1998). Among Puerto Ricans on the island, problems are considered to reside squarely in the family; it is this group that holds the responsibility for care of its members. Medical or mental health care is a last resort. This contrasts sharply with the social networks of other cultural groups on the US mainland, where Charles Kadushin (1966) documented the opposite. On the Upper West Side of Manhattan, larger weak and informal social networks, which he called the “Friends and Supporters of Psychotherapy,” were likely to encourage the regular and routine use of the formal mental health system for emotional problems. More broadly, in early network research in medical sociology, considering structure alone led to contradictory findings on the importance of networks in health behavior and a near dismissal of the approach (McKinlay 1972).

Proposition 5: Network ties are not always beneficial. The early linkage of networks with social support and social capital resulted in a bias toward researching the positive functions of sociality. Studies have found that having networks can, indeed, reduce stress, lead to major employment advantages, transfer key information for migrant group

survival, and even shape identity. However, social interactions can be positive or negative, helpful or harmful. They can integrate individuals into a community and, just as powerfully, shun them. Early on, the sociologist Edwin Sutherland, in his theory of “differential association,” saw interaction in deviant groups as the key to delinquency and crime. While it was later interpreted more as a learning theory of deviance, newer research has revived the role of network structure, content, and function in the criminal “career.” In particular, network density and cohesion appear to be crucial to the influence of peer ties translating into delinquency during adolescence (Haynie 2001). Certainly, theories of disease epidemics target the negative implications of contact (Colizza et al. 2006), and the search for the “dark networks” of covert or illegal organizations have also clarified the multivalent nature of network ties (Bright, Hughes, & Chalmers 2012).

Proposition 6: “More” is not necessarily better with regard to social ties. Following in part from the early studies which motivated the idea that networks are positive, there was a concomitant notion that it was only the absence of ties that was problematic. But as Durkheim (1951 [1897]) pointed out, too many network ties can be stifling and repressive. Similarly, strong ties are not necessarily optimal. They can provide the emotional support that individuals need; however, they can be limiting. As Granovetter’s classic work on the employment search demonstrated, “weak” ties often act as a bridge to different resources and information (Granovetter 1983). Further, the absence of ties around a person can represent an opportunity that can be exploited (Burt 1980).

Proposition 7: Networks across all levels are dynamic, not static, structures and processes. The ability to form and maintain social ties may be just as important as their state at one point in time. For personal networks, research suggests that membership turnover rates hover between 25% and 50%, while structural elements such as size tend to remain stable (Sutor & Keeton 1997; Wellman & Berkowitz 1997; Perry & Pescosolido 2012). As Moody, McFarland, and Bender-deMoll (2005: 1209) note, “... an apparently static network pattern often *emerges* through a set of temporal interactions.” Importantly, failing to identify the substantive nature of this “network churn” and relying only on aggregate measures of networks as a whole produces a distorted sense of stability (Sasovova et al. 2010). The underlying reasons for change mark important dynamics in the operation of social networks. For example, in the early

stages of a crisis, such as illness, it appears that support ties are activated and networks appear to swell; but, as the crisis continues, only network members that have a specific function and a close connection tend to be retained (Perry & Pescosolido 2012). In fact, Carrington and colleagues (2005) refer to the analysis of social networks over time as the “Holy Grail” of network research.

Proposition 8: A network perspective allows for, and even calls for, multi-method approaches. Any notion that there is only one way to approach understanding the nature, functioning, and effects of network ties is outdated and inefficient. There is no doubt that mathematical and quantitative research powerfully describes the structure of networks and documents whether their effects are significant or not, in a statistical sense. However, only by tapping into qualitative research can we describe the “on the ground” mechanisms of network process and functioning. While it is important to define classic types of network research, such as ego-centric or sociocentric as we do in Chapter 2, they are, in essence, ideal types on a methods spectrum. It is sometimes difficult, if not impossible, to cleanly classify a study as one or the other. And, as network science becomes more sophisticated over time, mixed designs are likely to be more commonplace.

Further, there is no standard way to collect network relationships, as both qualitative and quantitative approaches are relevant. Networks may be derived from a list on a survey where individuals are asked to name people they trust, admire, dislike, or with whom they share information. Alternatively, the information may be generated from observations of individuals’ behavior (e.g., who they talk to in their work group), “scraped” from the Internet, or recorded from archival sources. For example, in Bearman and Stovel’s (2000) research on how individuals were recruited into the Nazi party in the 1930s, autobiographical texts were used to identify the nature of network ties that translated into political action. In fact, as network research proceeds, the ability to draw in different kinds of network data, from different sources, will likely become an increasingly better match to the complexity of most systems and phenomena (Pescosolido et al. 2017).

1.5 MAJOR THEORETICAL TRADITIONS

There are a number of theoretical traditions in the social sciences that share a concern with connectedness, all of which have roots in Emile

Durkheim's classic body of work. Over a century ago, Durkheim (1951 [1897]) attributed patterns of suicide in Europe to levels of social integration and regulation. Social integration – or being bound to social groups – fosters feelings of belonging that provide a sense of purpose and identity. Likewise, social regulation – the ability of a group to guide and constrain its members – develops out of a sense of obligation to one's group and responsibility to conform to group norms. Durkheim theorized that high rates of suicide would be associated with excessively high and low levels of social integration and regulation, reasoning that moderate levels of embeddedness and obligation would provide a stabilizing degree of support and guidance without being inflexible and overly demanding.

Durkheim's ideas gave rise to a number of complementary perspectives that share a focus on social interaction and connectedness, including the social support, social capital, and social influence traditions introduced below. This discussion is not meant to be comprehensive, but it is important to have a basic understanding of the similarities and differences between these traditions, even though the boundaries may be fuzzy. Though they are not strictly distinct, they nonetheless represent different strands of research. They continue to use unique terminology and only sporadically cross-fertilize. Moreover, though these approaches have informed network theorizing and analysis, researchers drawing on them do not always employ network methods. Rather, these broad theoretical traditions are often used to develop more specific and testable network hypotheses. Thus, a decoupling of theory and method is important, as these are not necessarily network theories. Perhaps a better way to classify them, with respect to their role in network research, is as a set of mechanisms through which social networks are purported to influence individual and group outcomes (e.g., Berkman et al. 2000).

1.5.1 The Social Support Tradition

The social support tradition is among the most prominent and the most prolific in the social sciences. While some adherents claim a Durkheimian heritage (House, Umberson, & Landis 1988), others do not, citing stress research and Cassel's (1976) and Cobb's (1976) classic papers. As Thoits (1995) notes, social support is a psychosocial resource and has been documented to be a powerful influence, for example, in occurrence of and recovery from life problems. Emanating from a concern with actors' responses to stressful situations, this social reserve may either prevent or

buffer adverse events that occur in people's lives (Pearlin & Aneshensel 1986). Researchers often conceptualize four types of social support:

- *Instrumental support*: Ties that offer practical resources or assistance are said to deliver instrumental support.
- *Emotional support*: Ties that provide love, caring, and nurturing offer emotional support (Thoits 1995).
- *Appraisal*: This form of support targets network assistance in evaluating a problem or a source of aid (Pearlin & Aneshensel 1986).
- *Monitoring*: When network ties watch, discipline, or regulate the behavior of other social actors, the monitoring function is fulfilled (Pearlin & Aneshensel 1986).

While network scientists tend to think of social support as a "function" or even a type of content that may or may not be part of any network tie (Wellman 1981; Faber & Wasserman 2001), others think of the number of ties as a component or characteristic of social support (House, Landis, & Umberson 1988). While this may sound like a distinction in search of a difference, it is not. The starting point of research from different traditions shaped very unique approaches and meanings of data collection. Social support research came out of a psychological and social psychological framework, while network research derived from a structural framework, though the latter often borrows terminology and theoretical insights from the former.

Researchers in the social support tradition tend to ask study respondents whether they have enough support in coping with everyday life issues or critical events. Questions may target either perceived social support (i.e., the belief that love, caring, and assistance are potentially available from others; latent networks in the structural tradition) or received support (i.e., the actual use of others for caring, assistance, appraisal; Thoits 1995). Some research has documented that perceived support is more important than actual support received (House 1981; Turner & Marino 1994). Even more surprising, Cohen and Wills (1985) suggest that the simplest and most potent indicator is whether individuals report that they have a single intimate tie in which they can confide. In contrast, in the social networks tradition, the distinction between perceived or available support and actual exchanges or functions is a topic worthy of study in its own right, independent of any consequences for well-being. The identification of dyadic and network structural factors influencing a tie's transition from latency (i.e., available support) to activation (i.e., provided support) has been a major contribution of sociologist Barry Wellman and his students

(e.g., Wellman & Wortley 1989, 1990; Wellman & Gulia 1999a; Wellman & Frank 2001).

As Smith and Christakis (2008: 407) point out, social support studies “assess the quality and quantity of a person’s social ties.” In this approach, characteristics of those ties are conceptualized primarily as attributes of or resources available to the individual. While the structure of ties between individuals is not ignored altogether (Barrera 1986; Berkman 2009), the focus in this approach is on the sustaining qualities of social relationships (Haines, Beggs, & Hurlbert 2002). Researchers studying ego networks leverage insights from the social support tradition more frequently than those using a sociocentric approach. The support and ego network approaches share a similar view of “function” as either a name generator (in network tradition) or stem question (in the social support approach) or as something that can be asked of/about everyone in the social support group. Still, for researchers studying support from a network perspective, it is but one of a constellation of important factors, and the outcomes of support are often conceptualized as conditional on structural properties of networks.

For network researchers studying social support, the motivating issue is to identify how patterns of relationships in personal networks influence the flow of resources to an individual. For example, drawing on a network study of East Yorkers in Toronto, Canada, Wellman and colleagues (Wellman & Wortley 1989, 1990) explored how the size, composition, and structure of egocentric social networks affected the quality and quantity of support provided. They found that larger and more heterogeneous networks contained a higher percentage of supportive ties and provided access to a variety of different types of supporters (e.g., those that offer everyday assistance, emergency aid, companionship) relative to smaller, homogeneous ones. Also, networks that were more available, with a higher frequency of contact, contained a larger percentage of supporters of all kinds, suggesting that regular contact fosters supportiveness or provides an opportunity structure for support provision. Finally, findings from the East York study indicated that densely knit personal networks provided more emotional and material support than sparsely connected ones, reflecting the strong culture of obligation and reciprocity present in networks where most people know one another. These findings led to the conclusion that “a network is more than the sum of its ties” (Wellman & Frank 2001: 257), emphasizing the structural basis of support provision at the dyadic level.

1.5.2 The Social Capital Tradition

The concept of social capital was largely developed in the 1980s, and is one of the most “successful ‘exports’” from sociology to other social and health sciences (Portes 2000: 1). Portes argues that there are essentially two parallel social capital literatures operating at different levels of analysis: individuals and communities. Initially, work on the concept by Bourdieu (1990) and Coleman (1990) conceptualized social capital as resources that accrue to individuals or small groups as a result of network ties (Monge & Contractor 2003). Because individuals participate in social groups, they invest in and use the resources embedded in social networks. In contrast, others later defined social capital as an attribute of communities, characterizing the level of collective trust and civic participation at the level of neighborhoods, cities, or even nations (Putnam 1995) – forms of social capital that are typically less useful for researchers taking a social network perspective.

Much of the research that followed these theoretical developments has focused on identifying the kinds of resources that can be leveraged from social ties or networks, under which conditions, and to what ends. Following Bourdieu’s (1985) original conception of social capital as something to be cultivated and exchanged, substantial evidence has accumulated on access to job leads, novel information, or financial loans through personal or business connections (Portes 2000). The most well-known examples are Granovetter’s (1973) classic work on the role of weak ties in procuring valuable employment leads and Lin’s (1999a) research on social resources and socioeconomic status attainment. Moreover, social capital theorists in this tradition often talk about the costs and benefits of establishing ties, as well as how and why actors construct or maintain ties in the service of creating opportunities and resources (Baker 1990; Flap & Völker 2001). A related literature has centered on family ties and the forms of social (and cultural) capital that parents transmit to their children through socialization processes, including norms, values, and unconscious dispositions (Portes 1998). These can, in turn, be used in different contexts to gain access to material and social resources, including power and influence. Importantly, social capital in this vein is unevenly distributed across individuals and groups in society, and therefore is seen as a mechanism through which inequality is reproduced across generations (Bourdieu 1987; Lin 2000).

Another variation of social capital research follows Coleman’s developments, still focusing on individuals, relationships, and small groups.

Coleman (1988) was primarily concerned with *bonding social capital*, which takes the form of trust, social norms of reciprocity, cooperation, and participation that reside in relationships and small groups. This perspective therefore shares roots with many aspects of classical sociology and Durkheimian theory, drawing attention to the regulating and integrating functions of social structure (Portes 1998; Paxton 2002). Lin (2002), Putnam (1995), and others later distinguished Coleman's bonding social capital from *bridging social capital*. The former is typically cultivated in the context of strong relationships with similar others, while the latter refers to capital that accrues as a result of ties to dissimilar others through access to novel ideas, information, and perspectives.

The concept of bridging capital is closely linked to Burt's (2004, 2009) theory of structural holes. Burt reasoned that because individuals embedded in the same social groups are likely to be homogeneous, people who bridge two otherwise unconnected networks (thereby spanning a structural hole, or the absence of a tie) have a social advantage. Namely, Burt conceptualized access to unconnected and dissimilar networks as a form of social capital that is associated with exposure to novel information and ideas. In empirical research, structural holes have been linked to a variety of outcomes, including innovation (Ahuja 2000), performance (Zaheer & Bell 2005), flexibility and adaptation (Gargiulo & Benassi 2000), and good health (Cornwell 2009).

1.5.3 The Social Influence Tradition

Social influence, as a process or mechanism, has long been a mainstay in social network research. Marsden and Friedkin (1993: 127) define the term as "the social relations that provide a basis for the alteration of an attitude or behavior by one network actor in response to another." Thus, the social influence tradition is really a collection of theories that explain why people's attitudes, emotions, and behaviors tend to converge over time. Alternatively, social influence can be conceptualized as an umbrella term for a group of related but distinct social processes, including obedience, conformity, persuasion, peer pressure, and socialization – all of which have large scientific literatures associated with them. As a mechanism, it is of particular interest to network researchers because it provides a link between the structure and content of networks and the individual attributes of actors that are members of those networks.

In the network tradition, social influence is generally thought to be a function of two properties interacting over time, roughly corresponding

conceptually to the direction and strength of the “push” toward a particular outcome. First, the direction of influence depends on network content, or the attitudes, emotions, or behaviors flowing between network members. A researcher might be interested in whether a person’s network contains cigarette smokers, for instance – a factor that is among the strongest and most robust predictors of whether a nonsmoking adolescent will become a smoker (Simons-Morton & Farhat 2010). If two actors in a network are dissimilar with regard to the characteristic being studied (e.g., one is a smoker and the other is a nonsmoker, or one is a passionate political conservative and the other has no strong political leanings), there is potential for social influence over time. In the case of smoking, this influence might operate through active peer pressure, passive behavior modeling (i.e., learning by watching what others do), or because the nonsmoker adopts the peer group’s shared norms or values around smoking (e.g., the status benefits of smoking now outweigh the long-term risks).

Second, network researchers have been distinctively interested in social proximity as an indicator of potential for social influence, or the strength of the push toward similarity (Marsden & Friedkin 1993; Marsden & Friedkin 1994). For Burt (1987a) and others, social proximity has a structural basis, where greater proximity is reflected in the propensity for social interaction between two actors in a network. In addition to being useful for explaining direct social influence, such as the amount of time two people spend together, this conceptualization can account for indirect social influence processes, such as the diffusion of an innovation through a large network. In the case of indirect influence, proximity can be operationalized using the “number, length, and strength” of paths between actors, who may or may not be directly connected (Marsden & Friedkin 1993: 131). In the network tradition, therefore, social influence does not require purposive or conscious efforts to modify someone else’s behavior, nor is direct contact between two actors necessary for influence to occur (Berkman & Glass 2000).

Historically, another critical element of social influence research is social comparison – a process that has long been important in psychology and sociology (Erickson 1988). Particularly in cases where a person is experiencing uncertainty, they “obtain normative guidance by comparing their attitudes with those of a reference group of similar others. Attitudes are confirmed and reinforced when they are shared with the comparison group but altered when they are discrepant” (Marsden & Friedkin 1993: 129). In general, social influence is more likely to occur when the potential

source of influence is more interpersonally visible and socially relevant, and therefore a person who is a meaningful comparison point (Simmel 1950; Merton 1968[1949]). For example, to reduce conflict and bullying among adolescents, Paluck, Shepherd, and Aronow (2016) randomly selected small seed groups of students in a subset of schools to receive an anticonflict intervention, which they hoped would be disseminated to the student population at large. They found that the treatment effect was strongest when the seed group contained a “student referent,” defined as a person identified by a disproportionate number of peers as someone they spent time with. Policymakers and practitioners have become increasingly interested in practical applications of basic network science, especially in public health, developing resource-efficient interventions that rely on the social influence of “key players” (Borgatti 2006; Valente 2012).

In short, social influence is a complex and dynamic process involving multiple moving parts at the network and individual actor levels. As a result, a number of social network models and algorithms have been developed specifically for the analysis of social influence (Robins, Pattison, & Elliott 2001; Sun & Tang 2011). At the same time, theoretical and empirical advances in our understanding of social influence are not unique to network science. They have also emerged from other perspectives and methodologies, including experimental designs in social psychology, survey research, and qualitative, in-depth interviews (e.g., Huckfeldt & Sprague 1995).

1.6 CONCLUSION

This chapter begins and ends with the same point. There is no single network theory. However, there are basic concepts, a fundamental set of principles that guide theorizing, and analytic tools for egocentric and sociocentric network analysis. The process of doing network research begins with an understanding of a general framework that embraces dynamics, nonlinearity, and multi-level issues. However, the science of developing and empirically testing network hypotheses requires a deeper knowledge of network theory that is beyond the scope of this book, as well as a solid understanding of existing research on the social phenomena being studied. Here, our goal has been to lay the groundwork for subsequent chapters on ego network data collection and analysis by providing a basic introduction to the conceptual and theoretical foundations of this approach.

Sociocentric and Egocentric Approaches to Networks

Perhaps the most basic distinction in network research in the social sciences is between *sociocentric* and *egocentric* research designs. Sociocentric research starts with a set of actors and looks at the ties among all pairs of actors. When the data are collected by survey, every actor in a discrete universe or population is a respondent. The result is a single “complete” or “whole” network (or one for each kind of tie). In egocentric network research, a set of actors is sampled from a population. They then provide data about members of their network. The result is a set of separate networks, one for each ego.

The purpose of this chapter is to begin to map the landscape of network research, describing and differentiating sociocentric and ego network designs. We outline the basic goals and practical parameters of sociocentric and ego network designs, highlighting advantages and limitations of each. This introduction provides a foundation for the set of research design decisions addressed in subsequent chapters.

2.1 SOCIOCENTRIC NETWORK RESEARCH

While networks are typically defined simply as a set of nodes together with a set of ties that connect them (Wasserman & Faust 1994), what makes them interesting from a sociocentric point of view is that the ties interlink through shared nodes to form paths and, taken together, a structure. A fundamental tenet of sociocentric research is that structure matters. Just as the behavior of a molecule is determined not only by its constituent atoms, but by how they are connected to each other, so does the structure of a network affect its influence. An equally fundamental tenet is that within a structure, each node occupies a position, and this position also

has consequences. For example, a common view is that a person's position in a social network determines, in part, both the constraints and opportunities that the person will encounter (Brass 1984).

To capture the structure of a network – and the position of each node within it – we use a sociometric research design. In brief, this is done by (a) determining if the boundary of a population can be logistically demarcated, and selecting a complete set of nodes in that population, and (b) collecting the ties from each node to every other node in the population. Commonly, sociocentric data are collected by survey, in which case every node is both a respondent and is responded about by others in the network. One of the earliest published network studies was an observational study of interactions among a set of workers at the Hawthorne Western Electric plant in Chicago during the 1920s (Roethlisberger & Dickson 1939). The researchers took as their set of nodes all of the men working in what was called the Bank Wiring room. They planted an observer at the back of the room for several months' time, who recorded a variety of different kinds of interactions among the workers, including playing games with each other, having conflicts, and helping each other with their jobs. In addition, the observer recorded subjective impressions of affective relations, such as who liked whom. Figure 2.1 shows the

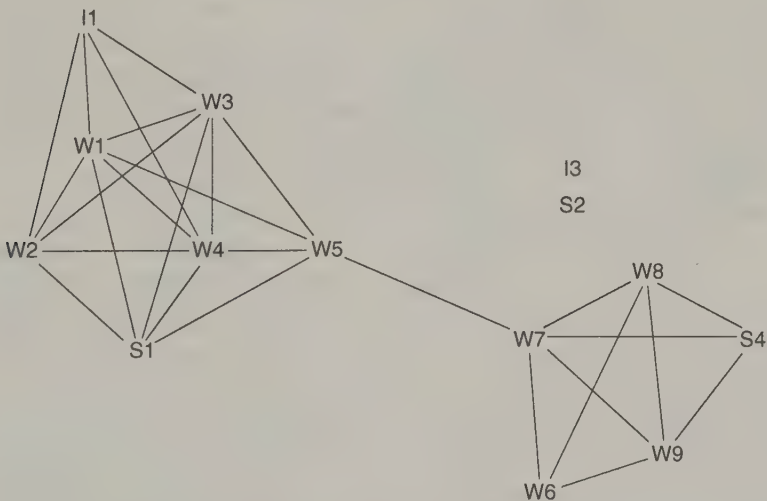


FIGURE 2.1 Example of sociocentric design: The bank wiring room games network
Reference: Kilduff and Tsai (2003)

games network, where actors are labeled “W” for wireman, “S” for solderman, and “I” for inspector. The structure of this sociometric network reveals a clear bi-partite structure of game playing. Only one pair of workers (W₅ and W₇) bridge this group structure, playing games with one another. Further, there are two individuals in the bank wiring room (I₃ and S₂) who do not play games at all.

An illustration of the advantages of a sociocentric approach comes from sociological research on sexually transmitted infections (STIs). Using whole network data on students in selected schools from the Add Health Study, Bearman, Moody, and Stovel (2004) examined the structure of adolescent sexual networks in “Jefferson High.” Using computer administration designed to increase privacy and decrease interviewer effects, the study provided students with a roster of all other students in the school. They were asked to indicate their romantic relationships. Results indicated that the partnering contact structure for adolescents resembled a “spanning tree,” which is a spindly structure in which there is only one path between any pair of points (see Figure 2.2). In other words, circular paths in which A went out with B who went out with C who went out with D, who also went out with A, were rare, possibly

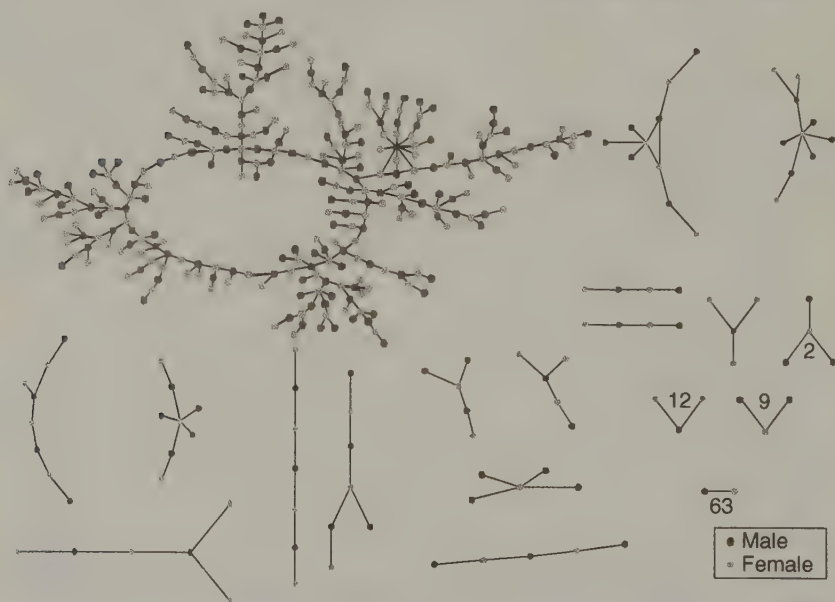


FIGURE 2.2 A sociogram depicting the structure of adolescent romantic or sexual networks from a sociometric study of sexually transmitted infections

Reference: Bearman, Moody, and Stovel (2004)

reflecting social norms about who is an appropriate partner (i.e., where “exes” of “exes” are off limits).

In contrast to densely knit core groups, spanning tree-type networks are easily broken by the removal of a node or tie. This has important implications for STI prevention interventions. Specifically, in the absence of a structural core, so-called “broadcast” interventions, which aim to reach all members of a population, will be most effective since a long chain can be broken at any point. Bearman and colleagues’ ability to view the structure of the entire romantic network led to important and novel insights. Namely, that the typical public health approach – to identify individuals with the greatest number of romantic partners and intervene there – is the wrong one.¹

As this example demonstrates, the obvious advantage of having sociocentric data is that the whole network structure (and each individual’s location in it) can be enumerated, described, and analyzed. A key consideration in sociometric research, then, is how to choose the set of nodes to study. In all cases, this is ultimately dictated by both the research question and what is feasible to obtain. In some cases, we are studying natural groups, defined emically (i.e., from the point of view of the participants themselves). For example, we may be studying neighborhood gangs. In these cases, setting the boundary of who belongs in the study may be conceptually easy, though typically not so in practice. What makes this task difficult is that group boundaries tend to be fuzzy and contested. There are core members, which are easy, but also peripheral members, including wannabees, novices, and people on the way out. In other cases, we are studying more arbitrary groups, defined etically (i.e., from the point of view of the researcher). For example, we might want to study the “music community” in a small town. Here, the boundaries are difficult to set even conceptually.

Why does this matter? Because the structure of the network – and the positions of individuals within it – changes with the boundaries. A given person may look very peripheral if only certain people are chosen for the study, but much more central if a wider net is cast. In general, a wider net provides more flexibility (because in the analysis we can choose subsets of nodes as needed), but makes it costlier to collect the data (see Chapter 3 on sampling). The hallmark of sociocentric survey data collection is the use of a roster – a list of people that the respondent is asked about. Respondents

¹ Under protocol design, given the age of the students, only “romantic” and not “sexual” relationships were collected with implications for STI interventions.

may be asked to identify people with whom they have particular kinds of relationships, for example, or how often and in what ways they interact. The key is that every person in the network is asked about every other person in the network (i.e., essentially a census). The longer the roster, the more unwieldy it becomes, necessitating strategies like grouping names by department or location. At some point, no strategy will help. The network is too big to allow for a sociocentric approach unless the data already exist naturally (e.g., the Twittersverse or email records).

In principle, from a purely methodological standpoint, every analysis that can be done on egocentric data can be done with whole network data, since each ego network is embedded in the full structure of the network.² However, collecting sociocentric data via survey (i.e., census) can only be accomplished when both the respondents and alters are identifiable to one another and to the researcher. In other words, alters and links between them cannot be anonymous, which may encourage social desirability bias and cause problems with institutional IRBs (i.e., review boards for protection of human subjects). Sociocentric data are also difficult to obtain; and in primary data collection, the burden on respondents is greater than the egocentric approach. Both the lack of anonymity and the high burden of sociocentric research can result in refusal to participate or survey noncompletion, reducing response rates, and the amount and quality of usable data. This is an important problem because structural analyses can be very sensitive to missing data (Borgatti, Carley, & Krackhardt 2006). Consequently, researchers conducting a sociocentric study tend to expend quite a bit of energy attempting to compel every person on the roster to participate.

Sociocentric research may also be limited by boundary specification problems. When studying the network structure of emergent groups, such as gangs, high-school cliques, terrorist groups, or business firms, it is important to get the boundaries right. However, boundaries are rarely clear and are typically contested. There are wannabees, newbies, retirees, apprentices, part-timers, temps, contractors, hangers-on, etc., that make it hard to draw a distinct line. The problem is easier when the sample is defined by the research question, as when studying 100 classrooms to test the hypothesis that the structure of a classroom friendship network

² In practice, there are a few caveats. First, the dependencies between the nodes means that we cannot use the same statistical techniques on ego networks drawn from whole networks as on egocentric ego networks (see Chapter 10). Second, because it is easier to collect, egocentric data tends to be much richer in terms of the number of different kinds of relations collected and the characteristics of ties between egos and alters.

affects the kinds of questions students are willing to ask. Even then, once a boundary is chosen, the data collection is effectively a census – we cannot sample randomly within our frame. As a result, the whole network design tends to limit the size of the networks we can study, at least if the data are collected by survey or other primary data collection method (e.g., observation). If secondary data are available, it is possible to analyze much bigger networks, such as all agencies in the US federal government. However, even in these cases, the research questions that can be addressed are constrained by the nature of information that exists. Moreover, this still leaves a fundamental problem: external validity.

The external validity issue is perhaps the most significant limitation of the sociocentric research design. If, in the above study of the US federal government, we find that more connected or centrally positioned agencies have greater influence, to what other network can this result be generalized? We cannot assume the results will hold for other organizations or even other governments. A typical sociocentric study consists of a single set of nodes – for example, the employees of an organization or the users of a particular gym. Suppose we conduct the analysis at the node level, as is the norm. For example, if an employee's centrality in the organization affects the probability of being promoted in the next time period, can we say that, among employees in general, centrality is associated with promotion? The difficulty is that we did not sample employees from the larger population of employees across multiple organizations. We only studied one organizational set. Since it is very plausible that other organizations might have different dynamics, it is imprudent to generalize. The accepted solution is to replicate the study in a variety of other organizations, and if the results are consistent, cautiously generalize to employees as a whole.³ Nonetheless, the ability to employ random sampling is a major benefit of an egocentric design.

2.2 EGOCENTRIC NETWORK RESEARCH

Egocentric research is focused on individuals and their immediate social environment. A fundamental tenet is that each person lives in a personal community – partially of their own creation and nearly unique to them – whose composition and structure have consequences. One goal of

³ Of course, in applied work, the generalizability issue is not a problem. If we are consultants trying to fix a dysfunctional organization, it is enough to study just the one organization.

egocentric research, then, is to predict ego outcomes from variables that describe how ego is connected to alters, characteristics of the alters, and characteristics of the pattern in which the alters are connected to each other. Underlying this goal is the theoretical proposition that individuals exist in a social context, and this context affects them. Another goal is to understand why egos have the ego networks that they do. For example, why do some people have personal networks in which many of the alters know each other, while others have very open networks in which the alters do not know each other. Underlying this goal is the theoretical proposition that historical time, geographical space, and social place shape the opportunities for and constraints on the nature of social interactions (Pescosolido & Rubin 2000).

In ego network research, we typically want to gather information about many nonoverlapping networks from the standpoint of the social actors situated within them (Marsden 2005). The data collected in ego network research also include detailed information about each person in ego's world, or each alter. Figure 2.3 depicts two ego networks. In these network diagrams, ego respondents are depicted as white circles, and alters (the people in their networks) are shown as black or gray circles, signifying kin and friends respectively. By definition, some kind of tie is present between ego and every alter in the network. Ties between the alters may or may not exist, or, if present, may not have been collected. Typically, an

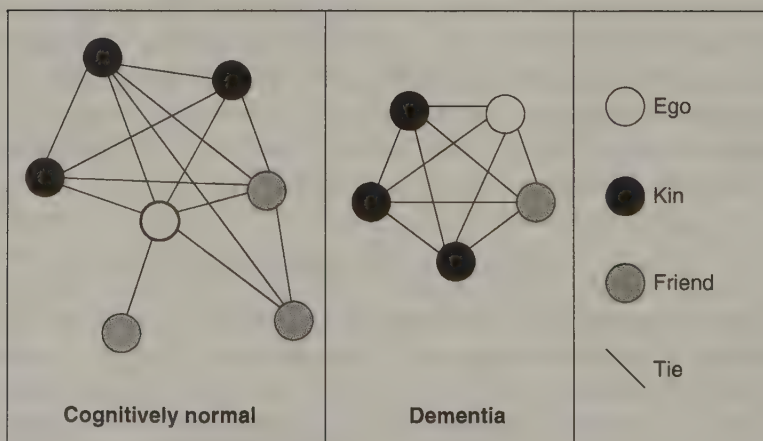


FIGURE 2.3 Egocentric social networks of patients with and without dementia
Reference: Perry et al. (2017)

interview instrument called the *name generator* is used to identify the alters in an ego's network (see Chapter 4), and another interview instrument, called the *name interpreter*, is used to gather information about characteristics of alters and relationships (see Chapter 5). This information can be used to construct measures of the structure, function, content, and strength of the network, and then employed to test network theories using regression (see Chapter 8).

The two network diagrams in Figure 2.3 typify the personal networks of older adults with dementia and those who are cognitively normal, and are based on a study of social networks and age-related cognitive decline (Social Networks and Alzheimer's Disease, or SNAD; Perry et al. 2017). Comparing these prototypical networks, one can see that respondents with normal cognitive status have egocentric networks that are larger, less densely connected, and have a higher proportion of friends, on average, than those with a diagnosis of dementia. These patterns imply two noncompeting mechanisms. First, having networks that are more homogeneous, insular, and involuntary may increase risk for cognitive decline and dementia (i.e., social causation). Second, as individuals become more cognitively impaired, their social networks may contract, with only the strongest ties remaining active (i.e., social selection). This example illustrates the kinds of cause and effect relationships that can be studied using egocentric research designs.

The SNAD study also highlights some of the advantages of egocentric data collection relative to a sociocentric approach. The population of older adults in the United States is large. Egocentric studies can take probability samples from this population and then generalize the results. In contrast, to draw conclusions about the entire population using a sociometric design, we would have to survey the entire population of older adults, which is not feasible. Importantly, ego network designs are more conducive to making inferences from samples to populations.

Even if we could conduct a sociocentric study of all older adults in the United States, this design would likely miss the point: we are probably not interested only in older adults' ties to other older adults, but also in their ties to the whole range of significant others, including children and grandchildren. Conducting a census of people in any one domain (i.e., the family, a workplace, a neighborhood, or a school) will omit important interaction partners of other types or functions. The alters depicted in Figure 2.3 are people with whom egos discuss important matters – an affiliation that is not confined to any one group, context, or location. In principle, confidants can be family members, coworkers, neighbors, classmates, etc. This is critical

because people typically form relationships through many different contexts (e.g., family, work, neighborhood) and shared activities (e.g., social clubs, team sports), and they may be the only point of intersection in their personal social networks (Crossley et al. 2015).

The most oft-cited ego network study is Granovetter's (1974) research on job hunting on the East Coast of the United States. He found that men who had only strong ties seemed to be disadvantaged in employment searches compared to men who had both strong and weak relationships. Weak ties, those that are infrequent and not of central importance to an individual (an old high school or college friend), are critical because they allow individuals access to other social networks and, as a result, to new and different information. Strong ties are likely to share similar knowledge and common access to opportunities. By tapping into outside networks, even in a transient and almost happenstance manner (e.g., running into someone in a grocery store), job hunters were able to gather new information. Granovetter's study illustrates the unique insights that ego network studies can produce. The starting point of his research was individual job seekers, and only through an enumeration of the diverse set of alters surrounding those egos could he identify the strength of weak ties.

In practice, of course, researchers do impose boundaries of various kinds in ego network designs (see Chapter 4), but in general these are much less limiting than in sociocentric studies. If the researcher's primary focus is on determining the implications of being embedded in diverse networks and relationships for individuals who simultaneously function in many overlapping social circles (Simmel 1955), ego network analysis is ideal. Moreover, in ego network studies, the researcher has considerable flexibility to define and redefine the boundaries of personal networks during analysis, as dictated by evolving and emerging research questions.

From a practical standpoint, it is worth noting that it is fairly easy to add an ego network module to an existing research project. In the case of Figure 2.3, ego network data were collected as an add-on to a large National Institutes of Health-funded cohort study of adults with early stage Alzheimer's disease. Collecting ego network data need not be the only focus of the research project, or even the primary focus. Moreover, the size limitations of a sociocentric approach do not apply to egocentric methods because (a) we can sample randomly from a large population, and (b) the number of alters mentioned by each ego tends to be fairly small (fewer than the number of egos), leading to data of very manageable

proportions. Assuming we have drawn a probability sample of egos, there is now a population to which we can generalize. Additionally, once the data have been properly formatted, ego network analysis can be conducted using standard statistical techniques and software packages. In contrast, sociocentric data typically require specialized software to analyze the data, and relatively esoteric statistical models for handling the non-independence of observations that is inherent in sociometric data.

Another practical advantage of egocentric research is that both the respondents and their alters can be anonymous, eliminating privacy issues and encouraging honest reporting. In sociocentric research, the researcher can offer confidentiality, but not anonymity. At the point of data collection, the researcher must know who is filling out the survey. This creates a number of problems. For example, the sociocentric researcher must convince respondents that their identities will not be revealed, whether on purpose or accidentally. Moreover, institutional review boards (IRBs) and ethics committees must also be convinced that the research subjects will be protected.

Egocentric network research does have limitations. In the survey format, egocentric data collection tools carry a heavy respondent burden compared to proxy measures of network size, support, or sociability (e.g., “how many friends do you have,” or a social support scale; see Section 2.3). This burden increases significantly with the size and complexity of the ego networks, the number of name generators, and the amount of information gathered about each alter and tie. That said, the limited number of alters for each ego permits us to ask far more detailed questions about each alter and their relationship with ego than would be possible in a whole network survey.

Another potential issue with egocentric designs is that all of the information about the alters, and the ties among them, are elicited from ego, and are therefore potentially inaccurate. We can ask ego whether each alter has had colorectal cancer screening, but in most cases the answers are likely guesses. However, when we are predicting ego’s choices as a function of social influence, it is ego’s perception of their alters’ choices that often influences ego, not the reality of the situation. Knowing the true characteristics of the alters matters in social capital studies. There, we posit that a person’s success is a function of the resources held by their alters. Whether they actually have the money or the expertise is more important than whether ego believes they do.

In the end, the key limitation of an egocentric approach may lie in the inability to map the broader social structure in which personal

networks are embedded. That is, while ego network data can tell us about local networks (and ego's position in those local networks), they cannot provide information about the larger and more complex structural context in which these local networks operate. Put differently, ego network methods are constrained to analyses of direct ties that are present. However, sometimes the most powerful indicators of an ego's social position may be indirect ties (e.g., friends of friends) or ties that do not exist (i.e., structural holes), which cannot be measured with an ego network design.

2.3 NETWORK RESEARCH BY PROXY

The hallmark of empirical research in networks, whether sociocentric or egocentric, is that data are collected on the specific individuals to whom actors are connected. For example, in a study about who discusses confidential matters with whom, the alters might be identified by pseudonyms. Yet, they are individually distinguishable from each other, and detailed characteristics of each alter are collected. In the analysis, however, the information might be aggregated to construct a variable such as the number of women or family members in which an ego confides. Since naming and collecting data on each alter individually is costly in terms of time and energy, some studies avoid this approach. Instead, respondents are asked for the higher-level variables directly: "How many women would you say that you discuss confidential matters with?" or "What proportion of your friends would you consider to be in the same ethnic group as you?"

This approach is common in the social support literature (Berkman 2009). In this tradition, the actual or perceived support resources available to individuals are assessed, and the amount of support is linked to a variety of outcomes (e.g., psychological distress, educational attainment). For example, the Multidimensional Scale of Perceived Social Support is a self-report instrument containing twelve items with a seven-point Likert response scale (Zimet et al. 1988). It includes questions such as, "There is a special person who is around when I am in need," and "I can talk about my problems with my family." While data are collected about the degree to which supportive others are present, those individuals are not distinguishable, and no additional information about the attributes of those alters or relationships, or the ties among them, is collected.

Similarly, there are a number of studies in the management literature that use network constructs but don't collect network data. For example,

a familiar concept in the network literature is group cohesion. The network method of measuring cohesion is to measure positive ties (such as friendship or trust) between pairs of teammates, and then count up the percentage of pairs that have the tie. The higher the value, the greater the cohesion. However, some studies simply ask team members about their subjective impression of the team's cohesion using a set of scale items. Others measure socio-demographic variables and make the assumption that the more similar people are in terms of their social characteristics, the greater the cohesion.

The advantages of not collecting individual-level network data are clear. Proxy measures are easy to administer, and take no more survey time than standard batteries designed to capture attitudes or personality characteristics. In addition, proxy questions are not threatening to respondents, who may balk at providing names, even if disguised through initials or nicknames. Such data can also be analyzed directly using standard statistical procedures, without having to construct variables from alter-specific data.

The disadvantage, of course, is relying on respondents to provide answers that they may not know, or may not be able to accurately convey with a Likert scale. Research suggests that proxy measures (e.g., "To what degree do your friends know one another . . . A lot, somewhat, or very little.") provide inaccurate approximations of the structure of linkages between members of a network, in particular (Burt 1987). They are also severely limited with respect to which structural conditions (e.g., structural holes) can be operationalized. In short, people seem to be largely unaware of the structure of their own networks. Moreover, while proxy approaches like the social support tradition share much in common with an egocentric network approach with regard to theoretical perspectives on social influence, the latter is more flexible and offers a greater number and variety of measures of social connectedness.

2.4 CHOOSING A RESEARCH DESIGN

The criteria for choosing a network research design fall into two categories: the data requirements for assessing a particular theoretical approach or answering a set of research questions, and the practical considerations involved in collecting the data. It should be clear at this point that both egocentric and sociocentric designs have important and often offsetting theoretical and practical strengths and limitations. These are summarized in Table 2.1. Here, we offer guidelines for determining

TABLE 2.1 *Comparison of sociocentric and egocentric research designs*

Issue	Sociocentric	Egocentric
Capture node's position	Yes	Yes, but limited to direct ties
Measurement of ego's indirect or absent ties	Yes	No
Random sampling	Not at node level; Whole networks can be sampled	Yes
Generalizability	Problematic for node and dyad level analysis; mitigated by replications	Yes
Anonymity	No, only confidentiality	Yes
Size limitations	Yes, for survey data	No
Missing data	Very problematic	Less problematic
Boundary specification	Usually one domain or context, such as a natural group	Can elicit alters from many domains or contexts
Alter-reported or ego-perceived alter data	Alter-reported; Ideal for social capital studies	Ego-perceived; Ideal for social influence studies
Aided vs unaided data collection	Typically aided, reducing recall errors	Unaided, vulnerable to recall errors
Data richness	Limited by burden on survey respondents	Lower burden allows more detailed questions

which considerations are most critical for successfully addressing one's research aims.

We begin with the theoretical considerations. Obviously, if the interest lies in mapping whole network structure, a sociometric design is needed. For example, suppose a researcher is interested in the performance of teams, and wants to know if teams with a centralized friendship structure are better or worse with respect to getting things done (i.e., team performance). The researcher would want to conduct a sociometric study of friendships within perhaps 200 teams, and correlate centralization with team performance metrics. Similarly, if a researcher is interested in examining the benefits of being along the shortest paths of an organizational network (e.g., potential control over flows of information), a sociometric study is in order. Therefore, the first key question to ask is, "Does my research question require that I measure global structure and/or position within the global structure?"

A second key question is whether the mechanisms or processes being studied suggest that egos and alters need to be drawn from different populations. For example, in Perry's (2017) SNAD study, discussed above, the interest was in older adults – this defines the populations from which egos are drawn. One aim lies in understanding the consequences of having dementia on a person's network and the subsequent ramifications of those network changes for cognitive and behavioral function or quality of life. Do the theorized processes involved suggest that the only alters of interest would be other older adults? This seems improbable, but if the answer were yes, then a sociometric study of a community of older adults would be appropriate. This approach would let us take advantage of the richness that results from having every node in the study be a respondent, including egos and alters. But if the answer is no, an egocentric study in which the alters could be just about anyone, including people of different ages, or in distant locations, should be considered.

A third key question is whether the phenomenon being studied (e.g., ego's sense of wellbeing) is theorized to be a function of alters' actual characteristics, or ego's perceptions of their characteristics. For example, how well a person weathers a financial crisis may be a function of the actual financial resources held by ego's friends and family. Ego's perceptions of extensive resources will not suffice. However, ego's probability of attending a social event may be a function of ego's perceptions of alters' intentions to go, regardless of their true intentions. This a key differentiating question because in an egocentric research design, we only have ego's perceptions. In contrast, in a sociometric design, we can still collect ego's perceptions of alter characteristics, but these can be supplemented with data on their actual characteristics, at least as defined by alters' self-reports.

With respect to more practical considerations, we have at least two critical questions to ask. First, do the respondents and/or their alters need to be anonymous? If yes, then we must use an ego network design because sociocentric data collection cannot be anonymous.⁴ Second, how many actors are involved? Assuming survey data collection (which we do throughout this discussion), the number of respondents in an egocentric design is just the number of egos, since only the egos are interviewed.⁵

⁴ It can, however, be anonymized after the data are collected, and the identities of those involved kept confidential.

⁵ Of course, any egocentric study can be carried out to additional steps (i.e., interviewing the alters that each ego named), but this is not a typical feature of an egocentric design.

In a sociocentric design, everyone is both interviewed and asked about (i.e., through a roster). Even if the central concern of the analysis is 100 specific nodes, everyone mentioned must be included in the data collection. Consequently, both the number of respondents and the size of the roster grow quickly. In general, ego network data collection is easier to conduct than sociocentric, particularly for large populations. For practical purposes, if research questions can be answered with either approach, an ego network design is preferable.⁶

2.5 CONCLUSION

The goal of this chapter has been to introduce social network research designs, and to differentiate sociocentric and ego network approaches. Each of these designs has distinct implications for the theoretical questions that can be addressed, as well as practical considerations that sometimes weigh heavily in methodological decisions. In the following chapter, we discuss the most critical decisions and considerations associated with collecting ego network data. Specifically, we address sampling strategies, data collection modes, and research ethics for ego network research.

⁶ A class of research that tends to fit this bill is social influence studies, in which we are trying to explain a person's adoption of a behavior as a function of mimicking the same behavior in peers. For example, if school children's choice of sneakers is determined by the choices of their classmates, we can study the question using either design, but if the school is large, it will be simpler to use the egocentric design.

PART II

Research Design for Ego Networks

Sampling, Data Collection Modes, and Research Ethics

An early and important decision in the research process, regardless of what is being studied, is the development of strategies for sampling and data collection. Many of the sampling issues associated with ego network research are general (e.g., representativeness), while others are more unique to egocentric methodology. Here, we provide an introduction to the common sampling designs used in ego network research, providing more detail where ego network methodology diverges from standard approaches (e.g., link-tracing designs such as respondent-driven sampling).

Another critical decision in any research project is the mode of data collection. Egocentric data is typically collected using surveys administered by an interviewer in person, over the phone, or, more recently, using self-administered surveys conducted online. In this chapter, face-to-face, paper, and phone modalities are compared in the context of ego network research. Issues such as social desirability, response quality, and interviewer effects are discussed for each mode of data collection. At the same time, we also provide an introduction to observational and archival methods for collecting ego network data, offering real world examples of each approach. Observational and archival methods have distinct advantages – principally the ability to directly access information about networks without self-report biases. However, they also have nontrivial limitations and special considerations, which are discussed in this chapter.

Finally, we address the question of ethics. Network research does not have a close fit with the origins and intentions of principles of human subjects research, such as those established by *The Belmont Report* (National Commission for the Protection of Human Subjects of

Biomedical and Behavioral Research 1979). Most notably, inherent in the egocentric approach is observational reporting of egos about their alters, raising the question of whether alters qualify as human subjects and are covered by informed consent policies.

3.1 SAMPLING METHODS FOR EGO NETWORK RESEARCH

Sampling methods refer to the procedures through which some members of a population are selected for inclusion in a research sample, while others are excluded. Collecting data on ego or personal networks proceeds in two steps: First, one of many different methods is implemented to draw a probability or convenience sample of ego respondents. Second, these ego respondents are asked to enumerate members of their networks using one or more questions called generators (e.g., “Name the people you are close to”). These generators ultimately determine the nonrandom sample of alters included in the research (see Chapter 4 for detail).

Sampling decisions always require trade-offs between the objectives of a study and the resources that are available. They are important in any research design. However, as Barrat and colleagues (2008) caution, network studies, in particular, require a critical eye that begins with an informed understanding of how data are gathered. For both egocentric and socio-centric network data, the key issue is drawing boundaries, as discussed in Chapters 2 and 4. What is the most useful boundary that encompasses the set of nodes crucial to the research? Importantly, no sample can be perfect given that influential and central alters can come from any of a vast array of domains of social life. Consequently, some significant network members will always be outside the sample (Borgatti et al. 2013). In the end, having a sample of cases with known properties and with a clear understanding of the limits of generalizability to the population of interest is critical. In this sense, the development and assessment of the quality of an egocentric dataset does not differ from traditional research methods.

3.1.1 Sampling Egos from a Large Population

One of the most common approaches for obtaining egocentric data in the general population uses a probability sample design that focuses on selection of egos. Indeed, the ability to use probability-based methods that yield representative samples of large populations is a significant advantage of the ego network approach over sociocentric network research. No special strategies for sampling ego respondents from large populations are

required – any of a number of different methods common in survey research can be used (e.g., simple random sampling, stratified sampling, cluster sampling). The key is to draw the sample of egos in a way that represents the population at large. These sampling methods are well developed and not discussed here at length since many classic references are available (e.g., Kish 1965; Cochran 1977; Kalton 1983; for a more formal treatment specific to networks, see Frank 2005).

While the cleanest approach is a simple random sample, resources often prohibit this, particularly in egocentric network designs that are already high-cost and high-burden. For example, in Wellman's (1979) study of primary ties among East Yorkers, a sample survey of 845 adults over eighteen years of age was drawn from a specific geographical area rather than the Canadian population as a whole to narrow the sampling frame. Similarly, the General Social Survey's initial module on "important matters" networks of the US population in 1985 (Marsden 1987) used a multi-stage area probability sample of census blocks to select US households. This covered about 97 percent of the resident population, excluding those living in college dorms and military quarters. The path-breaking National Health and Social Life Survey (NHSLS) also used this approach (Laumann et al. 1994) to test the theory that both social characteristics and social networks influence sexual behavior.

Since resource constraints often prohibit the use of pure random samples in network research (or otherwise), researchers need to carefully consider the biases created by other approaches. For example, at the block level, early GSS surveys moved to a quota design, where interviewers canvased for any respondents living in the selected block, while adhering to sex, age, and employment status quotas. This design, while substantially more resource-efficient, introduced potential biases associated with the overinclusion of people who happened to be at home when the interviewer came to their house. This source of bias requires correction to minimize error. More broadly, the basic point is that perfect designs are expensive, time-consuming, and often impractical. However, without a clear sampling plan which foresees and accounts for such potential biases, the conclusions derived from the data analyses cannot be evaluated by either the researchers themselves or their readers.

3.1.2 Targeted Sampling for Subgroups or "Hidden" Populations

For some research questions, a sample of the general population of egos is not possible or even desirable. Instead, sampling targets a narrow subset

of the population in a specific group or experiencing a particular event, episode, or condition, and then obtains the personal networks of those egos. In this case, finding an appropriate seed location from which to draw ego respondents is essential. For example, in the Indianapolis Network Mental Health Study (Pescosolido et al. 1998), the goal was to understand the nature, dynamics, and impact of social networks among individuals making their first contact with the mental health treatment system. In the INMHS, all individuals entering two major treatment facilities, one public and one private, were screened. If they fit the inclusion criteria, they were sent a letter of invitation to participate in the study and were contacted after that point by research staff. This continued until the target size of the sample or time frame for selection was reached. Given the rare nature of the event (i.e., while many people used each treatment system, most were repeat users rather than “first-timers”), the selection continued across a number of years.

Distinct sampling approaches for egocentric data collection may be necessary in these kinds of situations – that is, where potential respondents are difficult to locate from a practical standpoint or may even be intentionally concealing themselves. There are a number of factors that make a population relatively “hidden” from researchers. In such cases, there may be no meaningful sampling frame for identifying potential respondents, or the characteristic of concern may not be located in a particular site (e.g., recent immigrants or young widows). More traditionally, populations are classified as hidden because the behavior of interest is clandestine, illegal, and/or stigmatized (e.g., secret clubs, drug users, college students with sexually transmitted infections). Studies focusing on hidden populations are important because traditional sampling methods rarely capture enough of these kinds of respondents to address research questions about them.

Link-tracing designs are a common approach to sampling participants in hidden populations. These rely on the existence of a pattern of contact between members of the population (Frank & Snijders 1994), using “seed” respondents to identify and gain access to additional respondents known to the seed person. That is, “the social relation itself is used as a chain of connection for building the group” (Scott 1991: 59). Because of the ethical issues associated with seed respondents providing identifying information about other potential participants without their consent, respondents themselves typically recruit their alters into the study. Though this avoids breaches of confidentiality, which are especially problematic when studying stigmatized populations, this feature of snowball

sampling makes it difficult to accurately assess response rates and non-response biases (Heckathorn 2002).

Link-tracing designs have long been used to reveal the network structure of hidden populations, such as in Kadushin's (1968) early comparative study of power elites (see also Coleman 1958). However, the transition from link-tracing for studying social networks to link-tracing as a convenience sampling method led to a discussion of problems of inference associated with this approach (Erickson 1979). Notably, because the characteristics of the sample will be inherently biased by the selection of a particular set of seed respondents and their typically homophilous networks (Heckathorn 2002), seeds should ideally be randomly chosen. For example, a researcher could approach every *n*th person to walk through the door of a randomly selected group of sites (e.g., bars, support groups) frequented by members of a hidden population. Likewise, link-tracing designs can create biases associated with volunteerism (i.e., members of hidden populations who agree to participate, as well as their network members, are likely to differ systematically from those who do not). These methods are also disproportionately likely to include in the sample respondents with large networks and network members with particular kinds of affiliations (e.g., friends, kin) that are especially likely to be nominated by seeds (Erickson 1979). Also, areas of high redundancy can be suspect, leading Borgatti and coauthors (2013) to suggest that bridging ties be considered for further steps out to avoid capturing only one dense segment of a network.

More recently, researchers have engaged in efforts to differentiate distinct but related link-tracing approaches – and their associated strengths and weaknesses – that are often erroneously referred to as snowball sampling (Magnani et al. 2005). In a true snowball sampling approach, an initial set of seed respondents (sometimes referred to as indexes, Valente 2010: 42) is asked about their social ties, often of some particular type (e.g., people with whom the seed respondent did intravenous drugs). In turn, these alters are interviewed, and all of their alters are enumerated and subsequently interviewed (i.e., alters' alters). This is repeated until the researchers conclude that a saturation point has been reached. Snowball sampling is typically used to generate whole network data rather than egocentric network data. However, it is a useful starting point for discussion since alternative approaches are essentially adaptations of the original snowball sampling design.

A number of variations on snowball sampling, also characterized as link-tracing designs, have been used to address specific limitations of this

approach and are more conducive to the collection of ego network data. The chain link or chain referral design is a form of snowball sampling where initial seeds report their ties and nominate only one person for the next step outwards. The initial subjects serve as seeds through which wave one subjects are recruited; wave one subjects in turn recruit wave two subjects; and the sample subsequently expands wave by wave until the desired sample size is reached (Heckathorn 2011). Relative to snowball sampling, the chain referral method requires more steps out from the initial seed respondent, and a larger number of starter seeds to attain an adequate sample size. Consequently, the sample as a whole is generally more socially distant from seed respondents, and seed respondents themselves may be more diverse. Theoretically, this will increase the representativeness and the variation in the sample. Another advantage of the chain referral method is that the researcher can ask the seed respondent to refer a particular kind of alter (e.g., a woman, a person over the age of fifty, a drug user who has never been in treatment), which provides some degree of control in shaping the characteristics of the sample. This allows the researcher to select for specific attributes to achieve a sample that is believed to reflect the target population as a whole (Biernacki & Waldorf 1981).

A closely related approach often used in network research is the random walk design. This is a form of link-tracing where the seeds provide multiple network contacts, but only a subset (often one) of each seed's alters is randomly selected from the nominated ties. This method is used to obtain probability samples of large social networks in hidden populations or in cases where no known sampling frame exists. In this case, the sampling unit is the network as a whole rather than the individual, and the goal is to estimate properties of the network. Consequently, the random walk design is especially appropriate when the process or outcome being studied is clearly linked to structural characteristics of networks, such as disease transmission or diffusion of information. An advantage of the random walk design over some other link-tracing methods is that it introduces an element of randomness that permits the derivation of unbiased indicators and estimates of their precision (Heckathorn 2002). However, this requires a series of additional procedures that constitute what is often called respondent-driven sampling.

Respondent-driven sampling (RDS) methods were established largely in response to the criticism that link-tracing designs are essentially a form of convenience sampling and that biases associated with them prohibit any claims of representativeness (Heckathorn 1997). The key feature of

RDS, which is similar to chain referral or random walk designs, is that it is implemented in a way that allows for the calculation of selection probabilities. That is, seeds in the initial wave become limited recruiters in the next sampling stage. Specifically, each participant is given two to four coupons to enlist more participants under the following requirements: (1) documentation of who recruited whom; (2) rationing of coupons (three per seed is recommended); (3) information on personal networks is gathered and reported; and (4) recruiters and recruits must know one another through a preexisting relationship. When these conditions are met, long recruitment chains are likely.

The principal advantage of RDS is that Markov chain theory can be used to derive population estimates and sampling weights. Contrary to conventional wisdom, this allows bias from the convenience sample of initial subjects to be progressively attenuated at a geometric rate as the sample expands across waves. The final estimates from these procedures are asymptotically unbiased (see Heckathorn 1997, 2002; Salganik & Heckathorn 2004). The implication is that this sampling method could potentially become reliable when the number of waves is sufficiently large, even if the initial sample is not random. However, in cases where different groups (e.g., racial or ethnic groups) are sufficiently segregated (referred to as the inbreeding terms), the number of waves has to be large for each group. That is, when boundaries separating groups are impermeable, RDS should be used to draw samples from within, not across, groups. Additional and more detailed information about RDS designs and the calculation of estimates and sample weights is available (Salganik & Heckathorn 2004; Volz & Heckathorn 2008; Gile & Handcock 2010).

3.2 DATA COLLECTION MODES FOR EGO NETWORK SURVEYS

Following sampling decisions, another important consideration for network researchers using survey designs is the mode of data collection. Decisions about mode of survey administration have nontrivial consequences for data quality in any research study. However, the complex steps and procedures required to collect data on ego networks using surveys likely exacerbates the consequences of different administration modes.

Egocentric data collection through surveys is a high-burden task requiring multiple steps, including tools for generating a list of alter

names (i.e., name generators), a series of questions designed to elicit information about each alter named (i.e., name interpreters), and another tool for gathering information about ties between alters (i.e., density, or adjacency, matrix). The name generating process carries a particularly high cognitive burden for respondents, who are asked to process difficult and subjective questions (e.g., Who are you close to? Who do you talk to about important matters?; Bearman & Parigi 2004). Respondents must then scan a large cognitive network – upwards of 300 meaningful social ties or 1,500 acquaintances (Freeman & Thompson 1989; Killworth et al. 1990; McCarty et al. 2001) – for alters that meet the generator criteria, and return these to the interviewer. The name interpretation and matrix tasks are also burdensome because they are time-consuming and repetitive, potentially leading to cognitive shortcutting (e.g., providing the same response for successive pairs without processing the question). In general, the more cognitively complex and burdensome the survey instrument, the higher the stakes associated with methods for collecting the data.

Prior to discussing the advantages and disadvantages of face-to-face, telephone, and online administration modes, describing some of the most common problems affecting data quality in survey research is in order. Mode of administration affects the following conditions:

- *Social desirability bias* is the tendency for respondents to provide inaccurate answers to maintain or achieve a favorable impression of themselves. It may involve underreporting sensitive or negative behaviors and opinions (e.g., drug use) or over-reporting positive or prosocial ones (e.g., volunteering), both with respect to egos themselves and their alters. Because respondents are likely to want to appear well liked, socially active, and supported, social desirability can be an important source of bias in egocentric research.
- *Satisficing* refers to behaviors that reduce the cognitive burden of completing surveys, but adversely affect response quality. The respondent must carefully read or listen to the question, interpret its meaning, retrieve information from memory, and formulate and correctly return a summary judgment on the basis of that memory (Krosnick 1999). Many respondents cannot or are unwilling to exert this level of effort for the entire duration of a survey interview. Consequently, respondents often engage in various satisficing practices to complete the interview while reducing their cognitive burden, including executing all the cognitive steps required, but with less effort, or shortcutting some of the cognitive steps and providing

answers that are inaccurate but defensible. Satisficing is more common when the interview task is difficult, and is therefore highly relevant to ego network research. Examples include nonresponse and “don’t know” responses, or nondifferentiation (e.g., providing the same response to a name interpreter for all alters).

- *Interviewer effects* occur when variations in the way that surveys are administered systematically influence responses. Some interviewers are more enthusiastic, provide positive reinforcement, and prompt for valid responses. Others can seem bored or disengaged, move through the interview too quickly, and can even encourage satisficing or truncating the number of alters given (Marsden 2003; Eagle & Proeschold-Bell 2015). These differences in interviewers’ behaviors or attitudes can result in substantial variation in response quality. Interviewer effects are of particular concern when a multiple name generator approach is used or when the instrument is especially long or complex because both the interviewer and the respondent can become burned out and seek cognitive shortcuts.

Face-to-face, telephone, and online administration of egocentric network surveys have distinct advantages and drawbacks with respect to these potential problems.

3.2.1 Face-to-Face Administration

Because of the complexity and high burden on respondents associated with egocentric network surveys, face-to-face interviewing is the preferred and most common administration mode. Most of the classic ego network data were collected through face-to-face interviews. Barry Wellman’s East York study of community ties in Toronto, Canada was conducted in 1968 using face-to-face administration of paper surveys with 845 adults, and later in-depth interviews with a subset of the initial probability sample. A decade later, Fischer’s Northern California Community Study (Fischer 1977a), conducted in fifty communities in and around San Francisco, used face-to-face interviews to collect extensive ego network data from 1,050 individuals, recording responses on paper. More recently, studies such as the National Longitudinal Study of Adolescent to Adult Health (Add Health) have used face-to-face administration for the in-home interview, despite the availability and efficiency of web-based administration (Harris et al. 2009). The nationally representative Add Health study began in 1994 and is now entering its fifth wave of data collection. Most questions are read aloud in person by the interviewer, including the ego network

component. However, to acquire data about drug use, sexual behavior, and other sensitive topics, there are computer-assisted self-administered components. Multi-mode administration is ideal for long and complicated instruments with sensitive questions that are likely to cause social desirability bias. Though large-scale population or community studies such as these have produced invaluable insights into social behavior, they are extremely expensive and complicated to implement, often requiring collaboration between multiple academic and government institutions (Golder & Macy 2014).

Face-to-face administration continues to be the gold standard for ego network research because it is less cognitively challenging for the respondent than telephone or self-administered (e.g., web-based) surveys (Bowling 2005), easing the burden of egocentric data collection on respondents. Face-to-face administration requires only that the respondent use basic verbal and listening skills. While telephone interviews require the same skills, they have a greater auditory burden since non-verbal cues are unreadable. Moreover, respondents must remember response options since hand cards are unavailable and there is no other written text to which they can refer (Christian, Dillman, & Smyth 2008). Self-administered surveys, however, are the most burdensome (Bowling 2005) because the respondent must use visual, reading, and numeric skills, and must also rely on writing and keyboard skills (e.g., during the name generator task or open-ended responses), depending on whether the survey is pen-and-paper or web-based.

Interviews that are administered face-to-face, involving a back and forth between interviewer and respondent, offer a number of other important benefits. It is a significant advantage to have interviewers on hand to give directions and answer questions (Vehovar et al. 2008). Face-to-face interviews also perform better than telephone and online surveys with regard to satisficing because of the reduced social distance between the interviewer and respondent. Drawing on three national mode experiments, Holbrook, Green, and Krosnick (2003) compared levels of satisficing in face-to-face and telephone interviews. They measured satisficing using three outcomes: choosing a no-opinion response (e.g., “don’t know”), failing to differentiate between multiple items in a response scale, and acquiescing toward agreement regardless of content. They found that respondents were significantly less likely to engage in these satisficing behaviors in face-to-face interviews compared to telephone interviews. Research suggests that interviewers develop rapport with respondents in face-to-face interviews, but do so to a lesser

degree in phone interviews and not at all in online surveys (Drolet & Morris 2000). Because this rapport leads to better cooperation on complex tasks and respondent investment in the data collection process, satisficing is less likely to occur. Also, interviewers can pick up on nonverbal cues that respondents are losing focus or interest, and can respond with positive reinforcement or other techniques for drawing them back in (Shuy 2002). Finally, relative to web-based surveys, face-to-face (and telephone) surveys provide opportunities for probing no-opinion and nonresponse (Heerwegh 2009).

Of course, face-to-face data collection has disadvantages as well. Most notably, it is significantly more expensive and time-consuming than other modes of survey administration. Moreover, egocentric studies have been shown to be vulnerable to interviewer effects (Marsden 2003), which are eliminated when self-administered surveys are used. Face-to-face data collection is also more susceptible to social desirability bias compared to other modes, particularly when obtaining sensitive data (Tourangeau & Yan 2007). Consequently, network researchers have sought alternatives to face-to-face data collection, and conducted evaluations of the reliability and validity of data collected through other modes.

3.2.2 Telephone Administration

Telephone-administered interviews have been a popular mode of data collection for many decades (Holbrook et al. 2003). This mode is appealing to network researchers because it shares some of the advantages of face-to-face data collection (e.g., presence of an interviewer to answer questions and provide feedback), but at a substantially lower cost. For example, the Social Capital-USA (SC-USA) survey was collected via random digit dialed telephone interviews in 2004, with companion studies conducted in China and Taiwan (Lin, Fu, & Chen 2013). This national study of 3,000 currently or previously employed adults in the United States used position generators to evaluate access to social capital through egocentric networks. These data have been leveraged to answer important questions about network-based social capital, explaining the relative advantages of white men over women and other racial/ethnic groups in the labor market (McDonald, Lin, & Ao 2009; McDonald 2011). The SC-USA instrument was narrow in focus and fairly brief, taking an average of only thirty-five minutes to complete. Moreover, because the position generator is less complex and burdensome to administer than name generators (see Chapter 5), this instrument was ideally suited to telephone administration.

The principal advantage of telephone administration relative to the face-to-face mode is cost and efficiency. It is possible to complete a larger number of surveys more quickly and with less administration. It is easier to supervise interviewers when they are working in one location, leading to better standardization and quality control capabilities, and reducing interviewer effects. Additionally, surveys administered over the phone may be less susceptible to social desirability bias since there is more social distance between the interviewer and respondent (Kogovšek & Ferligoj 2005a).

However, telephone administration has a number of serious drawbacks. Random digit dialing – a feature that is critical to the efficiency of telephone interviewing – faces technology challenges (Kempf & Remington 2007). These include voicemail, caller ID, call blocking, and exclusive use of cell phones, which threaten population coverage, response rates, and participation rates. In particular, response rates for telephone surveys are very low (~50 percent, on average), and have continued to trend downward since the mid-1970s (Curtin, Presser, & Singer 2005). Finally, respondents being interviewed by phone could be engaged in various other tasks simultaneous to the interview (Holbrook et al. 2003). Multitasking siphons cognitive resources away from the interview task, threatening the quality of the data produced. Further, the advent of mobile technology means that area codes cannot be used to select respondents from a geographical area nor to geocode the respondent residence. This problem negates a major early advantage of telephone administration and requires that place data be added to the survey instrument.

Despite the challenges of telephone interviewing, this mode produces moderately reliable and valid ego network data. Kogovšek and colleagues (2002) compared the reliability and validity of egocentric data collected using face-to-face and telephone interviews. Contrary to their hypotheses, the authors concluded that the telephone administration mode with all interpreter questions asked by alter (i.e., rather than asking one question about every alter in turn) yielded better validity estimates than the face-to-face interviews. The authors speculated that these patterns are attributable to the sensitive nature of questions about personal relationships, arguing that respondents might feel more comfortable answering such questions over the phone. Additionally, for test-retest reliability, performance of telephone relative to face-to-face modes varied depending on the specific measure. For more cognitively demanding and objective name interpreters (e.g., frequency of contact), face-to-face administration

produced better reliability, while telephone interviews performed better for subjective measures that could be answered quickly (e.g., closeness). These comparisons regarding validity and test-retest reliability were later confirmed by the same research group using a meta-analysis (Kogovšek & Ferligoj 2005).

Research suggests that telephone administration of ego network surveys also produces fewer errors than web-based administration. Kogovšek (2006) compared web and telephone administration of an ego network module that included three name generators and nine name interpreters. She concluded that the test-retest reliability of network measurement is better when data are collected via the telephone, compared to web-based surveys. She attributed her findings to the presence of an interviewer answering questions and setting the pace of survey administration, reducing the likelihood of errors and satisficing.

In all, these studies indicate that telephone administration of ego network surveys may continue to be an acceptable replacement for face-to-face administration. However, as technology rapidly changes, researchers should give careful consideration to the use of traditional telephone approaches to data collection. It is not known how telephone administration would compare to the face-to-face mode when the social network module is only one component of a longer instrument, as is typically the case, or when the network module is particularly lengthy and complex. In these cases, face-to-face is likely to outperform telephone administration (Holbrook et al. 2003). Given these contingencies and the broader challenges associated with telephone interviewing (Bowling 2005), face-to-face remains the ideal mode when there are sufficient resources and time for an in-person interview.

3.2.3 Web-Based Administration

With an ever-increasing proportion of people gaining access to the Internet globally, many researchers are turning to web-based survey designs (Fricker & Schonlau 2002). The Internet has been a transformative development in social science research, with substantial advantages associated with greater economy of scale (Golder & Macy 2014). The ability to obtain very large samples quickly and at a relatively low cost is facilitated by companies providing access to web panels of survey participants. Some of these panels are opt-in, such as Amazon's Mechanical Turk, which provides an online global labor market for self-selected respondents paid per survey (Buhrmester, Kwang, & Gosling 2011). Others, such as the

Gallup panel, recruit individuals to web panels through random digit dialing or address-based sampling to achieve a nationally representative pool of respondents that is not compensated for participation (Rao, Kaminska, & McCutcheon 2010). Researchers can then pay to access these panels. For example, GfK, with its KnowledgePanel®, meets quality standards for federal and peer-reviewed studies. Using a complete list of US households and web-enabled technology, respondents in selected households without technological capability receive equipment and service connection; others receive monetary compensation. TESS, funded by the National Science Foundation, is a competitive, no-cost option open to graduate students, postdocs, and faculty, for survey-based social science research using the GfK panel.

Recently, researchers using ego network designs have begun to leverage such web panels. O'Malley and coauthors (2012) developed a web-based survey instrument for collecting information on up to eight alters using name generators adapted from the General Social Survey and National Social Life, Health, and Aging Project (Burt 1984; Laumann et al. 1995). They used a longitudinal, probability-based Gallup panel of households in the United States to access 6,000 randomly selected potential respondents. A total of 3,232 respondents agreed to participate (a 54 percent response rate), completing name generator, name interpreter, and adjacency matrix tasks, in addition to providing information on health and social behavior. Analysis of these data yielded important insights, suggesting that larger networks are associated both with engagement in prosocial behaviors and with better health status.

Web-based surveys have clear advantages, most notably the low cost and speed of data collection. They can be completed with little administration and at a fraction of the cost of face-to-face interviews. Moreover, web-based surveys eliminate the problem of interviewer effects (Manfreda, Vehovar, & Hlebec 2004), which can be particularly pronounced in ego network research because interviewers and respondents alike experience fatigue (Eagle & Proeschold-Bell 2015). This can lead interviewers to speed up the pace of the interview (compromising comprehension and cognitive processing) and to consciously or unconsciously give cues that truncate the number of alters named (Marsden 2003). Finally, web-based surveys are less vulnerable to social desirability bias than interviewer-administered surveys. Internet administration increases reporting of sensitive information, as well as reporting accuracy relative to telephone and face-to-face administration (Duffy et al. 2005; Tourangeau & Yan 2007; Kreuter, Presser, & Tourangeau 2008).

Nonetheless, web-based surveys have real disadvantages for egocentric designs. Most problematic is that there is no interviewer to guide the respondent through the complicated data collection task. Likewise, it is impossible to control the environment in which respondents complete web-based surveys, leading to errors associated with distractions, outside sources of information, or multitasking (e.g., checking Facebook while completing a survey; Rand 2012). Some respondents in web panels click randomly to efficiently complete a task – an extreme form of satisficing – necessitating precautions to identify and correct for these behaviors (Golder & Macy 2014). Moreover, missing data are a significant issue, with web-based surveys yielding high rates of nonresponse, “don’t know” responses, and incomplete interviews (Heerwegh 2009). These can be related to technical issues (e.g., incompatibility among different browsers, software versions, and Internet settings), which can arise during web-based surveys. However, incompleteness is more often due to respondent burnout, which is correlated with network size (Manfreda et al. 2004; Vehovar et al. 2008). Thus, when collecting data on larger networks online, there are likely to be biases caused by egos with larger networks abandoning the survey during the name interpreter or adjacency matrix tasks.

Matzat and Snijders (2010) conducted two studies comparing egocentric data collection via the web and face-to-face interviews. They found that respondents answered more “mechanically” in the online survey, and that this compromised the quality of network data. Specifically, there were high dropout rates during the network instrument in the web-based survey – 27 percent in the online mode compared to only seven percent offline. Moreover, satisficing behaviors were more common in the web-based survey. There were more missing data in the name interpreter section, and respondents were significantly more likely to engage in nondifferentiation in the web-based survey relative to face-to-face administration. Mode also affected network measures, with web-based surveys yielding smaller network size and larger density because respondents provided fewer names and tended to report that all alters were related. Overall, Matzat and Snijders (2010) concluded that traditional network tools (e.g., the name generator, name interpreters) are not effective in an online format, and that alternative techniques for collecting egocentric data using web-based surveys are needed to hold respondents’ attention and motivate them to exert cognitive effort.

Making the most of web-based surveys for egocentric research. While face-to-face interviewing is the preferred method for ego network research, it is

not always possible to use this mode of administration. Researchers have recently concluded that reasonably reliable and valid network data can be generated using self-administered surveys via the Internet (Coromina & Coenders 2006; Kogovšek 2006; Vehovar et al. 2008). In response to concerns described above, web-based surveys for collecting ego network data have begun to incorporate rich visual components and response methods designed to be more interactive and to keep respondents engaged (e.g., Hogan et al. 2016). Examples of software for web-based collection of egocentric data include EgoWeb 2.0, OpenEddi, Social Mirror, and netCanvas. Others, described below, have focused on improving web-based collection of ego network data to obtain the best possible results.

Published evaluations of web-based surveys for egocentric data collection suggest that it is critical to provide detailed and specific instructions for the name generator task (Manfreda et al. 2004). Respondents of web-based surveys tend to make errors in this section of the network instrument, in particular. Examples of common mistakes include alluding to multiple alters in one space, such as “parents” or “family,” typing multiple names in one box, or providing the same name multiple times. Such errors are relatively common (e.g., about eight percent in one study; Vehovar et al. 2008), and are difficult and time-consuming to correct at the data cleaning stage, but can be minimized with clear and effective instructions.

Other research indicates that the number of empty boxes presented in the name generator task influences the number of alters named (Manfreda et al. 2004). For example, a nontrivial percent of respondents will fill all the boxes because they believe this is what the researcher wants. Manfreda and colleagues (2004) found that when they supplied thirty spaces for names of alters from whom they could borrow things, 15 percent of respondents named thirty alters, though it would be highly unusual to have a lending network of this size. Vehovar and colleagues (2008) confirmed the strong relationship between number of boxes and network size by randomizing respondents to different conditions. Thus, use of a graphical interface for adding alters may be preferable to using empty boxes for obtaining alter names in a web-based survey. Alternatively, some suggest imposing a limitation on the number of alters that can be named in web-based surveys (Manfreda et al. 2004; Gerich & Lehner 2006), but this solution has its own drawbacks, discussed in Chapter 4. In short, serious consideration must be given to the formatting of any data collection instrument that is not face-to-face.

Additionally, the number of alters named by a respondent is strongly associated with the likelihood of missing data on the name interpreter

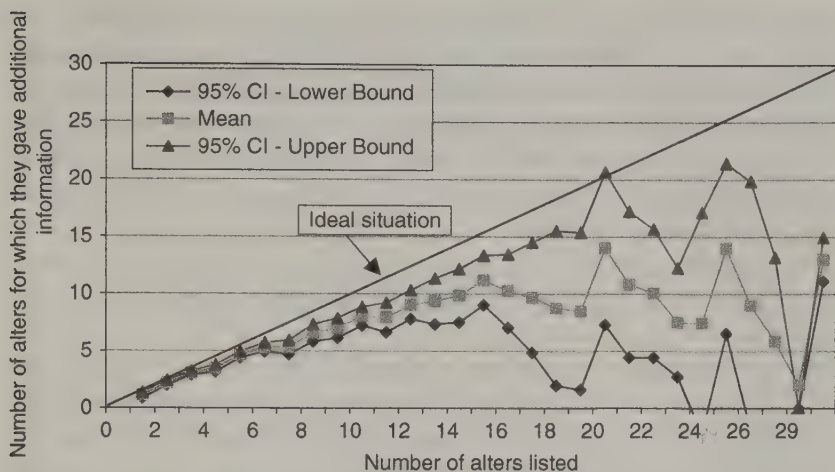


FIGURE 3.1 Name interpreter completion as a function of number of alters named in web-based ego network surveys (grayscale)

Note: From Manfreda et al. (2004)

task. Manfreda and colleagues (2004) found that respondents providing five or fewer alters mostly completed the name interpreter section, but nonresponse increased thereafter with each additional alter, as shown in Figure 3.1 (see also Vehovar, Manfreda et al. 2008). It is critical, therefore, to increase motivation to complete the name interpreter section of the interview, particularly for those with large networks. Alternatively, imposing a limit on the number of alters ego can name also reduces this problem (Manfreda et al. 2004; Gerich & Lehner 2006).

The organization of the name interpreter task also has important implications for data quality. Name interpreters can be administered by question or by alter. A meta-analysis of three network studies found that for web-based surveys, overall reliability and validity are stronger when responses are elicited by question (Coromina & Coenders 2006), contrary to findings for telephone administration of ego network tools (Kogovšek et al. 2002). Vehovar and colleagues (2008) found that dropout and nonresponse rates were significantly higher for the alter-wise compared to the question-wise format for name interpreters, suggesting that responding by question is less cognitively burdensome.

Finally, some visual features of web-based surveys, such as presenting multiple items in the form of grids, are not conducive to holding respondents' attention, and may reduce the level of cognitive processing

(Tourangeau, Couper, & Conrad 2004). This type of design format has been associated with increased item intercorrelation and nondifferentiation (i.e., not using all of the response categories available). For example, respondents may click mechanically without thinking about their answers. This format is often used for name interpreters and evaluation of alter-alter ties in online surveys, threatening the quality of these aspects of egocentric data collection.

In contrast, network visualization can be used to improve the survey experience and enhance data quality (Coromina & Coenders 2006). For example, Tubaro, Casilli, and Mounier (2013) developed a graphic interface that allows respondents to draw their networks directly, reducing the amount of time required to complete the name generator tool and some name interpreters (see Figure 3.2). Similarly, alter-alter ties can be elicited by having respondents draw lines between nodes containing alters' names (Fagan & Eddens 2014). These strategies reduce the length

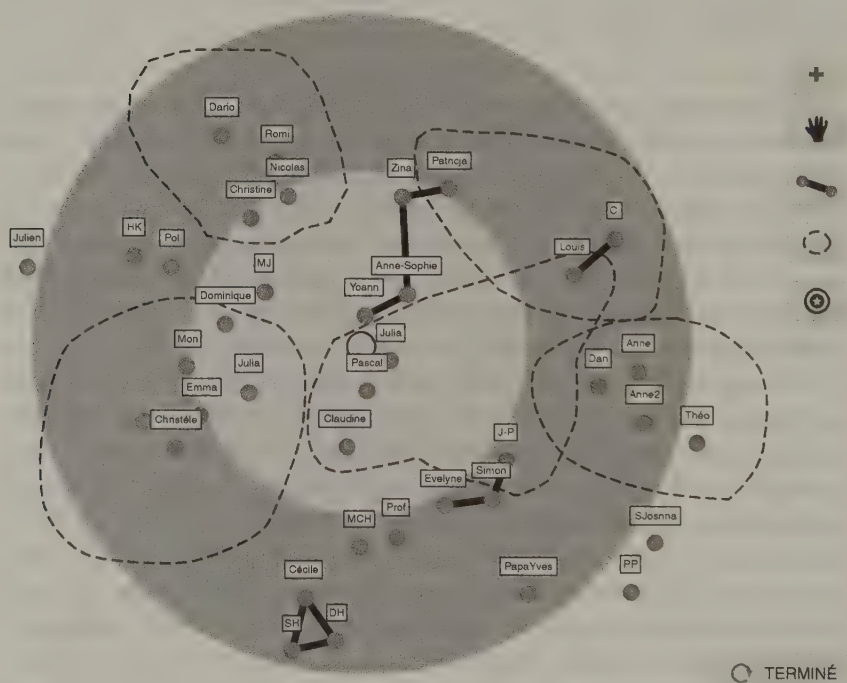


FIGURE 3.2 Online participant-aided sociogram for collecting egocentric network data (grayscale)

Notes: Buttons at right are used to add and move alters, create groups, designate alter attributes, and connect alters to one another; from Tubaro et al. (2013).

of the interview and the banality of adjacency matrix tasks that ask about every pair of alters (see Chapter 6). In addition to being more efficient and engaging, such tools provide the immediate gratification of a holistic view of respondents' networks, creating motivation to complete the network tasks. These tools are discussed in greater depth in Chapter 11.

3.3 OBSERVATIONAL AND ARCHIVAL METHODS FOR EGOCENTRIC DATA COLLECTION

Ego network data can also be generated using observational and archival methods, though these are used much less frequently than survey methods. Early anthropological work employed social network concepts and methods to synthesize complex patterns of relationships that were observed directly in the course of ethnographic field work (Barnes 1954; Mitchell 1969), constituting some of the foundational contributions to a social network perspective. In addition, archives have long been used to generate network data, and are becoming particularly valuable in the age of social media. The Internet constitutes a vast resource of publicly available electronic archives – such as Facebook or Twitter interactions – that are being leveraged to answer important research questions.

3.3.1 Generating Ego Networks through Observation

Social network data can be obtained by directly observing relationships or interactions between individuals, either as an unobtrusive observer or as a participant, usually for a substantial period of time. It is possible to generate reliable data that are unaffected by self-reporting, though they may be influenced by the observer's interpretation rather than respondents' perceptions in the survey format. These observed ties between actors, as well as characteristics of ties or individuals, can then be entered into conventional statistical software for analysis and visualization. Often, observational network data are enhanced with qualitative insight into the meanings or motivations of network structure or dynamics, including why ties form or dissolve, and how alters influence egos' attitudes, beliefs, or behaviors (Bellotti 2014).

An early example of the use of observation to generate network data comes from William Foote Whyte's *Street Corner Society* (1943). In this three-year ethnography, Whyte shadowed one man ("Doc") through his

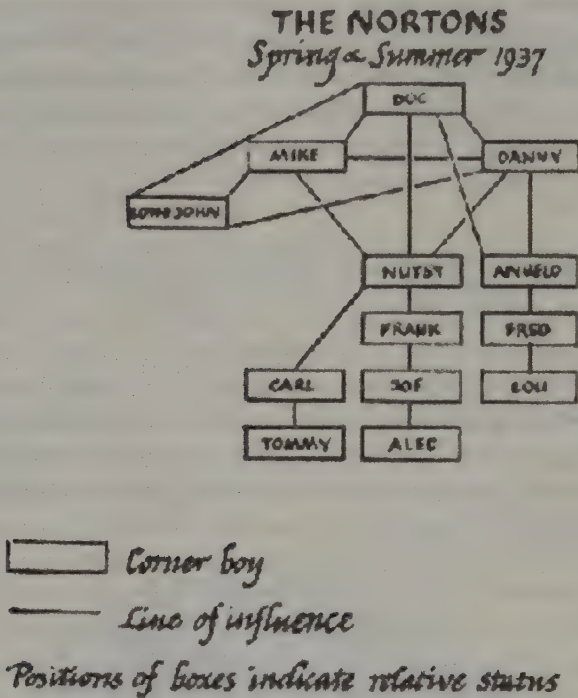


FIGURE 3.3 Doc's egocentric network, as observed by Whyte (grayscale, 1943)

everyday life in a Boston neighborhood that was home to working class first- and second-generation Italian immigrants. In addition to mapping Doc's egocentric network by recording his interactions (see Figure 3.3), he eventually expanded his observations to include a number of his alters' and alters' alters' interactions, constructing the network of the entire slum. As the leader of a group of criminally involved corner boys, and a bridge to other gangs and subgroups, Doc's life provided unique insight into the politics and organization of social life in a low-income urban ethnic community. More recently, Uzzi (1997) and Mische (2008) employed similar methods to construct the ego networks of women's garment firms in the New York City apparel industry, and networks of youth activists engaging in political participation in Brazil, respectively.

The principal advantage of observational methods is the naturalness of the data, which are not subject to many of the errors and biases inherent in survey research (e.g., social desirability, satisficing,

interviewer effects). In short, actual interactions are charted in real time and do not rely on people's descriptions or recollections (Marshall & Rossman 1995). Consequently, network data collected through observation may be more accurate; and, because they occur in natural settings, provide unique insight into the context and meaning of network ties or exchanges between network members (Bellotti 2014). In many cases, researchers gain valuable but unanticipated information from observations that would not have been included on a survey.

However, ethnographic observation of social networks also has limitations. Only a small number of networks can be observed or recorded, because doing so is extremely time-consuming. Further, bias is difficult to assess since it is not possible to measure how the observer might have changed subjects' behaviors or interactions. This is particularly true when participant observation is employed and the researcher is an identified or concealed member of the networks being observed (Merriam & Tisdell 2015). Likewise, researchers' observations may be biased by their own immersion in a particular community or culture, since the researcher must record and interpret what is observed. Finally, observations are limited to interactions – we cannot directly observe relational attitudes, such as who trusts whom.

3.3.2 Obtaining Ego Networks through Archival Research

The term “archival data” refers to any information that existed prior to a research project and was not produced for the purposes of research. Archival data takes many forms, including public records, legal documents, memoirs, or audio or video recordings. More recently, email chains and social media exchanges have been used to generate network data. Though not produced by the researcher, archival data are still open to subjectivity since the researcher chooses which documents or information to include and which to exclude (Brettell 1998).

Archives have primarily been used to generate sociocentric network data. A classic example is Padgett and Ansell's (1993) study of 15th-century Florentine families. Using archival data compiled by historians, including tax reports, employment data, and property records, the authors coded for nine types of relationships between powerful families (e.g., intermarriage, business ties, political relations). Using these data, Padgett and Ansell explained the Medici family's rise to political power, documenting how the Medici exploited their structural position vis-à-vis structural holes in a historical period of uncertainty. Another

classic dataset generated using archival sources is the Southern women affiliation network. These data were constructed using an ethnography (Davis, Gardner, & Gardner 1941) of social stratification in 1930s Natchez, Mississippi. The study included information about the coattendance of eighteen Southern women at fourteen social events ascertained via newspapers, guest lists, and interviews. These data have been used to develop new network measures and test theories about the structure of small groups and identification of group members as core or peripheral (Freeman 2003).

Ego network data have infrequently been obtained using historical archives, but research by Edwards and Crossley (2009) provides a recent example. Drawing on speeches, newspaper articles, and personal written correspondence, the researchers constructed the egocentric social network of the militant British suffragette Helen Watts between 1909 and 1914 (see Figure 3.4). Through a mixed method approach that combined formal structural analysis with insights from the content of written correspondence, the researchers identified alters that were key figures in Watts' radicalization, as well as alters who became involved in the suffragette movement as a result of their contact with Watts. Thus, they were able to demonstrate that activism is both a consequence of social network contacts and a cause of social network dynamics.

Archival methods have a number of advantages. Like data generated through observation, archival methods are not subject to self-reporting bias. From a practical standpoint, archival research can be very cost effective, since no data collection is required. Also, archival research provides opportunities to address unique empirical questions that cannot be examined any other way. For example, archives make it possible to explore historical events or periods from a network perspective, and to trace change (or stability) over time in how personal networks are structured or how they function. A researcher can also study sensitive or inflammatory topics and hidden populations more easily using archival data (e.g., what kinds of people joined the Ku Klux Klan; Fryer & Levitt 2012), since individuals are unlikely to report on these activities in a survey and may be difficult to access.

Yet, there are limitations associated with archival methods for egocentric research. Like observational methods, the time-consuming and laborious nature of gathering and coding archival data often constrain the number of networks that can be examined. Consequently, archival research often constitutes a case study that is difficult to replicate.

Moreover, since the data preexist the research project, key variables of interest may not be available, limiting the kinds of questions that can be addressed. Finally, because media reporting and the recording of even official data are conducted through human processes, they are subject to error and bias that must be considered (Gitlin 1980).

3.3.3 Archival Social Media Data

In the past decade, online social networks have been used to generate sociocentric and egocentric network data using, for example, contacts or exchanges on Facebook or Twitter. “Web scraping” constitutes a form of archival data collection because the researcher obtains relational data after the fact (even if very soon after) that was generated for a purpose other than research (i.e., socialization or information sharing online). Software for obtaining social network data online has been developed and used successfully in social science research. These software packages include NodeXL for scraping, analyzing, and visualizing various types of social media data in network form, and NameGenWeb for downloading Facebook friendship relations as egocentric networks (Hogan 2010; see detailed descriptions in Hansen, Shneiderman, & Smith 2010).

Brooks and colleagues (2014), for example, explored associations between the structure of Facebook friendship networks and perceptions of social capital in the online environment. They used a modified version of Hogan’s (2010) NameGenWeb to obtain online egocentric social network data from 238 university students. In addition to leveraging archival Facebook data, they collected self-report data from participants using a web-based survey. They found that Facebook networks are more compartmentalized than offline networks, and tend to be characterized by nonoverlapping cohesive clusters corresponding to different foci of activity (e.g., high school friends, college friends, family, coworkers). Consequently, relationships between structural properties of online ego networks and perceptions of Facebook social capital were opposite what would be expected based on findings from offline networks. In contrast, using similar methods and a much larger sample, Dunbar and colleagues (2015) found structural similarities between online networks through Facebook and Twitter and offline face-to-face networks. Specifically, they found that online personal communities have a layered core-periphery structure that mirrors offline network structure, and are similar with respect to the number of alters (e.g., 5, 15, 50, and 150) and frequency of contact with alters in each layer.

Online social networks have enormous potential for providing insight into natural interactions using web links, email threads, and other digital traces left behind by web users. Vast amounts of naturalistic data exist online, and can be scraped directly from online sources. This allows researchers to efficiently generate or obtain big network datasets that would be impossible to construct using survey methodology. In addition, because individuals' online networks through social media tend to be much larger than ego networks collected using name generators, they provide opportunities for examining structural properties that are not meaningful for smaller networks (Brooks et al. 2014).

However, online egocentric data have their challenges. When there is no respondent to report on the meaning of different ties or interactions, these may be misinterpreted. For example, what does "friendship" mean on Facebook (e.g., what resources does a Facebook friend provide?), and how does this vary from person to person (Lewis et al. 2008)? Moreover, online relationships typically involve less intensive contact and are easier to maintain, relative to offline ones, and therefore may have unique costs and benefits. This introduces questions about the generalizability of research on online social networks to relationships and interactions that occur in other contexts (Hogan 2008).

Additionally, web scraping raises ethical challenges that potentially violate the Common Rule and institutional review board (IRB) guidelines (US Department of Health & Human Services 1991). For example, it is difficult to obtain informed consent from individuals contributing data to Twitter or Facebook, and confidentiality or anonymity often cannot be guaranteed, even when sensitive information is being collected (Vitak, Shilton, & Ashktorab 2016). In an article aptly entitled, "But the data is already public . . ." Zimmer (2010) recounts a series of projects in which AOL, Facebook, and other social media sites provided access to data for research purposes in which individuals and organizations were easily identifiable. This revelation has three important implications: First, it underscores the fragility of the presumed privacy of potential research subjects on the Internet. Second, it reveals gaps in the professional and institutional protections for human subjects. Specifically, these breaches occurred despite the efforts of researchers to deidentify the data, after full review by the institution's IRB, and in spite of the implementation of "terms of use" for secondary data users. Third, these cases demonstrate that the use of public network data is fraught with difficulties, even as debates continue, expectations change, and individuals increasingly demand and gain control over restrictions placed on their own data (Bos

et al. 2009). This leads us to a discussion of ethical considerations in ego network research more broadly.

3.4 THE QUESTION OF ETHICS

Including human subjects in research raises important ethical issues. The problem for network science, as Charles Kadushin (2012) argued, is that social network research involving humans does not have a close fit with the origins and intentions of human subjects regulations. Specifically, the Belmont Report (National Commission for Protection of Human Subjects of Biomedical & Behavioral Research 1978) is out-of-date, directed toward medical interventions and deceptive research, and fundamentally out-of-line with the premise of network research. On the latter point, Kadushin argued that “the collection of names of either individuals or social units is not incidental to the research but the very point” (2012: 188). Further, network science must now confront big data and computational social science, where disciplines that lack a history of contact with human subjects face novel ethical issues unlikely to have been considered previously. To date, there has been little agreement among researchers about fundamental ethical and scientific issues, or about human subjects protection relevant to network research. However, as Morris (2004) concludes, the implications of human subjects protections for network research are profound.

3.4.1 The Belmont Report

Despite some of the issues specific to network studies, the basic principles of *The Belmont Report* apply to any research that seeks to understand behavior that involves humans in any way. Klov Dahl provides the essential definition of “human subject” from the Common Federal Rule: “a human subject means a living individual, about whom an investigator (whether professional or student) conducting research obtains: 1) data through intervention or interaction with the individual, or 2) identifiable private information” (Klov Dahl 2005: 127). When this is the case, the following principles from *The Belmont Report* apply:

- *Respect for persons* targets the autonomy of all potential human subjects and the requirement to treat them with courtesy and respect. This includes the use of informed consent, which asserts that researchers must be wholly truthful about the goals, procedures, or potential consequences of a research project.

- *Beneficence* refers to researchers' obligation to do no harm. They must maximize the benefits of the research for the participants, science, and society as a whole. At the same time, they are required to anticipate and minimize to the extent possible any risks to those who participate in the research.
- *Justice* reflects the requirement that research be administered using fair and reasonable procedures, and avoid exploitation of participants. It also ensures that both the costs and benefits of research for potential participants be distributed fairly and equally.

To these, Salganik (2017) has recently added another principle based on social changes since *The Belmont Report* was released, and on recommendations from *The Menlo Report* (Dittrich & Kenneally 2012).

- *Respect for Law and Public Interest* asserts that researchers must understand the relevance of existing laws, terms of service and contracts, and comply with the rules set therein. In addition, in accordance with transparency-based accountability, researchers must be clear about and take responsibility for goals, methods, and results throughout the research process.

Principles of *The Belmont Report* have been translated into different standards in each country. However, regardless of how they are implemented in practice, adherence to these principles reflects the position that some research is unacceptable, no matter the potential to contribute to scientific understanding (see Kadushin 2012 for a brief history of ethics violations in research).

3.4.2 Ethical Issues in Network Research

Focusing more narrowly on the ethics of social network studies, it is critical to have a general understanding of how the principles of *The Belmont Report* apply to network science research involving human subjects. However, the specific issues that arise during the course of planning and conducting network research are to some degree unique to each project, and to the standards of each institutional review board. That said, we provide ethical guidelines that distill Klov Dahl's (2005) requirements for social network studies down to five points. As he specifically focused on the network spread of infectious disease, these statements are just a springboard for a fuller discussion of ethical issues in egocentric research.

First, Klov Dahl (2005) argued that *no surgical, pharmaceutical, or other medical treatment can be withheld*. To generalize this point, social

network research cannot withhold any services or benefits to which individuals would normally have access. In research that involves any kind of intervention, a control group would have to meet the minimal standard of interventions that are typically provided (e.g., “treatment as usual,” standard education). This reflects the question of who benefits, as well as the mandate to do no harm. The issue of withholding benefits is becoming more relevant for the study of social networks, as researchers have begun developing and testing “network interventions” for the diffusion of beneficial information, innovations, or social norms through networks. This raises questions about how and when to offer “treatment as usual,” especially when the question at hand is whether the “treatment” ever reaches non-treated individuals or sectors of the network. Additionally, the question of who could potentially benefit from network interventions is sometimes ambiguous. For instance, outcomes may primarily benefit organizations rather than individuals (e.g., a study of the diffusion of innovations that increase the productivity of corporations), though an argument could be made that individual employees are indirectly affected.

Second, *network research based on personal interviews with primary participants requires voluntary and informed consent* (Klov Dahl 2005). While full reviews by IRBs and the requirement to obtain informed consent are standard in human subjects research, these can be waived under certain circumstances. For studies that involve data collection on the personal networks of individuals, informed consent is always required for the ego respondent. However, an issue that is central to egocentric research, in particular, and network science, in general, is how to classify alters or other members of the network (Bos et al. 2009). More specifically, observation and proxy reporting is inherent in the process of asking ego respondents to provide information about alters. Klov Dahl (2005) and others concluded that alters are, in fact, human subjects under the Common Rule, in part because private information about them is likely to be asked. This particularly applies to objective attributes of alters, such as their HIV status, rather than ego’s feelings about the alters. That said, there exists in the Common Rule a waiver of consent option if four conditions are met. These conditions are: (1) minimal risk such as those participants are exposed to in daily life or during the performance of routine physical and psychological examination; (2) inability to practicably conduct the research without the waiver or alteration of standard consent procedures; (3) no adverse effects on the ego’s or alters’ rights and welfare if consent is waived; and (4) any pertinent information about the

study is provided to egos and alters later, if appropriate. Because these conditions typically apply to alters in ego network studies, especially when no identifying information about alters is collected (e.g., using first names only, or pseudonyms), most researchers agree that alters do not need to be consented. Moreover, as Klov Dahl (2005) noted, seeking consent from alters could actually weaken the confidentiality protection for ego respondents, creating risk where none existed before. An important exception is link-tracing sampling designs for gaining access to hidden populations, described above. In this case, ego respondents must approach alters, who then contact the researchers themselves and give consent to participate in the study. While this may seem stringent, and does affect the completion rate among alters, ethical considerations override these concerns, just as they do in accessing egos.

Third, Klov Dahl (2005) argued that *effective means for the protection of the confidentiality of research data should be in place*. Consistent with this, many existing secondary datasets that contain network data (e.g., Add Health) have specific data-handling protocols for public use, including requiring that analyses be conducted on computers that are not connected to the Internet or to a network. For original data collection, it is important that any researcher and staff involved in the study complete necessary human subjects training and be included in an IRB application, and adhere to strict protocols regarding the sharing or storage of data or information about participants. Also, while network scientists often leverage visualization approaches for describing the spatial interrelationships among data points (Green et al. 2003; Lee et al. 2006; Lee et al. 2008), the presentation and publication of data with node labels has to be justified as not violating protections for confidentiality and anonymity. Well-known examples of identifiable actors in networks include historical figures like the Florentine families (Padgett & Ansell 1993), countries (e.g., Alderson & Beckfield 2004), and known terrorists and terrorist organizations (Asal & Rethemeyer 2015).

Fourth, and related to the above point, *ego and alter data should be deidentified at the earliest possible date, and no data retained beyond the end of a project should contain information permitting the identification of any particular ego or alter* (Klov Dahl 2005). This condition applies, in particular, when full names of egos and alters are collected. Often, researchers must obtain names or other information to identify the same alters over time, or to identify alters that appear in multiple ego networks. In these cases, participants' names must be converted to network node numbers as soon as possible. While attributes can continue to

be associated with both egos and alters, it is important to consider whether an individual can be identified on the basis of those attributes, even after traditional identifying information has been removed. For example, if there is only one male Latino teacher in a small school, color coding or otherwise demarking gender and race in the school's faculty network violates common standards of anonymity. Moreover, if the research team involves members of an organization or community, they should not have access to the identified files or to preliminary analyses that identify individuals. Along these lines, some researchers have devised "data dusting" procedures that alter very small aspects of the data that will not influence analyses or findings, but which mask the identification of individuals (Prithiviraj & Porkodi 2015).

Fifth, Klov Dahl (2005) suggested *that no identifying data should be shared outside the project without further IRB approval*. In this contemporary era of open-sourced data, funding agencies often require that data are made available at the end of a project. This raises ethical issues that need to be considered at the *start* of the research project rather than as an afterthought. For many kinds of administrative data (e.g., health records), part of the "consent to treat" includes clauses about the future use of data for research purposes. This model – where participants are informed during the initial consent process that their deidentified data may at some point be shared with other researchers – might be appropriate for basic science research, as well. In addition, all of the other points regarding protections for human subjects described above do apply once secondary data are obtained for analysis. In some cases (e.g., applied research), data are proprietary or protections cannot be guaranteed. When true, these data may not meet criteria for scientific publications since they cannot be shared with other researchers who seek to replicate or challenge findings. In network science, findings from data that cannot be shared have been published in the scientific literature, and the ethicality of this practice remains an open question.

In sum, since no human activity is risk free, researchers are required to assess and minimize risk, explain the risks that are present, and offer solutions in situations where risks become more than minimal (e.g., emergency protocols). It is likely that each research project brings unique concerns in terms of form and substance. In the end, the onus is on researchers to be clear about where their study lands regarding ethical parameters, but according to both Klov Dahl (2005) and Kadushin (2012), well-designed social network research should be able to meet the Common Rule. Going one step further, Sagalnik (2017) sees the IRB

regulations and approval process as a “floor” not a “ceiling.” That said, cutting-edge research will likely always encounter unanswered ethical questions.

3.5 CONCLUSION

This chapter has provided an introduction to basic considerations for constructing probability and targeted samples of ego respondents, and for using link-tracing designs to access networks of hidden populations. Our goal has been to provide an overview of basic sampling considerations, necessitating the sacrifice of depth for breadth. That said, we have provided references to more detailed discussions of the costs and benefits of various sampling approaches published elsewhere.

An additional aim of this chapter has been to review data collection modes for surveys measuring ego networks. Collecting such data is time-consuming and often costly with respect to respondent burden and project resources, increasing the temptation to use resource-efficient data collection modalities, such as self-administered online surveys. However, this can introduce specific biases that are more problematic in egocentric study designs than other types of research. In light of growing Internet usage coupled with the public’s decreasing willingness to participate in research studies, we have provided some guidelines for appropriate use of online ego network data collection. In addition to addressing these survey research methods, we have also provided brief descriptions of observational and archival methods for collecting ego network data, highlighting some of the benefits and drawbacks of these approaches.

Finally, in this chapter we have considered ethical questions pertaining to human subjects research, both broadly and with respect to issues that are unique to ego networks research. We have argued that ethical principles and requirements are guidelines that have to be tailored to the particulars of specific research designs, including egocentric research. In the end, it is not whether one faces ethical dilemmas in research but, rather, how one acknowledges them, collaborates with others for advice, and develops a well-defined set of procedures to protect those who are willing to participate in the advancement of knowledge that, in turn, may improve the conditions of life.

Methods for Eliciting Alters

A name generator is a question or set of questions, usually administered via survey, which is designed to elicit members of ego networks. While there are several different types of generator tools, name generators are among the oldest, most frequently used, and best understood (Bidart & Charbonneau 2011; McCallister & Fischer 1978). Consequently, the majority of this chapter discusses network boundary definition and issues of validity and reliability associated with name generators. At the end of the chapter, we present three additional methods for eliciting alters, all of which have narrower goals, but also a set of distinct advantages: resource generators, position generators, and methods for eliciting representative samples of whole networks.

4.1 NAME GENERATORS

The name generator approach has been in use by anthropologists since at least the 1950s, and was developed to conceptualize and measure social ties that crosscut bounded groups like kinship, tribes, or parishes. Elizabeth Bott (1955: 1957) developed one of the first iterations of the name generator for her in-depth study of urban families. She used prompts to elicit names of kin, friends, and neighbors, and then asked follow-up questions about their gender, occupation, method of meeting, and frequency of contact. Bernice Neugarten (Neugarten 1964; Tobin & Neugarten 1961) also employed a similar approach in her groundbreaking study of social interaction in older adults. Other pioneering ego network research using a name generator approach included Laumann's Detroit Area Studies (1969a), Fischer's Northern

California Study (1977a, 1982), and Wellman's East York studies (1979).

There are at least two essential components of network data collection using a name generator approach: (1) a question or series of questions designed to elicit the names of network members, called "name generators" (see Table 4.1); and (2) a series of questions asked about each alter identified by name, called "name interpreters." The former is the subject of this chapter, while the latter is the subject of Chapter 5. For example, perhaps the best-known name generator, which appears in the General Social Survey (GSS; Burt 1984), asks with whom respondents "discussed important matters." Similarly, Wellman's (1979) East York Study elicited "people outside your home that you feel closest to."

The name generator approach is conducive to collecting very detailed information about social networks. Name generators, together with name interpreters, are used to identify and subsequently produce data about each individual alter, permitting enormous specificity and flexibility with respect to analysis compared to employing proxy network measures (see Chapter 2). In part because they are so flexible, there is little consensus about the best name generators for eliciting members of ego networks, and hundreds of name generators appear in diverse academic literatures, from business and marketing to anthropology. Rather, there are different basic strategies for obtaining alters, all of which have trade-offs, most notably between accuracy, flexibility (or specificity), range, and efficiency. That said, there are a few name generator/ interpreter sets that have become somewhat standardized. Perhaps the most frequently used is the "important matters" name generator from the GSS (Burt 1984), which has been modified and expanded in many ways to suit different research agendas (e.g., Campbell & Lee 1991).

Estimates vary, but some research suggests that people are able to name up to 1,500 acquaintances (Freeman & Thompson 1989; Killworth et al. 1990). With respect to active networks, McCarty and colleagues (2001) discovered that the typical North American has about 300 meaningful social ties, while Bernard and colleagues settled on a range of 135–210 using a reverse small world experiment (Bernard, Shelley, & Killworth 1987). When disaggregated by tie strength, research suggests that, on average, total personal social networks include less than ten close confidants (mean of three), twenty people with whom they interact at least weekly, a few hundred semiregular interaction partners, and several thousand acquaintances (Bidart & Charbonneau 2011; Degenne & Forse 2004). In short, personal social networks are quite large and

TABLE 4.1 *Name generators (Pescosolido et al. 1998*)***Social Networks – Important Matters**

1. From time to time, most people discuss IMPORTANT MATTERS with other people, and we need people we can depend on for help. These can be family, friends, people who have been really helpful to you – what we are interested in are the ones that you are most likely to talk to about really important matters in your life, whether they live near you or far away.

Who are the people in your life with whom you discuss important matters?

Who are the people you can really count on?

► RECORD NAMES ON NETWORK FORM

2. Is there anyone who always wants to talk to you about your important matters in your life whether you want them to or not?

► RECORD NAMES ON NETWORK FORM

3. Are there people who are, in general, a burden to you, because they want to discuss matters important to them, even if you don't?

► RECORD NAMES ON NETWORK FORM

4. Have there been any important changes in this recently? In other words, were there people whom you used to discuss important matters with that you would not or no longer talk to about such things? These could be people whom you no longer feel close to, who might have moved or died.

Who are they?

► RECORD NAMES ON NETWORK FORM

Social Networks – Health

5. I'm interested in the people in your life that you talk to about health problems when they come up. Who are the people that you discuss your health with or you can really count on when you have physical or emotional problems?

► RECORD NAMES ON NETWORK FORM

6. Who are the people, whether or not you have mentioned them before, who are always talking about your mental and physical health and trying to get you to do things about them?

► RECORD NAMES ON NETWORK FORM

7. Are there people who are, in general, a burden to you because of their emotional or physical health problems and always want to talk to you about their problems, whether you want them to or not?

► RECORD NAMES ON NETWORK FORM

8. Have there been any important changes in any of these lists? In other words, were there people whom you used to discuss health issues with or that you would rely on for help that you would not or no longer talk to about such things? These could be people whom you no longer feel close to, who might have moved or died.

► RECORD NAMES ON NETWORK FORM

* Available at www.phenxtoolkit.org/index.php?pageLink=browse.protocols&id=211100

complex. Consequently, methods for limiting or focusing the list of alters included in an ego network are necessary. This focus or constraint is built into name generators.

The task of identifying an ego network – or establishing “operational boundaries on the interpersonal environment” (Marsden 1987: 123) – requires careful consideration. It is not as simple as asking a respondent directly who is in their social network. Research suggests that alters are often not easily or accurately recalled (Bell, Belli-McQueen, & Haider 2007; Brewer 2000), necessitating one or a series of prompts to identify network members. Moreover, defining network boundaries is complicated by the reality that different types of alters have distinct forms and degrees of influence depending on the social phenomenon or process of interest.

Decisions about how to elicit social ties have significant empirical consequences, including the ability to compare, generalize, and replicate findings. Direct comparisons of alters identified using different name generators reveal substantial differences across methods, particularly with respect to the number of alters named and characteristics of those alters (Campbell & Lee 1991). Consequently, choice of name generator may well be the most important methodological decision the researcher faces in the process of conducting a study of ego networks.

In the following sections, we lay out considerations for eliciting an optimal subset of alters from the universe of ego’s ~1,500 interaction partners. The first issue is *boundary definition* – what is the most appropriate generator or set of generators for obtaining the type of network that will have explanatory power in the target research area? Decisions about how to identify networks should primarily be based on the research questions at hand. *In other words, a generator should reflect the social processes that are theorized to underlie social influence or social capital in the researcher’s area or outcome of interest.* One of the most important considerations is the degree to which the network is expansive and more generalized, or smaller and more targeted; and if it is targeted, what factors should guide decisions about boundary definition? What are the trade-offs associated with different methods for limiting networks?

The second concern is the validity and reliability of the network data obtained from ego respondents using name generators. The concept of validity hinges on “the veracity of statements about the empirical world, and is related to the delicate operation by virtue of which mental abstractions of social actors are converted into external indicators” (Carpentier & Ducharme 2007: 4). In other words, do name generators

capture the structure, function, and composition of networks of alters as they exist in the real world? Do network measures capture what is intended? Here, the focus lies on issues of interpretation of name generators, interviewer effects, and question order effects. We also examine critical recall errors, describing the conditions under which alters are more or less likely to be forgotten. As an issue of reliability, this references the extent to which the network is consistent under test-retest conditions. Finally, we present strategies for improving validity and reliability in ego network research.

4.1.1 A Focused Approach to Defining Network Boundaries with Name Generators

As discussed in previous chapters, personal networks serve a wide range of functions, span diverse social contexts, and are relevant for nearly every outcome of interest to social or behavioral scientists. As Burt (1984) notes, “virtually every site for interpersonal relations and every reason for people getting together could be used to define name generator items.” Because there is little consensus about what constitutes the best name generator for a particular process or outcome, it is helpful to draw on the concept of functional specificity as a useful thought experiment when designing ego network research.

The functional specificity concept (Cutrona & Russell 1990; Wellman & Wortley 1990) draws on the idea that social interaction is often goal directed, and that people consciously, habitually, or subconsciously engage in selective activation of ties for particular purposes (Perry & Pescosolido 2015). Individuals tend to rely on various network members for only one or a few specific functions rather than counting on one person to fulfill all their needs. Networks are composed of many different relationships that vary in type of connection, intimacy, frequency of contact, proximity, and other characteristics. While certain associates may be primary in that they are broadly supportive, most networks contain specialized ties that are useful in a more limited (albeit still influential) way. Wellman and Wortley (1990) used a shopping metaphor to describe this process. They argued that in contemporary society, most of our social ties are “boutiques” rather than “general stores,” and that we must “shop” to obtain the goods and services that we need.

An extension of the functional specificity concept is that a particular kind of relationship or resource may be effective for one kind of task or problem, but not another (Sandefur & Laumann 1998). This has important

implications for selection of name generators. Namely, functional specificity represents a shift in the focus of network research from the question of “how many” or “how often” to the match between the problem or task at hand and the functional characteristics of alters. For example, if a researcher is interested in how social ties influence voting behavior, it might be more useful to have information about political discussion networks than kinship networks. In any event, asking who a person talks to about political issues may capture relevant family members, but requesting kinship networks may not yield any political discussants.

The principle advantage of a focused approach is the ability to simultaneously conserve resources and capture the most influential aspects of social networks with respect to the outcomes of interest. In other words, a carefully designed, targeted name generator can give a researcher more “bang for the buck” when there are strong constraints on survey time. Research suggests that characteristics of alters elicited using problem- or task-specific name generators are more predictive of matched outcomes than general or all-purpose name generators that produce a core support network (Perry & Pescosolido 2012). Thus, a researcher with limited resources or a directed interest in ego networks may not be sacrificing much with respect to explanatory power by taking the more focused approach. At the same time, a focused approach requires serious thinking about what kinds of ties matter for a particular phenomenon, a particular sphere of life, and a particular place. As such, it replaces a mechanistic approach with a decidedly conceptual one.

The primary disadvantage of a targeted approach – especially if only one name generator is employed – is the risk of not gaining access to information about influential social network members. In other words, it is possible that the alters elicited miss the mark with respect to the researcher’s goals. Laumann, Marsden, and Prensky (1989) call this the “partial system fallacy,” which occurs when the sample of network members obtained omits a subset of alters that are influential in the population. This might occur because of theory misspecification, leading to under- or overestimation of the importance of a particular kind of alter or network for the outcome of interest. Our theories about how and why networks matter may be wrong. Sometimes in science, the serendipitous findings are the most interesting and important. A more expansive approach to defining network boundaries provides a degree of insurance against having uninformative network data.

In other cases, the theory might be correct, but the name generator does not elicit the kinds of alters that were intended (i.e., errors in validity), or

may yield an incomplete roster of meaningful ties. This problem is often related to variation in the subjective interpretation of the name generator by respondents (see Section 4.4). At the same time, focused name generators may elicit *typical* interaction partners, but omit alters with whom interactions are transitory and happenstance, but nonetheless influential. For example, Small (2017) found that people often talk about personal problems with acquaintances or even strangers, often preferring to confide in others with similar experiences and without the complexities of longstanding relationships. These weak ties, though consequential, are unlikely to be elicited in response to a name generator asking who the respondents usually talks to or has talked to over a given period of time.

Relatedly, another limitation of the focused approach is that the data may not be as rich as those collected using a more expansive approach. Focused ego networks – particularly those defined by intimacy or support provision – are often de facto core networks. These are comprised largely of strong ties, where strength is defined as “a (probably linear) combination of the amount of time, the emotional intensity, the intimacy (mutual confiding), and the reciprocal services which characterize the tie” (Granovetter 1973: 1361). As a result, core networks tend to be homogenous, densely knit, and composed largely of kin and very close friends. When only alters comprising the core network are obtained, the researcher will have limited access to information about weak ties or alters which are dissimilar to the respondent with respect to characteristics like race, religion, or socioeconomic status, unless this is the explicit goal of the name generator (see Section 4.3). As such, the focused name generator approach often yields comparatively little variation in structural measures designed to capture phenomena like power, constraint, and hierarchy. For example, the National Social Life, Health and Aging Project (NSHAP; Waite et al. 2007) used a focused approach which elicited about 3.5 alters that are largely intimately related to the ego. With a tie density of 85 percent, most of the ties that were possible between alters were observed, on average, with minimal potential for measuring structural holes (Cornwell et al. 2009).

4.1.2 An Expansive Approach to Defining Network Boundaries with Name Generators

The goal of capturing an exhaustive census of an individual’s personal social network is unfeasible or, at best, impractical (i.e., the ~1,500 alters).

However, casting a wider net is necessary for capturing what are sometimes called “extended” or “peripheral” networks – that is, those alters that fall outside the small and intimate core networks discussed above. A more generalized and expansive approach is appropriate for exploratory research, when the mechanisms of social network effects are unknown, or when the scientific goal is to understand the general nature of social relationships in a place, organization, or time period. It is important to keep in mind that, for practical purposes, a large and diverse network can always be divided into functionally or contextually specific subnetworks in the analysis phase. For example, a researcher might extract coworkers from a large network for an analysis of job satisfaction, or family members and close friends for a study of emotional wellbeing. In short, a more expansive approach provides greater flexibility and opportunities for unanticipated discoveries, but there is a substantial trade-off with respect to respondent burden.

In addition, research by McCarty and colleagues (2007) demonstrated that collecting data about a substantial number of alters is necessary for computing accurate measures of network structure. They employed a name generator to collect a large number of alters based on knowing: “You know them and they know you by sight or by name. You have had some contact with them in the past two years, either face-to-face, by phone, mail or e-mail, and you could still contact them if you had to.” They found that this strategy elicits large networks containing up to forty-five alters. Moreover, their comparison of networks of different sizes suggested that obtaining a minimum of twenty-five alters is required for computing stable estimates of structural measures like density and centrality.

There are substantive reasons to employ an expansive approach, as well. Large and diversified networks like those collected using McCarty’s name generator are more likely to contain weak ties, which serve important functions. Weak ties are often characterized by less intimate relationships, relatively infrequent interaction, and one-dimensionality (e.g., being connected in one way, or serving only one purpose) – properties often referred to as relational weakness (Granovetter 1973). Ties can also be structurally weak, meaning that they are characterized by lower network embeddedness (e.g., a lack of ties to other members of the ego network). Frequently, relational and structural weakness are correlated, but this is not always the case (Marsden & Campbell 2012; Small 2017).

The most important role of weak ties is their ability to bridge otherwise unconnected clusters of nodes within networks. Because they are typically

not connected to other alters in an ego network (i.e., structural weakness), they are less confined to local social environments and thus more attitudinally (and often demographically) heterogeneous (Marsden & Campbell 2012). Consequently, weak ties have the potential to transmit novel information, or facilitate the diffusion of technical innovations or social movements (Centola & Macy 2007; Strang & Soule 1998). They can also increase exposure to communicable diseases or contribute to the spread of suicidal thoughts (Baller & Richardson 2009; Klov Dahl 1985). Thus, if a researcher is interested in social diffusion, or in contexts where access to novel phenomena is critical, a more expansive approach is preferable. Likewise, an ego's own role as a bridge between two or more unconnected alters (a position that is more easily measured in a larger and more diverse network) is linked to greater independence and individuality, which implies the power to mediate exchanges between others (Burt 2005; Cornwell 2009; Krackhardt 1999).

The Connected Lives Study (Carrasco, Miller, & Wellman 2008; Wellman et al. 2006) – which explicitly sought to include the peripheral ties of East Yorkers in Toronto, Canada – is illustrative of the more expansive approach to collecting ego network data. Respondents were asked to name alters to whom they felt very close and somewhat close, the latter being defined as people that are “more than just casual acquaintances, but not very close.” The resulting networks contained a median of twenty-three alters, nine of whom were very close and fourteen of whom were somewhat close. Very close and somewhat close alters were systematically different, underscoring the value of an expansive approach. Specifically, somewhat close alters were more likely to be extended kin, coworkers, neighbors, and people with whom ego was acquainted only through voluntary organizations or online. Capturing these weak ties has enabled Wellman and colleagues (2006) to address important questions about Internet communities, spatial location and sociability, and resource exchanges outside the context of strong ties.

The principal disadvantage of casting a wide net when eliciting alters is cost. Administering ego network data collection tools requires substantial time and increases respondent burden. Because a series of questions (i.e., the name interpreters) is asked about each alter named, the more complex and expansive a respondent's interpersonal environment, the more time required to complete the instrument. Estimates vary depending on the number of name interpreters, but a good rule of thumb when designing a study using name generators and interpreters is to plan on about three

minutes per alter (Burt 1984; Hamilton et al. 2011). Given that collecting network data is typically not the only (or even the primary) aim of much survey research, the decision to employ a more focused strategy is often a matter of practical necessity.

4.2 TYPES OF NAME GENERATORS

Social network scientists have suggested that there are four egocentric network approaches to defining who should be included in a network (e.g., see Marin & Hampton 2007 drawing from Milardo's 1988 discussion of family networks). These are the exchange, content (also, role-relation), affective, and interaction approaches (Milardo 1988; van der Poel 1993). The exchange approach defines network members as those who fulfill functions for ego, including providing helpful, instrumental, and informational support. Content targets types (e.g., neighbors, friends, or kin), or alters with particular characteristics (e.g., drug users). The affective approach looks to emotional evaluations, including name generators that ask for ties that are close, popular, or even disliked. The interaction approach defines networks on the basis of social contact, typically within a given period of time.

Two additional kinds of substance-free limitations can be imposed to constrain the number of alters named: time frame and size. These elements are often combined in one or multiple name generators. The distinctions between these approaches are sometimes ambiguous. In practice there may be substantial overlap in alters identified using different types of name generators (Bidart & Charbonneau 2011). However, this classification is useful for discussing the characteristics of ties and networks typically produced using these strategies.

4.2.1 Exchange-Based Name Generators

Networks identified by *exchange* elicit ties which provide access to particular resources or fulfill specific functions. The GSS "important matters" name generator is an example of an exchange-based name generator, eliciting ties with whom respondents discussed important matters. Often, multiple name generators tapping different types of resources are administered simultaneously (see also Section 4.3.1). For example, the E-neighbors project (Marin & Hampton 2007) included five exchange-based name generators (plus one intimacy-based generator), seeking to develop a simple but multidimensional tool for eliciting alters. As shown

TABLE 4.2 *An exchange-based strategy: Name generators from the E-neighbors project*

-
1. From time to time, most people discuss important matters with other people. Who are the people with whom you discuss matters important to you?
 2. Who from outside your home has recently helped you with tasks around the home, such as painting, moving furniture, cooking, cleaning or major or minor repairs?
 3. Suppose you need to borrow some small thing like a tool or a cup of sugar. From whom outside your household would you ask to borrow it?
 4. If you need to borrow a large sum of money, say \$1000, whom would you ask for help?
 5. Who are the people you really enjoy socializing with?
 6. Please list anyone who is especially close to you who you have not listed in one of the previous questions.
-

Source: Marin and Hampton (2007)

in Table 4.2, these name generators assessed the availability of instrumental, emotional, material, and informational resource exchange.

Exchange-based name generators are frequently employed to operationalize social capital and social support. They range from being very general and flexible (e.g., Who provides support when you need it?) to being very specific (e.g., From whom could you borrow \$1,000?). In fact, exchange-based name generators can be ideal for leveraging the explanatory power of functional specificity in networks and strategic activation of social ties toward particular ends when designed for this purpose. That is, the researcher can match a particular type of exchange or resource to the phenomenon of interest.

Podolny and Baron's (1997) study of social networks in the workplace illustrates the benefits of theorizing a priori the specific mechanisms through which networks influence outcomes, and using a series of targeted, exchange-based name generators to conduct an empirical test of the theory. Their goal was to determine how the presence of structural holes – or the absence of connections between alters – is related to mobility, with the mechanism being access to novel information that can be used to enhance job performance. They compared the influence of five different types of ties, drawing on the hypothesis that some kinds of information or advice are likely to be more valuable than others with respect to job mobility. Name generators were used to construct subnetworks defined on the basis of the following functions or resources provided: task advice, strategic information, buy-in (i.e., transmitting organizational identity

and norms), mentorship, and friendship. This series of name generators elicited networks that contained a diverse group of alters with respect to strength and characteristics of ties, position in the hierarchy of the company, and density of connections between alters, permitting an analysis of the interaction of structure and function in shaping advancement in the workplace.

Broadly, the authors found support for Burt's (1992) theory that large networks with sparse connectivity between alters promote upward mobility through access to novel information. However, they also demonstrated that for buy-in networks, where the functions are socialization and reinforcement of organizational identity and social norms, densely knit networks are beneficial. They concluded that "all structural holes are not of the same color; some are 'white holes,' propelling the individual upward through the organization and providing socioemotional benefits, while others are clearly 'black holes,' holding individuals at a particular rank in the organization and causing negative psychological consequences" (Podolny & Baron 1997: 689). In this case, the decision to capture variation in the functionality of ties (i.e., what they bring to the table) was critical, allowing the authors to extend and refine a prominent network theory.

In general, when choosing an exchange approach, multiple name generators that reference a variety of different kinds of specific functions or support resources are preferable (see Table 4.2). This strategy tends to elicit a larger number of alters that have greater range compared to employing one all-purpose name generator, and is more likely to elicit peripheral ties like neighbors and coworkers. For example, the important matters name generator elicits about two ties, on average, compared to eight alters per ego when a more defined exchange-based name generator approach is used (Hoffmeyer-Zlotnik 1989; Marsden 1990). Specificity in exchange-based name generators also improves network range relative to intimacy-based name generators. Van Sonderen et al.'s (1990) study of overlap between different kinds of name generators, exchange-based name generators obtained larger networks that were more representative of the network as a whole. Overall, 76 percent of alters elicited using an intimacy-based name generator were also present in an exchange network while only 46 percent of alters obtained with an exchange approach were included in an intimate network.

Importantly, exchange-based name generators may elicit alters who comprise a support network rather than a social network, depending how the name generator is worded. If so, the resulting network may not be

useful for predicting some kinds of outcomes (i.e., access to novel information), since certain relationships are unlikely to be represented. In the INMHS, for example, respondents were asked to name people they talked to about important matters and about their health. The resulting discussion networks were comprised of about three alters each, on average. Surprisingly, less than a quarter of these networks included a health professional, even though respondents were receiving medical treatment and coping with a health crisis at the time of the survey (Perry & Pescosolido 2010). Medical or mental health professionals almost never appeared in the more general important matters discussion networks, consistent with other research (Carpentier & Ducharme 2007). Consequently, using multiple name generators or prompting for specific kinds of relationships or alters is critical when these are theorized to be important for the targeted social process or outcome.

4.2.2 Content-Based Name Generators

Name generators tapping *content* elicit ties which have particular characteristics. That is, they target interaction partners with particular attitudes, attributes, roles, or shared experiences. Researchers might ask a respondent to name people they know with a particular political party affiliation, people they work with professionally, or household members. In his research on drug injectors, Friedman and colleagues (2006) relied on a content-based approach to generate alters in risk networks, focusing on social interactions that are associated with HIV transmission (see Table 4.3). Using a content-based name generator is rooted in the idea that an individual’s decisions and behaviors are influenced by their propensity and opportunity to act in a particular way. This, in turn, is

TABLE 4.3 *A content-based strategy: Name generators from the Social Factors and HIV Risk (SFHR) project*

In the past 30 days, who are the people who you...
1. Used drugs with
2. Had sex with
3. Live with
4. Are related to
5. Met socially or hung out with
6. Knew at work or hustled with

Source: Friedman et al. (2006)

dependent on shared activities, experiences, and opportunities presented through social interaction with network members in the domain in question.

Another content-based strategy is to limit social networks to the central locations where individuals interact with their network members (Carrasco et al. 2008). These might be geographical areas, such as a city or neighborhood; or alters might be identified on the basis of shared interaction at a specific location, such as a particular university, pub, country club, or shooting gallery (i.e., place where people engage in illegal drug use). For example, in the Nashville Neighborhood Study (NNS), respondents were asked to list “neighbors living in the nearest 9 or 10 houses” who they knew by name and had chatted with or visited at home (Campbell & Lee 1991). Locations can also be virtual rather than geographical, with name generators eliciting interaction partners on a particular social media site (e.g., Facebook), or in a topical chat room or online support group (Arnaboldi, Guazzini, & Passarella 2013). Because these name generators elicit alters for whom interactions are concentrated in a particular location, there is a high likelihood of capturing networks that are influential for outcomes or processes related to that social context.

The primary advantage of employing content-based name generators is the ability to hone in on specific kinds of interactions or social contexts that are theorized to be causally related to the phenomena of interest. By design, content-based name generators can elicit alters that are heterogeneous with respect to tie strength or other important characteristics (e.g., age, race, or ethnicity), permitting an investigation of a diverse set of alters that is nonetheless manageable in size and substantively and theoretically focused. For example, in their recent study of parents coping with pediatric cancer, Gage-Bouchard and colleagues (2015) used ego-centric methodology to elicit network ties to other families experiencing childhood cancer. They found that these experientially homophilous alters were critical sources of support because, from the perspective of current families facing pediatric cancer, they alone could offer empathy and valuable information about coping strategies and available resources.

Content-based name generators are almost always unidimensional, having been designed with functional specificity in mind. The trade-off, however, is that networks obtained are typically *not* multipurpose or adaptable to different problems or social processes. As a result, content-based name generators are appropriate when there is a strong theoretical foundation for the research. Researchers might hedge their bets and

couple a more targeted content-based generator with a more generalized and flexible one to maximize utility across the project.

4.2.3 Affect-Based Name Generators

Ego networks may also be defined on the basis of *affect*. Respondents nominate network members to whom they are close or have a strong attachment. Typically, this strategy is employed to elicit intimate (i.e., core) alters, which are often among the most influential. According to Burt (1984: 317), “Intimacy animates the primary ties through which interpersonal socialization operates to create normative pressures purported to define respondent opinion.” Not surprisingly, then, affect-based name generators are widely used. For example, among the name generators used in the National Social Life, Health, and Aging Project (NSHAP; Cornwell et al. 2009) were those eliciting alters who were “very important” or to whom they felt “especially close.”

Name generators that limit networks on the basis of affect have an important place in network science. Because intimate ties contribute to a respondent’s self-esteem, mastery, and sense of belonging, the number and qualities of these relationships have been linked to a wide range of measures of wellbeing and life chances (Hammer 1983; Thoits 2011). At the same time, intimate ties are often those with whom contact occurs most frequently, which provides opportunities for the exchange of resources, information, or ideas (Marsden 1987). Finally, intimate networks tend to be very densely connected. High-density networks constitute close-knit social contexts in which alters can share and compare information, coordinate support efforts, and pool resources. High network density is associated with more reliable and more frequent activation of informal resources (Haines, Hurlbert, & Beggs 1996; Hurlbert, Haines, & Beggs 2000). In short, affect-based networks fulfill a variety of important functions, and are nearly always influential because they serve as regular and critical sources of social integration and regulation. At the same time, the impact of core ties extends to a variety of different domains, from racial identity to health behavior, including virtually any area of social or behavioral research in which social psychological processes play a role.

A disadvantage of using a name generator based on affect is limited network size and range. The more intimate the name generator, the smaller the number of alters elicited. Wellman’s (1979) East York Study employed an affect-based name generator, obtaining a list of “people

outside your home that you feel closest to.” This name generator elicited 4.7 alters, on average. Likewise, the characteristics of intimate networks that confer predictive advantages (e.g., strong ties, high density, homogeneity) are precisely those that impose limitations on the kinds of questions that can be addressed. As noted in Section 4.3, affect-based name generators are ill-suited for eliciting alters that provide access to novel information or ideas, or for capturing important elements of structural position. In addition, affect-based core networks are fairly stable over time, particularly with respect to structure, function, or composition (as opposed to membership; Morgan, Neal, & Carder 1997). Thus, defining networks on the basis of affect alone may not be the best strategy for researchers interested in assessing consequences or predictors of network change, or studying the coevolution of networks and dynamic social processes.

4.2.4 Interaction-Based Name Generators

Name generators that define networks on the basis of interaction are designed to measure the level of sociability and/or social isolation (Bidart & Charbonneau 2011). This approach is particularly common in medicine, psychology, and epidemiology, where researchers are interested in studying how the frequency of contact with network members or the number of different social contacts influences physical and social health (Berkman & Syme 1979). This name generator strategy has been used to document relationships between social isolation or social interactions and a variety of indicators of morbidity and mortality, including physiological stress response, immune functioning, restorative sleep, and the spread of infectious disease (Cacioppo & Hawkey 2003; Mossong et al. 2008).

Generally, there are three strategies for data collection that fall under the umbrella of interaction-based ego networks. First, researchers are often interested in identifying alters with whom individuals have regular or frequent contact. For example, a name generator might read, “Who are the people in your life with whom you interact most often, either in person or by telephone or social media?” Alternatively, a researcher might ask with whom a person interacts “on a typical day.” This approach has been criticized because it is not clear what kinds of alters are being elicited. For example, there is surprisingly little overlap between interaction networks and alters that respondents identify as being important, raising questions about the validity of this method for

tapping social influence or integration (Marsden & Campbell 1984). Similar to time-limited name generators, interaction-based generators are also likely to omit strong and potentially influential ties with whom egos interact infrequently (see Section 4.2.5).

Second, researchers are sometimes interested in the overall number of contacts and the nature of the interaction that occurs, regardless of whether it is ongoing or a one-time meeting. In these cases, respondents may keep a contact diary over a given period of time (see Figure 4.1). In the diary, egos are asked to record the names of everyone they interacted with, “including all kinds of one-on-one contacts such as saying hello, chatting, talking, meeting, or send or receiving a message, that occurred face-to-face, over the phone, on the Internet, or by other means of communication” (Fu 2007: 199). Alternatively, names of alters may be drawn from a personal address book, email account, or telephone, and followed by a series of name interpreters, per usual. Because respondents have a concrete frame of reference and/or are reporting on social interactions in real time, the major advantage of this name generator strategy is its accuracy and access to weak ties and structural holes (when ties between alters are obtained). The principal limitation is the representativeness, and hence the generalizability, of the time period selected or the artifact (e.g., telephone address book) used to elicit names. Moreover, methods that elicit all social interactions are very time-consuming and costly, particularly with respect to respondent burden (e.g., contact diaries).

4.2.5 Comparing Strategies for Boundary Definition

Boundary definition strategies vary substantially. For example, within content-based name generators, a network of household members looks very different from a network of neighbors. Moreover, name generators often perform better when they are combined (e.g., the multiple name-generator strategy used in the E-neighbors project; Marin & Hampton 2007). As a general guide to selecting name generators, Table 4.4 lays out fundamental properties and assesses how different approaches crudely stack up.

Validity. Validity refers to the extent to which the obtained network measures what it is intended to measure, accurately capturing the targeted population of alters. Single-item exchange generators (e.g., important matters, social support) tend to have low validity because interpretation of terms like “important matters” or “social support” can be quite

Diary log

Please record the following information for *every* contact that you have made today, including all kinds of one-on-one contacts such as saying hello, chatting, talking, meeting, or sending or receiving a message, that occurred face-to-face, over the phone, on the Internet, or by other means of communication. Please start with a new page for each day.

	Name/Nickname of contact																			
A. Individual characteristics of the contacted person	Relationship to you (01–25)																			
	Duration of acquaintanceship (1–4)																			
	Gender (1,2)																			
	Age (1–7)																			
	Education (1–6)																			
	Occupation (three digits)																			
	Marital status (1–3)																			
	Current residence (three-digit zip code)																			
B. Contact situations	Form (1–4)																			
	Content (1–8)																			
	Duration of contact(1–5)																			
	Place (1–6)																			
	Audience (1–5)																			
	Expected in advance? (1–4)																			
	Who initiated? (1–4)																			
	Feeling (1–4)																			
	Contact again? (1–4)																			
	Significance to you(1–4)																			
	Significance to the contacted person (1–4)																			
	C. Your ties with the contacted person (1–4)	Frequency of face contact																		
Frequency of phone contact																				
Frequency of (e-)mail contact																				
Degree of familiarity																				
Familiarity with family																				
Importance to you																				
How well you like the person																				
D. Others	First time in log (1, 2)																			
	Sequence number																			

FIGURE 4.1 Picture of a diary log (grayscale)

TABLE 4.4 *Properties of networks elicited using different strategies for boundary definition*

Type	Example	Network properties ¹				
		Validity	Flexibility	Specificity	Range	Cost
Exchange	People with whom you discuss important matters	Low	High	Low	Low	High
Content	People you work with	High	Low	High	Varies	Varies
Affect	People to whom you feel particularly close	Moderate	High	Low	Low	High
Interaction	People with whom you have interacted recently	Low	High	Low	High	Low

¹ Assumes single name generator

variable (Bearman & Parigi 2004). Interaction-based name generators also have low validity because people are often unable to accurately recall all of their social contacts. Diaries and other strategies, when used, improve the validity of interaction-based generators. Additionally, affect-based name generators tend to have only moderate validity. Although people are largely able to accurately recall core networks, there may be some ambiguity in interpretation of terms like “friend” or “close.” Content and location-based name generators have high levels of validity because specificity tends to improve validity. However, if their goal is to elicit a very large number of ties, this compromises validity in this approach (see also Section 4.4).

Flexibility and Specificity. There are marked differences between name generator approaches in flexibility and specificity – two sides of the same coin. Exchange and affect-based networks are very flexible, meaning they elicit alters that are useful for studying a wide range of substantive areas and social processes. However, while the functional specificity of these kinds of measures has been criticized (Bearman & Parigi 2004; Perry & Pescosolido

2015), exchange-based networks can be designed explicitly to measure access to resources known or theorized to increase the likelihood of particular outcomes. Interaction-based networks may be the most flexible since most individuals come into contact with many different kinds of people in multiple contexts within relatively short periods of time. However, there is a trade-off in specificity. While networks defined on the basis of content are functionally specific and usually designed with a particular process or outcome in mind, the network data they produce will not likely be useful for studying other processes or outcomes.

Range. Network range refers to the extent to which the network is sparsely connected, heterogeneous, and large. Networks with high range can be said to provide access to diverse and novel resources, and are more likely to contain weak ties. Interaction networks have the highest range because of the diversity of people with whom individuals interact, but depend heavily on the type of connection being referenced. However, most content-based networks contain a moderate range of weak and strong ties, and are larger on average than affect-based networks. By contrast, both single-item exchange and affect name generators tend to elicit network members with low range. By contrast, interaction networks tend to have the highest range because of the diversity of people with whom individuals come into contact in a given period of time.

Cost. The amount of resources and respondent-burden required to complete the name generator, or cost, is directly related to range. The larger and more complex the network of alters, the more expensive it is to elicit. Consequently, exchange and affect networks are low burden, while interaction networks are very high burden. The cost of content-based networks varies in relation to the nature of the connection. For example, naming sex partners is a fairly low-burden task for most people, while naming people going to the same church or temple is high-burden.

4.2.6 Strategies for Limiting Network Size

Limiting the size of the network is sometimes necessary, typically for the purposes of conserving resources or reducing respondent burden. In these cases, ego networks can be further constrained by imposing time or numeric limits.

Time-limited name generators. Many name generators specify a particular time frame for interaction with alters (usually the past six months or

past year) in addition to using other methods for drawing boundaries around an ego network. This strategy serves the purpose of limiting networks to a manageable size without imposing additional substantive constraints (Campbell & Lee 1991). Specifying a timeframe is critical for longitudinal egocentric research (e.g., “who has loaned you money since our last interview?”). This makes it possible to document changes in the quality or frequency of interaction over time and has the added benefit of improving the accuracy and reliability of respondent recall (Blair, Czaja, & Blair 2013).

Time-limited name generators often provide incomplete information about alters with whom interaction may be infrequent, but nonetheless meaningful or consequential. Referred to as latent or inactive ties, these can provide access to important resources when mobilized. For example, an old well-connected fraternity brother might be the ideal person to help you find a new job (Granovetter 1973). Likewise, an annual beach trip with friends from college who are scattered geographically can provide an infrequent but regular, unique, and even foundational source of information and support. Research comparing networks obtained using the important matters name generator with and without a six-month time frame identified small but significant differences in network composition (Kogovšek & Hlebec 2005b). Considering the implications of omitting latent ties during the design phase of a project is critical.

Numerically limited name generators. Another standard method for constraining the size of an ego network is to impose a limit on the number of alters named by the respondent or recorded by the interviewer. This is cost-effective, but can produce error, particularly if the numeric limit is relatively small (e.g., five alters). Research suggests that the order in which alters are named is dependent upon factors like affective closeness or recency of contact (Brewer 1993) – biases which may or may not be consistent with the research goals. Since it is impossible to know who is being left out or why, truncating networks to an arbitrary size is an approach that should be used with caution (Bidart & Charbonneau 2011).

4.3 COMBINING MULTIPLE NAME GENERATORS TO EXPAND NETWORKS

Multiple name generators can be used effectively in combination to increase network size and range – a strategy pioneered by Claude Fischer in his

Northern California Study (1982). Most single name generators elicit only a few names. However, some research suggests that it is necessary to generate a minimum of five names to observe network effects (e.g., in organizational settings; Merluzzi & Burt 2013), and more are often preferable. The multiple name generator strategy typically entails combining exchange-, affect-, and/or content-based generators, ranging from broad to very specific.

For example, the INMHS (Pescosolido et al. 1998) gathered information about alters across a variety of domains: household, family, romantic partnerships, work and school ties, friends, acquaintances, enemies, people with common (i.e., mental health) problems, mental health professionals, important matters discussants, and health matters discussants. Each domain had a corresponding name generator, for a total of thirteen, with no limitations on the number or types of people respondents could name. This approach resulted in larger networks of people who had semiregular contact with respondents, including ties of great significance (e.g., best friends, spouses) as well as those with more peripheral, unidimensional, or even negative roles (e.g., neighbors, treatment providers, enemies). The mean number of unique alters elicited using this strategy was 15.5 (± 6.2), with a range of five to thirty-seven (Perry & Pescosolido 2015). Similarly, Burt and colleagues (2012) proposed an eight-item name generator instrument for network surveys in healthcare quality and safety research that can easily be tailored to different domains (e.g., workplace satisfaction and safety). This strategy yielded a mean of about nineteen alters, ranging in size from nine to twenty-eight, and resulted in networks with different levels of embeddedness, strength of tie, and function (see Table 4.5; see also Podolny & Baron 1997; Section 4.2.1).

Using multiple name generators has a number of advantages. First, from a methodological standpoint, this strategy yields more valid and reliable data because several types of relationships can be concretely defined, rather than using an abstract criterion that is subject to error (Burt 1984). For instance, it is more difficult to accurately name people who are available to provide social support than it is to recall who would loan you money, listen to you talk about your problems, or help with childcare in an emergency. Marin and Hampton (2007), who systematically compared multiple and single name generator approaches, found that multiple generators provide better estimates of network size, density, and composition.

Second, employing multiple name generators provides a methodological safeguard against omitting particular types of ties, in effect casting

TABLE 4.5 *Multiple name generator approach in health care domains*

From time to time, most people discuss important matters with other people THEY TRUST. These important matters may be personal, professional, or social.
The people with whom they discuss these matters may be family, work, friends, or advisors. Please list the 5 people with whom you discussed matters most important to you (in person, telephone, e-mail, etc.) over the last 6 months.
Think about the people with whom you spend your free time/leisure time. Over the last 6 months, who are the 5 people you have been with the most often for informal social activities (e.g., lunch, coffee, dinner, drinks, movies, sports, visits)? Please list the 5 people with whom you have been to the most informal social activities over the last 6 months.
Think about the people you talk to about quality improvement in caring for patients with pediatric inflammatory bowel disease (IBD). Please list the 5 people with whom you most often discussed Quality improvement (in person, telephone, e-mail, etc.) for pediatric IBD over the last 6 months.
Think about the people you talk to about treatments for pediatric inflammatory bowel disease (IBD) (e.g., people from whom you get information about new treatments, or with whom you discuss treatment decisions). Please list the 5 people with whom you most often discussed Treatment options for pediatric IBD (in person, telephone, e-mail, etc.) over the last 6 months.
Think about the people you talk to about routine clinical/operational matters in caring for patients in your practice. Please list the 5 people with whom you most often discussed Routine clinical/operational matters in caring for patients in your practice (in person, telephone, e-mail, etc.) over the last 6 months.
Think about the people you talk to about difficult cases in your practice. Please list the 5 people with whom you most frequently/most commonly discussed difficult cases (in person, telephone, e-mail, etc.) over the last 6 months.
Think about the people you talk to about matters pertaining to your profession or subspecialty – for example, people you talk with about new professional organizations, associations, research networks, journals, conferences. Please list the 5 people with whom you most frequently/most commonly discussed Matters pertaining to your professional development (in person, telephone, e-mail, etc.) over the last 6 months.
Think about the people who have been most important to you in helping you succeed and/or advance professionally over the course of your professional career. Please list the 5 people who have been most important to you in Helping you succeed and/or advance professionally over the course of your career.

a broader net for capturing social relationships. In other words, multiple name generator approaches are less vulnerable to partial system fallacy (Laumann et al. 1989), making it possible to tap into several different domains and cover most role relationships of influence. Moreover, research has demonstrated that this approach mitigates the

interpretation problem on the part of respondents. When Bearman and Parigi (2004) asked respondents what they talked about with “important matters” discussants, topics ranged from trivial news events (e.g., cloning headless frogs) to serious issues like finances and relationships. Thus, more subjective or open-ended name generators may fail to elicit influential alters – not because of recall error, but because of individual variation in the definition of terms like “friend” or “support.” A multiple name generator approach may provide better coverage of the network since name generators that are open to individual interpretation can be combined with more targeted and explicit ones (e.g., people living in the household).

Findings from NSHAP (Cornwell et al. 2009) highlight the importance of safeguarding against omitting particular types of ties. In the NSHAP interview, personal networks were elicited using a series of name generators and prompts. Respondents were first asked the “important matters” name generator from the General Social Survey, yielding a mean of about three alters. When respondents had previously reported being married or partnered, but did not name their spouse or romantic partner in response to “important matters,” the partner was added to the roster. Spouses and romantic partners were omitted by 15 percent of respondents. Next, respondents were asked to name anyone else who was very important or to whom they felt especially close. In response, over half of the people in the sample named an additional significant alter. Finally, respondents were prompted to add any remaining household members, and this elicited at least one additional alter for 15 percent of the sample. Thus, using multiple name generators and prompts can reduce the likelihood of omitting potentially significant alter roles (e.g., spouses) and regular interaction partners (e.g., household members).

Third, a multiple name generator approach provides opportunities to explore the influence of many important social phenomena. Understanding multiplexity, for example, requires specification of the numerous simultaneous ways that people are connected, or the different roles that alters fill. Studying fundamental social processes like tie formation (i.e., how relationships begin, or where they come from) and tie activation (i.e., who is called on for support or advice, and for which kinds of tasks or problems) also requires a multiple name generator approach. For instance, if the same alters tend to appear on coworker and friendship rosters, the researcher can make plausible inferences about both the workplace environment (e.g., it is likely collegial) and about the ego respondent (e.g., attachment to the job) that might explain important

outcomes (e.g., employee retention). Using different types of name generators to define the boundaries of the network affords richer and more nuanced distinctions in the study of interpersonal environments (Burt 1984).

The only disadvantage of using multiple name generators (beyond those of more expansive approaches more generally) is the need to integrate information across generators prior to administering the list of name interpreters. This may require interviewers or respondents to assemble a nonredundant list of alters during the interview itself. However, this task can be automated by computer-assisted interviewing tools, including those designed specifically for collecting ego network data.

4.4 ISSUES OF VALIDITY AND RELIABILITY IN NAME GENERATORS

Name generators elicit a sample of network members that fit a respondent's interpretation of the questions asked of them (Hammer 1984). The samples generated are likely not representative of the network as a whole, or even of the subnetwork (e.g., core network) the researcher intended to capture (Brewer 2000). In this section, we address the most problematic threats to validity and reliability associated with name generators in egocentric research.

4.4.1 Recall and Forgetting

Because free recall of network members is a cognitively challenging task, forgetting is a pervasive and nontrivial problem in egocentric research of all kinds. Cognitive psychologists have long used free recall exercises to study the organization of memory, finding that a person's prior experiences with the stimuli being recalled influence patterns of forgetting (Baddeley 2004). Of particular relevance to ego networks, psychologists note that the probability of object recall is a direct function of the depth of processing. That is, objects are more likely to be recalled when they are meaningful and related to other contexts or stimuli. Thus, recall of names in response to a name generator is unlikely to be either random or complete. Instead, people will more easily remember names of alters that are more meaningful or are linked in memory to other alters they have already named.

Estimates of the rate of forgetting vary substantially depending on the characteristics of the alters being elicited and of the particular name

generator used. Convergence of social network reports between two respondents can be used to assess forgetting, though discrepancies may be substantive in nature rather than methodological (Pescosolido & Wright 2004). The standard approach to measuring error is test-retest recall interviews, where respondents are asked to name alters on two occasions within a relatively short period of time. Any alter mentioned in one wave, but not the other, in response to the same name generator is assumed to be forgotten. Across a dozen studies reviewed by Brewer (2000), roughly 20 percent of affect and exchange-based networks were forgotten in test-retest conditions, though estimates are much higher for networks with larger range. This approach, however, likely underestimates the amount of forgetting since alters who were forgotten on both occasions cannot be identified.

Saliency. Several patterns on recall and forgetting have emerged from research. First, there seems to be a bias toward naming more salient alters (Brewer 2000) – those who are strong ties, those with whom respondents have more frequent interaction, and those they have known the longest (see also Small 2017). Van Groenou et al. (1990) found that the mean proportion of all personal network members forgotten in a single interview was 22 percent overall, but depended on the strength of tie. Forgetting was estimated at 6 percent for important people, 26 percent for relatives not mentioned as an important person, 31 percent for friends, and 42 percent for other types of ties. Likewise, Marin (2004) found that alters to whom respondents felt closer and those they had known longer were significantly more likely to be remembered.

Network structure. Structural properties of the network as a whole can also influence forgetting. Respondents are more likely to name alters that are highly embedded in the network. That is to say, alters that have ties to many other alters in the network, and those that are connected to other alters through a shared context or activity, are disproportionately less likely to be forgotten. For example, Marin (2004) found that the number of ties to other alters was positively related to the probability of being remembered. In contrast, alters who are bridging ties, because they provide access to people who are not already connected to the ego, are probably the most likely to be forgotten. At the network level, people with more densely connected networks remember a greater proportion of their alters, potentially leading to biases in structural network measures (Marin 2004).

In addition, the size of the network may influence recall errors. Ironically, the more alters (i.e., names) a person remembers, the more alters they are likely to forget (Bell et al. 2007; Brewer 2000; Marin 2004). Network size is likely always truncated in practice since those with larger networks are the most likely to omit people. Network size is probably the measure that is most affected by forgetting.

Specificity. The specificity of the exchange or interaction with alters also affects forgetting, where accuracy of recall is positively associated with greater specificity. For instance, Bell and colleagues (2007) found that the rate of forgetting was only 6 percent for thirty-day sexual partners and 18 percent for drug use partners, but 26 percent for close friends. Referencing specific functions, roles, or contexts when eliciting alters provides a more concrete cognitive anchor, leading to less forgetting in name recall tasks. Consequently, specific exchange and content-based name generators may have lower levels of recall error than those based on affect or frequency of contact. Further, recall is more accurate for behavioral compared to emotional exchanges (Claire et al. 2003; Kogovšek & Ferligoj 2005b).

Order effects. Research suggests that in addition to influencing whether an alter is elicited at all, recall error affects the order in which people are named (Brewer 1993). Memory operates according to associative properties, with objects that are linked in memory having a tendency to be recalled together. That is, when remembering extended kin, people are probably more likely to name all of the members of a nuclear family together (e.g., aunt, uncle, cousins). Alternatively, they might first name all of their aunts, then all of their uncles, etc. Clustering can occur around a variety of dimensions, including affiliation patterns (i.e., who associates with whom), chronology (i.e., people ego met during the same time period), location (i.e., alters who live in the same place), or shared context (i.e., coworkers). For example, in a study of students in a graduate program, Brewer (1993) found that recall of fellow students was strongly clustered by cohort, and that egos tended to recall alters in their own cohort first, followed chronologically by those in other cohorts.

Order effects are especially problematic when a limit is placed on the number of names a respondent can list (Campbell & Lee 1991), skewing the observed network toward core network ties and other alters that are associated with one another in memory. These patterns can potentially

bias structural measures, leading to ego networks that appear more densely connected than they are. In contrast, if a respondent is permitted to name as many associates as they wish, order effects in recall are relatively benign. Strategies for eliciting very large networks are an exception, however, since respondent fatigue can lead to biases in who is omitted.

Consequences of recall errors. As noted, network size is the measure most likely affected by forgetting. Other structural network properties, including ego network density, are more robust to recall errors but may still contain nontrivial levels of bias (Brewer & Webster 1999; Marin 2004). Compositional measures that are directly related to recall errors are compromised by default. For example, mean tie strength seldom accurately reflects the ego network as a whole since strong ties are more likely to be recalled. However, because alters' demographic characteristics do not predict forgetting, measures of demographic composition and homophily (e.g., percent women, percent same gender) tend to be relatively unaffected by recall bias (Brewer et al. 2005).

Measurement error associated with forgetting can have serious implications for predictive modeling where aggregated network measures are used as independent variables. Estimates of the effects of network characteristics on outcomes may not match true population values (i.e., bias), especially if the tendency to forget is correlated with status characteristics or other important covariates. Moreover, estimates for covariates may also be biased if network measures are incorrectly specified and, as a result, are not adequately controlled (Marin 2004). However, recall errors are more problematic under some conditions than others. Bridging alters, because they are usually weaker ties and are peripheral to the network, are especially likely to be forgotten (Brewer & Webster 1999; Marin 2004). Forgetting, then, is particularly apt to bias findings on processes like social diffusion, influence, and structural holes. In contrast, phenomena like social integration or support may be less affected. Finally, forgetting can have substantial real-world implications when the goal is to identify all of a person's network members for the purposes of developing a network intervention or contacting all interaction partners (e.g., when attempting to control infectious disease transmission; Brewer et al. 2005)

Improving recall. While completely eliminating biases and other problems associated with forgetting is unrealistic, strategies for minimizing recall error must be considered in the design phase. First, interviewers

can use nonspecific prompts to encourage respondents to search their memories for additional alters they might be forgetting (Brewer 2000). One common method is to simply ask, “Is there anyone else who [name generator content]?” after the respondent has indicated that they are finished. Another strategy is to read back the list of names to the respondent and then probe again for any missing alters. These strategies lead to modest increases in the number of alters named (Brewer et al. 2005).

Targeted recall cues or prompts are routinely used following name generators to increase the range or representativeness of the observed network. Prompting mitigates some validity and reliability risks associated with the name generator approach (Cornwell 2009). Targeted recall cues may also be used to shift attention to particular types of alters that might not otherwise come to mind. For example, Straits (2000) probed for negative ties following an affect-based name generator using the prompt, “These may include people that sometimes make you angry or upset.” A number of different types of targeted cues are possible, including friends who live far away, important people the respondent has not seen recently, or people the respondent interacts with frequently but to whom they are not particularly close.

A range of general prompting strategies can also be valuable. The list of alters itself can be an effective recall cue when respondents are prompted to name additional people who interact with or are connected to those already mentioned. Other researchers have used alphabetic (i.e., probing for names that begin with each letter of the alphabet), first-name (i.e., listing common first names and asking for alters with the same name), and location-based (i.e., probing using locations where people are likely to meet, such as school, work, or clubs) recall cues (Perisse et al. 2008). These general prompting strategies increase the number of alters named by about one-fifth (Brewer et al. 2005). When used together, general and targeted recall cues elicit networks that are 32 percent larger, on average, than those elicited without prompting.

Bidart and Charbonneau (2011) pioneered a promising approach using contextual cues in their study of young adults in France. They first elicited a list of social contexts (e.g., home, school, work, church) in which egos routinely moved. Then, they asked for the names of alters in each context that ego “knows a little better” or “talks to a little more.” From this large list of names, respondents identified strong ties, sources of resource exchange, and types of interaction. Though very time-intensive, this novel approach has potential to generate rich and

representative data on ego networks. Along the same lines, life history calendars have been used with some success to improve recall of alters (Das & Laumann 2010). This strategy involves first developing a chronological timeline of significant events (e.g., college, marriage, child birth), followed by probing for alters associated with each event and/or period in the life course. In fact, introducing a time frame, in general, is likely to improve the accuracy of recall (Bell et al. 2007; Campbell & Lee 1991).

4.4.2 Differential Interpretation of Name Generators

Different people often interpret name generators in distinct ways. If these variations are systematically correlated with status group characteristics or other important attributes, biases can result. Not surprisingly, a great deal of work on this topic has focused on the important matters name generator. Bailey and Marsden (1999) asked respondents what they believed was meant by “important matters.” While many respondents struggled to understand what was meant by the term, they often mentioned intimacy or frequency of contact rather than exchange or function. Moreover, as noted earlier, substantial heterogeneity was reported in the specific types of discussion topics that people considered to be important, ranging from serious and intimate subjects like personal health or finances, to relatively banal topics like politics and current events (Bailey & Marsden 1999; Bearman & Parigi 2004).

Of greater concern is Bearman and Parigi’s (2004) discovery that individuals talk to particular types of people about different kinds of important matters, a concept that they termed “role-topic” dependency. Wives tended to talk to their husbands about relationships, while husbands tended to talk about this same topic with friends. In other words, systematic differences in the interpretation of name generators across status groups like gender or social class can lead to correlations between social status and network characteristics that are a methodological artifact rather than a reflection of real group differences in sociability or connectedness. That is, respondents’ interpretations of the nature of the interaction that defines a name generator shapes the structure, function, and composition of the ego networks that are observed. Consequently, where there are cultural variations in the norms governing interaction (e.g., cross-national research), making inferences about the meaning of differences in network characteristics is problematic.

4.4.3 Interviewer Effects

The name generator/interpreter approach to collecting egocentric data is very time-intensive and burdensome for respondents and interviewers alike. Consequently, variations in the way that interviewers administer name generators may influence the number of alters elicited. Some interviewers may probe enthusiastically for additional alters, while others forgo such prompting or even subtly (or not so subtly) discourage more names. There is often variation in the degree to which interviewers positively reinforce respondents for providing more names (e.g., “That’s great, anyone else?”). Conversely, interviewers who seem bored or are unengaged might inadvertently truncate name recall, and some even actively discourage respondents from providing additional names. Finally, if interviewers are given no instructions in the event that respondents request clarification, there are substantial differences in the expansiveness of the definitions provided and the size of networks elicited (Marsden 2003).

Overall, correlations documented between interviewers and the number of alters elicited with name generators are large in magnitude, and may be a nontrivial source of error. Zero-order correlations of between 0.07 and 0.20 have been observed between interviewer and network size (Eagle & Proeschold-Bell 2015; Marsden 2003; Van Tilburg 1998). These correlations are of particular concern when a multiple name generator approach is used or when the instrument is especially long or complex. To minimize biases, interviewer training emphasizing minimal off-script verbal interaction, clear instructions for probing, and uniform responses to requests for clarification are critical. Of course, using a computerized, self-administered survey to collect network data eliminates the problem of interviewer effects entirely (Marsden 2003), but has other limitations, discussed in Chapter 3.

4.4.4 Question-Order Effects

Question-order effects, though not unique to name generator approaches, represent another major threat to validity, and should be considered seriously in the design phase (Schuman & Presser 1981). The location of name generators in a survey instrument or interview influences which alters are named. Questions that precede name generators influence their interpretation or prime respondents to remember particular alters (Bailey & Marsden 1999; Pustejovsky & Spillane

2009). For example, Bailey and Marsden (1999) found that respondents' definitions of what constitutes an "important matter" were significantly influenced by the preceding content of the interview. In a split sample research design, when respondents were administered questions about either family or politics prior to the name generator, their interpretation of the meaning of the question and the types of alters named differed.

In addition, when a series of name generators are administered sequentially, the order in which they appear affects who and how many alters are elicited. Pustejovsky and Spillane (2009) found that the number of alters listed in response to the first name generator effectively created a ceiling, reducing the number of names elicited by name generators that followed. They also identified "question-scope redefinition," where a preceding name generator or set of interpreters primed a respondent to answer a subsequent name generator in a particular way. For example, first asking ego to name workplace contacts would likely increase the number of coworkers named as friends in a subsequent name generator, biasing measures of multiplexity.

Because these question-order effects can cause substantial biases in results, researchers should pay careful attention to survey design, especially when a multiple name generator approach is used. There are several effective strategies for minimizing question-order biases (Pustejovsky & Spillane 2009). First, administer all name generators prior to asking any name interpreters, and ideally at the beginning of a survey. Second, respondents should explicitly be instructed to include alters they have named before, where applicable. Third, if possible, consider randomizing the order of name generators across interviews, particularly if accurately measuring network size is a primary objective. Fourth, if randomization is not feasible, place name generators that are likely to yield more alters earlier in the instrument to avoid satisficing. Likewise, administer more open or subjective name generators (e.g., "who are the people you are close to?") prior to asking about specific, content-based ties (e.g., "who lives in your household?"), because the latter are less likely to be influenced by the former.

4.5 OTHER TYPES OF GENERATORS

In the past twenty years, several qualitatively different strategies for delineating ego networks have emerged. Developed largely in response

to the methodological or substantive limitations of name generators, these include position generators, resources generators, and reverse small world (RSW) approaches. In addition, participant-aided sociograms represent a visualization-based method for eliciting names. However, because the primary advantages of sociograms relate to the collection of information about alters and relationships between them, they are discussed in detail in Chapters 5 and 6.

4.5.1 Position Generators

The position generator was developed in 1986 by Nan Lin (Lin & Dumin 1986) to measure social capital accessible through the occupations of network members. Research on social support and ego networks (using name generators) had been used to test theories of social capital, but had been limited to linking actual receipt of resources to instrumental ends. Lin was particularly concerned with identifying access to resources available through social ties, whether those resources were activated or not. Measuring access to resources via a name generator would require a complete census of all of a person's alters, from strong ties to acquaintances. Even if it was feasible to obtain an accurate total network this way, asking follow-up interpreters to elicit information about a range of resources available (e.g., knowledge, money) clearly is not. The position generator was developed as a relatively low-burden tool for measuring the presence of social capital rather than specific relationships in a network. Moreover, the goal of position generators is to be "content free" (Lin, Fu, & Hsung 2001). That is, data are collected without regard to substantive application or functional specificity.

The theoretical foundation of position generators is the idea that the occupational prestige of network members is a reflection of the social resources accessible through those relationships, and of the degree of influence exerted by them (Lin 2002). Thus, a position generator presents a list of twenty or so occupations, ranging in prestige from lower blue-collar (e.g., laborer, server/bartender) to upper white-collar (e.g., lawyer, small business owner). The respondent is asked whether they know anyone having this occupation. If so, they are asked a series of follow-up questions (i.e., interpreters) about the alter in that occupation (see Table 4.6). Interpreters can vary depending on the goals of the research (see Chapter 5). However, all position generators at minimum collect information about the strength of the tie, often measured by the type of relationship (e.g., family member, friend), duration of relationship, or

TABLE 4.6 *Position generator*

Here is a list of jobs (show card). Would you please tell me if you happen to know someone (on a first-name basis) having each job?

Job	1. Do you know anyone having this job?	2. How long have you known this person (no. of years)?	3. What is your relationship with this person?	4. How close are you with this person?	5. His/her gender	6. His/her job
Job A						
Job B						
Job C						
Etc.						

*If you know more than one person, think of the one person whom you have known the longest.

Source: Lin (2002)

closeness. These data are then used to construct measures of (1) highest accessed prestige (because access to high-prestige alters yields the highest returns); (2) range in accessed prestige (because diversity of resources is beneficial); and (3) number of different positions accessed (again, to measure diversity in access to resources).

Position generators have a number of important advantages. The position generator takes less time to administer than most name generator instruments, alleviating respondent burden. Also, position generators are firmly rooted in a well-developed theory of social capital (Lin 2002), making it possible to modify this approach for the study of virtually any society (Van der Gaag & Webber 2008). Consequently, comparative and cross-national research employing the position generator approach has been especially fruitful. Finally, because this method was developed specifically to test theories of social capital, it is ideal for studying issues of status attainment, mobility, and labor market opportunities.

Position generators, like name generators, are still subject to problems with reliability and validity. Errors may be especially likely if people are only vaguely cognizant of their associates' occupations, such as when alters are acquaintances (Laumann 1969b). People sometimes classify professional relationships as personal ones, introducing a confounding effect of ego's occupation (Van der Gaag & Webber 2008). Also, respondents may reply in the affirmative to questions about knowing someone in

an occupation that is similar to the occupation of a known alter (e.g., teacher versus teacher's aide), though the specific occupations might have differing levels of occupational prestige.

Additionally, position generators may be less useful for particular kinds of research questions. Specifically, access to social resources through occupational prestige is not relevant to all types of social capital, let alone to all of the functions and mechanisms of social networks more broadly. Thus, compared to name generators, it is a more restrictive approach. Some researchers question the utility of position generators for measuring social capital that provides expressive benefits (i.e., the influence of emotional support on well-being). They recommend name generators or resource generators in these cases because "such outcomes are less obviously a result from access to prestige-rich positions; there is no reason to believe that network members in more prestigious occupations are also more directly supportive in expressive actions" (Van der Gaag, Snijders, & Flap 2008: 3). In other words, the effects of receiving emotional support should not vary depending on whether the person providing it is a janitor or an engineer. Likewise, the position generator does not accurately account for the value of resources available through people not in the labor market (e.g., homemakers, students, retirees), while other approaches do. For these reasons, Van der Gaag and colleagues (2008) warn against using a position generator approach in studies of health or wellbeing. Position generators should not be employed when the functional specificity of network mechanisms is of principal concern, except in the areas of instrumental social capital and status attainment directly targeted in Lin's (2002) theory.

4.5.2 Resource Generators

Resource generators are also used primarily for measuring social capital, and were developed in large part as a response to some of the limitations of position generators (Van der Gaag & Snijders 2005). Unlike position generators, resource generators explicitly assess the diversity of different kinds of social capital accessible through personal social networks. In other words, specific resources are measured directly rather than inferring access to resources through alters with differing levels of occupational prestige.

The organization and administration of the resource generator is similar to that of the position generator. A list of around 27 concrete types of

social resources is presented to the respondent (see Table 4.7). For each, they are asked, “Do you know anyone who . . . (social resource)?” For instance, “Do you know anyone who knows how to fix problems with computers?” If the respondent answers in the affirmative, they are asked about their relationship to that alter (i.e., acquaintance, friend, or family member). The interviewer records only the alter with the strongest relationship if multiple alters with a particular skill are known. Role relationship is intended to capture whether the respondent can actually leverage the skill or resource for their own goals. Presumably, social capital is more accessible the closer the relationship. Theoretically, additional name interpreters (e.g., measuring tie strength) could be administered, though this is an underdeveloped area of research.

Van der Gaag and Snijders (2005) argue against simply summing the number of resources to obtain a measure of total access, reasoning that social capital is not an all-purpose good. A particular kind of social capital might be beneficial in one context but not another. Consequently, they scaled resources into latent classes of social capital. In the Netherlands, these were: (1) prestige and education-related social capital (e.g., “Has knowledge of literature”); (2) political and financial skills social capital (e.g., “Is active in a political party”); (3) personal skills social capital (e.g., “Can speak and write a foreign language”); and (4) personal support social capital (e.g., “Can give advice about conflict at work”). However, subsequent research in other countries produced very different latent classes. For example, in the United Kingdom, there were four factors (i.e., domestic resources, expert advice, personal skills, and problem solving resources; Webber & Huxley 2007), and three factors in the United States (i.e., expert access, problem-solving, and personal support; Foster & Maas 2014). Resources appear to be highly contextual and culture-specific.

In sum, the resource generator approach has a number of advantages. Its aim is to investigate social capital in a population without being constrained to a particular social context or domain, much like the “content free” aspiration of the position generator. However, because many different types of social capital are measured, the resource generator combines the potential for functional specificity afforded by the name generator approach with the advantages of comprehensiveness and access to latent resources associated with a position generator. That is, one of the key benefits of the resource generator tool is its emphasis on productivity, specifically how social capital helps individuals advance and achieve some kind of instrumental goal. The resource generator also permits an analysis

TABLE 4.7 *Resource generator*

Resource Generator - Us									
"Do you currently have access to someone who..."		If yes, access through...							
		Immed. family	Wider family	Friend	Neigh.	Colleague	Acquain.		
1	Knows how to fix a car								
2	Give advice on using a personal computer								
3	Has a professional occupation								
4	Is an elected official								
5	Works at City Hall								
6	Can sometimes employ people								
7	Knows a lot about government regulations								
8	Has good contacts at TV/radio/ newspaper								
9	Give advice about money problems								
10	Give advice on problems at work								
11	Help dispose of bulky items								
12	Help with small household jobs								

13	Do your shopping if you are ill								
14	Provide care for a serious health condition								
15	Lend large sum of money								
16	Lend small sum of money								
17	Give career advice								
18	Provide a place to stay for a week								
19	Discuss politics								
20	Give sound legal advice								
21	Give a good job reference								
22	Can babysit others' children								
23	Help find someplace to live								
24	Watch home or pets while away								
25	Be there to talk about the day								
26	Owens a car								

Source: Foster and Maas (2014)

of “which part of the social capital is responsible for which effects, and under what conditions (Van der Gaag & Snijders 2005: 2).” For example, the measurement of the role of the person through whom a particular type of resource is available captures ease of access, as well as the potential benefit for instrumental ends. A friend is probably more likely to help you paint your house than an acquaintance. However, strong ties are not always the most helpful ones. Having access to a job lead may be more beneficial when it occurs through an acquaintance than a family member, as Granovetter (1973) documented, in contrast to some other forms of social capital. In other words, the detailed inventory of social resources indexed is ideal for examining whether social capital is goal- or context-specific.

Relative to position generators and, particularly, to name generators, less work has been done to assess the reliability and validity of resource generators. However, the instrument seems to perform well in test-retest conditions, and displays good convergent/divergent and predictive validity (Webber & Huxley 2007). More broadly, there are questions about the degree to which people can accurately evaluate their access to social capital, and this may vary systematically across different social groups. For example, people who activate their social capital more frequently likely make better predictions about availability. But because resource generators tap into different aspects of social capital than position generators, and less is known about this approach generally, Van der Gaag and Webber (2008) recommend including both instruments, if possible, to measure access to social capital.

4.5.3 Generators for Drawing Large and Representative Samples from Total Networks

Two methods for eliciting representative samples of egocentric social networks are the reverse small world (RWS) method and first-name cued recall. These methods aim to accurately estimate network size and obtain a limited number of compositional measures that represent the network as a whole rather than a subset of the network.

RSW is based on Milgram’s (1967) groundbreaking small world experiments, which identified remarkably short path lengths for social networks in the United States. Individuals in Kansas or Nebraska were asked to generate acquaintance chains to people in Boston, Massachusetts, with specific characteristics (e.g., location, occupation). Chains of correspondence were formed by asking respondents to send a letter to someone they

knew who might know the targeted individual. The recipient of the letter then followed the same procedure, and so on, until the letter reached the target. The small world experiments inspired the phrase "six degrees of separation."

The reverse small world method of name elicitation draws on Milgram's (1967) research design. Provided with characteristics of 500 targets, respondents are asked to name people with whom they would start a small-world chain of correspondence (Killworth & Bernard 1979). A random sample of about sixty alters is drawn from the 500 names provided, and a set of name interpreters is asked of each. These responses are then used to estimate characteristics of the network as a whole. The RSW method is extremely time-consuming to administer, but is intended to represent the entire ego network rather than a specific subset of the network.

First-name cued recall (McCarty et al. 1997) shares RSW's goal of drawing representative samples of alters from ego networks, but is much more efficient to administer. A predetermined list of up to fifty common first names is used to cue respondents to name alters. When a first name on the list matches someone the respondent knows, the name is added to the roster. This process is repeated until the respondent names fourteen alters, or the fifty names are exhausted. Finally, a series of name interpreters is asked about each alter.

These tools do have some notable advantages. With respect to obtaining an accurate and representative estimate of network size and composition, RSW and first-name cued recall are probably the best methods available. Unlike other generators, there are no associative biases in first-name cued recall (Brewer 1997). Moreover, they are excellent for obtaining information about peripheral networks. In fact, there is little overlap between the alters elicited using the important matters generator and RSW or first-name cued recall (Bernard et al. 1987) methods. Yet, the connection between alters elicited using these methods and instrumental ends is tenuous, particularly if the influence of core networks is important. Consequently, these methods are inappropriate for research seeking to establish relationships between network characteristics and individual outcomes. However, they are optimal for answering questions about patterns pertaining to the size or characteristics of the network as a whole (e.g., which sociodemographic characteristics of egos are associated with the percent of the network comprised of kin; McCarty et al. 1997).

4.6 CONCLUSION

As this chapter demonstrates, deciding how to go about eliciting alters and where to draw the boundaries around ego networks involves a number of important considerations. Accuracy, flexibility, range, efficiency, and feasibility must be balanced. Perhaps the most significant concern lies in recognizing that the characteristics of the observed network are a product of the generator approach that is employed. The generator selected defines the boundaries of the network, and therefore strongly influences whether respondents' networks are large or small, densely knit or sparsely connected, or contain active supporters or latent ties. The nature of the interaction or social capital that defines the generator – and even the respondents' interpretations of them – shapes the structure, function, and composition of ego networks.

Eliciting all of the influential alters in the lives of egos, and therefore capturing a representative snapshot of the network as a whole, is not feasible. Consequently, a substantial proportion of this chapter has been devoted to discussing the types of networks likely to be obtained with different approaches. We have emphasized that measurement in ego network research – particularly around name generators – is a critical issue that needs to be considered carefully. Issues like role-topic dependency and boundary definition pose a threat to the validity of findings when researchers overstate their theoretical or substantive implications. For example, in response to data suggesting that a sizeable minority of Americans have no one with whom they discuss important matters, social scientists speculated about growing anomie and social isolation in contemporary society (McPherson, Smith-Lovin, & Brashears 2006). In fact, other research suggests that about half of these “isolates” probably simply had nothing important to talk about, and different measures of social isolation directly contradict the idea that Americans lack community (Bearman & Parigi 2004; Fischer 2009). This type of attribution error illustrates why caution is called for when discussing the meaning of results from ego network studies. While the same caution applies to whole network studies, such issues are less often front and center.

Methods for Gathering Data about Alters

After eliciting the list of names (alters) from respondents (egos), a series of questions called name interpreters are asked about these alters. Name interpreters are versatile; they may be used with position or resource generators, though they are typically associated with name generators. Data on alters and ties collected using name interpreters can then be used to compute a variety of measures of personal social network composition and structure (Chapter 7). In this chapter, we describe standard and tailored name interpreters, provide guidelines for constructing egocentric instruments, and discuss issues of data quality associated with measuring characteristics of alters and ties from ego's point of view.

5.1 TYPES OF NAME INTERPRETERS

Name interpreters are questions asked about (typically) every alter elicited through name generators. Together with name generators, they form the basis of data about characteristics of alters, properties of relationships or exchanges between egos and alters (i.e., ego-alter ties), and connections between alters (i.e., alter-alter ties). Some characteristics of alters are used to determine the content of the network. Information about relationships or exchanges are used to measure factors like network strength and function. Connections between alters provide the basis for structural measures of personal social networks like density (see Table 5.1 for definitions).

A fairly standard set of name interpreters exists that are nearly always included in research on personal social networks. These capture some of the most essential and influential characteristics of alters and relationships,

TABLE 5.1 *Types of network measures and locus of data*

Type of measure	Definition	Locus of data
Content	Characteristics of the alters, including resources, beliefs, and practices, whether or not they are accessible	Alter
Strength	The quality and intensity of bonds between ego and alters within the network	Ego-alter tie
Function	Types of exchanges, services, or supports provided or accessible through ties to alters	Ego-alter tie
Structure	The presence and patterns of linkages between actors in a social network, including ego's position vis-à-vis alters	Alter-alter tie

such as how the ego and alter are connected. Knowing, for example, that an alter is a spouse versus a neighbor is an indicator (albeit imperfect) of other aspects of the relationship, such as tie strength. Consequently, these standard measures are typically predictive of a broad range of outcomes. Other standard measures include alter gender and highest degree earned, frequency of contact, closeness of the relationship, and support functions.

As with name generators, name interpreters can be more general and multifunctional, or can be tailored to the specific mechanisms underlying the social process or outcome being studied. Standard measures are typically augmented with one or more name interpreters that are targeted to the goals of the project. This strategy is particularly useful when a more expansive name generator approach has been adopted. Here, name interpreters can be used to differentiate subnetworks for the purposes of specific research questions. For example, if research questions target social influence effects on uptake of novel technologies, name interpreters on alters' use of and attitudes toward technology would be theoretically appropriate. These data could then be used to compute measures of network behavior and culture. The rationale for using a focused strategy for choosing name interpreters mirrors the functional specificity perspective applied to name generators (see Chapter 4). The advantages and limitations of this approach with respect to flexibility, cost, and other issues are also similar, and should be carefully considered during the research design phase.

5.1.1 Content: Obtaining Information about Alter Characteristics

Data on alter characteristics obtained via name interpreters are used to measure the content of ties and ultimately the composition of the network.

Content is essentially the sum total of resources, practices, and ideas that are held in or by the network, whether or not they are accessible by an ego (see Table 5.2 for examples). This includes sociodemographic and other status characteristics and group memberships, which can be used to provide insight into ego's active or potential access to economic, social, and cultural capital (Lin 1999a; see Chapter 4). Alters' level of education, for example, is useful for constructing measures of average or highest access to prestige, which might be used to predict ego's ultimate educational attainment or career success. Alternatively, the educational range of alters can be used as an indicator of the diversity of resources available to ego. Likewise, researchers often measure alter gender because norms of social interaction and responsibility for caregiving differentially shape the types and amounts of support provided by and to men and women. Men are largely the beneficiaries of instrumental and emotional support in relationships with women rather than the givers (Wellman & Wortley 1990; Umberson et al. 1996). Similarly, women are disproportionately likely to engage in social regulation, urging loved ones to engage in healthful and prosocial behavior (Umberson, Crosnoe, & Reczek 2010). Thus, gender composition has important implications for the social support available and the amount of social influence leveraged by the network. Other measures of alter social status and group membership include age, race, and religion.

Content also refers to the knowledge, behaviors, and cultural characteristics (i.e., ideas, attitudes, values) flowing through the network (Wasserman & Faust 1994). When conceptualized in this way, network content provides information about the direction of a push toward a specific outcome. For example, if the majority of a person's network does not smoke, there is likely to be social pressure to become or remain a nonsmoker, while the opposite may be true for someone embedded in a network of smokers. Likewise, alters' knowledge is indicative of the types and amount of information that may be present in the network, though whether it is accessible can depend on characteristics of relationships and the structure of the network. For instance, the Cross-National Election Studies (Beck, Dalton, & Huckfeldt 2000) contain a name interpreter asking how much each alter knows about politics. Someone whose networks are very knowledgeable about politics, on average, may be more informed about political candidates, and therefore more apt to vote in primary elections. Measures of knowledge or cultural content are nearly always tailored to the specific research questions at hand, and are designed to tap into social or cultural processes or resources in a particular domain, such as health, politics, or education.

TABLE 5.2 *Examples of name interpreters for obtaining information about alters*

Name interpreter	Source
1. Is [NAME] male or female?*	GSS
2. As far as you know, what is [NAME]'s highest level of education? Is that less than high school, high school diploma, a college degree, or an advanced or graduate degree?*	CNES
3. Is [NAME] Asian, Black, Hispanic, White, or something else?	GSS
4. How old is [NAME]?	GSS
5. What is [NAME]'s religious preference? Is it Protestant, Catholic, Jewish, some other religion, or no religion?	GSS
6. Are you economically better off, same as, or worse off than [NAME]?	MLSFH
7. Is [NAME] generally a Republican, Democrat, Independent, or what?	GSS
8. Generally speaking, how much do you think [NAME] knows about politics? Would you say a great deal, an average amount, or not much at all?	CNES
9. Does [NAME] trust doctors to take care of people's problems? Would you say a lot, most of the time, or not very much?	INMHS/PhenX
10. Is [NAME] on public aid?	UPFLS
11. To the best of your knowledge, has [NAME] ever been arrested for a violent or gun-related crime?	CGP
12. Based on what you know about how the AIDS virus is transmitted, how would you rate [NAME]'s chance of acquiring AIDS virus infection? Would you say high, low, or is s/he already infected?	SFHR

* Standard name interpreter; Notes: GSS = General Social Survey (Smith, Marsden, & Hout 2011); CNES = Cross-National Election Studies (Beck, Dalton, & Huckfeldt 2000); MLSFH = Malawi Longitudinal Study of Families and Health (Kohler et al. 2014); INMHS/PhenX = Indianapolis Network Mental Health Study and NIH PhenX Toolkit (Pescosolido et al. 1998); UPFLS = Urban Poverty and Family Life Studies of Chicago (Wilson 1987); CGP = Chicago Gun Project (Papachristos et al. 2012); SFHR = Social Factors and HIV Risk Study (Morris & Rothenberg 2011)

5.1.2 Tie Strength and Function: Obtaining Information about Ego-Alter Ties

Name interpreters are also used to obtain data on tie strength and the function of relationships embedded in a network. *Strength* is an important measure in egocentric research, capturing the intensity and duration of

bonds between an ego and alters within the network (Marsden & Campbell 1984). Others define tie strength with respect to durability, arguing that strong ties are characterized by commitment to maintaining a relationship and the ability to do so despite internal and external challenges (Shi et al. 2009). If network content provides information about the direction of a push toward a particular outcome, then network strength is indicative of the force or magnitude of that push. Also, as noted in Chapter 4, the presence of strong ties is an indicator of social integration and regulation, while weak ties can signify access to novel resources or a position of relative power in the network.

Tie strength is operationalized in a number of different ways, the most common of which are frequency of contact, duration of relationship, and emotional attachment or intimacy (Granovetter 1973; see examples, Table 5.3). While these measures are typically correlated, they need not be, and may reflect unique properties of relationships that have distinct effects on outcomes. For example, when making everyday decisions about health, such as what to eat, regular interaction partners may be most influential. However, the decision to undergo major surgery (e.g., bariatric

TABLE 5.3 *Examples of name interpreters for obtaining information about relationships and exchanges*

Name interpreter	Source
1. Which of the following best describes [NAME]'s relationship to you? (CHECK ALL THAT APPLY; PROMPT IF NEEDED: So this person is your. . .); HAND CARD OR LIST TYPICALLY INCLUDES: spouse, ex-spouse, romantic/sexual partner, parent, parent-in-law, child, step-child, brother or sister, grandchild, other relative of yours, other in-law, friend, neighbor, coworker or boss, minister, priest or other clergy, psychiatrist, counselor, or other therapist, caseworker/social worker, housekeeper or home health care provider, and other (SPECIFY)*	NSHAP
2. How close do you feel to [NAME]? Would you say not very close, somewhat close, very close, or extremely close?*	NSHAP
3. How often do you see or talk to this person? Would you say almost every day, at least once a week, at least once a month, or less than once a month? IF RESPONDENT ASKS, SAY THAT TALKING OVER TELEPHONE, TEXT, AND PERSONAL EMAIL (I.E., EMAIL BACK AND FORTH BETWEEN THE TWO OF YOU) MAY BE INCLUDED.*	GSS

TABLE 5.3 (cont.)

Name interpreter	Source
4. What does [NAME] do for you? Check all that apply* – Listen to you when you are feeling down or upset – Tell you they care about what happens to you – Give you suggestions when you have a problem about what you could do, where you could go, or who you could talk to – Help you with daily chores, like shopping for food, cooking, fixing things, cleaning your home, or taking you places you need to go – Help out when you are short of money by giving or loaning you money	INMHS/ PhenX
5. How long have you known [NAME]? Would you say less than one year, one to three years, three to five years, five to ten years, or more than ten years?	GSS
6. Does [NAME] hassle you, cause you problems, or make life difficult? Would you say a lot, sometimes, or not really?	INMHS/ PhenX
7. Rate the strength of your relationship with [NAME] on a scale from 1 (acquaintance) to 10 (best friend).	SFHR
8. Do you typically turn to [NAME] for information or knowledge on work-related topics?	PES
9. Does [NAME] go to the same church as you?	MLSFH
10. Is [NAME] among those you talked to about sexually transmitted infections?	MLSFH
11. Does [NAME] live in your neighborhood?	CGP

* Standard name interpreter; Notes: NSHAP = National Social Life, Health, & Aging Project (Waite et al. 2007); GSS = General Social Survey; INMHS/PhenX = Indianapolis Network Mental Health Study & NIH PhenX Toolkit (Pescosolido et al. 1998); SFHR = Social Factors & HIV Risk Study (Morris & Rothenberg 2011); PES = Petrochemical Engineers Study (Cross & Cummings 2004); MLSFH = Malawi Longitudinal Study of Families & Health (Kohler et al. 2014); CGP = Chicago Gun Project (Papachristos et al. 2012)

surgery for weight loss) may be strongly affected by long-standing and intimate ties with whom interaction is less frequent. Research by Marsden and Campbell (1984) suggested that closeness or intensity of a relationship provides the best indication of tie strength, while frequency and duration of contact are more loosely related. An advantage of collecting information about tie strength is that these variables are predictive of a broad range of outcomes in virtually any social, political, or economic domain.

Name interpreters may also be used to obtain data about the *function* of ties – or the types of exchanges, services, or supports accessible through ties to alters (see Table 5.3). Network function is essentially a measure of what, how often, and how much alters and egos do for one another. Unlike content, which taps resources that are present but may be inaccessible, function is a measure of perceived or actual access to those resources. It can also include measures of how ego is connected to alters, since different types of relationships (e.g., friend, spouse, sibling) tend to have distinct sets of prescribed or normative functions.¹ For example, neighbors are useful for picking up mail while on vacation, but would be unlikely to loan large sums of money. In addition, name interpreters can also be employed to gather data on dysfunction, including conflict, hassling, violence, or burden. These so-called “negative” ties are often found to be more influential than positive or supportive ones, particularly in research on well-being, performance, and network dynamics (e.g., Rook 1984; Labianca 2014).

Function may be “built in” to networks by virtue of the particular name generators used to elicit alters, but can also be determined (or further specified) using name interpreters. For example, the PhenX Social Network Battery (Pescosolido et al. 1998) prompts respondents to name people they talk to about important matters and health matters, eliciting alters who function as discussants. Respondents are then asked whether alters fulfill a series of five additional support functions (e.g., instrumental, emotional, informational) using a series of name interpreters. Making distinctions between specific functions is important because particular types of support can be more or less effective for different outcomes.

5.1.3 Procedures for Administering Name Interpreters

Procedurally, name interpreters may be administered question-wise, alter-wise, or by roster. In question-wise format, one question is asked about each alter or a subset of alters in turn before moving to the next question. For example, an interviewer might ask, “Is [NAME 1] male or female? Is [NAME 2] male or female? And what about [NAME 3]?” ... etc. until the gender of all alters has been provided. Alter-wise format requires that

¹ The connection between ego and alter is sometimes used as a proxy for tie strength. However, using survey data to compare measures of tie strength, Marsden and Campbell (1984) found that the type of relationship is only weakly related to other indicators of strength, including closeness.

all name interpreters be asked of one alter prior to moving to the next alter. A typical script might be: “Is [NAME 1] male or female? How old is [NAME 1]? How close are you to [NAME 1]? . . .” etc. until all questions have been asked of the first alter. Some research suggests that the question-wise format is faster and produces higher quality data than alter-wise administration (Vehovar et al. 2008), while other research suggests that data reliability and validity is slightly better when alter-wise administration is used (Kogovšek et al. 2002). This discrepancy may be related to the cognitive complexity of the task at hand, with question-wise administration being superior for simpler items (e.g., reporting on alter gender, race, duration of relationship) and alter-wise for questions that are more subjective and difficult to evaluate (e.g., closeness, conflict, frequency of contact).

A special case of question-wise administration is the roster method. Here, a list of all alter names provided is shown to the ego respondent. Then, he or she is asked to choose alters that meet a particular criterion. For example, an interviewer might ask, “Which of these people you mentioned lives in your household?” or “Who, among the people on this list, would be able to loan you a large sum of money if you really needed it?” This approach is most effective when the criterion (1) applies to only a minority of alters on the roster, (2) aims to collect a lot information about alters without the tedium of a question- or alter-wise approach; and (3) has a binary response set. However, there may be a trade-off in accuracy. Alters on the list who fit the criterion might be missed. The longer question-wise format is preferable for questions with more complicated responses.

5.2 THE ACCURACY OF DATA ON ALTERS AND EGO-ALTER TIES

Standard survey design considerations (e.g., question ordering, number of response categories) apply to name interpreters, but egocentric research raises additional concerns. Specifically, much of the data obtained through name interpreters requires proxy reporting rather than self-reporting, which is the provision of data by one person on behalf of another. This is a source of nonrandom error in network data. Further, evaluation of relationships between an ego and alters is subject to a similar set of biases. Limitations associated with different kinds of name interpreters are key to facilitating survey design and helping to identify possible sources of error in ego network research.

However, accuracy in ego reporting is not always the goal of egocentric data collection. For researchers studying the adoption of new smart phone technology, for example, ego's *perceptions* of how many alters use this technology may be more predictive of ego's behavior than the actual number of alters that use it. In general, if researchers are interested in processes like social influence or diffusion of attitudes and behaviors, ego's perceptions about alters are often the causal factor, and accuracy is less relevant.

5.2.1 Issues in Proxy Reporting on Characteristics of Alters

Proxy reporters use different response strategies than self-reporters. Self-reporters tend to provide data that is detailed, based in experience, and biased by efforts to maintain a positive presentation of self (Sudman et al. 1994). In contrast, proxy reporters' perceptions of others are often grounded in their own experiences, behaviors, and beliefs. This creates convergence in self-reported ego and proxy-reported alter characteristics that is partially attributable to methodological artifact (Blair, Menon, & Bickart 1991; Epley 2008).

Relationship between ego and alter. The accuracy of proxy reports by ego respondents also depends, to some extent, on the nature of the relationship between ego and alter. Broadly, egos report more accurately on stronger core ties compared to weaker and more peripheral ones. Thus, the quality of proxy reporting is positively related to the amount of social interaction between ego and alter (Sudman et al. 1994). For example, more frequent contact predicts accuracy of reporting on friends' group identities (Reysen, Hall, & Puryear 2014). Among cohabitants, coworkers, and social group members, egos with more frequent interaction provide better assessments of others' communication patterns (Romney & Weller 1984). Along the same lines, proxy reporting is more accurate in the context of multiplex and intimate relationships, perhaps because these alters are more important or salient to the ego, or because more is known about them (Kogovšek & Ferligoj 2005).

Other research indicates that the accuracy of proxy reporting varies depending on how the ego and alter are connected. As part of a study of HIV status disclosure, Shelley et al. (2006) examined which types of relations know twenty-six different characteristics, ranging from blood type to marital status. They found that parents tend to know the most about respondents (25.5 items, on average), and that friends (12.3 items) know more than acquaintances, who know very little (only 5.1 items, on

average). On balance, individuals appear to be capable of producing fairly accurate reports about their spouses, children, siblings, and friends, depending on the type of data obtained, while reporting on other kinds of kin, coworkers, neighbors, and other weak ties is subject to substantial error (Williams & Thomson 1985; Nelson et al. 1994; Shelley et al. 2006; Triplett 2013; Reysen et al. 2014).

The nature of information being collected. The accuracy of proxy reporting also depends on the nature of the information being elicited. Respondents are capable of correctly reporting on alters' status characteristics, including age, educational attainment, and religious group membership (Laumann 1969b; Nelson et al. 1994). Most egos know alter occupation, employment status, marital status, age, and sexual identity (Shelley et al. 1995). However, people tend to know less about other alter characteristics such as travel habits, problems, health status, political party, and participation in social organizations.

Not surprisingly, when data are directly observable, the accuracy of proxy reporting is greater. For instance, comparing indicators of health, Nelson et al. (1994) found that proxy reporting of height, weight, BMI, smoking status, and alcohol use ranged from very accurate (intraclass correlation = 0.94) to fairly accurate (intraclass correlation = 0.82). However, less visible health indicators, such as use of hormone replacement therapy or diagnosis of hypertension were reported much less accurately. Along the same lines, in a study of Kenyan villagers, White and Watkins (2000) found that women reported observable alter data like number of children very accurately, but were much less likely to know whether their alters used contraception – a behavior that is not observable and often not discussed with others. Importantly, ego respondents answered this question even when they were uncertain, projecting their own contraceptive behavior onto others in the absence of information about alter behavior.

Proxy reporting is perhaps most problematic in cases where the characteristic is stigmatizing or non-normative. This pattern may be due to communication bias, or the tendency of people to selectively disclose non-normative attitudes or behaviors to those most likely to share their views. When people are asked to report on the private behaviors or attitudes of others, their inferences are dependent on the disclosure patterns of their social ties, which conform to their own views and experiences (Kitts 2003). For example, focusing on abortion and mis-carriage, Cowan (2014) found that private or stigmatizing information is shared selectively with like-minded or sympathetic members of social

networks. Thus, people are unlikely to be able to accurately report on alter characteristics that are private or stigmatized, and when they do, egos provide responses consistent with their own worldview (see also Shelley et al. 1995, 2006, on HIV-positive individuals).

Egos are also often inaccurate when evaluating others' attitudes, beliefs, or values. Focusing on friendship pairs among urban men, Laumann (1969b) identified considerable error in proxy reporting of friends' political party preferences. More recent research by Goel, Mason, and Watts (2010) suggested that when people are able to accurately identify friends' general and issue-specific political views, this is largely an artifact of homophily. That is, people with similar opinions are more likely to form relationships, and egos have a tendency to project their own views onto their alters, inflating the accuracy of proxy reporting on political attitudes. Consequently, there may be systematic biases in the accuracy of reporting on attitudes and opinions such that people with minority status or views are more likely to make errors. Moreover, when people are asked to identify the views of someone whose perspective differs from their own, they tend to rely on stereotypes or other idiosyncratic information about their alters, producing additional non-random error (Epley & Caruso 2008; Goel et al. 2010). For example, ego might wrongly assume that a friend does not support same-sex marriage because that friend voted republican in the previous election. In short, it may be problematic to expect egos to provide correct information about alter characteristics that are unobservable or subjective, particularly if this involves behaviors or views that are sensitive or non-normative.

5.2.2 Issues in Reporting on Characteristics of Relationships

Respondents' perceptions of relationships are typically more accurate than proxy reporting on characteristics of alters because ego has experience as a direct participant. Nonetheless, research suggests that there is sometimes disagreement between two parties about properties of a relationship. The extent to which this occurs is dependent on characteristics of individuals, relationships, and the type of information being obtained. In general, factors that affect the accuracy of ego reporting on alter characteristics also influence concordance, or similarity, of data collected from two people in a dyad.

Relationship between ego and alter. Relationship quality appears to influence whether two members of a dyad agree about support provision and

other exchanges. For instance, spouses with higher perceived marital quality are more likely to agree about whether supportive and unsupportive behaviors occurred in the context of the relationship (Christensen, Sullaway, & King 1983; Norton & Manne 2007). This may be due to “sentiment override” – the tendency of perceptions of specific behaviors to be shaped by global assessments of relationship satisfaction. Likewise, higher levels of intimacy, self-disclosure, and communication are associated with stronger agreement among dyads (Coriell & Cohen 1995; Gant, Calsyn, & Winter 1999). Thus, errors in reporting may be systematically biased such that people in weak or dissatisfying relationships provide poorer quality data on tie characteristics.

The nature of information being collected. Concordance is higher for observable, objective, and specific characteristics and behaviors (e.g., duration of relationship, sexual activity) than for more diffuse or subjective ones (e.g., conversation topics, relationship quality; White & Watkins 2000; Harvey et al. 2004). For example, accounts of emotional, informational, and financial support between dyads are more concordant than companionship and material support (Ferligoj & Hlebec 1999; Kogovšek & Ferligoj 2005). However, this pattern may be explained by the types of alters that are likely to provide emotional, informational, and financial support – namely, strong ties. In addition, agreement may be stronger for negative or unsupportive behaviors than positive ones, perhaps because negative events are more memorable (Norton & Manne 2007). Also, dyads are more likely to agree about whether a behavior or exchange happened than how often it happened, and reports become more discordant over time, consistent with forgetting (Bell, Montoya, & Atkinson 2000). Together, this research suggests that name interpreters focusing on the occurrence of concrete behaviors and exchanges that are time-limited (e.g., in the past six months) are likely to yield the most reliable data on ego-alter relationships.

5.2.3 Respondent Fatigue

When an individual’s social network is large, answering many questions about alters, particularly if they are cognitively complex, can be tiresome. Consequently, the probability of error may increase with network size and the number of name interpreters. In fact, respondents are likely to abandon the name-interpreting task altogether after about fifteen alters (Manfreda et al. 2004). Research also suggests that having information

on about fifteen to twenty alters is sufficient for producing reliable and valid measures of ego networks (McCarty et al. 1997; Hogan et al. 2007). Consequently, if large networks are obtained, drawing a random sample of alters for the name interpreter task and evaluation of alter-alter ties (see below) may be a useful strategy for reducing respondent burden (Marsden 1993).

5.3 MEASURING TIES BETWEEN ALTERS

In ego network research, respondents are often asked to report about the nature of connections between alters they have named. This is critical because alter-alter ties form the basis of structural measures of ego networks. Network *structure* is defined as the pattern of linkages between actors in a social network. Many different measures of network structure can be computed using information on alter-alter ties (see Chapter 7). Structural measures operationalize important social constructs like power, constraint, and the presence of structural holes.

Two basic methods for eliciting alter-alter ties for the purposes of capturing network structure are adjacency matrices (also called density matrices) and personal network mapping. The former asks about the relationship between every pair of alters, while the latter allows respondents to identify ties between visually depicted alters.

5.3.1 Adjacency Matrices

Table 5.4 presents a picture of an adjacency matrix from the PhenX Toolkit Social Network Battery. The diagonal is empty because ties between an alter and him/herself remain unmeasured since they have no relevant meaning. The matrix is also symmetric, reflecting undirected (i.e., reciprocal) ties between alters. Ties can be unvalued (i.e., binary), signifying presence or absence of a relationship between two alters, or they can be valued to denote the strength or nature of the tie.

A series of name interpreters asking about the relationship between each pair of alters is used to complete the matrix. Examples of common alter-alter name interpreters are presented in Figure 5.3. Typically these ask about the closeness or level of contact, such as whether the pair of alters are strangers, acquaintances, or know one another well. However, it is possible to ask about particular types of exchanges or relationships to create subnetworks. For example, the Social Factors and HIV Risk

TABLE 5.4 *Adjacency matrix from PhenX Toolkit Social Networks Protocol*

Please think about the relations between these people you told me about earlier. Some of them may be total strangers in the sense that they wouldn't recognize one another if they bumped into each other on the street. Others may know each other a bit, or they may be especially close. First, think about (1) and (2). How close are they to one another:

1 = Very close

2 = Sort of close

3 = Not very close

4 = Don't know each other

8 = I don't know

	2	3	4	5	6	7	8	9	10
1	_____	_____	_____	_____	_____	_____	_____	_____	_____
2		_____	_____	_____	_____	_____	_____	_____	_____
3			_____	_____	_____	_____	_____	_____	_____
4				_____	_____	_____	_____	_____	_____
5					_____	_____	_____	_____	_____
6						_____	_____	_____	_____
7							_____	_____	_____
8								_____	_____
9									_____

Notes: Indianapolis Network Mental Health Study and NIH PhenX Toolkit (Pescosolido et al. 1998)

Study (Morris & Rothenberg 2011) measured sexual and drug use relationships between and among ego and all alters, in addition to gathering information on alters' HIV risk behaviors and seropositive status. These data were then used to quantify potential exposure to HIV or other communicable infections through sexual contact and needle sharing associated with intravenous drug use, which depends, in part, on the structure of the ego network.

While the adjacency matrix is the most common strategy for capturing alter-alter ties, it is extremely burdensome. If respondents report on the undirected relationship between every pair of alters, the number of comparisons increases nonlinearly with network size (McCarty et al. 2007). The formula for the number of comparisons is:

TABLE 5.5 *Examples of name interpreters for measuring alter-alter ties*

Name interpreter	Source
How well do [NAME 1] and [NAME 2] know each other?*	GSS
What is the likelihood that these two people talk to each other when you are not around?	DANL
That is, what is the probability that these two people have a relationship independent of you? Is it not at all probable, somewhat probable, or very probably?	
To your knowledge, do [NAME 1] and [NAME 2] know each other? Yes or no?	SFHR
Do [NAME 1] and [NAME 2] have sex with each other?	
Do [NAME 1] and [NAME 2] do drugs with each other (get high together)?	
Finally, I would like to briefly ask you how well all the people you told me about know each other. Are [NAME 1] and [NAME 2] confidants, just friends, acquaintances, or do they not know each other?	MLSFH

* Standard name interpreter; Notes: GSS = General Social Survey; DANL = Dynamics of Actors and Networks across Levels (Lubbers et al. 2010); MLSFH = Malawi Longitudinal Study of Families and Health (Kohler et al. 2014); SFHR = Social Factors and HIV Risk Study (Morris & Rothenberg 2011)

$$\frac{n^2 - n}{2}$$

Thus, the number of alter-alter evaluations required for a small network containing five alters is only ten, but this number increases to forty-five for ten alters and 190 for twenty alters. Figure 5.1 depicts the relationship between number of alters and pair evaluations, demonstrating the non-linear effect of increasing network size.

A more efficient method for completing an adjacency matrix does not require asking about each pair of ties. Similar to the roster method described above, an alter's name is listed at the top of a page or screen, and all other alters are listed below. Ego is then asked to identify any alters that meet the specified criterion, for example, "Who would [NAME 1] go to for advice?" This provides a good alternative for respondents reporting a large number of alter-alter pairs, or for online self-report surveys that are vulnerable to satisficing (see Chapter 3). However, the need to scan a (potentially large) list of alters to look for matches may compromise accuracy.

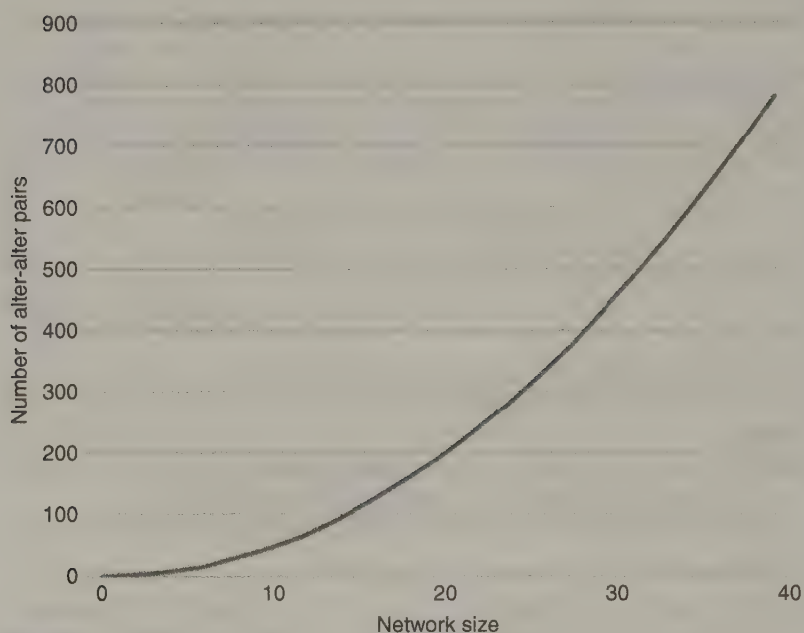


FIGURE 5.1 Relationship between network size and number of evaluations of alter-alter pairs

Methods for reducing respondent burden associated with systematically evaluating ties between pairs of alters, including randomly selecting a subset of alters, have been explored. Structural measures (e.g., density) are sensitive to overall network size and the fraction of ties sampled (Anderson, Butts & Carley 1999; Costenbader & Valente 2003). Some minimum number or proportion of alter-alter ties must be obtained to preserve data quality and the ability to compare across ego networks. McCarty, Killworth, and Rennell (2007) have suggested that a random subset of ten to twenty alters sampled from a larger list is sufficient to capture certain structural patterns. Other tested alternatives include proxy measures that ask respondents to estimate the density of connections between named alters with a single item (e.g., How many of your friends know one another? Would you say that all your friends know one another, most, only a few, or none?). However, these provide unreliable estimates of network structure that correlate weakly with direct measurements using adjacency matrices (Burt 1987a).

While adjacency matrices are a high-burden method for eliciting alter-alter ties, they are ideal for capturing a broad range of structural

properties of personal social networks. Because the matrix method asks about each alter-alter pair, the structure of the network emerges from patterns of dyadic relationships. Therefore, the resulting structure is not influenced or constrained by respondents' cognitive biases about the organization of their own networks (McCarty et al. 2007). In other words, it is simpler for respondents to accurately report on relationships between all pairs of alters than to identify relationships from among a large list of alters, or to respond to complex questions about the overall structure of their networks. However, because reporting on alter-alter ties is tedious, respondents may become fatigued or even drop out of the exercise (particularly when using web-based surveys), compromising the quality of the data (Manfreda et al. 2004). Consequently, randomly sampling a subset of alters for the adjacency matrix task may produce the most reliable estimates of network structure when the network contains more than about twenty alters (McCarty et al. 2007).

5.3.2 Mapping and Visualization Methods

Network mapping and other visualization tools can also be used to elicit information about ties between alters. In network mapping, egos draw their personal networks rather than using name generators and/or interpreters to elicit alters and information about them. These may be drawn freeform or using computer software, and may be based on observation or self-report. Typically, alters' names are depicted on paper or on a screen, with lines denoting ties between alters. Some researchers use concentric circles or other tools to aid respondent recall of network members and their connections to one another and to ego (Antonucci 1986). The participant-aided sociogram method (Hogan et al. 2007) uses name generators to elicit names of alters and asks egos to place alters who know one another in closer proximity on a large sheet of paper. These distances can then be quantified to represent a valued tie (where lower values signify closer relationships) between alters. Using this approach, the entire process of identifying alter-alter ties in networks with an average size of twenty-four alters takes about fifteen minutes. Moreover, egos report this as an enjoyable and informative exercise.

Newer technologies for collecting network data, including alter-alter ties, electronically using visualization are quickly gaining momentum because of their efficiency and ease of use. For example, the open-source program OpenEddi (see Figure 5.2) is an interactive online tool for obtaining data on large or small networks whether egocentric and

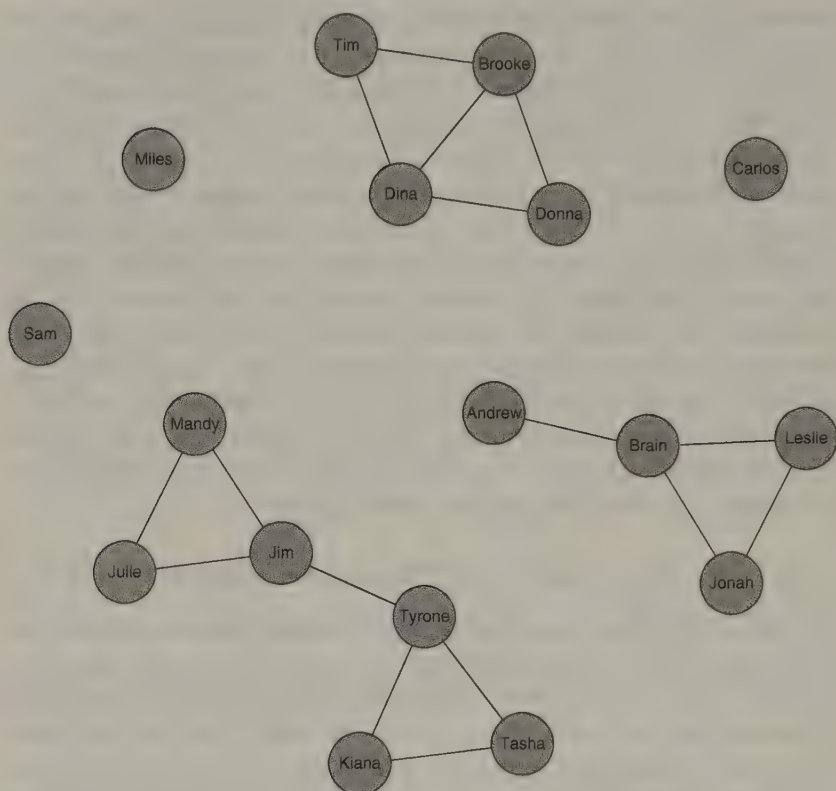


FIGURE 5.2 Web-based software designed to facilitate reporting of ties between alters

Source: OpenEddi, Fagan, and Eddens (2016)

sociocentric (Fagan & Eddens 2014). Following a series of customizable name generators and interpreters, alter nodes are depicted on the screen and respondents are asked to draw lines with their mouse (on a computer) or their finger (on a tablet) between nodes that have a particular type of connection, defined by the researcher. The nodes move closer together into groups once connected, enabling the user to see the structure of their network developing as they complete the task. This method for eliciting ties between alters is user-friendly and very efficient. Moreover, the dynamic interactive features may improve the accuracy of reporting, particularly for alters embedded in cliques. Alters more likely to be directly connected become closer in proximity once they are indirectly connected through other alters. These and similar programs for electronic

network data capture are efficient and intuitive. They may be the only practical option for obtaining data on the structure of large ego networks without sampling a subset of alters.

5.4 CONCLUSION

Most data for ego network analysis is generated by name interpreters, collecting information about alters, characteristics of relationships between an ego and alters, and alter-alter ties. These data form the basis of measures of the content, strength, function, and structure of ties in an ego network. Like name generators, name interpreters may be specific to a particular research question or outcome, or they may be more general and multipurpose. Standard survey design considerations apply to name interpreters, but issues like the accuracy of proxy reporting and information biases raise additional concerns unique to egocentric research. In particular, balancing data quality and efficiency is critical since name interpreters and methods for evaluating alter-alter ties are extremely burdensome for respondents and interviewers.

PART III

Analyzing Ego Networks

Visualizing Network Data

Network visualizations facilitate the process of identifying patterns in ties between alters, and provide insights about ego's position in the network. In other words, network diagrams help researchers to develop hypotheses. Visualizations also provide an effective means of communicating findings to others. This chapter begins with a discussion of a typical ego network diagram, focusing on patterns and elements that are typically of interest. A more technical discussion of methods for creating and embedding information in the diagrams with visualization software follows. Finally, the chapter ends with specific applications of network visualization, including the use of diagrams to collect network data and to depict network change.

6.1 A SIMPLE NETWORK DIAGRAM

Figure 6.1 shows a standard ego network diagram. Ego (node 22) is located in the center, and, by definition, has ties to everyone else in the network. Ties among the alters are also depicted, and the diagram reveals a somewhat clustered structure with multiple groups. Ego links everyone together, but there is another node (17) that also serves as a bridge between other alters. In this network, that node is ego's husband, and the cluster at the bottom right is the husband's family. The cluster at the top left is ego's family. The two nodes at the top are people that ego knows through People for the Ethical Treatment of Animals (PETA), where she volunteers. Finally, the group at the right are work friends. Figure 6.2 adds all of this information as annotations drawn on top of the diagram.

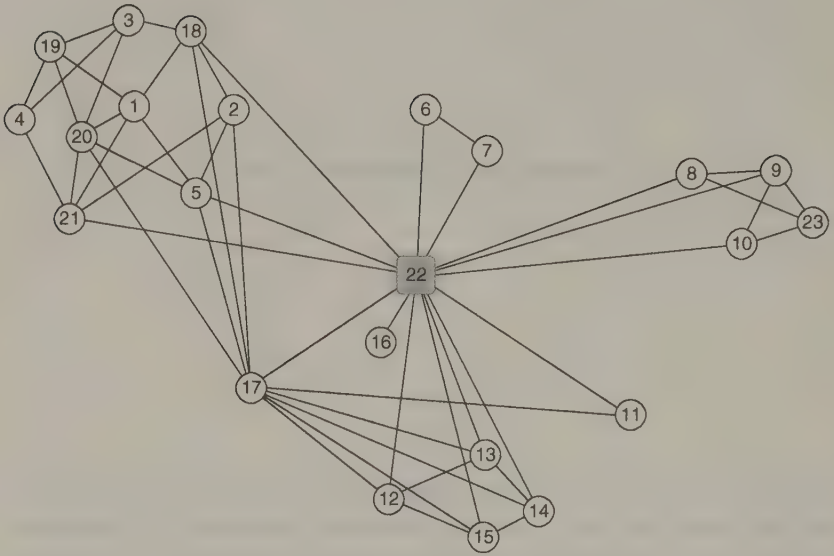


FIGURE 6.1 Simple ego network diagram

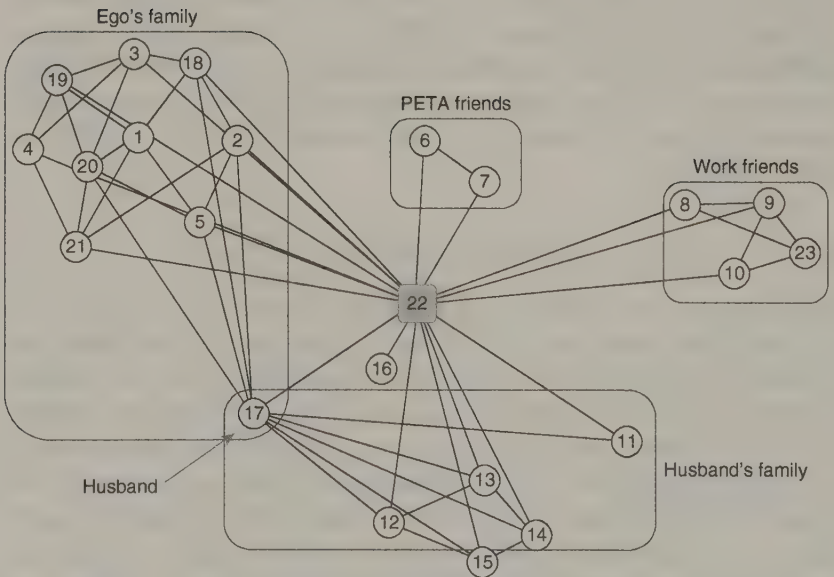


FIGURE 6.2 Simple ego network diagram with annotations

The annotated network diagram, which combines the topology of the network with attributes of the alters, is quite informative. At a single glance, the viewer can learn quite a bit about a person's life. Ego appears very family oriented. Her work friends are fewer, and she has not involved them in her family life. Even her husband has no connection to her work friends. Her PETA friends are also segregated from other types of alters. Her involvement with that organization may represent a unique interest or a community of respite. The network seems highly compartmentalized, though this hypothesis must be confirmed through comparisons with other ego networks.

Indeed, a key reason to visualize networks is to detect patterns and generate hypotheses. Consider Figure 6.3, in which the nodes (including ego) have been colored by political party. The light nodes are Republicans, and the dark nodes are Democrats. Ego is a Democrat, along with her work friends and her husband's family, but her own family is Republican.

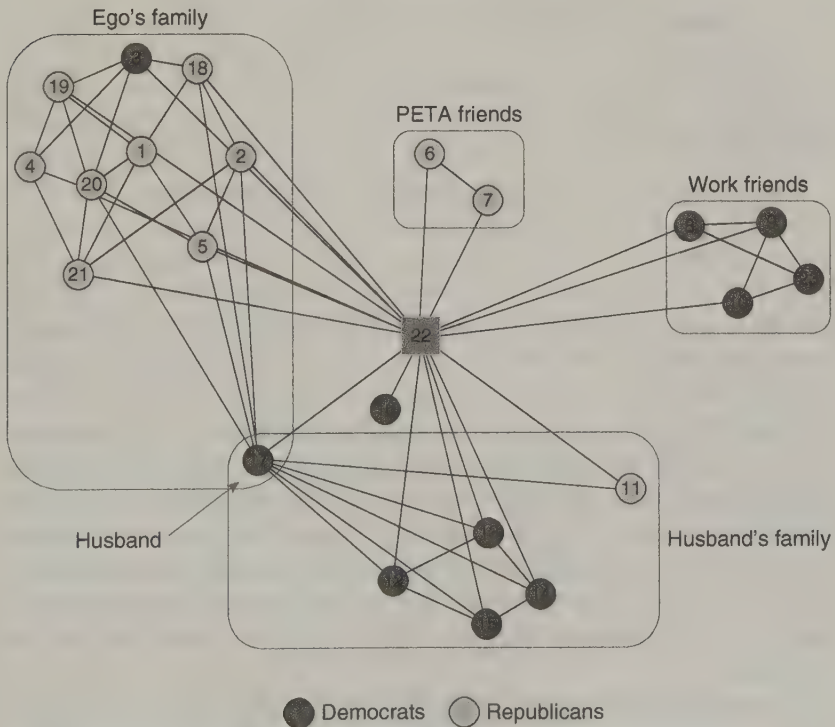


FIGURE 6.3 Simple ego network with political affiliations

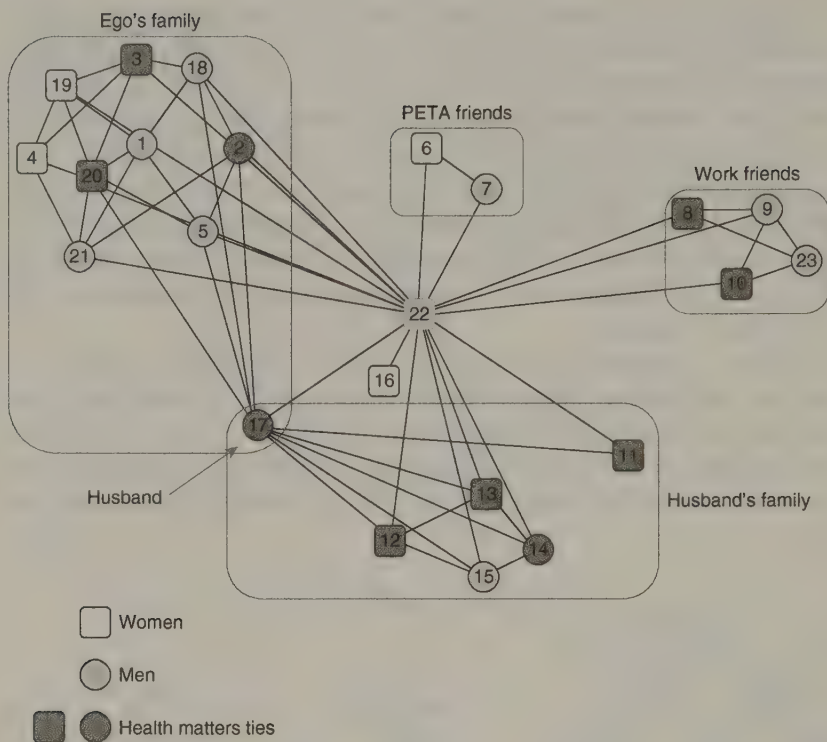


FIGURE 6.4 Simple ego network with gender and type

Ego likely avoids talking politics with her family. Political differences might also explain the paucity of ties between ego's husband's family and ego's family.

Figure 6.4 replaces political party information with two alter attributes. Shape represents gender (squares are women), while color indicates whether ego discusses health issues with the alter (dark is yes). The pattern seems clear – ego is much more likely to discuss health matters with other women than with men. However, it is important to note that this is just an impression based on one network, and should be tested using the entire sample of ego networks, ideally with the multilevel models discussed in Chapter 8.

Up to this point, we have implicitly assumed that the only consumer of a network diagram is the researcher, who is looking to detect patterns and generate hypotheses. However, one of the most powerful

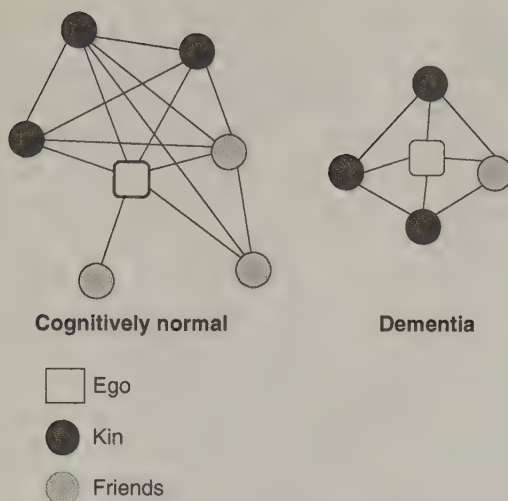


FIGURE 6.5 Networks of individuals with and without dementia

ways to use an ego network diagram is to discuss it with the respondent whose data it represents. They are in a unique position to explain patterns observed in the diagram, such as the apparent tendency to discuss health matters primarily with other women. The researcher can use the network diagram as a starting point for collecting additional rich qualitative data.

Another powerful way to use network diagrams is to compare different kinds of egos. For example, patients with dementia have characteristically different networks than people who are cognitively normal (Chapter 2). To illustrate this graphically, a composite or prototypical network diagram can be used to illustrate the typical network of an ego with a particular characteristic. Figure 6.5 shows two network diagrams which visually communicate two consistent differences between the personal networks of cognitively normal people and those of people with dementia (Perry et al. 2017). The network representing people with dementia is smaller and has proportionally fewer friends than the one representing those who are cognitively normal.

6.2 EGO IN OR OUT?

As discussed in several chapters of this book, the analysis of ego networks may or may not include ego. Visualization is no exception. On the one hand, having ego present in the center of the diagram is a powerful

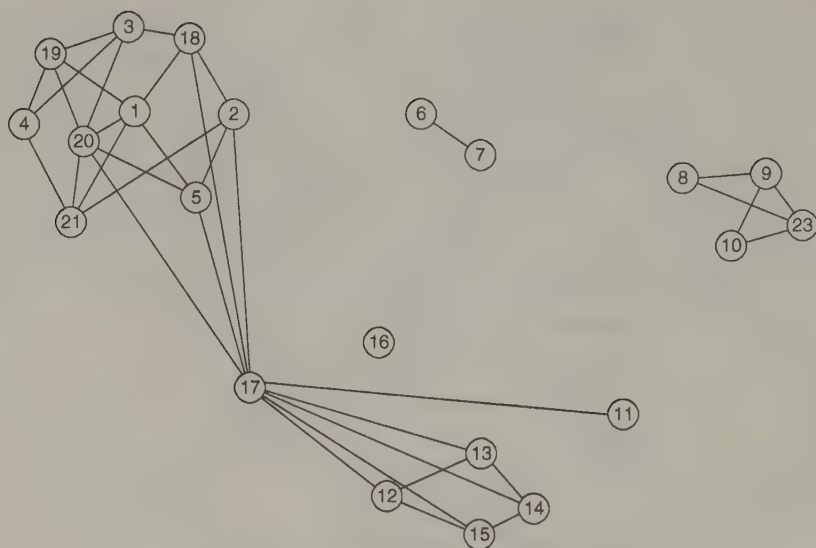


FIGURE 6.6 Simple ego network with ego excluded

reminder that the data depict an ego network, and provides a visual anchor that facilitates comparison across different ego networks. On the other hand, including all those ego-alter ties adds clutter, making the pattern of ties among alters less evident. Figure 6.6 depicts the same network shown in Figure 6.1, but without ego. This visualization emphasizes the disconnection between ego's different groups of alters. It also highlights the husband's role as a bridge between the families, which are otherwise disconnected. Finally, it highlights the one friend of ego that is also a friend of her husband (node 11).

An important consideration when examining diagrams of ego networks – which has bearing on the ego-in/ego-out question – is that frequently the ties between ego and alter are not of the same type as the ties between alters. In general, the ties among the alters are all of one type and tend to be very general (e.g., do they know each other? Do they communicate regularly?). The ties between ego and alter may be all of one kind, but they can also be different, as when one alter was named in a family name generator and another in a coworker name generator. When the ties among alters are systematically different from the ties between ego and alter, excluding ego from the diagram can prevent misunderstandings about the nature of alter-alter ties. Alternatively, alter-alter ties can be drawn differently, as discussed in Section 6.3.

6.3 WORKING WITH NETWORK VISUALIZATION SOFTWARE

The annotated diagram in Figure 6.2 is, for most purposes, more useful than Figure 6.1. The difference is the incorporation of additional data about the nodes and their relationships. Several software packages offer ways to incorporate these data into the visualization automatically, and any software capable of visualizing networks can be used to draw ego networks – the program does not have to be specifically designed for egocentric data. Figures 6.1 through 6.4 were drawn using Visone (Brandes & Wagner 2004), which is largely used for whole networks. An advantage of Visone is the ease of creating networks visually, without having to construct a data file first. Three other well-known network visualization packages are Pajek (Batagelj & Mrvar 1998), known for its ability to handle very large networks, Gephi (Bastian, Heymann, & Jacomy 2009), known for its versatility, and NetDraw (Borgatti 2002). The chief advantage of using Netdraw is that it is closely integrated with UCINET (Borgatti, Everett, & Freeman 2002), which is convenient for those conducting most of their quantitative analyses in that program.

When using a whole network visualization program with ego networks, there are two main strategies. The first is to create a separate data file for each ego network. While simple, it does raise the question of how to efficiently draw and store hundreds of ego networks. Most programs have batch or scripting languages that can automate the process. Table 6.1 shows a batch file for NetDraw that reads, draws, and outputs diagrams for three networks. Diagrams are saved as Windows metafiles – vector-based graphics that can be scaled up or down without loss of resolution, and which can be edited in PowerPoint or other programs. The second strategy is to join all of the ego networks into a single, large data

TABLE 6.1 *NetDraw batch commands
for drawing three ego networks*

```
Loadvna ego1.vna
Colornodesbyattr gender
Savemetafile ego1.emf
Loadvna ego2.vna
Colornodesbyattr gender
Savemetafile ego2.emf
Loadvna ego3.vna
Colornodesbyattr gender
Savemetafile ego3.emf
```

TABLE 6.2 *NetDraw batch commands to read a galaxy file and draw two individual ego networks*

Loadvna galaxy.vna
Drawego index001
Runlayout
Drawego index012
Runlayout

file, forming a giant network of unconnected components that correspond to ego networks. That done, the entire network can be drawn, forming what we term a “galaxy diagram” (Figure 6.7). Each ego network appears as an independent “solar system,” unconnected to any other solar system (Maya-Jariego & Holgado 2005; Maya-Jariego & Armitage 2007). Alternatively, in NetDraw it is possible to cycle through each ego network one at a time, either via on-screen controls, or via the batch language (see Table 6.2).

In addition to whole network visualization programs, there are a number of software packages designed specifically for ego network visualization. This includes all of the software specifically designed to collect egocentric network data, such as Egonet (McCarty 2003), VennMaker (Kronenwett & Schönhuth 2011), OpenEddi (www.openeddi.com/),

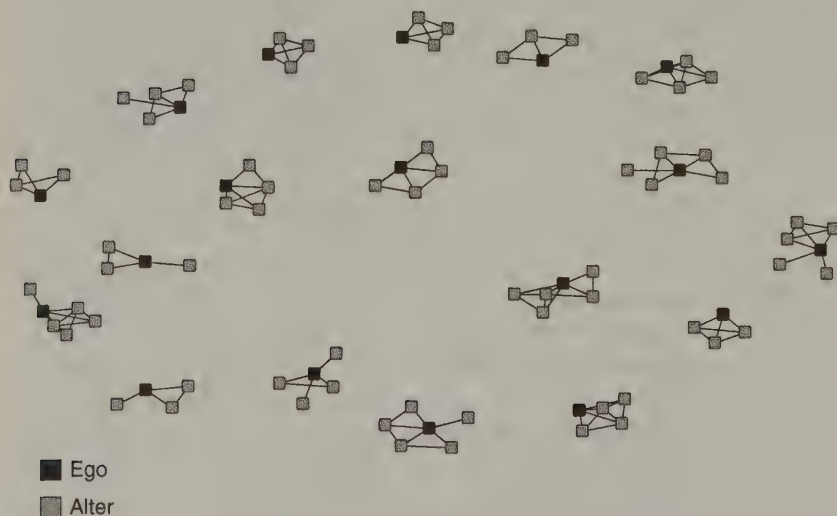


FIGURE 6.7 Galaxy diagram produced by NetDraw

and netCanvas (Hogan et al. 2016). In addition, there is E-Net (Borgatti 2006), which is not designed for data collection, but has the ability to visualize ego network data.

6.3.1 Alter Attributes

Most network visualization software has the capability of showing the values of alter variables by hovering over or right-clicking on a node. For example, Figure 6.8 shows a screenshot from NetDraw, where the user has right-clicked on the node at the top center of the figure. This opens a table of attribute values for that node. This feature facilitates interactive manipulation of a network, and has the advantage of showing the totality of an alter's characteristics simultaneously. For presenting network data in a publication or presentation, however, the interactivity is not helpful. Additionally, seeing all characteristics of one node often impedes comparison across all nodes. For comparing alters, it is typically more effective to visually depict just two or three characteristics at a time, such as gender or age.

One approach to comparing alter characteristics visually is to embed the information in the symbols used to represent each node in the diagram. For example, we can represent gender using the shape of the symbol, such as using circles for women and squares for men. Color is especially effective if

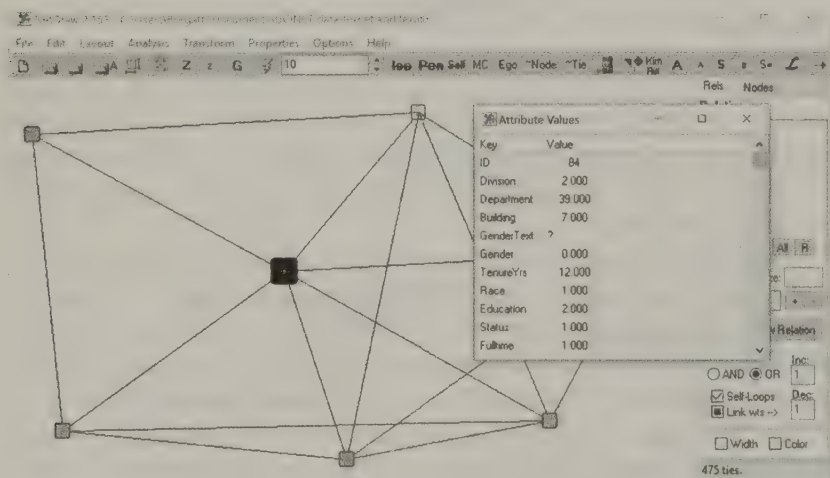


FIGURE 6.8 NetDraw screenshot* (grayscale)

* User calls up all attributes of a node by right-clicking on it

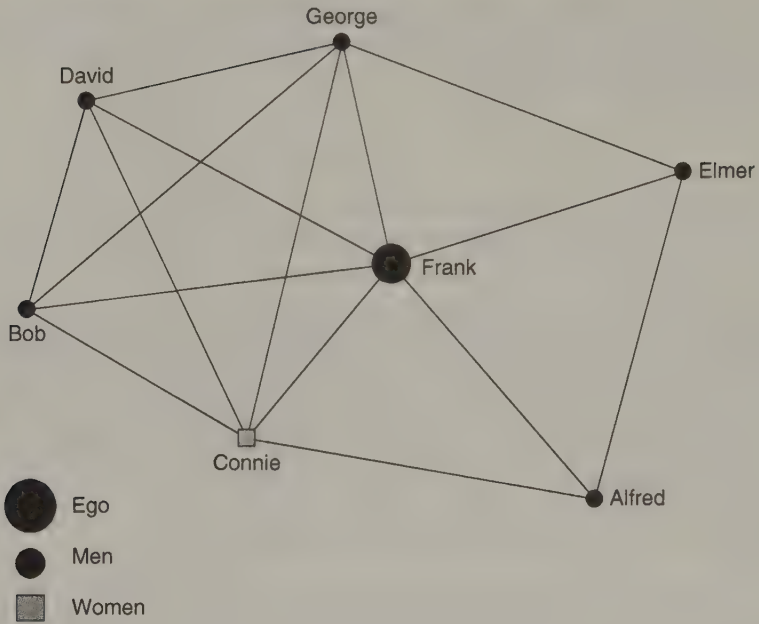


FIGURE 6.9 Ego network with gender

the publication medium allows it. In Figure 6.9, men are represented by dark circles and women are light squares. Frank, the dark circle in the middle, has a highly homogeneous and homophilous network, with only one woman in his personal network. In general, visual properties like shape and color are effective for representing categorical variables, whereas properties like size, color gradients, or thickness of the rim around a symbol are effective for continuous variables. Figure 6.10 uses symbol size to represent the age of each person.

In sociocentric research, labels are often used to present the names of nodes, when assurances of confidentiality are not violated. In egocentric research, the names are typically less important. In this case, the text labels may be used to identify alter characteristics by writing the values of the node attribute. For instance, we can identify each alter by age or occupation. Figure 6.11 uses text labels to indicate each node's occupation, including ego's. Of course, text labels can also be different sizes, colors, and fonts, and all of these may represent different node attributes. However, it is easy to overload the viewer with too much visual information. It is best to limit the number of meaningful alter attributes embedded in a diagram to about three (Brandes, Kenis, & Raab 2006).

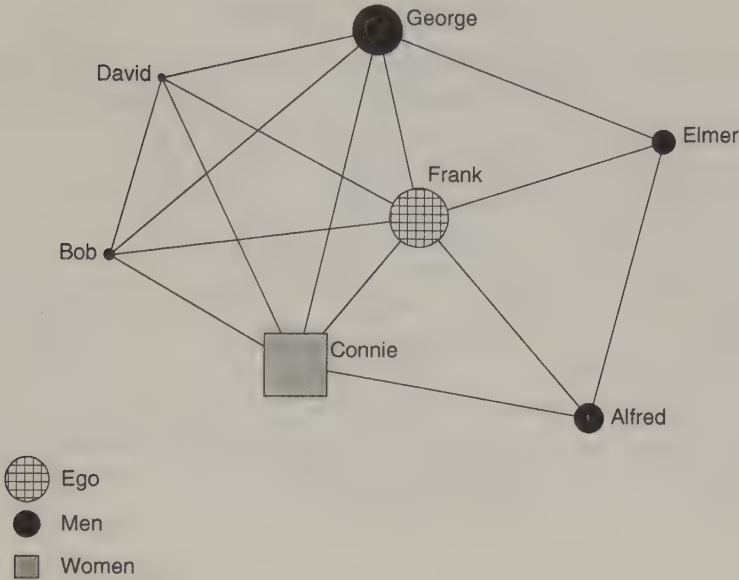


FIGURE 6.10 Ego network with age* and gender
* Larger symbol indicates older age

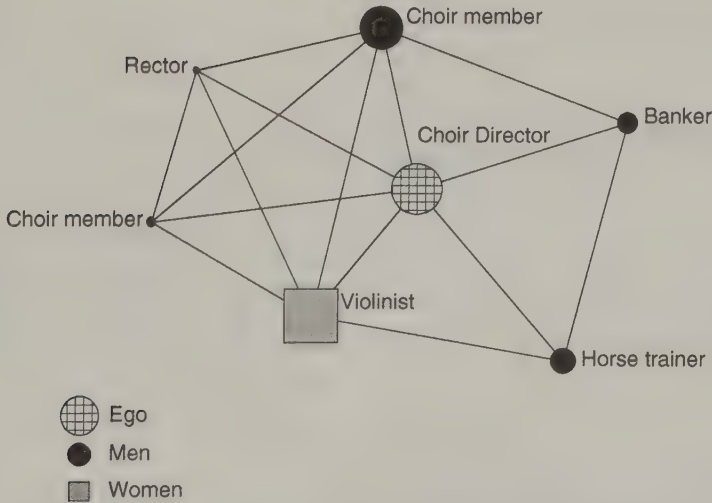


FIGURE 6.11 Ego network with age*, gender, and occupation
* Larger symbol indicates older age

6.3.2 Relational Attributes

As noted elsewhere, the relationship between two nodes can be multiplex. In part, this refers to the different kinds of ties that can bind people (e.g., neighbors, friends, kin, and coworkers). It also refers to different dimensions of a given type of tie. For example, friendship ties vary in years of duration, emotional intensity, and conflict, among other attributes. All of these properties can be represented in a network diagram.

For depicting categorical relational attributes, the obvious visual properties are color, dashed lines, and text labels. For instance, we might use blue for friends and red for coworkers. For pairs of nodes that are both friends and coworkers, a third color, such as violet, could be used to indicate that both kinds of ties are present. Alternatively, multiple types of lines, such as solid or dashed, can be used. Figure 6.12 shows marriage ties and business ties between Florentine families during the Renaissance, with the Medici

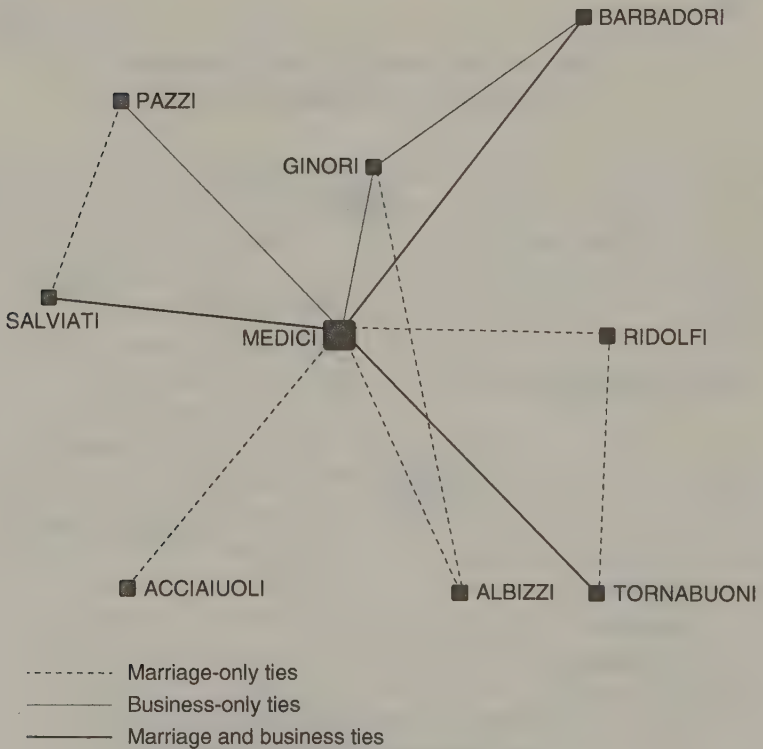


FIGURE 6.12 Ego network of the Medici family

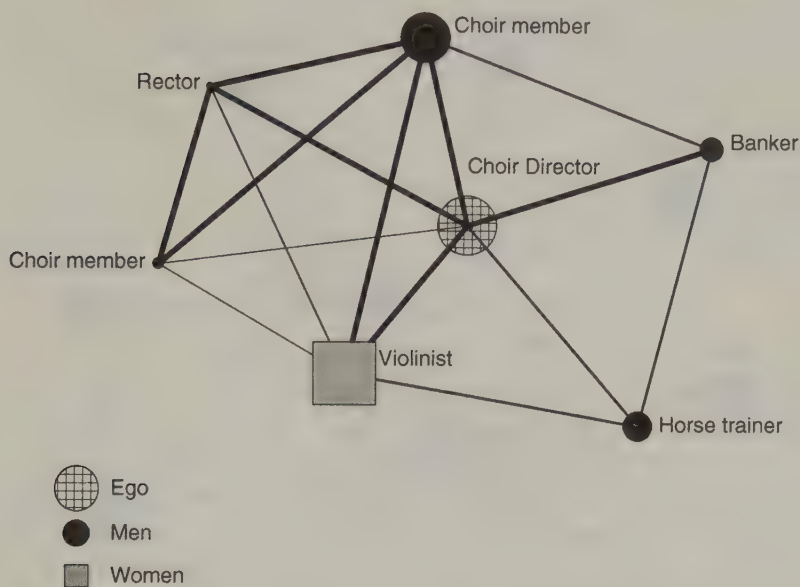


FIGURE 6.13 Ego network with age*, gender, occupation, and tie strengths**
 *Larger symbol indicates older age **Affective strength indicated by thicker lines

serving as the focal node (Padgett & Ansell 1993). Dashed lines indicate marriage ties only, thin solid lines indicate business ties only, and dark thick solid lines indicate relationships where both ties are present.

For continuous relational attributes like strength of tie, the width of the line is effective, although very thick lines can be aesthetically unappealing (see Figure 6.13). Gradients from light to dark can also be useful if the lines are wide enough that gradients are distinguishable. Finally, edge labels can be used, in written form, to indicate the strength of tie. Figure 6.14 shows an example. Note that in these data, ties among alters are directed, so there can be a line from alter A to alter B, and a separate one from B to A, and these need not have the same strength.

6.3.3 Layouts

The layout of a network diagram refers to the positions of nodes. Software packages principally designed for whole networks typically use a graph layout algorithm to locate nodes in the diagram space. Many of these algorithms take a force-directed approach in which connected nodes attract each other while disconnected nodes repel. Well-known examples

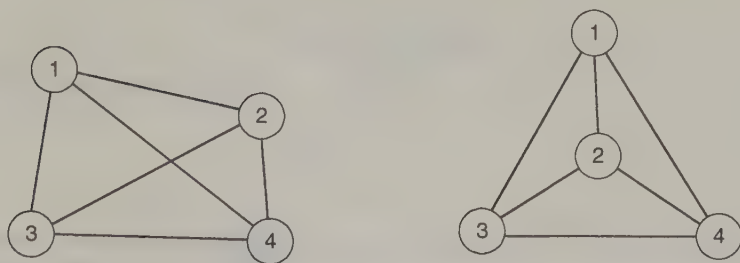


FIGURE 6.15 Two representations of the same network

Figure 6.15 shows two drawings of the same network. They are equally valid, since the exact same connections are present in both. However, the graph on the right gives the misleading impression that Node 2 is more central than the others. Consequently, interpreting any particular visualization without careful attention to supporting quantitative analyses can be problematic.

When applied to ego network data with the ego included, algorithms have a strong tendency to place ego near the center of the picture, but this is not guaranteed. Programs designed specifically for visualizing ego network data typically do not leave this up to chance and fix ego in the center of the picture (or leave ego out entirely). The layout criteria may be slightly altered so as to be governed primarily by the relations with respect ego. For example, several programs use concentric circles to define bands that indicate a certain level of closeness to ego, based on amount of interaction, closeness of tie, or spatial distance. Nodes placed in the inner bands have a closer relation with ego than those placed in the outer bands. Subject to the constraint of keeping nodes within their appointed bands, the algorithm then distributes them to reflect social proximities to each other, along with aesthetic criteria such as avoiding one node obscuring another. Other ego net programs establish sectors or pie slices that define larger categories of nodes. For example, all work friends are placed in one sector and all school friends in another. This feature is characteristic of software developed to collect the data using visualization tools (see Section 6.5).

6.4 ABSTRACT VISUALIZATION OF NETWORK CHARACTERISTICS

An alternative approach to visualizing ego network data is to depict network characteristics rather than the networks themselves. This is often

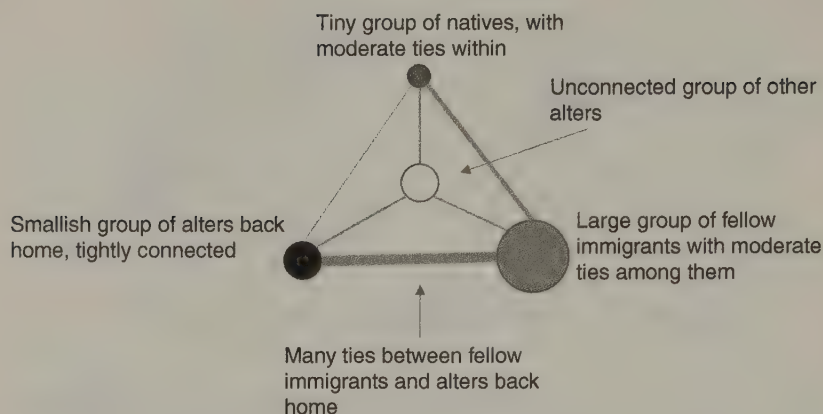


FIGURE 6.16 Classes of network ties among Argentine immigrants
Adapted from Brandes et al. (2008)

more space-efficient, making it easy to compare ego networks on chosen characteristics. For example, in a study of Argentine immigrants in Spain, Brandes and colleagues (2008) defined four classes of alters: native Spaniards, fellow Argentine immigrants, Argentines in Argentina, and a residual class of all others. For a given ego network, they counted ties between these classes, creating a new reduced network known as a block-model, or an image graph. What was then drawn is this reduced model rather than the actual ego net. An example is shown in Figure 6.16. In this figure, locations are fixed so the top node always represents native Spaniards, the bottom right node is fellow Argentine immigrants, the bottom left node is alters back in Argentina, and the middle node is all others. This facilitates comparisons across egos. Node size reflects the number of alters in each category. In Figure 6.16, ego's biggest group of alters is fellow immigrants, and the smallest group of alters is native Spaniards. Line thickness represents the extent of ties between the two groups of alters, and darkness of nodes represents the density of ties within a group.

Although it requires time to internalize what the different features of this visualization represent, these diagrams are extraordinarily efficient for presenting complex information. Combined with their small size, this makes them very effective for showing multiple ego networks side by side for comparison. Figure 6.17, borrowed directly from Brandes and colleagues (2008), displays each individual ego net in the dataset. This method is quite versatile. For example, Maya-Jariego, Holgado, and Florido del

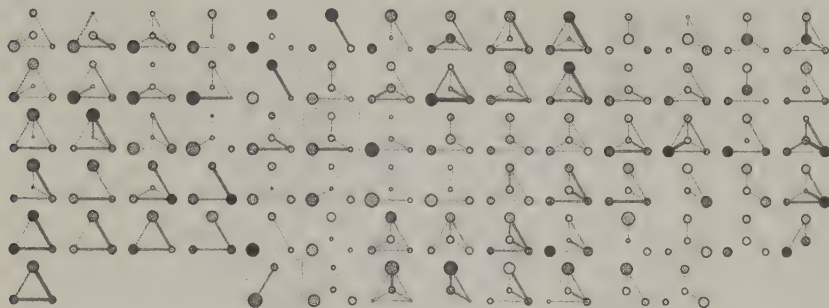


FIGURE 6.17 Comparison of characteristics of immigrants (grayscale)

Source: Brandes et al. (2008)

Corral (2016) used this technique to compare the networks of Atlantic and Mediterranean fishermen.

Scaling methods, such as multidimensional scaling (Kruskal 1964) or correspondence analysis (Greenacre 1984), can also be used to depict patterns in ego network data. Burt, Hogarth, and Michaud (2000) used this approach in their study of differences in the ego networks of French and American managers. Respondents completed a nine-item name generator that elicited alters in multiple domains – discussing personal matters, socializing, boss or subordinate relationships, discussing leaving the firm, highly valued ties, difficult ties, and alters needed for buy-in. In addition, for each alter, egos reported emotional closeness, frequency of contact, duration of relationship, and knowing the person prior to joining the firm. After dichotomizing ordinal variables with multiple categories, a correlation matrix of all variables was produced. This process was completed separately for each group of managers. Finally, each correlation matrix was submitted to multidimensional scaling for visualization. The results for the French managers are shown in Figure 6.18a, and for American managers in Figure 6.18b. In the diagram, the proximity of two characteristics corresponds to their co-occurrence among managers' alters. For example, “valued” and “buy-in” are near each other in the maps because managers tended to value coworkers from whom they needed to obtain buy-in.

In both Figures *a* and *b*, there is a negative to positive dimension running horizontally from left to right. Further, there is a vertical dimension that corresponds to instrumental (top) and expressive (bottom) relations. Interestingly, among French managers, expressive ties were found to be their relationships of longest duration (10+ years), whereas

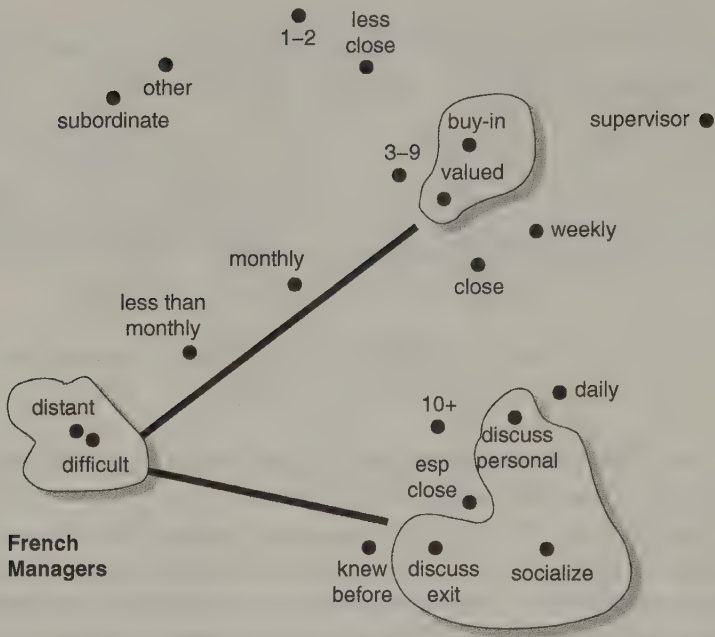


FIGURE 6.18A Multidimensional scaling of alter characteristics for French managers

Source: Burt et al. (2000)

for the Americans, work ties were those of longest duration. Similarly, the contacts that French managers knew prior to joining the firm were personal ties, whereas for the Americans they were work ties. Based on other findings of the study not reported here, Burt et al. (2000) concluded that social capital functions the same way for French and American managers, but the etiology is different. Americans are anchored on their work contacts, and come to a firm with long-standing relationships already established. French managers arrive with less social capital and must build it as they go.

6.5 VISUAL DATA COLLECTION THROUGH NETWORK MAPPING

Network mapping has long been used in genetic and clinical research to visualize families and other important associates (McGoldrick, Gerson, & Petry 2008). Anthropologists have also employed mapping

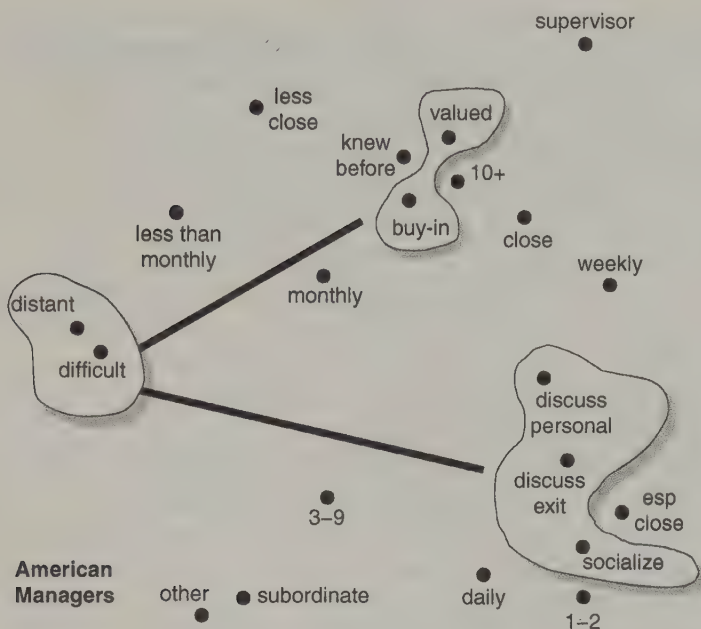


FIGURE 6.18B Multidimensional scaling of alter characteristics for American managers

Source: Burt et al. (2000)

techniques to understand networks of kinship, exchange, warfare, and other social phenomena (Borgatti et al. 2009). Traditionally, personal network maps are drawn freeform on the basis of observation or self-report. Names of network members may be written on paper, with lines denoting ties between alters, circles indicating subgroups, and overlapping circles signifying linkages or bridges between subgroups. Alternatively, some researchers use concentric circles or other tools to aid respondent recall of network members and their connections to one another and to ego (Antonucci 1986). Others take a more hands-on approach, using paper, colored pens, post-it notes, plastic discs, and figurines to elicit qualitative and quantitative data about network structure and influence (Schiffer & Hauck 2010).

Network visualization methods for data collection have a number of significant advantages. First and foremost, these tools are efficient and intuitive, and may be the only practical option for obtaining data on the structure of large ego networks without sampling a subset of alters. Having a visual representation of ego's network structure during the

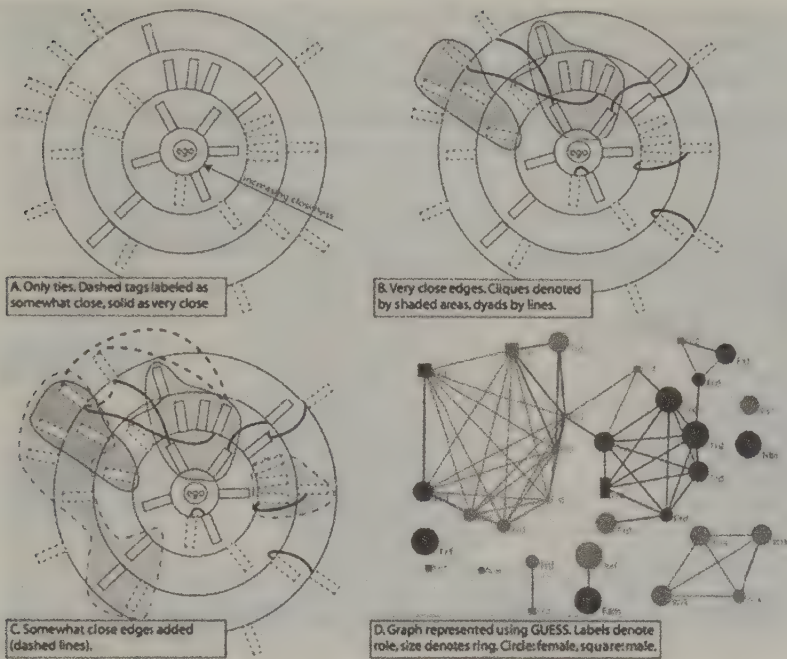
data collection process also provides opportunities to question respondents about interesting patterns or anomalies. A researcher could ask why a particular alter is an isolate, how alters embedded in a clique know one another, or why a specific alter serves as a bridge between two components of the network. The primary limitation of this approach is that the network elicited using mapping will closely adhere to the network structure available in memory (McCarty et al. 2007a). This is subject to numerous recall and associative biases (see Chapter 4). For example, isolates are more likely to be omitted because they are not linked in memory to other alters or groups. Moreover, respondent-defined groups and structures tend to adhere to patterns of association or interaction (e.g., family, work, social clubs) that may not reflect the multiplexity of connections among alters.

Participant-aided sociograms, developed by Hogan, Carrasco, and Wellman (2007), offer a useful network mapping technique that is interactive and does not require computers (see also Kahn & Antonucci 1980; Schiffer & Hauck 2010). The process begins with a set of name generators and prompts that are designed to elicit both strong and weak ties. These names are written on small tags. Ego respondents are given a large piece of paper with four concentric circles on which to arrange them. Because circles represent closeness, respondents are asked to place their closest associates on the inner circle and work outward. Alters who know one another are also placed in close proximity. To elicit ties between alters, respondents are instructed to first identify cliques (i.e., groups in which all members are very close to one another) by drawing a circle around alters. Respondents then draw lines between closely connected dyads not in cliques.

A similar approach is reported by Schiffer and Hauck (2010). They have developed a toolbox called Net-Map that uses figurines, paper, adhesive cards, and markers to collect network data. Figurines of varying sizes are used to represent differential status or influence. The advantage of this and the Hogan et al. (2007) approach is that these methods are very tactile and low tech – they can be used almost anywhere. However, these approaches are also cumbersome, and results have to be manually coded and entered into a computer.

6.5.1 Software for Visual Data Collection

Visualization software for data collection has existed since the 1990s, when Freeman (1994) published a computer program, MAP, for



- a. Arrangement of alters.
- b. Drawing of strong ties – cliques first, followed by edges.
- c. Drawing of somewhat close ties – cliques first, followed by edges.
- d. Graph as coded and represented using the GUESS Network Visualization software (Adar 2006).

FIGURE 6.19 A participant-aided sociogram in four stages of completion (grayscale)

Source: Hogan et al. (2007)

collecting a person's perceptions of the relationships among people around them. The program instructed respondents to move nodes around in a two-dimensional space such that the closer the perceived relationship between them, the closer the respondent should put them on the map. For each respondent, the program outputted the distances between alters, as well as information about the trajectory of each node – how often was it moved or who its neighbors were after each move. The trajectory information provided a unique indication of which nodes were difficult for the respondent to place.

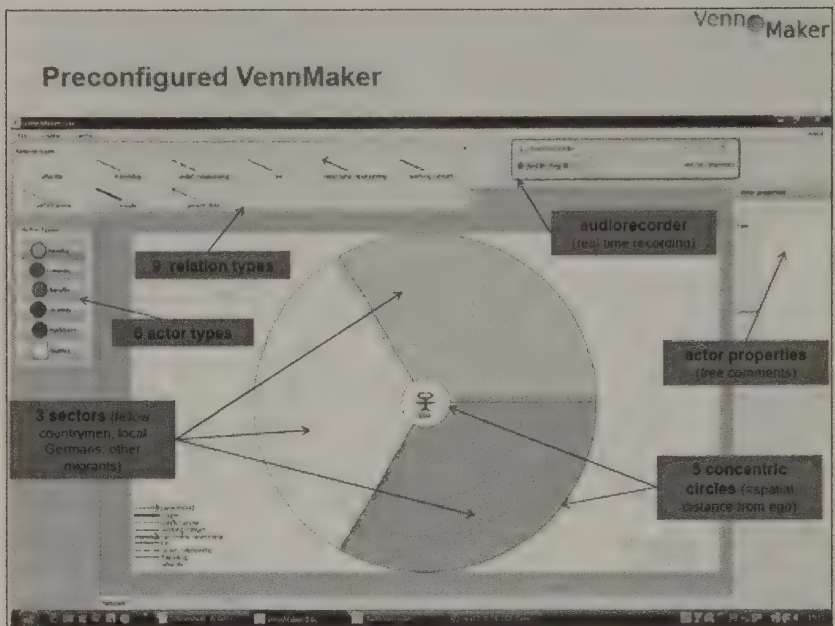


FIGURE 6.20 Screenshot, VennMaker (grayscale)

Source: Image taken from VennMaker manual, www.vennmaker.com

Since then, numerous programs have been developed. Among the best known is a desktop software package, VennMaker¹ (Kronenwett & Schönhuth 2011). Following the long tradition in ego network analysis described above, the program features a circular layout with ego at the center along with concentric circles defining social or spatial distance from ego. Sectors, like pie slices, indicate general categories of alters, as in work, home, or school. Respondents place their alters exactly where they want them in this space. They are then asked to draw lines connecting actors, with different color lines representing different kinds of ties. The program includes an audio recorder to capture what a respondent says about their network as they draw (VennMaker screenshot, Figure 6.20). Another similar program is netCanvas (Hogan et al. 2016), which has versions for both standard computers and handheld devices.

Finally, a web-based open source program, OpenEddi, has been developed by Fagan and Eddens (www.openeddi.com/). The survey is conducted

¹ www.vennmaker.com

on the web, thereby running on any platform, but requiring an Internet connection. Once all of the names and characteristics of alters have been elicited from the name generators and interpreters, the program constructs a network diagram in which all alters are displayed. The respondent then drags a line from one to another alter to indicate the presence of any kind of tie requested. The program has many modes, but in one mode the program continuously runs a graph layout algorithm as the respondent adds ties, maximizing readability. The limitation of this strategy is that the respondent cannot deliberately place a node in a particular position.

6.6 NETWORK DIAGRAMS AS AN INSTRUMENT FOR CHANGE

In applied settings, showing people their own networks can have a powerful transformative effect. In a consulting engagement in which one team within a global firm was not living up to expectations, Cross, Borgatti, and Parker (2002) conducted a network analysis and brought the group together to view the network diagram. The effect was powerful. First, the nature of the problem became immediately clear to participants. The group had split into two subgroups based on the skill sets they were able to proffer clients. One person, the “star,” possessed both skill sets, and was used by both groups when their clients needed the other skill set. He was connected to both groups. This dual demand created a bottleneck, slowing deliveries and creating stress. Second, the diagram inspired the will to change, allowing participants to see themselves as part of the problem (i.e., not having collaborative ties with partners that had the other skill set), and providing ideas for action. Nine months later, a second network analysis showed a fully integrated network.

In executive coaching settings, it is common practice to show managers diagrams of their own networks. Almost invariably, managers find this illuminating. One reason is that it enables the manager to see the forest as well as the trees. The diagram shifts ego’s perspective so that larger patterns are both more visible and more salient. With respect to social capital, people often have a good idea of what kinds of resources they can mobilize through their social ties. However, they are less aware of structural patterns, such as which members of their networks are key bridges between clusters (Molina, Maya-Jariego, & McCarty 2014).

Ego network diagrams may also be used in executive coaching to help managers align their work networks with professional goals (Baker

1990; Wilburn 2016). For example, Wilburn's Leader Network Diagnostic tool (www.leadernetworkdiagnostic.com/) starts by having executives draw their work networks, grouping people by areas (e.g., departments, locations) and putting those they are closest to toward the center of the diagram. The true diagram is then compared with several hypothetical ones representing different kinds of problem networks, such as having too few ties to key areas, or relying exclusively on a close set of colleagues. The tool guides managers to identify network changes that align with their goals.

6.7 VISUALIZING NETWORK CHANGE

An important use of visualization is to depict how a person's network changes over time with respect to characteristics like network size, shape, and composition. For example, Erin (2015) studied conversion of American Christians to Islam, examining how this transition altered their networks. In most cases (at least among whites), those who converted tended to lose large parts of their network, not by choice, but because of rejection by family and friends. Figure 6.21 shows the pre-conversion network for one respondent, Fred, while Figure 6.22 shows his postconversion network. The diagrams clearly demonstrate that Fred lost quite a few non-Muslim friends after conversion, who were replaced with new alters, including several Muslims.

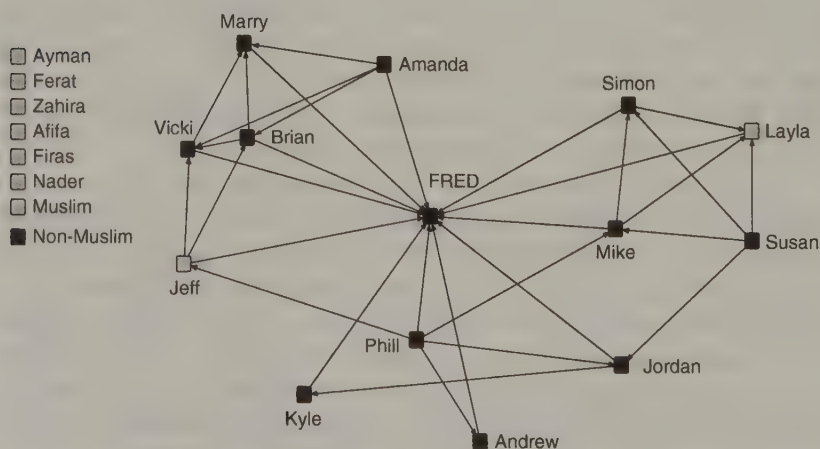


FIGURE 6.21 Fred's preconversion network

Source: Erin (2015)

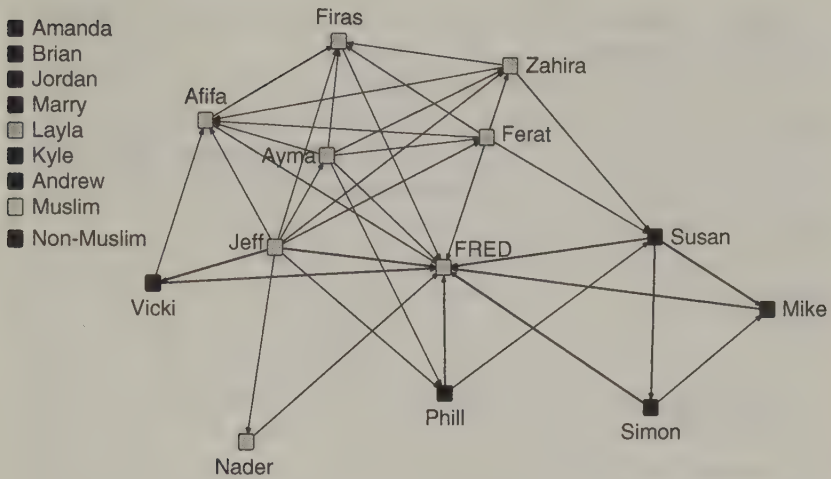


FIGURE 6.2.2 Fred's postconversion network

Source: Erin (2015)

Another approach to depicting change combines old and new ties in the same diagram. In Figure 6.2.3, ego is Russ. Blue lines are lost ties (to Bert and Steve) that Russ had at Time 1 but not Time 2. Red lines refer to new ties that were not present at Time 1 but were acquired by Time 2 (i.e., Bill and Holly; Bert and Gery). Finally, gray lines represent

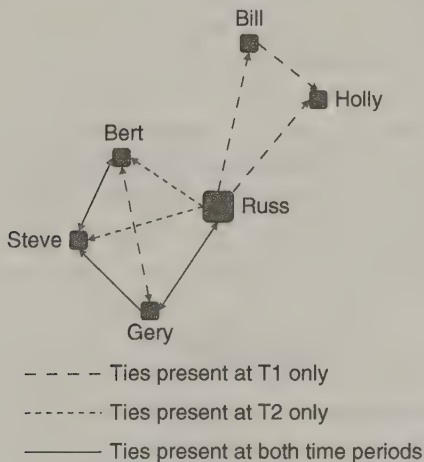


FIGURE 6.2.3 Visualization of ego network change

persistent ties such as the tie from Russ to Gery and the relationship between Steve and Bert.

A few network visualization software packages (e.g., Visone and Sonia) are able to create animations to depict change. Given a series of time slices, the programs create a visual representation of each slice, rotated to maximize similarity. Movements of each node are then interpolated between time periods to create a smooth animation.²

The approaches presented so far for visualizing change are well-suited to examining a few representative cases. However, to visualize change for a large number of egos, approaches that depict network characteristics rather than the networks themselves are preferable. As discussed in Section 6.4, Brandes et al. (2008) reduced ego networks to a standardized blockmodel form that enhances comparability. Lubbers et al. (2010) demonstrated how this approach can be used to depict network change. Figure 6.24 shows twenty pairs of blockmodel diagrams representing ego net characteristics of twenty-five immigrants at two points in time (left is Time 1, right is Time 2). Recalling that the size of the top node represents the number of alters who are natives, it is easy to scan each pair of diagrams to see which egos are acquiring an increasing number of alters. For instance, Lubbers et al. (2010) noted that the third ego from the top, second column, seemed to be successfully integrating with the native population, while others remained relatively isolated from natives and maintained strong ties to alters in the home country and to other immigrants in the local context.

Another approach to visualizing ego network change is provided by Wu et al. (2016). Like the Brandes et al. (2008) approach, the diagrams represent network characteristics rather than egos and alters. Figure 6.25 depicts the magnitude of change in an ego network. The diameter of the circle as a whole indicates network size (useful when showing multiple egos side by side). The darkness of the color of the center circle indicates the density of ties among alters. The size of the quadrants in the outer ring mark the degree of change in tie strength since the previous wave of data collection. This same chart, generated at multiple points in time, can be linked by a bar whose thickness represents the number of alters in common across adjacent time points (Figure 6.26). This approach can effectively present a large amount of complex longitudinal data.

² Details are provided on the website for Visone: [http://visone.info/wiki/index.php/Collections_\(tutorial\)](http://visone.info/wiki/index.php/Collections_(tutorial))

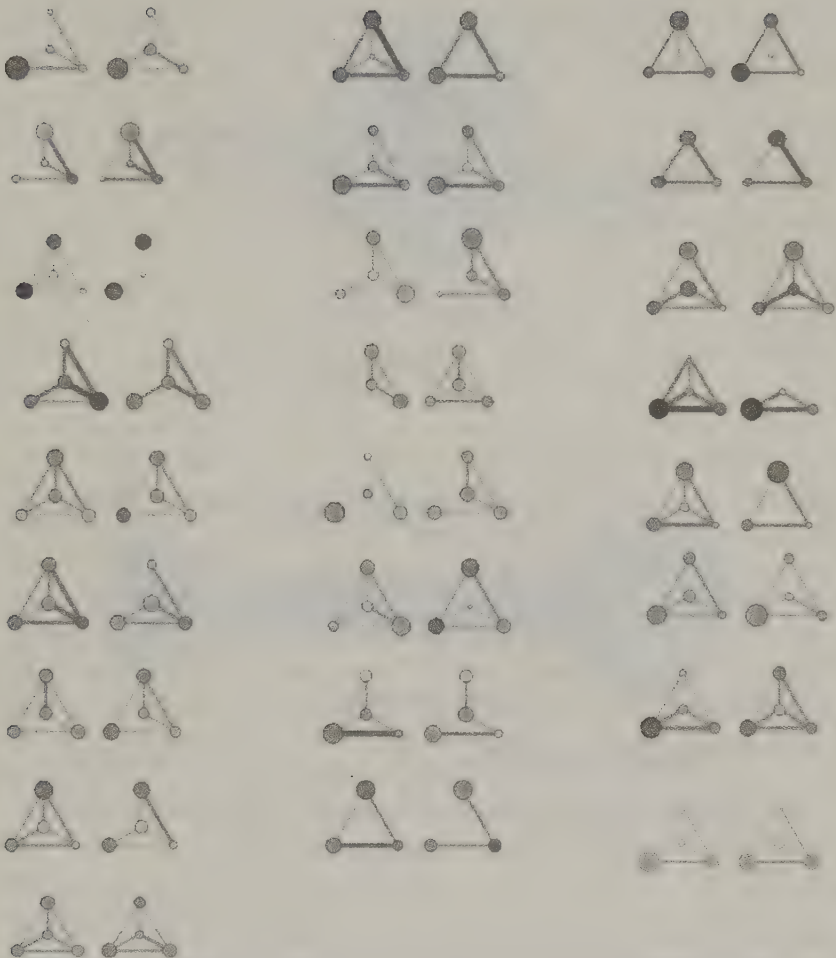


FIGURE 6.24 Stylized representations of selected characteristics of twenty-five ego networks at two points in time (see Figure 6.17 for interpretation of symbols) (grayscale)

Source: Lubbers et al. (2010)

6.8 CONCLUSION

In many ways, methods for visualizing ego networks are similar to those for visualizing sociocentric networks. The underlying techniques and principles are the same. However, there are also differences, particularly with respect to the purposes of visualization. In ego networks, it is not

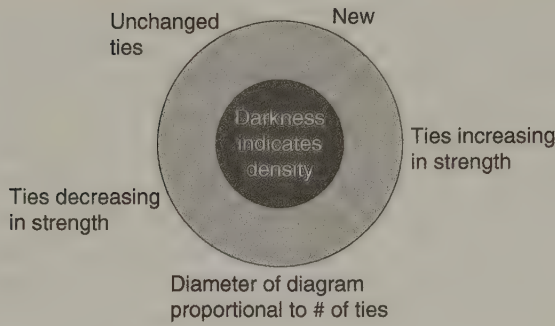


FIGURE 6.25 Magnitude of change in an ego network

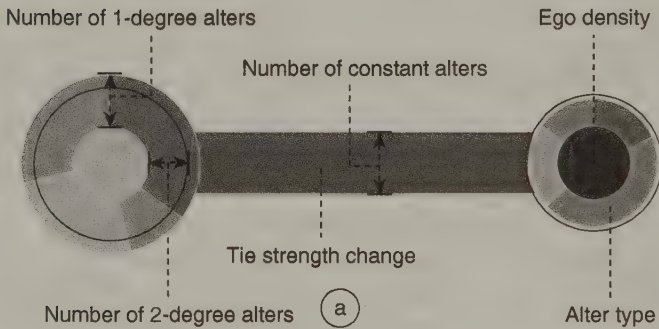


FIGURE 6.26 Characteristics of an ego net changing over time

Source: Wu et al. (2016)

uncommon to use visualization for data collection. This is rare in whole networks. Further, in whole networks, a network diagram is essentially a presentation of the full dataset. The diagram contains the same information, at the same level of granularity, as a data matrix. In ego networks, each respondent generates a unique network. For most studies, it is neither feasible nor particularly helpful to present a diagram of each person's ego network. Instead, modal networks are presented to illustrate key findings, or other visual representations depict key network characteristics rather than the ego networks themselves.

Ego Network Composition and Structure (with Ann McCranie)

This chapter focuses on the definition, utility, and computation of variables describing ego's network. The chapter is organized into three broad sections based on the kinds of data needed to compute these measures. The first section addresses ways to capture the number and nature of relationships between egos and alters in the network, including concepts like average strength of tie. The second section presents measures that make integral use of alter characteristics, typically gathered in the name interpreter stage of the survey, such as multiplexity and homophily. The final section discusses measures that assess the pattern of relationships among the alters, commonly called network structure. This section concludes with the Gould and Fernandez (1989) measures that make use of not only ties among the alters, but also alter and ego characteristics.

7.1 EGO-ALTER TIES

This section discusses three ways of summarizing the number and kinds of ties that ego has with alters – network size, multiplexity, and tie strength.

7.1.1 Network Size

Perhaps the most basic ego network measure is network size – the number of people to whom ego is connected. In whole network settings, this is typically called degree. Ego network size is often seen as an indicator of concepts like social integration, social capital, social

activity, or potential for social support (Vaux & Harrison 1985; Lin 1999a; Berkman & Glass 2000; McPherson et al. 2006). It has been linked to many outcomes, with larger networks typically having more positive effects on, for example, social status and mobility (Campbell, Marsden, & Hurlbert 1986; Marsden & Hurlbert 1988), physical and mental health (Haines & Hurlbert 1992; Smith & Christakis 2008), and life satisfaction (Taylor et al. 2001). Of course, having more negative ties, such as relationships with people who hinder your work, or cause you stress or harm, is generally associated with negative outcomes (Sparrowe et al. 2001; Labianca & Brass 2006). However, even with positive ties, larger networks can have negative consequences because there is substantial cost associated with maintaining a large network (Antonucci, Akiyama, & Lansford 1998). Evolutionary anthropologist Robin Dunbar (1992), for example, has argued that humans suffer from information overload if their number of meaningful social ties exceeds 150. In short, more is not always better.

Network size is usually defined as the number of distinct alters named by ego. For example, Figure 7.1 depicts an ego network with seven alters. Although network size is a simple concept, there are a few subtleties involved in its measurement. First, many egocentric research designs limit the number of alters that an ego can name. As a result, assertions about network size must be made with caution. The claim that humans are connected to 6.7 other humans, on average, is problematic if respondents have been constrained to name no more than ten people. Network size is useful for comparing within studies, but not across different

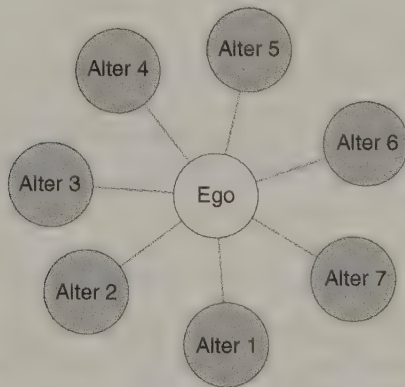


FIGURE 7.1 Ego network (seven alters, no alter-alter ties)

studies. Variations in numeric limits and name generators used to elicit ties typically invalidate cross-study comparisons of network size.¹

Second, in general, network size refers to the total number of unique individuals named by ego, across all types of ties. Thus, alters are not double-counted when they appear in response to different name generators. Additionally, there are reasons to choose not to count all types of ties. For a given research question, we might count only alters mentioned in response to positive name generators (e.g., excluding alters that ego dislikes). Indeed, we can compute a separate network size variable for every kind of tie, as in counting the number of alters that provide emotional support, or the number that provide material aid. A multiple regression of ego outcomes on these different network sizes would likely reveal some to be better predictors than others, depending on the specific outcomes considered.

7.1.2 Multiplexity

Multiplexity is a concept that refers to the multidimensionality of interpersonal relationships (Kapferer 1969). For example, it is common for people who work together to develop friendships and begin socializing outside of the workplace (Feld 1981). An alter may be ego's family member, but also a coworker in the family-run business. Or, thinking in terms of functions, an alter may provide emotional support, lend money, drive ego to doctor appointments, and have ego over to dinner. In this sense, a relationship between two people can be decomposed into a (possibly unique) configuration of analytically distinct types of ties.

In general, multiplex ties are thought to be affectively stronger, more durable, and often more supportive than unidimensional ties (Granovetter 1973; Wellman & Wortley 1990). This creates greater motivation to maintain them and may also influence frequency of contact (Park, Lee, & Kim 2012). For example, alters identified as both family members and friends by respondents are more likely to be described as very close, provide more support functions, and have more frequency contact relative to alters that are only friends or family (Bush, Walker, & Perry 2016). Moreover, egos with more multiplex ties in their networks report higher self-esteem, better psychological adjustment, and greater satisfaction with their social networks and relationships, on average (Mesch & Talmud 2006).

¹ The exception, of course, are studies specifically aimed at discovering how many people respondents know (e.g., Pool & Kochen 1978).

Because certain configurations of ties or functions are caused by the same underlying conditions, the distribution of multiplexity is not uniform. That is, certain combinations occur more often than others. Hence, one potential analysis is to compare the observed frequencies with the expected frequencies under a model of independence of relations. This can be done at the level of the entire sample, but also for different classes of ego, such as men and women. Further, some of the more common combinations can be related to ego outcomes in multivariate models. For example, is the presence of love-hate relationships within an ego network associated with greater stress and anxiety? Are men and women differentially likely to have “frenemy” relationships?

Another way of using multiplexity is as a measure of tie strength, as noted above. We can measure the extent of multiplexity as a simple count of the number of ways that ego and alter are related. If ego and alter 1 are connected by just two kinds of ties while ego and alter 2 are connected by five kinds of ties, we regard the overall relationship between ego and alter 2 as stronger than the one between ego and alter 1. Although, if the researcher has a framework for weighting different kinds of relations (e.g., being a neighbor is worth three points but being a coworker is only worth one point), a finer-grained measure of strength could be computed as the weighted sum of the different relations that bind ego to each alter. This could provide the basis for a number of relationship and network hypotheses. For example, ego’s relationship with alter 1 may be more likely to dissolve over time, relative to the relationship with alter 2 (but see Lizardo & Pirkey 2014 for a contrary view). Ego might also be more constrained and influenced by alter 2 than alter 1 because there is more at stake in their relationship.

7.1.3 Tie Strength

Tie strength captures the intensity and duration of bonds between an ego and alters within the network as well as the frequency of interaction (Marsden & Campbell 1984). It has been linked to a range of outcomes in empirical research, with both weak and strong ties having benefits, depending on the nature of the social phenomenon considered. The presence of strong ties is often seen as an indicator of social integration and regulation, while weak ties provide access to novel resources or indicate that ego has a position of relative power in the network (Granovetter 1973).

Given a measure of tie strength between ego and each alter, an obvious measure is the average strength of tie that ego has across all alters. For

example, when limiting the alters to a particular neighborhood or community, average tie strength might be interpreted as a measure of ego's community engagement. The dyadic variable used to operationalize tie strength could be the multiplexity measure discussed above, or, it could be based on other data collected from the respondent. For example, the name interpreters may have included a set of valued questions such as, "How long have you known this person? (in months/years)," "How close would you say are?," or "How often do you see this person?" A separate average could be calculated for each interpreter.

Tie strength could also be dichotomized or trichotomized to construct each relationship as a strong or weak tie (or neither). The latter disentangles the measures from network size, which is important if both size and strength are included as covariates in a regression model. For example, the number of strong ties may predict rate of recovery from illness. However, since the raw frequency is correlated with network size, it is unclear whether the strength of tie or the number of ties is driving the association. In these cases, controlling for network size or using proportion instead of number is recommended.

Other measures of central tendency, such as modes, medians, maximum values, and trimmed means, may be preferable to using the mean. For example, to weather a personal crisis, perhaps having just one close friend is enough. Having more close friends may provide little additional value compared to the difference between having no close friends and one close friend. In this case, the maximum could be used to measure the strength of the closest tie.

The standard deviation (or any other measure of dispersion) of tie strength can be used to measure relational diversity. We might hypothesize that coping with the varied eventualities of life is optimized by having a portfolio of tie strengths. Strong ties, while valuable for their reliability and their motivation to help, may not offer a fresh perspective. As discussed throughout this book, strong ties with people who are very similar to ego are not necessarily good sources of alternative perspectives. In contrast, weak ties can be quite different from ego, providing new information that is beneficial in a variety of contexts (e.g., among entrepreneurs). Here, dispersion may be more interesting than the mean.

7.2 ALTER ATTRIBUTES

This section discusses ways of summarizing the number and kinds of alters that egos have in their network, including the diversity of those alters.

We group the discussion into three topics: network composition, ego-alter similarity (i.e., homophily), and heterogeneity.

7.2.1 Composition

In discussing network size and tie strength, we introduced the notion of separate counts for different kinds of ties – how many strong ties are present or how many people provide emotional support. Similarly, we can also distinguish between different kinds of alters based on individual attributes of those alters. Such measures may reflect the content of the network, or the material and nonmaterial resources that are present, as well as knowledge, behaviors, and cultural characteristics (e.g., ideas, attitudes, values) flowing through the network (Wasserman & Faust 1994). Assessing the composition of an ego network is a fundamental task: How many women are in ego's network? How many African-Americans? How many doctors? How many millionaires? How many smokers?

Network composition is important because of the potential consequences for ego's own behaviors, attitudes, and outcomes. Perhaps the most studied class of consequences is some variation of performance, achievement, or reward. A fundamental concept in social network theory is social capital – the notion that one's connections are an important determinant of success, whether it is status attainment, positive health outcomes, income, or power. In particular, social resource theory (Lin 1999a) posits that the benefit of others' resources transfer through network ties. An entrepreneur benefits from having friends who are millionaires, corporate lawyers, and experts in enterprise-level information technology. People who have ties to members of city council are in a much better position to affect what happens in their communities than people who don't have that kind of access. Their network tie may be in a position to offer advantageous insider knowledge of how city council works and even directly press their cause in deliberations. Persons undergoing a health crisis may be much better off if their circle of friends includes people with medical training or individuals who have beliefs consistent with modern medicine (Freidson 1970; Perry and Pescosolido 2015).

However, having particular kinds of alters or ties in your network can also have negative consequences. Having dependents, whether they are young children, aging parents, or sick friends, can have a constraining effect on ego's options, potentially causing stress and even affecting health.

Similarly, alters' behaviors can influence egos' attitudes or behaviors. Christakis, Fowler, and colleagues (Christakis & Fowler 2007, 2008; Fowler & Christakis 2008; Rosenquist et al. 2010) examined a wide range of phenomena, finding that obesity, smoking, alcohol consumption, and happiness appear to spread from person to person.

Composition measures can also be used to examine broader patterns of social interaction in society. Using the General Social Survey of Americans, McPherson, Smith-Lovin, and Brashears (2006) found that while the number of both kin and non-kin members of discussion networks declined from 1985 to 2004, the decrease in non-kin networks was more pronounced. This finding was used as evidence that American society is experiencing an overall decline in civic life and social capital, consistent with Putnam's claims (2000). More importantly, McPherson et al. also found that more educated individuals reported a higher percentage of non-kin in their discussion networks, and that this relationship was stronger in 2004 than in 1985. If socioeconomic stratification of social network composition is increasing over time, the implications for inequality in American society are profound.

Composition measures for categorical alter attributes. For categorical attributes, composition can be examined by counting the number of alters that possess a given trait. For example, a count of the number of alters that belong to different political parties can be used to operationalize the nature and degree of political influence. As discussed earlier, these counts can be recorded as frequencies or proportions. The frequency or proportion of Democrats in the ego network then becomes a variable attached to ego, as the does the frequency or proportion of Republicans. Figure 7.2 depicts an ego network with two types of alters: Democrats (white nodes) and Republicans (gray nodes). This particular network has four Democrats (not including ego) and two Republicans. The proportion Democrat is 0.67 (4/6), which we could use to try to predict whether an ego intends to vote for a Democrat in a presidential election.

When the categories relate to specific resources that an ego can draw on, frequencies, rather than proportions, are typically of interest. More friends who are capable of helping you move boxes are better than fewer, regardless of what proportion they are of one's ego network. When the phenomenon being studied is social influence, proportions are probably more relevant. If more than half of ego's network leans Democrat, the effect is potent, the absolute number isn't as important. This is a well-known finding in political persuasion (Lerman, Yan, & Wu 2016).

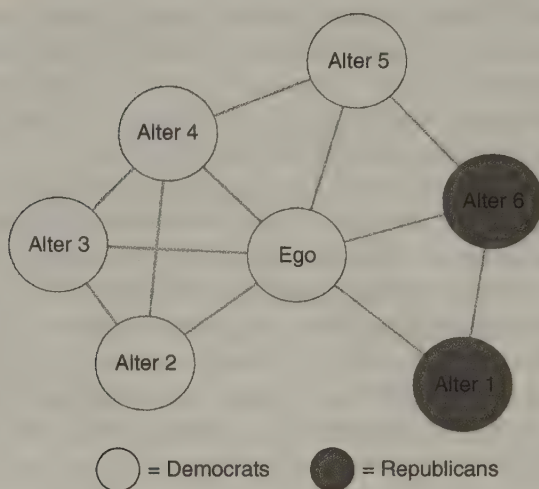


FIGURE 7.2 Ego network identifying political party

Composition measures for continuous alter attributes. For continuous alter attributes, the composition of the ego network is typically summarized using a measure of central tendency, such as the mean or median. For example, if the median income of an ego's alters is in the hundreds of thousands, they probably live in a very different world than an ego whose alters have a median income in the tens of thousands. However, when it comes to assessing an ego network's potential for supporting ego, the maximum value (i.e., the income of the richest person in the network) may be more relevant. If ego is starting a business, it may not matter that ego has a lot of poor friends who reduce the median. The important factor is the presence of a few rich friends who can invest in the business. An operational compromise between the maximum and the median is the value of a certain percentile. For example, we might operationalize ego's resource-richness as the income of the alter who is at the eightieth percentile among ego's friends.

7.2.2 Ego-Alter Similarity

One of the most robust findings in social network research is the tendency for people to have positive ties and interactions with people who are similar to themselves on socially significant attributes, including demographic variables, behaviors, interests, and attitudes (Lazarsfeld &

Merton [1954]).² For example, the 1985 General Social Survey showed that people overwhelmingly discuss important matters with people who are similar to themselves with respect to gender, race, age, education, and religion (Marsden 1987).

Ego-alter similarity can be useful for imputing unobserved characteristics of ego. For example, it is possible to predict with high accuracy whether someone is a sexual minority or not by looking at their Facebook friends (Jernigan & Mistree 2009). If the friends include several people who identify as a sexual minority, the probability that ego is a sexual minority increases substantially, even if ego's own Facebook page makes no mention of their sexual orientation. Likewise, the police intuitively reason that if a person associates with known criminals, there is a very good chance they are criminals themselves – guilt by association, in line with Sutherland's (1955) differential association theory. Under an assumption of ego-alter similarity, examining ego's alters allows cautious inference about ego's unmeasured characteristics.

Theoretically, at least three classes of mechanisms can lead to the outcome of ego-alter similarity. The first is preference, also known as homophily³ (Byrne 1961). Homophily is, quite literally, liking of one's similars – the tendency to *prefer* those similar to oneself on socially significant attributes, such as race, gender, education, religion, and social class. Heterophilous choices are less common, with the notable exception of romantic relationships, which tend to be heterophilous on gender. However, romantic relationships still tend to be homophilous with respect to other variables, such as race and socioeconomic factors (Rosenfeld 2008; Schwartz 2013). In a review of network effects on inequality, DiMaggio and Garip (2012) concluded that homophily is a ubiquitous feature of social life and in some cases, exacerbates inequality by increasing advantages to certain privileged groups. McPherson and colleagues (2001) also provided a review of the many ways in which homophilous choices with respect to race and ethnicity, age, religion, education, occupation, and gender can limit individual social worlds and opportunities. Homophilous networks insulate egos from outside influences and ideas, reinforcing in-group behaviors and biases. At the same time, homophily is identity-affirming, fostering a sense of comfort and

² Ego-alter similarity is sometimes discussed as assortativity or assortative mixing, or, in certain contexts, as network auto-correlation.

³ Some researchers use “homophily” as a catch-all term to describe ego-alter similarity. Others (e.g., McPherson, Smith-Lovin, & Cook 2001) exclude the influence mechanism and refer to the preference and availability mechanisms as two kinds of homophily.

belongingness and facilitating greater ease of communication among those that share backgrounds.

The second mechanism leading to ego-alter similarity is availability – i.e., the kinds of people with whom ego has an opportunity to form ties. Growing up in a neighborhood that is predominantly white probably leads to having mostly white playmates, even if children have no particular preference for whites. The same effect is often seen for coworkers in organizations and occupations, where nursing and elementary education skew heavily toward women, while construction skews toward men. In a study of college students, for example, Small and Sukhu (2016) found that individuals ego asked for help regarding academic problems were not necessarily the best suited, but were contacted because they happened to be available.

The third mechanism is influence. Repeated interaction tends to result in individuals becoming more similar over time. For example, spouses become more similar with respect to what foods they prefer, what recreational activities they enjoy, and how they interpret current news events (e.g., Schimmack & Lucas 2010). Of course, influence applies mainly to attitudes, decisions, and behaviors. No matter how much two people interact, ascribed statuses (e.g., biological characteristics like age or skin color) will not change over time to become more similar.

To summarize, we have discussed three mechanisms. Of these, preference and availability represent mechanisms through which similarity causes ties to form. In contrast, influence is a model of how ties cause similarity. Thus, preference and availability differ from influence in the direction of causality. Statistical measures of ego-alter similarity can sometimes distinguish preference from availability, but, at least with behaviors and attitudes, they cannot distinguish these from influence, since this is an issue associated with the direction of causality.

Similarity measures for categorical alter attributes. For categorical alter attributes such as race or gender, the most obvious measure of ego-alter similarity is a simple count of the number of alters that have the same trait as ego (e.g., male, tennis player, Catholic). Typically, this is expressed as a proportion – the number of similar alters divided by the total number of alters. This is generally known as the “proportion homophilous” measure, although it would be less misleading to use “proportion similar to ego.” An alternative measure is the E-I index (Krackhardt & Stern 1988), defined as the number of alters different from ego (external ties *E*) minus the number of alters the same as ego (internal ties *I*), divided by the

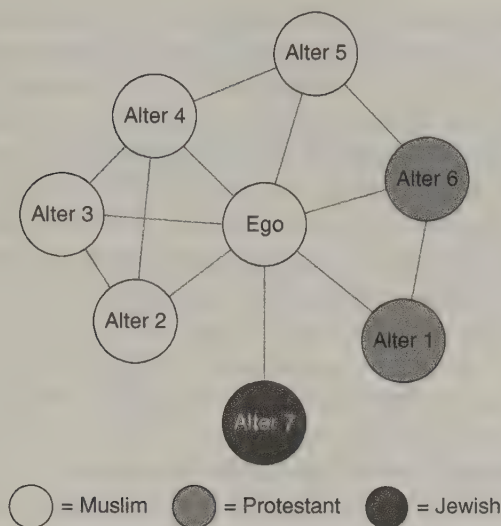


FIGURE 7.3 Ego network identifying religious affiliation

number of alters (see Equation 1). This is a reverse measure of homophily, since a larger number indicates greater heterophily. Because it is a linear transformation of the proportion homophilous measure, the correlation between the two is a perfect -1.0 .

$$EI = \frac{E - I}{E + I} \quad (7.1)$$

Figure 7.3 depicts an ego network with three different types of alters, in this case different religions: Muslim (white nodes), Protestant (gray nodes), and Jewish (black nodes). In this network, ego is Muslim, so the proportion similar to ego would be 0.57 ($4/7$). The E-I index is -0.14 . The measures indicate a slight tendency toward having ties with fellow Muslims, but, again, preference cannot be inferred. Indeed, if ego lives in a Muslim community, having only 43 percent Muslim friends would likely indicate strong *heterophily*.

In egocentric research, the only way to determine the underlying preference is to adjust the measures to take into account the demographics of the population from which ego draws their contacts (e.g., the workplace or the neighborhood). One way to accomplish this is to treat the distribution of races (or whatever categories are of interest) in the community at large as the expected values in a null model of no homophily/heterophily.

Then, comparing this with ego's choices, we can construct a measure of departures from the null model using the familiar chi-square statistic, and then normalize it to calculate the Pearson's Phi statistic. Specifically, Step one calculates expected values. If a given ego has named twelve alters and their neighborhood is 75 percent white, the expected values – under a model of no preference – are nine white alters ($0.75 * 12$) and three African-American alters ($.25 * 12$). Step two is calculating the chi-square statistic (Equation 2). Suppose that, in reality, ego has named ten whites and two African-Americans. These are the observed values. Computing $(observed - expected)^2 / expected$ for each category, and summing these $[(10-9)^2/9=.11 \text{ plus } (2-3)^2/3=.33]$ produces a sum of .44, the chi-square value. Step three is calculating Phi, which runs between zero and one (see Equation 3).⁴ Dividing chi-square by twelve (N, the number of observations) and taking the square root yields .192, a very modest level of homophily. If ego had named nine whites and three African-Americans, the value of Phi would have been zero, indicating no preference (i.e., no homophily).

$$\chi^2 = \sum_k \frac{(O_k - E_k)^2}{E_k} \quad (7.2)$$

$$\phi = \sqrt{\frac{\chi^2}{N}} \quad (7.3)$$

Similarity measures for continuous alter attributes. To assess ego-alter similarity with respect to a continuous variable, such as age, one approach is to calculate the sum of squared differences between ego's age and the age of each alter. The problem with this, however, is that this measure tends to increase with the number of alters, making it difficult to compare across egos with networks of different sizes. To address this limitation, the measure is modified by taking the mean squared difference instead of the simple sum. In addition, we take the square root of the sum to return the values to the original units. The resulting measure – called average Euclidian distance – is similar to a standard deviation around ego (see Equation 4). It can be interpreted as the typical difference between ego's age and each alter's age. In the formula, k indexes alters, a_k is the age of alter k , and e is age of ego.

⁴ Note that if the observed value for ego's own category is smaller than the expected value, we can multiply Phi by -1 to indicate heterophily.

$$\frac{\sqrt{\sum_k (a_k - e)^2}}{n} \quad (7.4)$$

For example, if a thirty-year-old ego has three alters aged twenty-five, thirty-two, and forty, average Euclidean distance would be calculated as follows:

$$\begin{aligned} \frac{\sqrt{(25 - 30)^2 + (32 - 30)^2 + (40 - 30)^2}}{3} &= \frac{\sqrt{25 + 4 + 100}}{3} \\ &= \frac{\sqrt{129}}{3} = \frac{11.4}{3} = 3.8 \end{aligned}$$

It is important to note that, while average Euclidean distance is comparable across egos, it is not comparable across attributes. This is because it is unbounded, and its scale is determined by the attribute being measured. A score of 3.8 is small with respect to age, but would be large with respect to number of children.

7.2.3 Heterogeneity or Diversity

Homophily, discussed in Section 7.2.2, refers to the similarity of ego to alter, whereas homogeneity concerns the similarity of alter to alter; in short, the degree of diversity within an ego network. If a male has only female friends, he exhibits strong gender heterophily but no heterogeneity. If he has only male friends, then his network is characterized by both homophily and homogeneity.

Heterogeneity has often been used to operationalize the “range” of a network. Range describes the degree to which ties within an ego network cross social, institutional, or organizational boundaries (Burt 1983). The more people from different groups an ego has in their network, the greater the range of nonredundant information, experiences, skills, and support they can access (Campbell et al. 1986). For example, Rodan and Galunic (2004) examined the association between knowledge heterogeneity in managers’ ego networks in the workplace and managerial performance and innovation. Having access to ties with diverse forms of knowledge (i.e., content diversity) was predictive of overall and innovation performance. However, the link to innovation was stronger, suggesting that access to diverse forms of knowledge is key to idea creation and implementation.

Heterogeneity (or the lack of it) also describes structural patterns of social interaction that can maintain or undermine social integration and equality. McPherson and colleagues (2006) found that sociodemographic heterogeneity of discussion confidants was significantly lower than would be expected based on population averages for age, gender, race, and education, which has implications for the social segregation of culture, ideas, and resources. Lewis et al. (2008) extended this idea to online networks, demonstrating substantial ethno-racial heterogeneity in college students' Facebook and housing networks. Women, students with lower socioeconomic status, and members of racial and ethnic minority groups had higher ethno-racial heterogeneity relative to their advantaged, white, male peers. This implies that the latter may maintain advantage in career mobility through social closure. That is, although women, minorities, and low-SES students had the potential to access cultural, informational, and attitudinal diversity and its career benefits, they were largely excluded from the networks that propel upward mobility into middle- and upper-class careers.

Measures of heterogeneity for categorical attributes. The most common measure of network heterogeneity for categorical data is Blau's index (also known as Herfindahl's index or Hirschman's index). The formula is given in Equation 5:

$$H = 1 - \sum_k p_k^2 \quad (7.5)$$

In the formula, p_k is the proportion of ego's alters that fall in category k . When all alters fall in just one category (e.g., all white), then the index equals 0, indicating perfect homogeneity for a male ego. The maximum value occurs when there are equal numbers of alters in each category. For example, if the alter attribute is gender and we target two genders, then Blau's heterogeneity reaches its maximum score when there are equal numbers of each gender. This maximum is 0.5 for two categories. The measure does not reach 1.0 because the maximum score is $1 - 1/k$, where k is the number of categories. As the number of categories increases, so does the maximum value. When Blau's measure is normalized to range from 0 to 1 (by dividing by the maximum possible value [$1 - 1/k$]), the resulting measure is known as the index of qualitative variation or IQV (Agresti and Agresti 1978: Equation 6):

$$IQV = \frac{\left(1 - \sum_k p_k^2\right)}{\left(1 - 1/k\right)} \quad (7.6)$$

While the normalized measure may appear preferable, there is reason to choose Blau's index instead. If the heterogeneity of alters is interpreted as diversity of inputs, then regarding a person's network with five kinds of alters as more diverse than one with two kinds of alters seems reasonable, even if in both cases the alters are uniformly distributed across all available categories. For instance, having an equal number of Protestants and Catholics in a network probably has different implications for religious tolerance than having equal numbers of Protestants, Catholics, Jews, and Muslims.

Returning to Figure 7.3, we have an ego network with three categorical attributes. To calculate Blau's index, the squared proportions of alters in each of the three religious categories are summed, yielding $1 - (0.57^2 + 0.29^2 + 0.14^2)$ or 0.58. This is close to the maximum for an attribute with three categories, which is $(1 - \frac{1}{k})$, or 0.67, suggesting strong heterogeneity in this network. The IQV in this case is Blau's index (0.58) divided by the maximum value for a three-category attribute (0.67), yielding 0.85.

Measures of heterogeneity for continuous attributes. For continuous alter attributes, heterogeneity can be measured using the standard deviation (Campbell, Marsden, & Hurlbert 1986). For example, the heterogeneity in education of ego's alters is calculated by taking the standard deviation of years of education across all relevant alters. Higher values indicate more heterogeneity.

7.3 ALTER-ALTER TIES

In this section, we consider measures that describe the structure or pattern of ties within an ego network. As such, all of these measures require data on ties between alters, whether binary or valued. There are two classes of measures. One class essentially treats an ego's network as a small socio-centric network⁵, and uses standard network-level concepts, such as density and fragmentation, to construct ego-level measures. These measures are geared toward describing the shape or cohesiveness of the ego network, and are calculated by excluding egos and their ties to alters. The other class of measures, typically referred to as brokerage measures, essentially capture centrality in ego's own network. These include Burt's

⁵ See Chapter 10 for some important differences.

(1992) structural hole measures as well as ego betweenness (Everett & Borgatti 2005). Measures in this class are calculated with ego included.

7.3.1 Density

Perhaps the simplest and most fundamental structural characteristic of an ego network is density – a measure of the degree of connectedness of alters in a network.⁶ Network density can be related to the availability of social support and the presence of a strong social safety net, consistent with arguments about the benefits of triadic closure (Kazak & Marvin 1984). Likewise, high density has been conceptualized as a component of bonding social capital, or close-knit ties to homogeneous others, which foster shared social norms and cooperation (Putnam 2001; Lakon, Godette, & Hipp 2008). Density is often seen as an indicator of normative pressure to conform and the capacity of alters to collectively influence egos (Bott 1957; Mayhew 1968). For example, in their study of fertility decisions in Kenya, Kohler, Behrman, and Watkins (2001) found evidence that ego network density is an indicator of the magnitude of social influence in women's choices regarding use of modern contraception. Specifically, when there was a low proportion of contraception users in the network, high density reduced the odds of using modern contraception. However, if a high proportion of alters was using contraception, high density enhanced a woman's odds of also using contraception. In short, the interaction between network composition and structure followed a canonical pattern in which composition provides the direction of the influence, while density provides the strength of the pressure to conform (Pescosolido 2006).

Personal networks with low density have been found to provide a diverse set of coping strategies that foster adaptation and resilience in the face of adverse circumstances, such as divorce (Hirsch 1980; Wilcox 1981). Consistent with Burt's argument, low density is associated with less information redundancy in a network and greater novelty (Burt 2000). When everyone knows one another, they tend to have similar views, at least when it comes to subjective phenomena, such as making sense of recent news. A person would ideally have both the bonding benefits of being part of a tight group and the bridging benefits of ties to other groups. In Granovetter's "strength of weak ties" argument (1973),

⁶ In the physics literature, ego net density is also known as the node-level clustering coefficient.

the key idea was that acquaintances were less likely to be connected to one's other contacts and therefore more likely to possess unique information about job opportunities. In other words, a more loosely connected network may be more advantageous, as it provides opportunities to span different social groups and have access to novel job leads.

In physics and other recent network science research stemming from that tradition, ego network density is known as the clustering coefficient. In many networks, the average clustering coefficient is quite high – higher than the density of the network as a whole – indicating that most individuals live in pockets of high density. For example, in a study of online networks, Ugander, Karrer, Backstrom, and Marlow (2011) found that Facebook ego networks have average clustering coefficients of around 0.14 (ranging up to .50), which is about 500,000 times denser than the Facebook network as a whole.

Finally, ego network density can also be conceptualized as transitivity – the property embodied by the saying “a friend of a friend is a friend.” Consider an ego with two alters. If there is a tie between the alters, the triad is said to be transitive, or closed. Hence, the number of ties among ego's alters is equal to the number of triads involving ego that are transitive.

The density of an ego network is defined as the number of ties among the alters divided by the number possible. For undirected ties, the number possible is the number of unordered pairs of alters, which is $N(N-1)/2$, where N is the number of alters. For directed ties, the number possible is the number of ordered pairs, which is $N(N-1)$. Ego, and ties with ego, are not included in the calculation. This yields two formulas for density, where T is the number of ties present (see Equations 7 and 8):

$$\text{For undirected ties : } \frac{2T}{N(N-1)} \quad (7.7)$$

$$\text{For directed ties : } \frac{T}{N(N-1)} \quad (7.8)$$

The network depicted in Figure 7.4 is sparsely knit, with only three of forty-two possible ties between alters present. The density is $(2 \times 3)/(7 \times (7-1))$, or 0.14.

An ego network (excluding ego) can be represented as an $N \times N$ adjacency matrix of 1s and 0s, indicating which alters have a tie to which other alters. Figure 7.5 presents the adjacency matrix for the network depicted earlier (Figure 7.4), with missing values placed in the main diagonal.

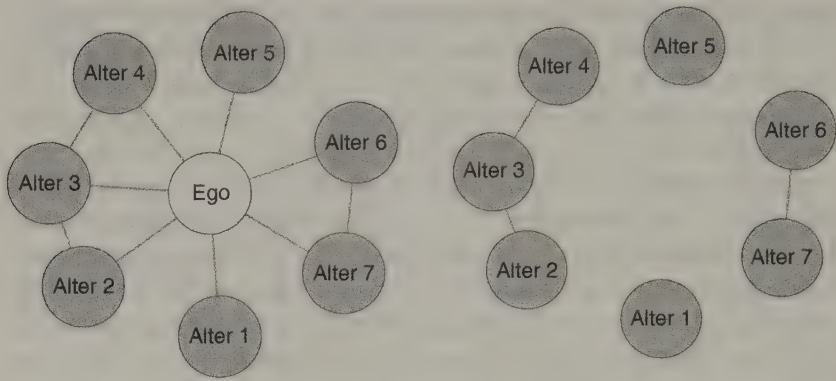


FIGURE 7.4 Ego network with and without ego

	A1	A2	A3	A4	A5	A6	A7
A1	0	0	0	0	0	0	0
A2	0	0	1	0	0	0	0
A3	0	1	0	1	0	0	0
A4	0	0	1	0	0	0	0
A5	0	0	0	0	0	0	0
A6	0	0	0	0	0	0	1
A7	0	0	0	0	0	1	0

FIGURE 7.5 Adjacency matrix matching network in Figure 7.4

By representing the network in this way, density can be calculated by taking the average value of all nonmissing cells in the matrix. This is because the sum of a set of 1s and 0s is the same as counting the number of 1s. What makes this particularly useful is that it neatly generalizes to valued data. The strengths of tie among the alters (rather than 1s and 0s) are used, and density for a valued network is defined as the average tie strength.

7.3.2 Components and Fragmentation

A component is a maximal fragment of a network in which every node can reach every other node through some path. “Maximal” here means that

the fragment is as large as possible without violating the condition of connectedness. In an ego network (minus ego), the existence of multiple components means that ego's personal network has separate groups of alters that have no connection to each other except through ego (or possibly through nodes outside ego's empirically observed world). For example, the components may correspond to work friends, high school buddies, and neighbors. The presence of multiple components is usually negatively correlated with density, since by definition there can be no ties between members of different components. Figure 7.4 presents a network with four components. One component is made up of three ties, another of two ties, and there are two isolates.

The number of components in an ego network determines the extent to which ego's alters have opportunities to communicate with one another, or conversely, the degree to which they are socially isolated and compartmentalized. An ego with multiple components has "limited liability." If, for example, some unsavory information about ego surfaces in one component, it is unable to spread to the other alters since they are not connected. The disadvantage, however, is that alters in multiple components cannot coordinate with each other to help ego. For example, if ego experiences a crisis and is in need of intensive support from alters, the fragmentation of ego's network would prevent coordination across alters. In essence, the safety net will likely be limited to the one group of alters that is aware of the problem.

One measure of fragmentation is to simply count the number of components in an ego network. However, since larger ego networks can have more components, it would be better to construct a measure that takes network size into account. One such measure is the component ratio, defined as the number of components minus one divided by the number of nodes in the ego network (excluding ego) minus one (see Equation 9):

$$\text{Component ratio} = \frac{(C - 1)}{(N - 1)} \quad (7.9)$$

The ratio achieves its maximum value of 1.0 when every alter is unconnected to every other, and achieves its minimum value of 0.0 when the ego network forms a single component. The network depicted in Figure 7.4 has a component ratio of $(4-1)/(7-1)$ or 0.50.

One difficulty with the component ratio is its insensitivity to component size. That is, it does not differentiate between an ego network in which every alter, except one, is in one component and a network with

two equal sized components. In the first case, almost everyone can reach everyone else. In the second, many alters are disconnected from many others. The component ratio is the same in both cases, but the second network is clearly more fragmented. This difference has obvious implications for the degree of diffusion or contagion possible in the two networks.

A solution to this problem is the fragmentation index. This measure is very much like density, except that instead of counting the proportion of alters that are directly connected, it counts the proportion that are connected by any path, no matter how long. The adjacency matrix of ego's network (minus ego) is replaced with the reachability matrix. The reachability matrix R has a 1 in a given cell (a , b) if node a can reach node b by some path, and is 0 otherwise. Fragmentation is calculated as 1 minus the average of R :

$$F = 1 - \frac{\sum_{i \neq j} r_{ij}}{N(N-1)} \quad (7.10)$$

The minimum value of the fragmentation index is 0, and occurs when every alter can be reached by every other alter. The maximum value is 1, and indicates that every alter is an isolate, with no ties between any alters.

Table 7.1 presents the reachability matrix for the network in Figure 7.4 (compare to the adjacency matrix, Figure 7.5). Only two additional cells receive a value of 1, because alters 2 and 4 are connected indirectly through alter 3. Here, fragmentation is equal to $1 - (8/42)$, or 0.81. This high fragmentation value reflects the low level of density and proportionally large number of isolates in ego's network.

TABLE 7.1 *Reachability matrix matching network in Figure 7.4*

	A1	A2	A3	A4	A5	A6	A7
A1	0	0	0	0	0	0	0
A2	0	0	1	1	0	0	0
A3	0	1	0	1	0	0	0
A4	0	1	1	0	0	0	0
A5	0	0	0	0	0	0	0
A6	0	0	0	0	0	0	1
A7	0	0	0	0	0	1	0

7.3.3 Burt's Structural Hole Measures

Burt (1992) introduced the concept of structural holes as a source of social capital. Narrowly speaking, a structural hole is the absence of a tie between alters in an ego's network. The existence of a structural hole implies the potential for ego to serve as a broker between the two alters. An ego network rich in structural holes provides ego with at least two types of advantages: informational benefits and social control benefits. These advantages, in turn, enable ego to achieve goals and receive rewards. Structural holes have been linked to a number of outcomes, including early promotion and corporate profitability (Burt 1992), compensation (Seibert, Kraimer, & Liden 2001), good ideas (Burt 2004), innovation (Ahuja 2000), knowledge transfer (Abbasi, Chung, & Hossain 2012), individual performance (Cross & Cummings 2004), and health (Cornwell 2009).

The argument for informational benefits posits that alters that have direct or indirect ties between them are more likely to share opinions and information. Therefore, the more an ego's contacts are unrelated to each other, the more likely they are to be able to provide ego with novel input that is nonredundant with input from other contacts. This enables ego to make better decisions, whether at work or with respect to their own health. In addition, if ego is connected to two groups that are otherwise unconnected, ego is in a position to solve problems in one group by bringing solutions already discovered by the other group, thus providing additional value to the first group. This argument can be extended to any kinds of resources that might flow along network channels.

There are multiple arguments for social control benefits, as well. First, if ego is a conduit for resources to flow from one part of the network to the other, then ego is in a position to punitively turn off the flow, or, in the case of information, to subtly filter or color the information passed along. The same is true for positive social control, facilitating resources or not. Second, an ego in a structural hole can divide and conquer. In an organization which conducts salary negotiations with employees separately, the organization can play one employee against the other, often with considerable advantage. But if employees engage in collective bargaining, the structural situation reduces to a dyad consisting of just the organization and the union. Third, as Simmel (1950) noted, when alters are actively in conflict, the brokering position presents opportunities for mediating conflicts and gaining advantages through such arbitrations. Fourth, structural holes enable autonomy. If the alters don't

communicate with each other, ego's presentation of self can differ across groups. In contrast, connected alters constrain ego to be consistent, if not necessarily truthful.

The converse of structural holes, at least when the number of alters is held constant, is triadic closure, or transitivity. A triad in an ego network is closed if the two alters have a tie to each other (Simmel 1950). Granovetter (1973) famously argued that when ego has strong ties to alters, there is a high probability that the alters will develop at least a weak tie, producing closure. Coleman (1988) argued that closure leads to the development of shared social norms that effectively guide the actions of an individual. It also fosters interpersonal trust and a sense of obligation to others in the group, which can promote feelings of belongingness at the individual level and cooperation and cohesion at the group level. For example, tight connections among the parents in a neighborhood ensure that all parents can be relied upon to supervise each other's children.

Coleman's and Burt's arguments may seem contradictory, with the former arguing that closure conveys greater social capital, while the latter suggests structural holes do so. However, both argue that closure constrains behavior. The difference is social context. In the settings that Burt studied – managers in organizations – constraint was associated with lower performance. In the settings Coleman studied – children and parents – constraint was advantageous.

Effective size and efficiency. Burt (1992) has introduced a number of measures related to structural holes. The first of these is effective size, which measures the number of nonredundant alters to whom ego is connected. For binary data, effective size is equal to network size minus the average number of ties that each alter has within the ego network, excluding ties to ego.⁷ The formula for effective size is:

$$\text{Effective size} = N - \frac{\sum_j d_j}{N} = N - \bar{d} \quad (7.11)$$

In the formula, N is network size, d_j is the number of ties that alter j has within the ego network (excluding ties to ego) and $\bar{d}$ is the average of d_j across all alters. In Figure 7.4, network size is seven. Alters 1 and 5 have no ties to other alters. Alters 2, 4, 6, and 7 are each connected to one other alter. Alter 3 is connected to two alters. The average of these numbers

⁷ Effective size can also be calculated with valued ties. The formula is given in Burt (1992).

$(0+0+1+1+1+1+2)$ is 0.9, with effective size being $(7 - 0.9) = 6.1$. If no alters were connected to each other, the average degree among the alters would be zero, and the effective size would be $7 - 0 = 7$. In this case, the effective size is equal to network size. Conversely, if all alters were directly connected to each other, the average degree would be six (i.e., each alter connected to six other alters), and the effective size would be $7 - 6 = 1$. The measure indicates that, in essence, ego is connected to just one “pot” of information.

All else being equal, the larger the ego network, the greater the effective size, as effective size is a positive function of network size and a negative function of the number of ties among alters. These two elements are represented as the two terms in the formula. As discussed in Section 7.3.4, this also holds true for Burt’s other measure of structural holes – constraint. This is because in regressions involving measures of structural holes, we are often tempted to control for network size. However, since network size is a key component of Burt’s conceptualization of structural holes, controlling for it deletes this element from the measure. This leaves average alter degree, which is just network density multiplied by $N-1$. Controlling for network size in a regression analysis would not change the R-squared. But the interpretation of the coefficient for effective size would have to be altered because the variable is no longer effective size; it is just a transformation of density.

Burt (1992) also defined efficiency, which is effective size divided by N . This is not a measure of structural holes per se, but a measure of social or political skill. It indicates the extent to which a person is choosing ties wisely with respect to maximizing social capital per unit of relational energy. Burt and Ronchi (2007) described a program to train executives to understand the value of structural holes; an obvious measure of the effectiveness of the training would be an increase in efficiency post training relative to a control group.

Constraint and hierarchy. The most commonly used measure of structural holes is Burt’s constraint index. It is an inverse measure of structural holes in the sense that larger values correspond to fewer holes. Intuitively, the measure assesses the extent to which ego is connected to few others, and simultaneously the extent to which these few others are not independent of each other. Although similar to effective size, historically it has proved the better predictor of ego outcomes, such as success in business.

To compute constraint, the first step is to generate a score for each alter which measures the extent to which that alter constrains ego. The alter

constraint score combines two elements: (1) how redundant the alter is with ego's other alters, and (2) the extent to which ego is dependent on the alter (meaning that ego has few other alters). Constraint scores are summed for each alter to obtain an overall score for ego.⁸ The formula for the constraining effect of each alter is given in Equation 12:

$$c_j = \left(p_j + \sum_q p_q p_{qj} \right)^2, q \neq j \quad (7.12)$$

In the formula, j refers to the alter whose constraining effect on ego is being calculated, and q refers to another alter in the ego network. The term c_j represents the extent that j constrains ego, while p_j represents the strength of ego's tie with alter j , divided by the total tie strength with all alters. Conceptually, this reflects the proportion of ego's relational energy that is invested in alter j . For binary data, this is $1/N$, where N is the number of alters. The formula gives a higher score (more constraint, fewer holes) for smaller networks, since p_j is larger to the extent that ego has a small network. Like p_j , the term p_q is ego's investment in alter q . Again, for binary data, this is $1/N$. Finally, p_{qj} is the strength of alter q 's tie with alter j divided by the total tie strength q has with all others in the ego network (including ego). Equation 13 provides the simplified formula for binary data:

$$c_j = \left(1/N + 1/N \sum_q p_{qj} \right)^2, q \neq j \quad (7.13)$$

The formula can be reduced even further by factoring out the $1/N$ terms, but the contribution to clarity is minimal. Once c_j scores have been computed, they are summed to obtain ego's overall constraint, as shown in Equation 14:

$$\text{Overall constraint} = \sum_j c_j \quad (7.14)$$

The reduced formula is similar to the formula for effective size. The first term in the simplified formula is the exact reciprocal of network size. The second term corresponds to network density. As with effective size,

⁸ Overall constraint approaches zero as network size goes to infinity (and no alter is connected to any other), but can be larger than one in very small ego networks in which every alter is connected to every other.

controlling for network size when regressing ego outcomes on the constraint measure is problematic.

Burt (1992) also introduced another measure, hierarchy, which is an extension of constraint. It quantifies the degree to which an ego network is constrained by a single alter. This measures inequality in the distribution of how each alter constrains ego. Strong hierarchy represents a situation in which one of ego's alters is nearly as well connected to ego's other alters as ego is. Burt (1998) found that experienced male managers benefitted from low constraint (many structural holes), but female managers and young males often advanced more quickly under higher constraint, but only if hierarchy was also high. He argued that female managers and young males do not have the legitimacy of older males. As a result, they were more likely to advance when they had a "sponsor." In this situation, female and young male managers "borrowed" the network of the sponsor. The result was a high constraint network, due entirely to having a sponsor connected to all other alters.

In the formula for hierarchy (Equation 15), C is ego's overall constraint, with other symbols retaining their previous definitions.

$$\text{Hierarchy} = \frac{\frac{c_j}{C/N} \ln\left(\frac{c_j}{C/N}\right)}{N \ln(N)} \quad (7.15)$$

7.3.4 Ego Betweenness

In sociocentric network analysis, a popular and populous class of measures is centrality. Centrality is an aspect of a node's position in the network, often described as measuring node importance, power, influence, or status (Borgatti et al. 2013). One particularly well-known measure of centrality is betweenness (Freeman 1977, 1979). Betweenness can be loosely described as the number of times that a node needs a given node (in our case, ego) in order to reach another node by the shortest possible path. Looking back to Figure 7.2, the shortest path from alter 2 to alter 6 goes through ego, so ego "gets a point" for that. Ego also gets a point for being between alter 2 and alter 1. But alter 2 doesn't need ego at all to reach alter 3 or alter 4, so ego gets no points there. When there are multiple equally short paths, ego is given partial credit depending on what proportion of those shortest paths go through ego. For example, to get from alter 2 to alter 5, there are two equally short paths. One of them passes through ego, and ego gets half a point for that path.

The formula for betweenness is given in Equation 16, where k refers to ego, g_{ij} is the number of shortest paths that connect alter i to alter j , and g_{ikj} is the number of shortest paths from i to j that pass through ego. Ego's betweenness is zero when all alters are directly connected to each other, as no alter needs ego to reach any other alter by the most efficient path. Ego's betweenness achieves its maximum value when the ego network looks like a star with ego in the middle. Moreover, the bigger the star (i.e., the more alters), the greater the betweenness score, similar to Burt's structural hole measures. Indeed, ego betweenness serves as an alternative measure of structural holes. It is particularly appropriate when the interest is in power and control, because it captures the extent to which alters are dependent on ego to reach the others.

$$b_k = \sum_{i,j} \frac{g_{ikj}}{g_{ij}} \quad (7.16)$$

7.3.5 Gould and Fernandez Brokerage

While effective size and constraint are the most commonly used measures of structural holes, simpler measures exist. The most obvious is a count of the number of pairs of alters that do not have a tie to each other. This taps the frequency with which ego serves as a broker between two alters, capturing both size and density elements of Burt's measures. In addition, however, it allows us to examine brokerage between different kinds of alters. Suppose there is a categorical attribute that describes both egos and egos' alters, such as what kind of organization they work for in their community: a business, a government agency, or a church/voluntary service organization. Each person belongs to one of these three kinds of organizations, and to the extent that egos span structural holes, they may be brokering between different combinations of types of organizations.

Assuming directed data, this yields five different possibilities (Figure 7.6), which are regarded as different structural roles (Gould and Fernandez 1989). In this figure, ego is B; A and C are alters. Membership in the same organization is depicted with circles around the nodes. The first type of brokerage, the coordinator role, occurs when ego brokers between two alters who work for the same kind of organization that ego does. In the second type, the representative role, ego receives a tie (e.g., a flow of resources) from an alter working for the same kind of organization, but sends to another alter in a different kind of organization.

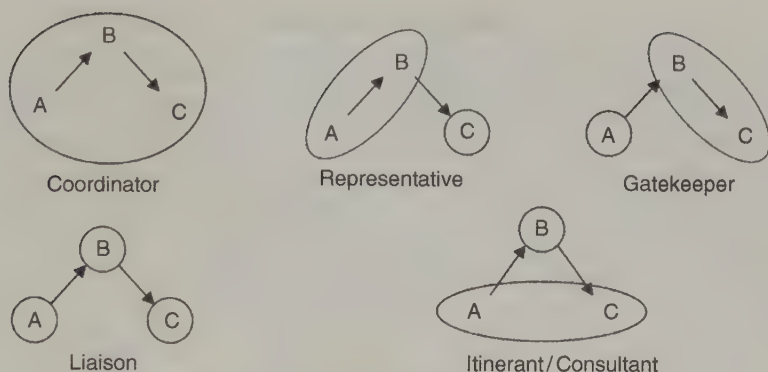


FIGURE 7.6 Gould and Fernandez brokerage types

The third type, the gatekeeper role, is a mirror image of the representative role. Ego receives a tie from a different kind of organization but sends to an alter in the same kind of organization. For undirected data, these two roles are the same. The fourth type is the itinerant or consultant role. This is where ego brokers between alters in the same kind of organization as each other, but different from ego's kind. Finally, the fifth type is the liaison role, in which all three players belong to different kinds of organizations.

The Gould and Fernandez approach provides a finer-grained way of looking at bridging of structural holes. Instead of lumping together situations where ego is between two alters that aren't directly connected, their approach divides brokerage roles into five distinct types of boundary spanners. In similar fashion, racial or ethnic categories, types of ties (e.g., family, friends, and coworkers), or gender can replace organization types as long as they are mutually exclusive.

To use the Gould and Fernandez typology in a research study, the number of times ego brokers in each of the five ways is counted, yielding five variables for directed ties (four for undirected ties). In turn, these can be used as independent or dependent variables in multivariate modeling. In addition, the sum of the five variables for a given ego equals the number of structural holes that ego spans. Looking at Figure 7.2, if color refers to one of two organizations the nodes could belong to, ego serves as a coordinator for two pairs of alters (3-5 and 2-5) and a representative/gatekeeper for seven pairs (4-6, 3-6, 2-6, 1-2, 1-3, 1-4, and 1-5). Thus, ego's coordinator brokerage value is two and ego's representative/gatekeeper value is seven, with a total brokerage value of nine.

7.3.6 Alter Centrality

Alter centrality can be measured by calculating a centrality score for each alter in the ego network (with ego excluded). One reason for calculating alter centrality is to understand the overall effect of the alters on ego. For example, Kennedy and coauthors (2013) studied sexually active homeless men in Los Angeles. Each ego named female alters he had had sex with in the previous year, and was then asked to indicate contact ties between these alters. They found that alters' centrality was positively associated with ego having unprotected sex with that alter. The authors concluded that high centrality reflects an alter's level of embeddedness in the ego's network, and this embeddedness, in turn, builds trust and willingness to engage in riskier sexual behaviors. Similarly, in a study of erectile dysfunction, Cornwell and Laumann (2011) argued that when a man's partner is more central in the man's ego network than he is himself, it puts the man in a position of reduced autonomy, greater dependence on the partner for social contact, and lack of privacy. All of this, they argued, mitigates against feeling masculine in a traditional sense, and leads to erectile dysfunction.

In both studies, the mechanisms purported to drive the relationship between alter centrality and ego outcomes were social psychological. Therefore, how ego perceives the alter's structural position is the key factor, and the alter's actual position is less relevant. In contrast, when studying a mechanism like social capital, which relies on the true ability of ego to leverage resources from an alter's structural position, centrality measures must reflect accurate and complete data on each alter's direct and indirect ties. In a given ego network, an alter might appear peripheral, but only because of the absence of data on alters' alters. In reality, that alter might be just one intermediary away from every alter in ego's network. In this case, ego's perceptions of alter centrality would be inaccurate, but, again, this is not problematic if the hypothesized mechanism is social psychological. In the Cornwell and Laumann (2011) study, for example, ego's perception that the partner was more central in ego's network than ego himself distressed ego and caused erectile dysfunction. In the Kennedy et al. (2013) study, ego's awareness of alter's connectedness in the network fostered trust, which increased ego's willingness to take risks with that alter.

7.4 E-NET FOR GENERATING EGO NETWORK MEASURES

All measures introduced in this chapter can be generated in traditional statistical software, such as Stata, SAS, or SPSS. However, while some are

quite trivial to compute using programming code (e.g., proportion same as ego), others, such as constraint, are difficult and time-consuming to create “by hand.” E-NET – a free software program designed specifically for the organization and analysis of ego networks (Borgatti 2006) – greatly simplifies the task of generating ego network measures. Designed by the creators of UCINET and available for download at www.analytictech.com, E-NET can read data in multiple formats, and has a built-in function to export measures to other software packages. The measures can then be used in regression models or other analyses in any statistical software package (see detailed user instructions for E-NET in Halgin & Borgatti 2012).

E-NET accepts data in three different formats. Ego network researchers most commonly use the Excel-based column-wise format (also known as “wide” format) because all statistical and electronic data-capture software packages are capable of exporting data as Excel files. In this format, ego attributes are located in columns, with one row for each ego. Alter variables are also contained in columns, repeated for each alter and numbered sequentially (e.g., *aage1*, *aage2*, *aage3*, *asex1*, *asex2*, *asex3*). Variables containing information about ties between alters are named using the following format: <variable name><alter id number>-<alter id number> (e.g., “*friend1-2*” indicates whether alters 1 and 2 are friends, “*knows4-6*” indicates whether alter 4 and 6 know one another). Ties between alters may be binary or valued, with a blank space indicating missing data. If these naming conventions are used, E-NET will automatically recognize the different kinds of variables (i.e., ego, alter, or alter-alter ties) when an Excel file is imported. If the data are not named using these conventions, the user can manually identify what kind of variable each column represents.

Table 7.2 provides an example of data in column-wise format and uses the naming conventions for automatic variable recognition in E-NET. Variables “*egoid*,” “*age*,” and “*female*” are characteristics of ego, as indicated by the absence of a numeric suffix. Variables *aage1*, *atie1*, and *aclose1* are attributes (age, type of tie, and closeness, respectively) of alter 1. Variables “*aage2*,” “*atie2*,” and “*aclose2*” are the same attributes for alter 2, where the root indicates the variable name and the numeric suffix indicates alter number. The last three columns, labeled “*frnd1-2*,” “*frnd1-3*,” and “*frnd2-3*,” indicate friendship ties between alters. The variable *frnd1-2* contains information about a non-valued, undirected tie between alters 1 and 2. For ego 1 (i.e., first row of data), alters 1 and 2 are not friends, but alter 3 is friends with both alters 1 and 2. If directed ties

TABLE 7.2 *Data in column-wise format for importing into E-NET*

egoid	age	female	age1	atie1	aclose1	age2	atie2	aclose2	frnd1-2	frnd1-3	frnd2-3
1	28	0	18	4	2	22	3	1	0	1	1
2	36	1	45	1	1	46	1	3	0	0	1
3	21	0	33	3	1	63	1	2	1	1	1
4	45	1	27	2	3	43	5	2	0	1	0
5	51	1	31	1	1	19	3	1	1	1	1

are measured, the data would contain two variables representing potential ties between each pair of alters. Variable “frnd1-2” would indicate whether alter 1 sends a tie to alter 2, while “frnd2-1” would indicate whether alter 2 sends a tie to alter 1. Alter-alter tie data can also be valued (e.g., closeness on a Likert scale 1–4) rather than binary.

Once data are in column-wise format, they can be imported into E-NET. Column-wise is the default data format, but E-NET also accepts row-wise or “long format” data as vna files, as well as sociocentric data as UCINET system files. Once imported, E-NET automatically reads the number of records (i.e., cases) and variables. A variable window (see Figure 7.7) provides options for manually identifying ego variables, tie, and alter variables (i.e., “Ego-Alter Ties and Perceptions”), and alter-alter ties. Alternatively, users can toggle the “Auto” button and variables will be automatically moved to the appropriate boxes (if named according to the aforementioned conventions).

After data have been imported, variables will appear in three different tabs. The “Egos” tab (see Figure 7.8) presents variables that contain information about ego, where each row is one ego. The “Alters” tab displays data on characteristics of alters and ties between egos and alters (i.e., data typically collected with name interpreters). In the “Alters” tab, each row is an alter, so every ego will have a number of rows that corresponds to the number of alters. Finally, the “Alter-alter ties” tab contains information about ties between alters, where each tie that is present between alters is represented by a single row.

The major benefit of using E-NET is the ability to quickly generate network measures that would be difficult to calculate using variable coding syntax in traditional statistical software programs. A complete list of measures available in E-NET is presented in Table 7.3. Any measures that are generated will appear in the “Measures” tab (see Figure 7.9) and can be exported in Excel format for use in regression models or other analyses using Stata, SAS, or other statistical programs. To compute measures in E-NET, the user clicks on the “Analyze” menu at the top of the main window. Three menu options will compute ego network variables using attributes of alters or ties between ego and alters: “Composition,” “Heterogeneity,” and “Homophily.” For each option, users are required to identify categorical (e.g., alter gender, type of tie) and continuous measures (e.g., age, years of education). Subsequently, a standard set of measures is computed and added to the “Measures” tab.

Measures of structural holes can be generated by navigating to “Analyze | Structural Holes.” A window appears which requires the

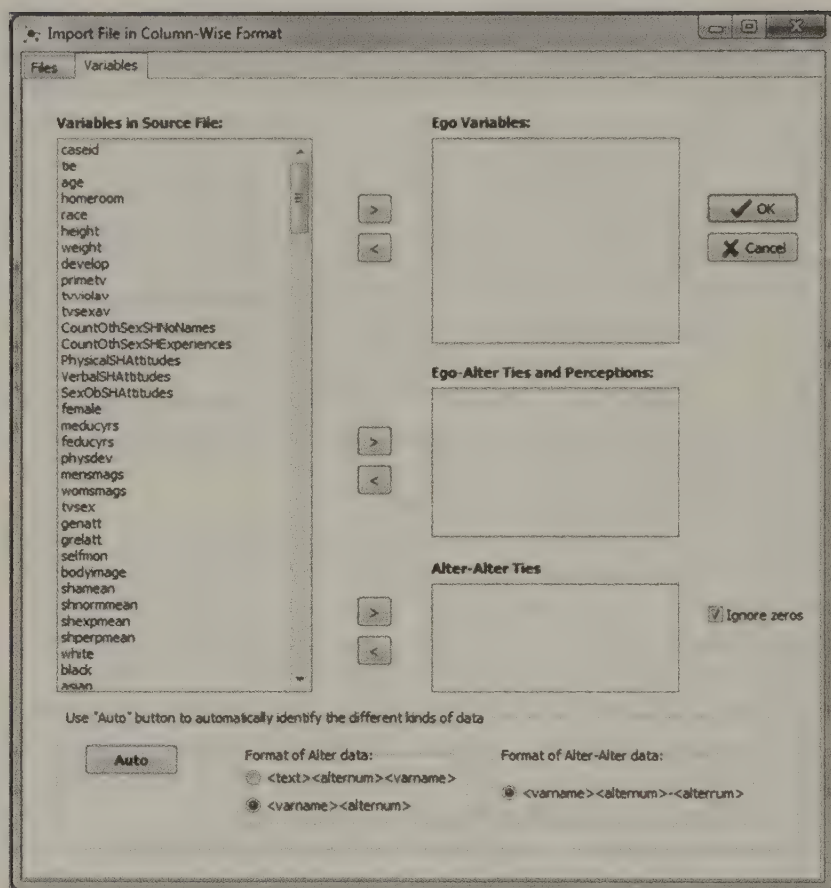


FIGURE 7.7 Screen shot of “File | Import | Column-wise | Variables” window in E-NET (grayscale)

user to identify the alter-alter tie variables that will be employed in calculating these measures. In some studies, multiple types of ties between alters may have been elicited. Alter-alter ties may be binary or valued, reflecting tie strength. In addition, there is an option to identify an ego-alter tie variable in cases where the same tie strength question was used to determine ties between alters and ties between ego and alters. If no such ego-alter tie variable exists, calculations will be made under the assumption that ego has equivalent present ties to each alter. A set of measures of structural holes is computed and added to the “Measures” tab.

TABLE 7.3 *Ego network measures currently computed by E-NET*

Measure	Variable name	Example
<i>Composition</i>		
Percentage	<varname>:<label>	Percent women (sex:female)
Mean	<varname>:Avg	Mean yrs of educ (educ:Avg)
Minimum	<varname>:Min	Minimum yrs of educ (educ:Min)
Maximum	<varname>:Max	Maximum yrs of educ (educ:Max)
Sum of all values	<varname>:Tot	Sum of all alter yrs of educ (educ:Tot)
<i>Heterogeneity</i>		
Blau's index	<varname>:Het	Racial heterogeneity (race:Het)
Agresti's IQV	<varname>:Iqv	Racial heterogeneity (race:Iqv)
Standard deviation	<varname>:SD	Age heterogeneity (age:SD)
<i>Homophily</i>		
Proportion same as ego	<varname>:SameProp	Proportion same religion (relig:SameProp)
E-I index	<varname>:E-I	Religious homophily (relig:E-I)
Average Euclidean distance	<varname>: AvgEucDist	Income homophily (income:AvgEucDist)
<i>Structural holes</i>		
Degree	SH:Degree	—
Density	SH:Density	—
Effective size	SH:Effsize	—
Efficiency	SH:Efficiency	—
Constraint	SH:Constraint	—
Hierarchy	SH:Hierarchy	—

Note: Measures can be exported to standard statistical software packages.

In the “Visualization” tab, users can scan through network diagrams for each ego. Figure 7.10 depicts a screen shot from E-NET of an ego (center node) with five alters and ties between them. The network diagrams can be configured as a wheel and spokes with ego at the center of the layout, or with a spring embedding layout that positions nodes that are socially distant farther away in physical space (see Chapter 6). E-NET also provides a data filtering capability, allowing users to select which egos, alters, and ties are included in variable construction and visualizations. For instance,

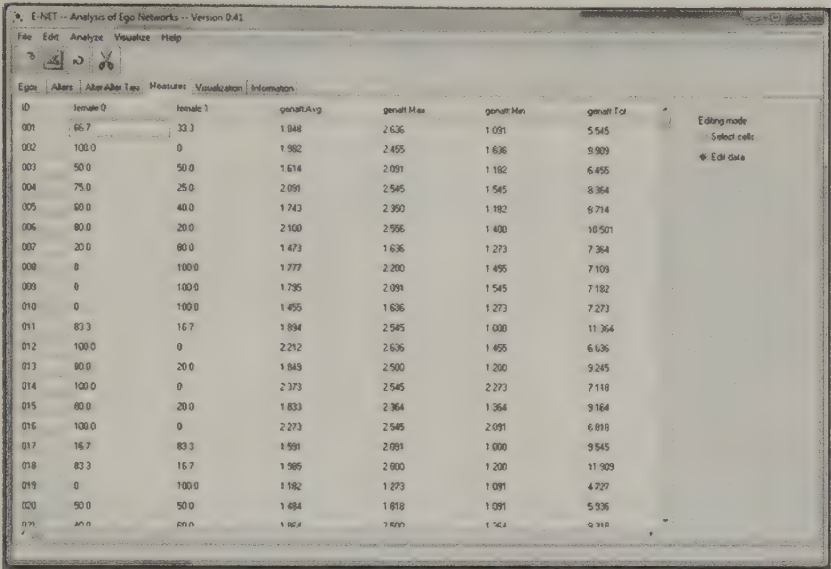


FIGURE 7.9 Screen shot of “Measures” tab view in E-NET (grayscale)

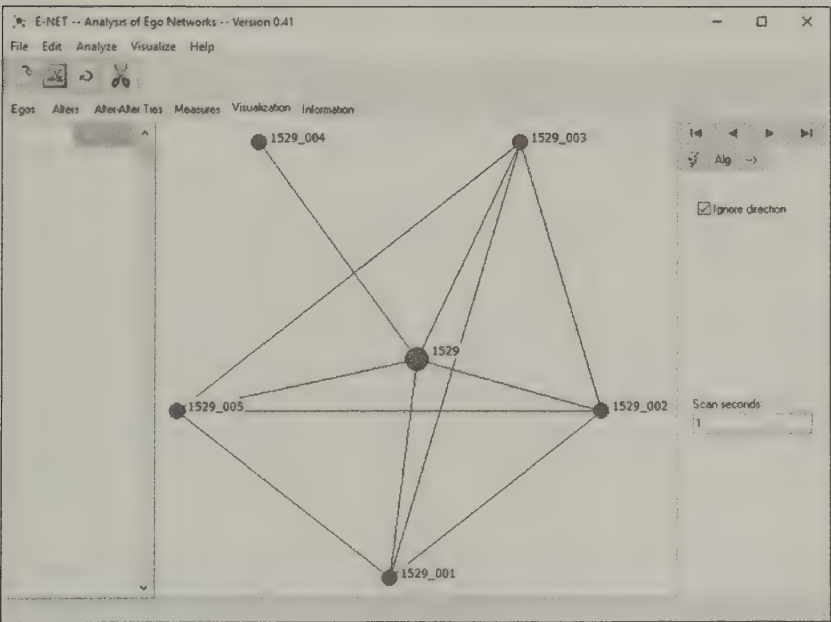


FIGURE 7.10 Screen shot of “Visualization” tab view in E-NET

conducting an analysis of older adults which compares retired and employed egos, any egos under the age of 55 could be filtered out.

7.5 CONCLUSION

In this chapter, we have presented measures that characterize the composition and structure of an ego network. Many of the measures discussed, such as counting the number of components in an ego network, are purely structural in the sense that they ignore ego and alter characteristics, considering only network configurations. Others, such as ego network composition measures, ignore structure and focus exclusively on alter characteristics. Finally, a few, such as the Gould and Fernandez brokerage measures, combine both structure and alter characteristics. Most of the measures discussed can be calculated using the E-NET software package. These measures can then be used to test a variety of research questions about individuals, relationships, and networks, as discussed in Chapter 8.

Multivariate and Multilevel Regression Models for Ego Networks

The goal of collecting ego network data and creating the measures discussed in Chapter 7 is to test hypotheses. As a starting point, networks can be conceptualized as independent variables, dependent variables, or both. Further, research questions can be examined at multiple levels of analysis, specifically, for ego networks, the focus may be on predicting characteristics of egos or characteristics of alters. Any discussion of statistical modeling of ego networks requires a clear explanation of these classifications.

After laying this foundation, basic and more advanced regression modeling techniques for ego network analysis are introduced. Ego network analysis using standard multivariate regression models is reviewed, focusing on when and why to employ this approach. Topics include limitations of these simple models and common violations of regression assumptions that are of particular concern in ego network analysis. Next, a discussion of multilevel modeling (MLM), increasingly common in ego network research, outlines the merits and basic logic of this more advanced approach, followed by additional detail about how these models work. Finally, a step-by-step guide to multilevel model building in the statistical software package Stata is provided.

8.1 NETWORKS AS PREDICTORS AND OUTCOMES

According to Borgatti and Halgin (2011), the landscape of any network research, whether sociocentric or egocentric, can be understood by examining the role of network processes. At a basic level, network variables can

TABLE 8.1 *Typology of theoretical approaches to the study of social networks*

Independent variable	Dependent variable	
	Non-network variable as outcome	Network variable as outcome
Non-network variable as antecedent	A. Not a network theory	B. Theory of networks
Network variable as antecedent	C. Network theory	D. Network theory of networks

Source: Borgatti and Halgin (2011)

be predictors (i.e., independent variables), outcomes (i.e., dependent variables), or both. Table 8.1 presents a two-by-two table where the independent variables and dependent variables are classified as either network-based or not network-based. This creates four cells, representing four kinds of theory and modeling.

The first quadrant (cell A) targets research in which neither the explanatory variables nor the outcome variables are network-based. This research is outside the purview of network science. Cell B identifies research in which the outcome variables are network-based, but the explanatory variables are not. This is research that seeks to develop a “theory of networks,” and focuses on explaining the nature of networks, how they form, and how they change using non-network processes or mechanisms. For example, a theory of networks might argue that distinct stages of the life course are associated with systematic differences in the nature of a person’s family and peer networks.

Cell C, network theories, reverses the causal direction, focusing on non-network outcomes as functions of network-based mechanisms. Outcomes could be any of a wide range of phenomena to which networks have been applied – the workings of the brain, the response to illness, how the population votes, whether or not individuals or cities recycle, or whether organizations thrive or fail, among dozens of others. In each case, the researcher posits that ties between nodes shape outcomes; that is, they are the engine of action. Two canonical types of network theories are social capital theories and diffusion theories (Borgatti & Halgin 2011). Social capital theories essentially attempt to explain individual achievement, reward, or well-being as a function of a person’s network ties. Diffusion theories seek to explain how networks influence the spread

of outcomes, from why people hold the beliefs, attitudes, and behaviors that they do, to how and where epidemics peak.

Finally, in Cell D, *both* the phenomena being predicted and the key explanation lies in networks. The notion that broader network dynamics at the organizational level filter down to change the networks of individuals would constitute a “network theory of networks.” For example, in *Everything in Its Path*, Kai Erikson (1976) documented how the network structure of living arrangements set by FEMA in the wake of the Buffalo Creek flood in West Virginia fundamentally decimated the pre-existing community-based network ties of the individuals who were displaced by the event. Altered housing networks disrupted the functions of personal social networks, impeding the typical assistance functions of community ties in disasters and prohibiting individuals’ ability to recover.

This distinction between network theory and theory of networks applies to both sociocentric and ego network research. That is, the consequences of network processes (i.e., network theory) can be examined using a sociocentric or an ego net research design. In a sociocentric design, career success might be hypothesized to be a function of having an advantageous position in an organizational trust network. In an ego network design, career success might be hypothesized to be a result of the resources a person has access to through their personal social ties, whether inside the organization or not.

8.2 LEVELS OF OBSERVATION

In addition to determining whether networks are antecedents or outcomes, ego network processes can be measured at two different levels of observation, at minimum, to start – the ego level and alter level. Level of *observation* refers to the data point being analyzed, and is sometimes referred to as the unit of observation, whether years, people, neighborhoods, or nations. Typically, the level of observation of greatest interest is the dependent variable, since this determines the types of statistical models that are appropriate.

In the framework of hierarchical modeling, the lowest level of observation in cross-sectional ego network research is the alter level. Here, the variables have a value for each alter within an ego network. Variables can be attributes of the alters, such as their age or gender, or characteristics of the tie (i.e., relationship) between ego and the alter, such as whether they are close, discuss matters of health with each other, or interact frequently. Variables observed at the alter level come from the name interpreter

component of a survey, where ego is asked about each alter or their relationship with each alter.

The next level of observation in egocentric research is the ego level. Here, the variables have a value for each ego. They can be attributes of egos, such as voting behavior or test performance, or they can be characteristics of ego's networks, including density or gender composition. A key distinction between ego and alter levels of observation is that every alter attached to a particular ego gets the same value for ego-level observations. This is analogous to all students at a university having the same value for school graduation rate. Likewise, there is one value for network size that applies to all alters, and each alter, whether a man or a woman, is nested in a network with the same proportion of women.

A peculiarity of egocentric research is that alter-level observations are often used to calculate ego-level observations. This is called aggregation, or aggregating alter attributes up to the ego level, as with alter gender and gender composition of the network. Ego level refers to the ego level of observation, because each alter gets the same value for an ego-level aggregated variable.

Another potentially confusing aspect of ego network research is that the level of *observation* may not match the level (or unit) of *analysis*. The former concept describes the data, while the latter refers to the target of research (i.e., what is being studied; Babbie 2015). For example, we might want to draw conclusions about countries (e.g., the efficacy of welfare state policies) through an analysis of data collected from individual residents of five different nations. In this case, the level of observation is *individuals*, while the level of analysis is *countries*. In ego network research, the goal might be to predict voting behavior (individuals), development of trust and intimacy (relationships), or racial diversity of peer groups (networks). While hypotheses might be at any of these levels of analysis, there are only two levels of observation – egos and alters (Figure 8.1).

Levels of *analysis* are important for formulating research questions and testing theory. However, levels of *observation* determine the kinds of statistical analyses that are possible. For this reason, the focus here is on levels of observation. In egocentric research, there are usually two levels of observation (ego and alter), but there may be more. When ego networks are collected at multiple points in time, there are ego, alter, and time observations. Likewise, when ego networks are nested in a higher-level context like neighborhoods, there are neighborhood, ego, and alter

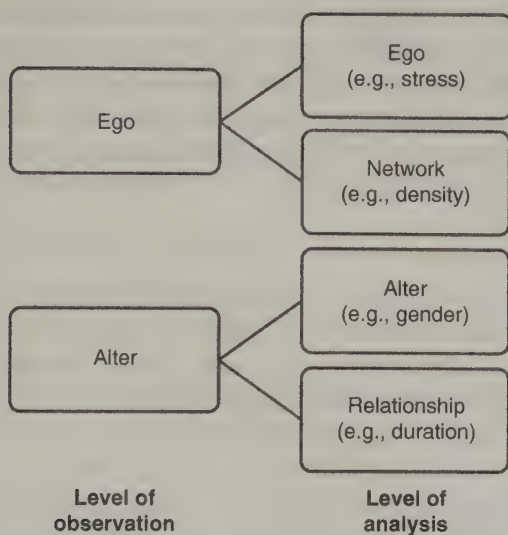


FIGURE 8.1 Levels of observation and analysis in ego network research

observations. We begin with a discussion of multivariate regression models where the dependent variables are ego-level observations.

8.3 ANALYZING EGO-LEVEL NETWORK DATA WITH STANDARD REGRESSION MODELING

When observations are at the ego level, standard regression modeling techniques are appropriate. That is, to predict ego happiness as a function of the average happiness of alters, ordinary least squares (OLS) regression is appropriate (assuming other standard regression assumptions, such as linearity, have not been violated). Likewise, OLS can be used to determine whether the educational attainment of egos influences the density of their networks. Along the same lines, regular binary logistic regression is appropriate for dichotomous outcomes at the ego level. In each of these cases, each ego has one, and only one, value for the independent and dependent variable.

8.3.1 Review of the Linear Regression Model

While knowledge of OLS regression is widespread, reviewing the model facilitates a comparison with more complex multilevel models introduced later in this chapter. An OLS regression model has two major

components. Part I is called the model for the means and is composed of fixed estimates. “Fixed” means that there is a single estimate that applies to every observation. This is the part of the model of greatest interest to researchers – the coefficients reveal how the outcome varies, on average, as a function of predictors. The intercept β_0 and slope β_1 in the regression equation below are fixed estimates:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (8.1)$$

Part II of an OLS regression is the model for the variances and includes residuals. The residual term ε_i is random, meaning that each observation has its own value. The error term represents the difference between the prediction (a function of the fixed estimate) and the actual value of that observation. Typically, researchers make assumptions about the residuals (e.g., that they are normally distributed), and conduct diagnostic tests (e.g., residual plots) to determine whether these assumptions have been violated. Residuals (i.e., how “off” the prediction is) are usually not interpreted or reported, but they play an important role in the calculation of estimates. Namely, the goal of OLS regression is to calculate optimal estimates of β_0 and β_1 such that the sum of the squared differences between the values of the dependent variable y predicted by the regression equation and the actual values of y are as small as possible. Since this is a linear regression function, the slope and intercept can be interpreted as the average effect of x on y , and the average value of y when x is 0, respectively.

8.3.2 Common Model Violations Caused by Ego Network Data

Ego-level network data can be treated like any other data analyzed with standard regression techniques. However, testing for common model violations is very important because these are often more pronounced in the context of ego network research. For example, multicollinearity occurs when two or more variables in a regression model are highly correlated. When substantial multicollinearity is present, the standard errors and p-values associated with coefficients are suspect. Which variables among those that are correlated are statistically significant is quite random. Multicollinearity is common in egocentric research because measures of ego networks are often correlated in practice. In particular, variables like the network’s average closeness, supportiveness, frequency of contact, and duration of relationship tend to be closely correlated. All of these are antecedents or consequences of intimacy.

Further, egocentric survey designs are susceptible to common method variance, or systematic variance among variables measured using the same method (Podsakoff et al. 2003). In other words, someone more comfortable with the name generator/interpreter format might answer those items differently than someone who finds them cognitively burdensome. This potentially creates correlations that are due to the method of measurement rather than to the constructs the method was designed to capture. Consequently, when conducting analyses involving ego network variables, multicollinearity should be evaluated and addressed, usually by removing collinear variables until the level of multicollinearity is acceptable.

Nonlinearity and skewness are also commonly observed in ego network data. Several standard network variables are nearly always skewed (i.e., not normally distributed), including network size (often positively skewed) and average closeness or tie strength (often negatively skewed), particularly in research designs that elicit core networks (see Figure 2.3). Taking network size as an example, most egos will have a core network size of between three and five alters. However, a handful of egos with very large core networks create a long tail in the distribution. Skewness is typically associated with nonlinearity and often with errors that are not normally distributed; both are violations of OLS regression assumptions.

Nonlinearity can also have important implications for theory testing. The prediction for the dependent variable is assumed to be a linear function of the values of each independent variable, plus the random error term. However, when a variable like network size is skewed, larger networks often provide diminishing returns. For instance, having one additional friend when the network only contains two friends may have a substantial effect on ego's happiness. However, if ego already has a dozen friends, adding one more friend may have a smaller effect on happiness. Skewness and nonlinearity can often be corrected with variable transformations, and/or the addition of squared or cubed terms to the standard linear model (i.e., a polynomial model), as described in detail in most introductory texts for multivariate regression (e.g., Lynch 2007).

Finally, the nature of egocentric data collection often leads to heteroscedasticity. A basic assumption of the OLS regression model is that the variance of the error term is constant, and is not dependent on the value of any independent variables. In other words, a scatterplot with a fitted regression line would depict data points scattered relatively evenly on

either side of the line for the entire length of the line. When they are not, this creates heteroscedasticity, biasing standard errors and making the determination of statistical significance unreliable. Ego network data are susceptible to heteroscedasticity because people with large networks (i.e., egos who name many alters), are less likely to report accurately on the attributes of their alters and ties among them (see Chapters 4 and 5). A person with one or two friends will be more likely to accurately report on the political party identification of those friends than a person with twenty friends. There is less information to recall, and less information to actually know. In this case, the errors (e.g., differences between the real value of percent of alters who are Democrats and the value predicted), will be greater for those with larger networks. The observations will be more spread out around the fitted regression line at higher values of network size compared to lower values of network size. Because many network variables of interest are correlated with network size, heteroscedasticity is often a problem. This model violation is easy to assess using variance inflation factors and trivial to correct with standard error adjustments (see Hayes and Cai 2007).

8.3.3 An Illustration of the OLS Regression Model: Extroversion and Likeability

A concrete example of how OLS regression works may be a review for most readers. However, this example will illustrate and clarify concepts that are integral to understanding the more complex models introduced later in this chapter. Suppose we are researchers interested in explaining what makes people like one another on first impression. We conduct an egocentric survey, asking respondents to name the three most recent people they met for the first time. Respondents are instructed to rate each person's likeability on a scale from 1 to 10. These data contain some number of egos who have named up to three alters each. We might create a dependent variable that measures the perceived likeability of the "new ties" network as a whole, taking the mean likeability rating for each ego's alters. Each ego would have one value for their network, making the level of observation the ego.

Now suppose that we want to add an independent variable. Because the level of observation for the dependent variable is the ego, OLS regression would be an appropriate strategy for modeling the relationship between an explanatory variable and mean likeability of the "new ties" network. We might hypothesize that egos who are extroverts have more favorable

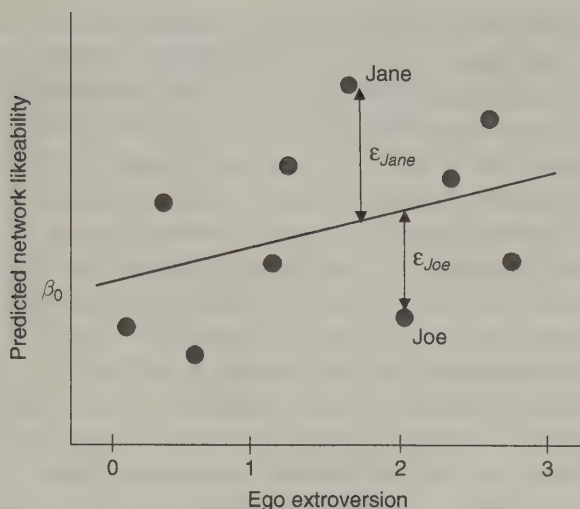


FIGURE 8.2 Scatterplot with fitted OLS regression line for the effects of ego extroversion on network likeability

first impressions, rating people they have just met as more likeable than egos who are more introverted. Extroverts tend to be outgoing, sociable, and gain energy through interaction, while introverts are more introspective and must expend energy to interact with others.

Figure 8.2 presents a scatter plot of ten observations, with mean likeability of the “new ties” network on the y-axis and ego extroversion on the x-axis. Each observation is an ego, including Jane and Joe, whose errors are illustrated by dotted lines. Jane rates her “new ties” network more positively than does Joe. In each case, however, the prediction indicated by the fitted regression line does not reflect the actual data. It contains error. In Jane’s case, she rates her network as more likeable than we would predict on the basis of her level of extroversion – her random error term will be positive. Joe rates his network as less likeable than we would predict – his random error term will be negative.

In this model, variation between egos (in both our independent and dependent variables) is used to model the relationship between ego extroversion and subjective first impressions of people they have just met. The slope is positive, consistent with our hypothesis. The higher people score on an extroversion scale, the more favorably they rate the mean likeability of their network, relative to other egos. Because the level of observation is the ego, the comparison is between egos – one ego compared to another ego.

Because data were collected from the ego respondents about three people they have recently met, there is another way to conduct this analysis. Instead of predicting the mean likeability of the *network* of alters (an ego level of observation) we could predict *each individual alter's* likeability rating (an alter level of observation), ignoring that each alter is connected to a particular ego. This strategy would have a number of advantages. It would capitalize on the richness of data at the alter level. By aggregating likeability to the ego level, a lot of information about alters is lost – most notably all the interesting variation in likeability ratings for a given ego's new ties (i.e., variation within egos). Further, if we wanted to examine relationships between two alter-level variables (e.g., alter attractiveness and alter likeability), aggregating both of these variables to the ego level creates an ecological fallacy. That error occurs when data at the level of a group are used to generate conclusions about individuals. Finally, by analyzing the data at the alter level, we could examine interactions between the ego and alter levels of observation. Perhaps the likeability evaluations of extroverted egos are more strongly influenced by alter attractiveness than are those of introverted egos. The latter group's ratings might be more affected by alters' internal traits. It would not be possible to test this hypothesis using ego level data.

While there are many reasons to conduct the analysis of alter likeability at the alter level, standard OLS regression cannot be used to predict alter level observations. Doing so would violate an OLS regression assumption about error terms. Specifically, the errors, or residuals, are expected to be independent of one another. If alter observations associated with a particular ego are more similar to each other than to the alter observations of other egos, this creates dependence. Violating the assumption of independence yields regression models that are inefficient and/or biased, potentially causing the researcher to make inaccurate conclusions about the data. Using a special class of regression called multilevel modeling, or variance components modeling, allows the researcher to analyze observations at the alter level without violating regression assumptions.

8.4 ANALYZING ALTER-LEVEL NETWORK DATA WITH MULTILEVEL MODELING

Often, researchers are interested in predicting characteristics of alters or of relationships between an ego and their alters. For example, a study of micro-interactions in the workplace might address who defers to the boss, where the boss is ego. This is not a research question effectively answered

by analyzing ego-level observations, especially if characteristics of alters or of the relationships between egos and alters are important predictors. We might hypothesize that women are more likely to defer to the boss, and in this case our independent variable, alter gender, is observed at the alter level. Likewise, we could hypothesize that alters who work more frequently with the boss are less likely to defer to her. Again, the predictor is measured at the alter level. However, the boss's age may matter, with alters being less deferential to younger bosses. In short, predicting variation in alter-level observations as a function of variation in both alter- and ego-level observations simultaneously is often of interest, as are interactions between levels. It is possible to test all of these hypotheses with multilevel models.

8.4.1 Why We Cannot Use OLS Regression to Predict Alter-Level Outcomes

OLS regression is suitable for outcomes observed at the ego level. However, ego network research often focuses on features of alters and relationships, such as why egos develop cross-racial ties, or what influences the development of intimacy. In these cases, the answers may depend, in part, on ego's characteristics or the characteristics of ego's network. For example, the amount of support that any given alter provides to ego likely depends on factors such as ego's need for support or the average amount of support provided by the network in general. Alters embedded within one ego network provide a level of support that is likely more similar to one another than to alters nested in a different ego network.

This within-network similarity is called dependence, or clustering, and leads to a violation of the OLS regression assumption that errors are independent. In this circumstance – when the outcome of interest is observed at the alter level – a multilevel modeling approach is necessary to avoid producing biased estimates and standard errors. Dependence is a hallmark of hierarchical data structures, where some set of observations is nested in another class of observations (see Figure 8.3). In the case of ego networks, alter observations are nested in ego observations. In the parlance of MLM, the lower level of observation is alters (Level 1) and the higher level is egos (Level 2). Specifically, Jen, Pat, and Sam are Level 1 alters nested in Level 2 ego Lawrence.

There are two sources of dependence in hierarchical data, including, but not limited to, ego network data. That is, there are two distinct reasons why the variation between ego networks is likely to be greater

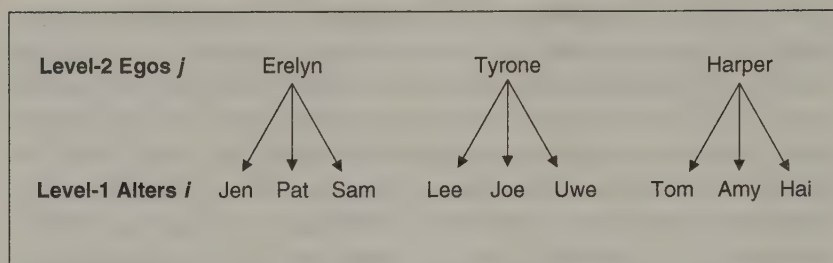


FIGURE 8.3 Hierarchical data structure for egocentric network analysis

than the variation within ego networks (Raudenbush & Bryk 2002). First, a hierarchical data structure often results in mean differences in the dependent variable across Level 2 observations. In the context of ego network research, alters nested within a particular ego are likely to differ, on average, from those nested in a different ego. In predicting social capital available to ego, data would reflect that some egos have networks with higher mean levels of education than others, perhaps as a function of ego's own educational attainment, occupation, family background, etc. Consequently, assigning an intercept for each ego that reflects mean educational differences of alters across clusters (e.g., ego networks) would be more useful than having one intercept to represent all the alters in the dataset. Each ego's intercept would be based on their own alters' educational attainment, more accurately representing the baseline level of social resources an ego can access.

A second source of dependence in network data lies in differences in the effects of alter-level predictors across ego networks. For example, the effect of frequency of contact between an ego and alter on ego's evaluation of the closeness of that relationship may vary from ego to ego. Some people easily maintain affectively strong bonds with friends and relatives that they rarely see or talk to, while others require more frequent contact to sustain closeness. Differences across egos in the slopes of the frequency of contact variable, measured at the alter level, are expected. In this case, it is ideal for each ego to have their own slope to represent the relationship between frequency of contact and closeness among their own alters. This provides a more accurate reflection of the effect of frequency of contact for any particular ego than estimating the slope using all of the alters' data simultaneously and forcing the slope to be the same for all egos.

Ignoring these two forms of dependence – intercept and slope dependence – associated with the hierarchical structure of ego network

data has negative implications for statistical models. Multilevel models account for this dependence by including the ego-specific intercepts and slopes described above, removing the threat of model violation in the process.

8.4.2 Additional Advantages of Multilevel Modeling

Multilevel models have a number of other benefits, providing opportunities for testing research questions that cannot be addressed with conventional regression models. Most ego network research involves determining how characteristics of the network context influence egos' behaviors, attitudes, or outcomes. When predicting an ego level dependent variable, information about alters or ties must be aggregated to the ego level of observation and traditional regression models are appropriate. However, quite often in ego network research, ego level observations are also used to make inferences about alters or relationships. For example, the gender composition of a network (e.g., percent women) may be positively related to the mean level of supportiveness of the network. This correlation does not actually test whether women, as individual alters, provide more support than men, even if we are tempted to draw that conclusion. But to claim this is to commit an ecological fallacy, as described above. Aggregate findings on gender composition cannot be translated directly to the actions of men and women at the individual level.

Another disadvantage of aggregating all alter-level data to the ego level of observation is that we can only speculate about what is happening at the alter level. Moreover, a lot of potentially informative heterogeneity among alters is being ignored. Wellman and Frank (2001: 235) note that "People who provide support are not homogeneous grains of sand . . . Hence, when analyses of social characteristics look only at the attributes of what aggregated heaps of individuals 'possess,' they neglect variation in which kinds of alters provide support." In other words, properties of the network as a whole do not adequately capture properties of individual alters, nor do they capture how relationships influence how alters behave, how ties develop, or to whom they connect.

MLM offers the ability to estimate coefficients at the ego and alter levels, while simultaneously controlling for variables measured at other levels of observation (Wellman & Frank 2001). In contrast, aggregation to the ego level makes it impossible to tease out the relative effects of processes occurring at different levels. This introduces the possibility that alter or relationship effects will be misattributed to network effects or vice

versa. Wellman and Frank's multilevel study of support provision in East York (2001) – where they found independent alter-level and network-level effects of gender – illustrates this difference. Specifically, female alters were more likely to provide support relative to male alters, on average. In addition, networks composed of higher percentages of women were also disproportionately supportive compared to networks with fewer women. This suggested that gendered network contexts potentiate or constrain the amount of support provided by the network as a whole. An ego-level analysis could not have distinguished between these social processes, which have very different implications for social resources and wellbeing.

Multilevel modeling also holds the ability to test cross-level interactions – interactions between variables observed at the alter and ego levels. MLM permits explicit modeling of interactions between all levels of observation, in addition to more conventional within-level interactions. The social forces of individual agency and the more structural constraints or opportunities afforded by the social network context in which ties are embedded are often important theoretical distinctions. In contrast to traditional regression modeling, MLM allows the researcher to explicitly model the complex contingencies that develop from the interplay of ego, alter, tie, and network characteristics (Wellman & Frank 2001).

8.4.3 What Is a Multilevel Model?

Multilevel models are a class of statistical models designed to accommodate hierarchical and longitudinal data structures. A number of variations within this family of models exist, and they have many different names, including mixed models, random effects models, fixed effects models, variance components models, hierarchical linear models, and growth curve models (if data are longitudinal). All of these terms refer to specific types of multilevel models for multilevel data structures. They share many basic assumptions and capabilities.

Classic multilevel research conditions include students nested in schools, people nested in countries, and employees nested in organizations. The higher level can be conceptualized as the context in which lower-level entities operate. Context tends to influence micro-processes occurring at the lower level. For example, expenditures per student at the school level (i.e., the amount of funding provided to schools) are likely to affect the academic performance of students nested in those schools. Well-funded schools have higher paid teachers, better learning tools

and technologies, and fewer students per classroom. Students in poorly funded schools do not have these advantages, and likely perform worse, on average. In short, different schools produce different outcomes. Consequently, hypothesizing that students nested in the same school will perform more similarly to one another than to any random student in another school, on average, is reasonable. Multilevel models adjust for this *dependence* – or similarity between observations nested in the same context – by adding one or more extra random variables that are estimated using the sample data.

Three conditions need to be met to use MLM for ego network analysis (Snijders, Spreeen, & Zwaagstra 1995). First, as noted above, the dependent variable must be measured at the alter level (i.e., Level 1). If a researcher is interested in predicting a characteristic of ego (e.g., health, employment outcomes, social movement participation) or of the ego network as a whole (e.g., density, heterogeneity), MLM is not appropriate because there is no variation at Level 1. That is, all alters have the same value for ego's health or the density of the network.

Second, observations at the ego level must be independent of one another. In other words, the researcher would ideally have a probability sample of ego respondents that are not linked to one another. Dependence at the ego level violates the assumption of independent errors – the very same assumption that MLM is meant to address, but cannot fix in this case. This assumption still applies to the highest level in a multilevel model, which is often the ego level in egocentric network research. However, if ego dependence is patterned by design, such as sampling residents from neighborhoods, the researcher can fit a three-level model with alters, egos, and neighborhoods.

Third, personal networks of egos cannot overlap. This overlap occurs when the same alter appears in two or more ego networks. In practice, this happens occasionally and at random when sampling from one context or community, and can be safely ignored. However, when overlap is pervasive, as when sociocentric methods are used to collect data, MLM is not appropriate. A separate class of models for egocentric analysis of whole networks exists (Chapter 10). However, at times, the distinction between egocentric and sociocentric data is fuzzy. For example, members of an ethnically homogeneous immigrant community in one metropolitan area might be recruited via snowball sampling. When egos are asked to name friends and family members, there is a strong likelihood that egos will be named as alters, and that the same alter will appear in multiple networks. If alters named to multiple ego networks can be matched based on name or

other characteristics, a special type of MLM called cross-classified models may be used (see Section 8.4.13). However, if alters cannot be uniquely identified, a cross-classified MLM will not work and a hierarchical MLM will likely produce biased results, depending on the extent of network overlap. In this case, aggregating all alter-level data to the ego level and employing standard regression models represents the best solution.

8.4.4 The Random Intercept Model

In OLS regression, the only random variable is the residual.¹ “Random” here means that each observation gets their own value. OLS fits a regression line through the data points in a way that minimizes residual variation (Equation 8.1). In an OLS regression with a dependent variable at the alter level, each observation (alter) would have a residual ε_i equal to its observed value minus its fitted value (Equation 8.2). The residuals for alters nested within the same ego would likely be dependent, or clustered.

$$y_i = \beta_o + \beta_1 x_i + \varepsilon_i \quad (8.2)$$

$$\varepsilon_i = y_i - \hat{y}_i \quad (8.3)$$

In the random intercept model, dependence (i.e., clustering) is explicitly modeled by splitting the residual ε_i into two random components, one for each level. The random intercept model with one independent variable, adapted from Raudenbush and Bryk (2002) can be written as:

$$y_{ij} = \beta_o + \beta_1 x_{ij} + \zeta_j + \varepsilon_{ij} \quad (8.4)$$

In this equation, j represents the Level 2 (i.e., ego level) identifier, while i is the Level 1 (i.e., alter level) identifier. The formula is unchanged relative to Equation 8.1, except that the single random residual has been replaced with two random components. In ego network research, there is an ego-specific random intercept ζ_j , and an alter-specific residual ε_{ij} . The ego-specific random intercept ζ_j is constant across all alters, while the alter-specific residual ε_{ij} represents the deviation of each alter’s observed value from the ego-specific random intercept ζ_j . Figure 8.4 presents an example where one ego has three alter observations.

A variable’s variance quantifies how much the observed data points vary around the regression line. The larger the residuals, the higher the

¹ While each observation does have its own value for the independent variable x , this is not an estimate, or model output. It is a model input.

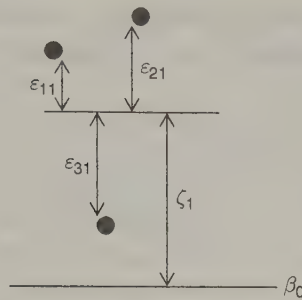


FIGURE 8.4 The random intercept model splits up the intercept variance into within- and between-cluster differences

variance. The difference between the OLS regression model and random-intercept MLM lies in the way the variance is handled. The latter splits the variance into two types (Hoffman 2015): (1) between-ego variation, or Level 2 inter-cluster differences ζ_j , which describe how egos differ from other egos; and (2) within-ego variation, or Level-1 intracluster differences ε_{ij} , which describe how alters nested within the same ego differ from other alters in that same network. Comparing Equations 8.1 and 8.3, each observation differs from the overall mean β_0 by a total residual ε_{ij} , which is the sum of two residual components, ζ_j and ε_{ij} . This is where the term “variance components model” comes from. MLM permits an examination of both within- and between-ego heterogeneity simultaneously.

8.4.5 Interpreting the Random Intercept Model

One of the first elements of the model to interpret is how much dependence exists in the data. The intraclass correlation (ICC), or ρ (rho), measures the variance within clusters as a function of total residual variance (i.e., both within-cluster and between-cluster variance). It can be expressed as between-cluster heterogeneity or within-cluster homogeneity. These are two sides of the same coin. The ICC is generally interpreted as a measure of within-cluster correlation (Raudenbush & Bryk 2002). The ICC ranges from 0 to 1, where higher values indicate more correlation within clusters, similar to Pearson’s r . However, because there can never be less than zero variance, the ICC, unlike Pearson’s r , cannot be negative. The ICC represents a standardized way of expressing the amount of dependence due to cluster mean differences. The larger the ICC, the more problematic it

becomes to ignore dependence. As the ICC approaches 0, estimates from a random intercept model look more similar to OLS regression estimates, where clustering is ignored.

As the above suggests, hierarchically structured data may, in actuality, exhibit a negligible degree of dependence (ICC ~ 0). This question can be answered using a likelihood-ratio test of the random intercept model. The null hypothesis is that the random intercepts are equal to 0. This would only be the case if the residuals were independent. If the null hypothesis is rejected at $p < .05$, this suggests that a multilevel model is needed to account for ego-induced dependencies. If the likelihood-ratio test is not significant, OLS regression is appropriate to analyze alter level observations.

The components of the multilevel model are typically estimated using maximum likelihood, though other estimation procedures are possible (see Rabe-Hesketh & Skrondal 2008 for estimation details). The fixed estimates are the intercepts and slopes. The fixed overall intercept β_0 is a weighted mean of the cluster means. The weighting process means that some clusters provide more information toward the overall intercept than others. Clusters weighed more heavily are those with lower within-cluster variance and a higher number of Level 1 observations. In the case of egocentric research, egos whose alters are more similar to one another and egos with more alters (i.e., larger networks) are weighed more heavily. The fixed intercept is interpreted the same way for OLS and multilevel models. That is, it is the predicted value of the dependent variable when all independent variables are equal to 0. Likewise, fixed coefficients for the effects of independent variables on the dependent variable are interpreted identically to OLS (i.e., the predicted change in the dependent variable associated with a one-unit increase in the independent variable).

Finally, ego-specific random intercepts are treated as random components in MLM (Rabe-Hesketh & Skrondal 2008). They are usually not interpreted. However, estimated values for the random intercepts ζ_j for each ego can be obtained. This process would be similar to obtaining predicted residuals in an OLS model. The predictions can then be used for MLM diagnostics, visualizing models, and for examining particular ego networks. For instance, knowing a particular ego's "true" intercept value, which would be equal to $\beta_0 + \zeta_j$, might be particularly useful in applied settings (e.g., working with patients or clients).

8.4.6 An Illustration of the Random Intercept Model: Attractiveness and Likeability

Returning to our fictional study of factors that influence likeability on first impression, we have data in which egos named up to three people they had recently met for the first time. Clustering within ego respondents is likely, such that some egos rate people as more likeable, on average, than others. This might be because some egos tend to be particularly generous (or harsh) in their first impressions.

The alter observations of two fictional egos – Jane and Joe – are presented in Figure 8.5. Jane has named three new ties, Don Juan, Bob, and Ann. She has rated Don Juan as her most likeable new associate, and Ann as least likeable. Though Jane rates people she meets as more likeable, on average, than other egos (all of her observations are above the overall mean β_0), there is substantial within-cluster (i.e., within-network) variation. Joe named two new associates, Amy and Sue, both of whom he rates as less likeable, on average, than other egos rate people they have recently met. Joe also has less within-cluster variation than Jane does, meaning Joe's alter observations are more tightly clustered around

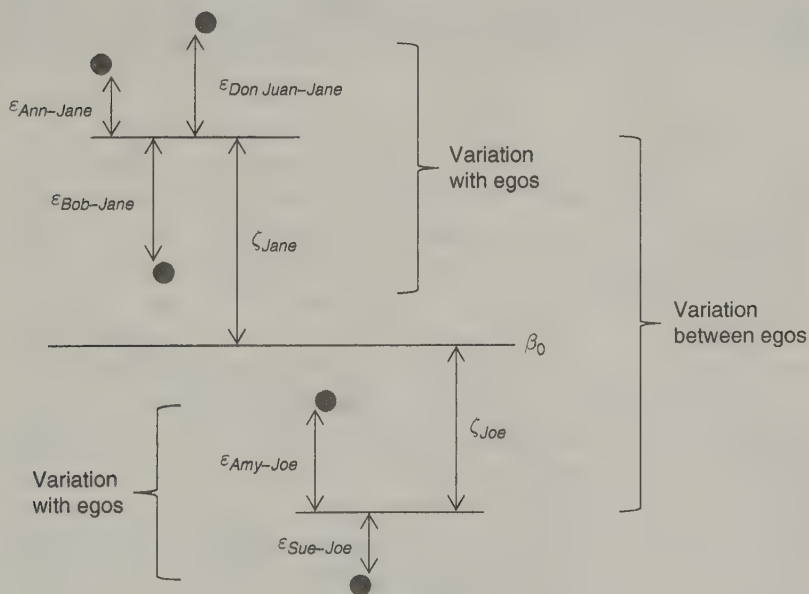


FIGURE 8.5 Illustration of a random intercept regression model predicting egos' perceptions of alters' likeability

his intercept relative to Jane. Also, the variation between clusters is greater than variation within clusters. This means that egos' likeability ratings tend to be more different from other egos' than their ratings of their own associates are from each other. In other words, there is evidence of dependence.

As before, we could add an independent variable to the model. Having asked egos to report how physically attractive they found each new alter, we hypothesize that alter attractiveness will be positively associated with egos' ratings of alter likeability. Figure 8.6 presents a scatter plot of Jane's and Joe's alter observations, with likeability on the y-axis and attractiveness on the x-axis. In this case, variation within egos reflects whether new associates who are perceived to be more attractive are given higher likeability ratings by a particular ego relative to his or her other alters. Variation between egos reflects whether egos with more attractive new associates also rate their alters as more likeable, on average, compared to how other egos rate their alters.

Using a standard OLS regression model, each respondent would have the same intercept. It would be equal to the grand mean (i.e., the mean of all pooled observations, ignoring clustering). Using MLM, both Jane and Joe get their own random intercept ζ_j that reflects the difference between their cluster mean and the grand mean. Jane's line appears above the

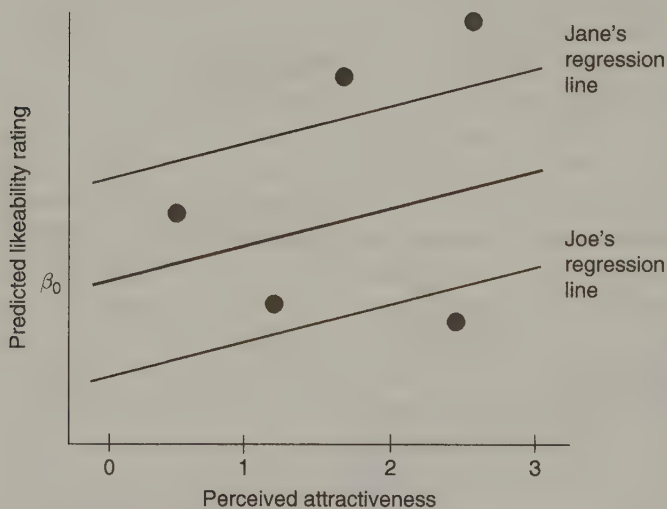


FIGURE 8.6 Scatterplot with fitted random intercept regression line for the effects of alter attractiveness on alter likeability

grand mean in Figure 8.6 because she has more favorable first impressions than others do, on average. Joe's line appears below the grand mean because he tends to rate new associates as less likeable than average.

Jane's and Joe's lines are parallel. In the random intercept model, only the intercept is allowed to vary from ego to ego. The slope is constrained to be constant. Consistent with our hypothesis, the slope is positive, meaning that the more attractive an alter is perceived to be, the higher ego's evaluation of their likeability.

8.4.7 The Random Coefficient Model

The two sources of dependence common in hierarchical data (see Section 8.4.1) are (1) mean differences between clusters, or intercept differences, and (2) differences in the effects of alter level predictors across ego networks. For example, the effect of ego-alter similarity in political ideology on frequency of political discussions between ego and alter may vary from ego to ego. Conflict-avoidant egos are just not willing to have political discussions with friends and family members of different political parties. Others have little interest in talking politics with people who agree with them, preferring opportunities to "convert" others to their way of thinking. Here, differences across egos in the slopes of the variable measuring ideological similarity would be expected. Ideally, these slope differences should be reflected in our analytic model. The random intercept model, above, accounts for the first kind of dependence – mean differences across egos in the frequency of discussion. That is, egos' propensity to discuss politics, in general. The random coefficient model accounts for the second kind of dependence. Each ego has their own estimated random slope component representing the unique relationship between similarity in political ideology and frequency of political discussion among their own alters.

The random coefficient model is a straightforward extension of the random intercept model. The latter is based on the premise that each Level 2 ego needs its own random intercept to account for dependence in Level 1 alter observations. Yet, the effect of any independent variable x is assumed to be constant across egos. But if the effect of x on y differs across clusters, or egos, a different approach is required. For example, alter age may be strongly related to the amount of support provided for some egos, but the effect might be less pronounced among other egos. Reducing some of this between-cluster variation in coefficients can be accomplished by controlling for relevant ego level variables, such as

ego's age or the age distribution of the network. However, omitted variables that are unmeasured will continue to be a source of unexplained between-cluster variation in coefficients. In this case, modeling dependence using a random coefficient is critical. Calculating the coefficient using the entire population of observations, without accounting for within-group clustering, could produce biased estimates.

In addition to a random intercept (Equation 8.3), the random coefficient model adds a cluster-specific random slope (Equation 8.4). The formula, adapted from Raudenbush and Bryk (2002), is the same as the random intercept model, except that the two random variables have been replaced by three. There is an ego-specific random intercept ζ_{0j} , an ego-specific random coefficient ζ_{1j} , and an alter-specific residual ε_{ij} .

$$y_{ij} = (\beta_0 + \zeta_{0j}) + (\beta_1 + \zeta_{1j})x_{ij} + \varepsilon_{ij} \quad (8.5)$$

The ego-specific random components are constant within egos, while the alter-specific error term ε_{ij} varies within egos and across alters. Figure 8.7 provides an example where one ego has two alter observations. Here, the independent variable has a larger effect for this ego compared to the population-averaged coefficient. Also, this ego has an intercept that is greater than the grand mean.

The difference between OLS regression and random coefficient MLM again lies in how the residual variance of the dependent variable is split up (Hoffman 2015). In the random coefficient model, variance is split into three types: (1) between-ego intercept variation (ζ_{0j}), which reflects how egos differ from other egos when the value of x is 0; (2) between-ego

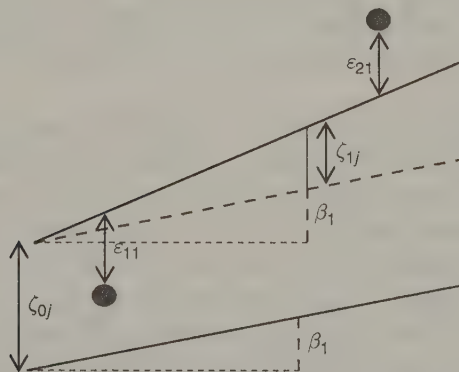


FIGURE 8.7 The random coefficient model

Source: Rabe-Hesketh and Skrondal (2008)

slope variation (ζ_{ij}), which reflects how the effect of x in one ego differs from the effect of x in other egos; and (3) within-ego variation (ε_{ij}), which indicates how alters nested within the same ego network differ from other alters in that same network. With the random coefficient model, cluster-specific differences in intercepts and slopes are estimated. The model relaxes the restriction that regression lines for different egos must be parallel.

8.4.8 Interpreting the Random Coefficient Model

Like with the random intercept model, the intraclass correlation ρ in the random coefficient model can be interpreted as a measure of within-cluster correlation (Raudenbush & Bryk 2002). Also important in random coefficient models is the correlation between random intercepts and random slopes, or ρ_{21} . This correlation describes whether having a higher or lower than average intercept is related to having a higher or lower than average slope. For example, we might want to predict the frequency of health discussions using alter closeness. For egos who discuss health more frequently than average, being close to any given alter may have a more modest effect on health discussion compared to those who rarely discuss health in their network. In short, some egos are more private, and therefore more discriminating, when deciding with whom to discuss health. Here, a negative correlation between random intercepts and slopes would be observed, because egos with a larger than average intercept exhibit a smaller than average effect.

Determining whether a random slope is needed is a critical first step prior to interpreting estimates. A likelihood-ratio test determines whether an alter level independent variable has a unique effect on the dependent variable across different egos. The null hypothesis is that all the random slopes are equal to 0. A rejected null hypothesis indicates that a random coefficient is required.

Again, the components of the random coefficient multilevel model are typically estimated using maximum likelihood. As in the random intercept model, the fixed overall intercept β_0 is a weighted mean of the cluster means. In addition, the fixed overall coefficient β_1 for any independent variables that have been assigned a random slope are also weighted means. Some clusters provide more information toward the overall intercept and slope than others; those with lower within-cluster variance and a higher number of Level 1 observations are weighed more heavily. The fixed intercept and slope are interpreted the same way in a multilevel

model that they are in an OLS model. However, these estimates are not the same as those obtained when pooled alter observations are used in estimation procedures without accounting for clustering. In practice, they may be similar.

Finally, though ego-specific random intercepts and slopes are treated as random variables and not usually interpreted (Rabe-Hesketh & Skrondal 2008), estimated values for the random intercepts ζ_j and slopes ζ_{1j} for each ego could theoretically be obtained. Similar to predicted residuals in an OLS model, predictions for random components can be used for MLM diagnostics, visualizing models, and for examining the “true” effect of some independent variable in specific ego networks.

8.4.9 An Illustration of the Random Coefficient Model: Variation in the Effect of Alter Attractiveness

Returning to the fictional research on first impressions, the effect of alter attractiveness on likeability may vary across egos. In other words, attractiveness may be a stronger factor in likeability for some egos relative to others. Egos who are single, for example, may be more attuned to physical attractiveness than those who are married, influencing the degree to which physical beauty determines likeability. Likewise, egos in particular locations or occupations might tend to encounter more attractive alters, desensitizing them to the influencing effect of beauty. If alter attractiveness varies systematically across egos, there are likely to be within-cluster differences in the effect of alter attractiveness on egos’ first impressions. In addition to clustering due to mean similarities in likeability for alters nested in the same ego, this pattern introduces dependence in the effects of attractiveness on likeability.

Jane’s and Joe’s alter observations are presented in a model that includes a random intercept and a random coefficient (Figure 8.8). From the random intercept model, findings indicated that Jane rates her new associates as more likeable, on average, than other egos, giving her a positive intercept bump. Here, the effect of attractiveness is also larger among Jane’s alters compared to the population-averaged coefficient. Consequently, her random slope is positive. The total estimated effect for Jane is equal to the overall effect β_1 plus her cluster-specific random coefficient ζ_{1j} . Joe’s within-cluster regression line has a smaller slope compared to the average effect, so his random coefficient is negative. Further, if the pattern observed for these two Level 2 egos is any indication, a positive correlation is expected between random intercepts and

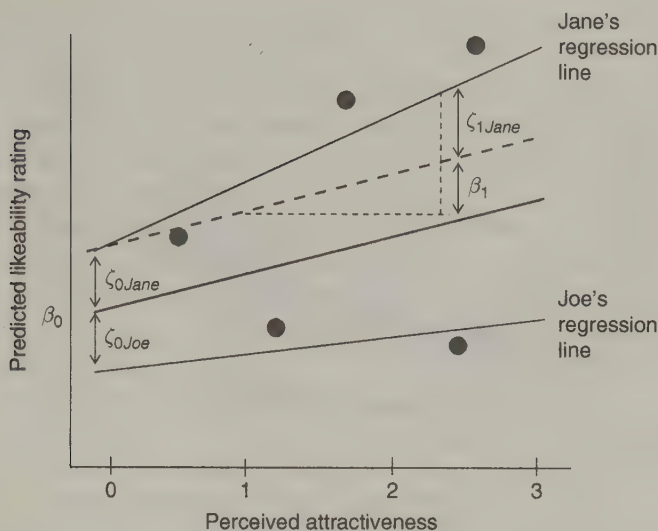


FIGURE 8.8 Scatterplot with fitted random coefficient regression line for the effect of alter attractiveness on alter likeability

slopes. That is, the more likeable an ego rates new associates, on average, the stronger the effect of alter attractiveness on likeability.

The estimated effects for all the ego respondents in our fictional sample are displayed simultaneously in Figure 8.9. Both the intercept and the coefficient are random. That is, each ego has their own intercept and their own sloped line based on their nested alter observations. The overall intercept β_0 and slope β_1 (depicted with the heavy, solid line) can be interpreted as weighted averages of each ego's intercept and slope.

8.4.10 Cluster Confounding: Distinguishing Individual and Network Effects

In the study of first impressions, each alter's likeability score is likely a reflection of both ego and alter characteristics. Part of each alter's rating is due to ego attributes that are the same across alters. For instance, an ego's own traits – extroversion, age, gender, etc. – may affect ratings of alter likeability. At the same time, each alter's likeability rating is also affected by attributes that are unique to each alter or relationship. Perceived likeability may vary as a function of alter's personality, body language, and attractiveness. Similarly, characteristics of the relationship might also be important, including the shared

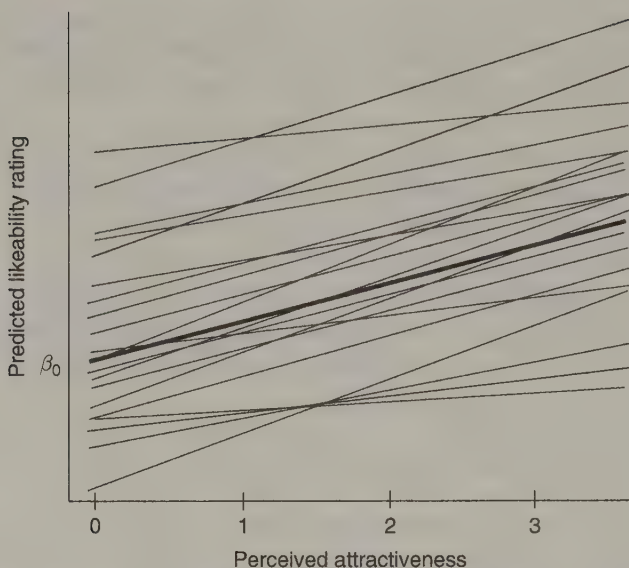


FIGURE 8.9 Individual regression lines, random coefficient regression model predicting alter likeability

context in which ego and alter met (e.g., business or pleasure), the length of their first conversation, or the presence of others. In short, both ego level and alter level characteristics contribute to the observed value of an alter level observation.

These patterns can lead to omitted variable bias. Specifically, if omitted Level 2 (i.e., ego level) characteristics are correlated with Level 1 (i.e., alter level) independent and dependent variables, the coefficient for the Level 1 variable will be biased. In other words, the estimate will not reflect the true effect of the Level 1 covariate. For example, suppose ego is in the fashion industry. Ego's occupation may be correlated both with the attractiveness of new people ego meets and with the likeability of new associates. Consequently, the estimated effect of alters' attractiveness on likeability produced by the model will actually be the combined effect of ego occupation and alters' attractiveness. Given this, failing to control for ego's contribution to the effect of alter attractiveness – here, ego being in the fashion industry – may lead to the wrong conclusion about the real effect of perceived attractiveness on first impressions. Our estimated coefficient could be artificially inflated or suppressed, relative to the true population parameter. Moreover, oftentimes, ego's characteristics that might bias the

estimate of alter level variables cannot be controlled because they are not measured.

This scenario violates an important multilevel modeling assumption. Level 1 covariates (i.e., variables observed at the alter level) must be uncorrelated with the random intercept. This is often a problematic assumption because every Level 1 covariate varies both within and between clusters. All Level 1 variables contain information about alters and about egos (or ego networks). Applied to the network context, knowing the age of an alter, for instance, tells us something about the alter and about the age composition of the network. This creates a source of bias called cluster confounding (Hoffman 2015).

Another problem with cluster confounding, in addition to creating bias, is the messiness of interpretation. Including one variable that reflects both within- and between-ego variation – a process often called pooling – essentially assumes that within-cluster and between-cluster effects are equal. They are blended into one coefficient that is presumed to accurately reflect two effects that may, in reality, be quite different. For example, including one variable for alter age assumes that the effect of the age composition of the network is equal to the effect of alter's age on some dependent variable. This is problematic if the true values of these effects are different, because the estimated coefficient will be a blend of between-ego and within-ego effects that is difficult to accurately interpret (Hoffman 2015). Consequently, cluster confounding creates problems of interpretation: Is this a between-cluster or within-cluster effect? What is our comparison group – other egos or other alters? In theory, between- and within-cluster effects could be opposite in direction, leading to a gross misinterpretation of a blended coefficient.

A relatively easy solution to cluster confounding, ideal for ego network data, is to add a contextual effect. Simply adding the cluster mean of the Level 1 variable to the model and leaving the Level 1 variable as is adjusts for Level 2 variation. The cluster mean is already a familiar concept, having used it to generate compositional variables (e.g., the mean closeness of the network) when aggregating to the ego level. Returning to the likeability illustration, adding a contextual effect separates out the influence of having a network with high or low average attractiveness (a variable that is likely to be correlated with ego level attributes like occupation) from each individual alter's attractiveness. This is important because this network context could affect the true value of the alter level effect of attractiveness. The within-ego effect of attractiveness would reveal whether egos rate attractive people they have just met as more likeable,

relative to other less attractive alters they have encountered. A between-ego effect of attractiveness would reflect whether egos who tend to meet more attractive people, on average, have more favorable first impressions than other egos.

The addition of a contextual effect is consistent with network theory, in addition to solving the problems associated with cluster confounding. Contextual effects test whether the network variable has any significant influence over and above the alter level variable. This approach determines whether the network context matters, in line with the contention that a “network is more than the sum of its ties” (Wellman & Gulia 1999b: 83). For example, it is important to know whether having an ideologically homophilous network affects the likelihood of political discussion (perhaps reflecting circulation of redundant political ideas), separate from whether being ideologically similar to any given alter influences the frequency of political discussions. The contextual effect allows this distinction.

Contextual effects should be added for each covariate that is measured at the alter level. However, they can be dropped from the model if nonsignificant. In fact, if the contextual effect is not significant, then the within-ego and between-ego effects of the variable are the same – something worth noting when interpreting coefficients.

8.4.11 Cross-Level Interaction Effects

The ability to model cross-level interactions is one of the most important advantages of using MLM. Cross-level interaction models address fundamental questions in social network science, including how social contexts, including networks, shape processes occurring at the alter level. For example, a researcher studying recent widows might be interested in determining whether relationship conflict in ego’s network (Level 1) affects support provision (Level 1), but varies depending on network density (Level 2). In densely knit networks, the negative effect of conflict on support may be attenuated (i.e., weaker) because dense networks tend to hold stronger proscribed support norms and demand more accountability. These kinds of questions are only testable with cross-level interaction models.

In a cross-level interaction, a Level 1 variable is interacted with a Level 2 variable. An interaction is supported when the effect of some alter level variable depends on the value of some ego level variable. In other words, the effect of one independent variable is larger or smaller at different levels of another. Returning to the fictional likeability study, a hypothesized cross-level interaction between ego gender and alter attractiveness

could be tested. Specifically, alter attractiveness may have a larger effect on likeability among men relative to women.

Cross-level interaction effects are estimated with the inclusion of a multiplicative interaction term. Conventions for modeling interaction effects (i.e., moderation) are similar in MLM and OLS regression. An interaction is statistically significant if the p-value for the coefficient associated with the interaction term is $<.05$. Interactions involving one or more categorical variables are straightforward to interpret. The effect (intercept change or slope) is provided at each level of the categorical moderator. An interaction between two continuous variables is more complicated. One strategy – the dummy variable method – is to split the moderator into two or more binary variables (e.g., tertiles) and interpret as above. Another approach is the simple slopes method, where the slope for the independent variable of interest is calculated holding the moderator at low and high values (e.g., at the mean and one standard deviation above and below the mean).

Since interaction terms are simple products, either the Level 1 or Level 2 variable in the term can be viewed as the moderator (Aguinis, Gottfredson, & Culpepper 2013). The Level 2 variable will more frequently be conceptualized as the moderator in network research, implying that the network context changes the effect of some alter level variable. However, this decision should be based on theory (see Jaccard & Turrissi 2003 for estimation details).

Two important considerations for modeling cross-level interaction effects need to be addressed (Aguinis et al. 2013; Preacher, Curran, & Bauer 2006). First, it is important to model a random coefficient for the alter level variable being moderated. A cross-level interaction allows the researcher to determine whether Level 2 variation in the slopes of Level 1 variables is attributable to a particular Level 2 variable. In other words, the cross-level interaction model is a natural extension of the random coefficient model. Constraining the effect of the Level 1 variable to be equal across Level 2 units makes little sense. The cross-level interaction model with a random coefficient is identical to the random coefficient model (Equation 8.4) with the addition of a Level 2 covariate and an interaction term. β_3 is the coefficient for the cross-level interaction.

$$y_{ij} = (\beta_o + \zeta_{oj}) + (\beta_1 + \zeta_{1j})x_{1ij} + \beta_2 x_{2j} + \beta_3 x_{1ij}x_{2j} + \varepsilon_{ij} \quad (8.6)$$

A second consideration is how to treat cluster confounding in cross-level interaction models. Both the alter level variable and the associated

contextual effect are included in the model and interacted with the Level 2 moderator, resulting in two interaction terms: one cross-level interaction term and one intralevel (i.e., Level 2 $\times$ Level 2) interaction term. In the above example focusing on the moderation of the effect of relationship conflict (Level 1) on relationship closeness (Level 1) by network density (Level 2), the following covariates would be included:

- Relationship conflict (Level 1 main effect)
- Network conflict (Level 2 contextual effect; i.e., cluster mean)
- Network density (Level 2 main effect)
- Relationship conflict * Network density (cross-level interaction effect)
- Network conflict * Network density (intralevel interaction effect)

However, if neither the contextual effect nor the intralevel interaction effect are significant, these can be safely dropped from the model.

8.4.12 Multilevel Modeling Assumptions

Several assumptions underlie linear multilevel models. Like OLS, the functional form of the model must be correctly specified. If the relationship between an independent and dependent variable is nonlinear, polynomial terms are necessary. Further, multicollinearity and the effects of any outliers should be assessed.

Three other assumptions pertain to the residuals and random components. First, the Level 1 error term (ε_{ij}) and all Level 2 (and Level 3, etc.) random intercepts and slopes are assumed to be normally distributed with mean of zero. All statistical software packages include the capability to predict residuals at every level, and the distribution is easily produced with a histogram. Additionally, standard descriptive statistics can be used to determine whether the mean is zero, or close to zero.

Second, Level 1 residual variance is assumed to be constant within and between Level 2 units. Heteroscedasticity in the Level 1 errors is assessed by plotting standardized residuals against the linear prediction. A fan, hourglass, diamond, or other shape depicting nonconstant error variance indicates a violation of this assumption. However, heteroscedasticity problems are easily corrected using robust standard errors.

Third, Level 1 and Level 2 residuals are assumed to be uncorrelated. Correlation between residuals is frequently attributable to crossover relationships (Aguinis et al. 2013), which occur when Level 1 outcomes are influenced by more than one Level 2 unit. When some characteristic of an

alter or relationship is influenced by multiple egos, usually because the alter is a member of more than one ego's network, this assumption is violated. The degree of correlation can be assessed with a basic Pearson's correlation after predicting residuals. If there is a modest amount of correlation, an unstructured covariance matrix may be used to relax the assumption of uncorrelated residuals. Some suggest always using the unstructured option because the assumption is so frequently violated. Statistical software packages can easily accommodate this (Hoffman 2015). However, when there is a substantial amount of crossover, the ability to explicitly model this unique form of dependence is useful. This can be achieved with cross-classified MLM (Section 8.4.13).

Violations of these assumptions may lead to model misspecification, threatening the veracity of findings. Violations can cause biased standard errors, which lead to inaccurate conclusions when hypothesis testing. Other options, including the unstructured covariance matrix, which permit relaxation of some of the assumptions regarding residuals, are detailed elsewhere (Culpepper 2010; Raudenbush & Bryk 2002; Skrondal & Rabe-Hesketh 2004).

8.4.13 Special Cases and Additional Considerations

Extensions and variations on the random intercept and coefficient models introduced above are summarized here in brief. More detailed information is available elsewhere (e.g., Hoffman 2015; Rabe-Hesketh & Skrondal 2008; Raudenbush & Bryk 2002).

Three-level multilevel models. Sometimes, egocentric network research is conducted in distinct locations or social contexts, resulting in clustering at a higher level. For example, egos may be sampled from a number of different schools, neighborhoods, cities, or countries. This higher level may influence ego or alter level variables, creating dependence. If so, a three-level multilevel model may be appropriate (Figure 8.10). For example, Volker and Flap (2007) conducted an analysis of how neighbors in the Netherlands come to be included in personal social networks using a three-level MLM design. Information about alters was collected from egos who were drawn from seventy-two municipalities, the third level of observation. This approach provided an opportunity for complex multilevel theory testing and development. Merging publicly available data on municipalities allowed the authors to test whether human capital (e.g., average income) and opportunities for meeting people in the neighborhood (e.g., presence of public parks and schools) increased the likelihood that neighbors would

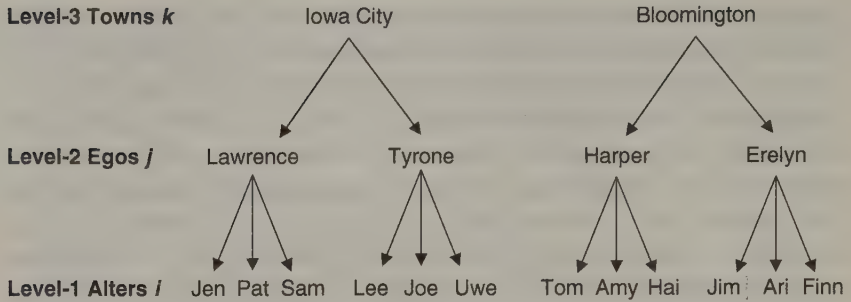


FIGURE 8.10 Hierarchical three-level data structure for egocentric network analysis

appear in personal social networks. Their dependent variable at Level 1 was the relationship between an ego and an alter – specifically, whether the alter was a neighbor. Characteristics of municipalities (e.g., presence of public schools), egos (e.g., having children at home), and alters (e.g., gender and marital homophily) were found to have independent effects on the formation of relationships to neighbors.

The three-level MLM is a straightforward extension of the two-level model. In the three-level random intercept model, dependence (i.e., clustering) is modeled by splitting the residual into three random variables, where each level gets its own random intercept. Suppose data consist of alters nested in egos nested in neighborhoods, as in the Dutch study above. This model (Equation 8.6), adapted from Raudenbush and Bryk (2002), includes k , the Level 3 (i.e., neighborhood-level) identifier; j , the Level 2 (i.e., ego level) identifier; and i , the Level 1 (i.e., alter level) identifier.

$$y_{ijk} = \beta_0 + \beta_1 x_{ijk} + \zeta_{jk} + \zeta_k + \varepsilon_{ijk} \quad (8.7)$$

The formula is the same as the two-level random intercept model (Equation 8.3), except that the single random intercept has been replaced with two – one for each higher-level nested context. The neighborhood specific random intercept ζ_k is constant across all egos and alters in a given neighborhood. The ego specific random intercept ζ_{jk} represents the difference between each ego's observed value and the Level 3 random intercept. It is constant across all alters nested within an ego. The alter specific error term ε_{ijk} is the difference between each alter's observed value and the ego specific random intercept ζ_{jk} .

The cross-classified model. A common threat to the assumptions of MLM is crossover, which occurs when two or more types of clusters are not

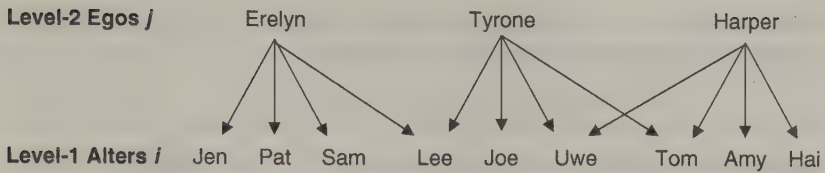


FIGURE 8.11 Crossover data structure for egocentric network analysis

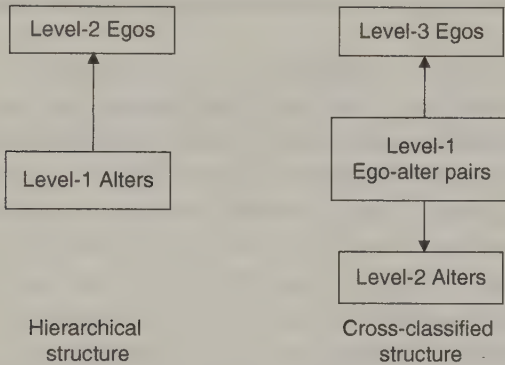


FIGURE 8.12 Hierarchical versus cross-classified multilevel model

hierarchical. In ego network research, this often occurs when egos are recruited through respondent-driven sampling (see Chapter 3). In this case, an ego already in the study provides access to other egos. Not only is there a pre-existing tie between egos, but there is non-negligible overlap in egos' social networks. Further, a person may serve as both an ego and an alter, making these two clusters nonhierarchical. This crossover structure is depicted in Figure 8.11, where egos Tyrone and Harper share two alters, Tom and Uwe.

When there is a substantial degree of crossover in egocentric networks, ego-alter pairs (Level 1) can be conceptualized as being simultaneously nested in two higher-level units – alters (Level 2) and egos (Level 3). In this structure, clustering multiple ties within the same alter creates dependence that would otherwise go into the residual (see Vacca, Stacciarini, & Tranmer 2016). In the cross-classified model, Level 1 variables are characteristics of pairs, or relationships, which are theoretically driven by both ego and alter level attributes. The cross-classified random intercept model (Equation 8.7) is very similar to the three-level hierarchical model, with two random intercepts. However, in the cross-classified model, the subscript notation for the random intercepts does not reflect a hierarchical

structure. That is, ζ_j is not nested in ζ_k . The Level 1 random components do reflect nesting in both higher-level clusters.

$$y_{ijk} = \beta_0 + \beta_1 x_{ijk} + \zeta_j + \zeta_k + \varepsilon_{ijk} \quad (8.8)$$

The random coefficient model could include random slopes for Level 1 variables at the ego, alter, or both levels. Such decisions should be theoretically based. Moreover, contextual effects should be considered at both the ego and alter levels, but can be dropped if assessed and found to be nonsignificant.

Vacca and colleagues' (2016) research on social isolation and wellbeing among Latino immigrants to the rural United States provides an example of a cross-classified data structure for egocentric networks. Recruiting through churches and community-based events, they obtained data on 1,780 alters from eighty-nine egos. Using a matching procedure to identify crossover ego-alter pairs on the basis of alter name, gender, ethnicity, and age, they identified substantial overlap in ego networks in their rural communities. Although most alters were connected to only one ego (seventy-seven percent), most egos named at least one alter that was tied to other egos.

Vacca and colleagues' (2016) cross-classified model included a random intercept for egos and alters, explicitly modeling clustering at both levels. Predicting help-seeking for financial problems, they found that characteristics of alters, egos, and relationships contributed to patterns of tie activation. The cross-classified model provided a better fit than the hierarchical MLM, significantly reducing between-tie (or within-ego) residual variation. Moreover, in the model that ignored crossover, the fixed estimates (i.e., coefficients) tended to be smaller in magnitude with larger standard errors at the alter, tie, and ego levels. This research provides proof of concept and support for using a cross-classified model, where appropriate. Another closely related type of clustering is multiple membership. In this case, alters are hierarchically clustered within egos, but alters may appear in multiple networks (see Mo and Wellman 2016).

MLM for categorical outcomes. In ego network research, outcomes at the alter- or tie-level are often binary, ordinal, or categorical. Predicting the likelihood of a tie to a particular type of alter, such as a family member, or determining whether various kinds of resources (e.g., advice) are provided in the context of a relationship, for example, are standard concerns. In such cases, a multilevel logistic regression model, which estimates the conditional probability of some binary outcome, is necessary. Perry and Pescosolido (2015) employed multilevel logistic regression to study tie

activation. The binary outcome measured whether respondents discussed health (1=yes; 0=no) in the context of their personal relationships. The probability of tie activation for health discussion was higher in the context of kinship and more intimate relationships, and in conversations with alters who had been diagnosed with similar health problems (i.e., experiential homophily). These patterns suggested that egos tend to activate ties according to the functional specificity hypothesis, choosing to talk about health with alters that understand their situation or that are most likely to help.

In the random intercept logistic regression model (Equation 8.8; Raudenbush & Bryk 2002), the right side of the equation is nearly identical to the linear random intercept model (Equation 8.3). The log odds are linearly related to the fixed and random components. The only difference is the absence of a Level 1 error term ε_{ij} .

$$\text{Log odds} = \ln\left(\frac{P_{ij}}{1 - P_{ij}}\right) = (\beta_0 + \zeta_j) + (\beta_1)x_{ij} \quad (8.9)$$

The residual error term is undefined in all logistic models because these estimate a probability rather than a concrete value. In other words, this difference between linear and logistic models is not unique to MLM. Interpretation of logistic regression models, including MLM, typically uses odds ratios because the log odds are not particularly meaningful. In addition to multilevel binary logistic regression, there are multilevel versions of ordinal and multinomial logistic regression models, as well as multilevel poisson and negative binomial regression for count outcomes.

8.4.14 Step-by-Step Guide to Hierarchical Multilevel Model-Building

This section illustrates the standard progression of multilevel model-building using an egocentric network dataset. The data come from Claude Fischer's Northern California Community Study (Fischer 1977a), which is publicly available through the Inter-University Consortium for Political and Social Research (ICPSR). This study was conducted in 1977 to examine the transformation of community in modern urban life, and was the basis of Fischer's (1982) classic book, *To Dwell among Friends*. Personal interviews were conducted with 1,050 adult residents of fifty Northern California communities in and around the San Francisco Bay area.

The NCCS allows examination of predictors of reliance on network members for support. Ego respondents were asked to indicate which alters

in their network had been activated for each kind of support. The dependent variable *support* is a sum of six different kinds of support, ranging from help around the house to discussing personal matters. After truncating the sum to eliminate small cell sizes at four or more, the dependent variable ranges from 0 to 3. Independent variables include alter gender *altgen* at Level 1 and ego gender *egofem* at Level 2 (1=female, 0=male). In addition, *EGOID* contains unique ego respondent identification numbers, while *PLACE* contains unique identifiers for the fifty communities in which data were collected.

Managing nested data in Stata. Hierarchical data differ from the typical single-level data structure. Most often, data are stored in an $n \times m$ matrix in which rows are cases or observations and columns contain variables, or attributes of those observations. Analysis then consists of correlating columns, with an emphasis on explaining variation in one column (i.e., the dependent variable) using variation in multiple other columns (i.e., independent variables). However, data must be structured in *long form* for multilevel analysis in Stata. In long form, each row is a Level 1 observation, and Level 2 cluster attributes are embedded in columns, but do not vary across Level 1 observations. In other words, both Level 1 and Level 2 variation is accounted for in columns.

In ego network research, when data are in long form, each row represents an alter, and both ego and alter level variables are stored in columns. In Table 8.2, the variable *egoID* and *alterID* represent unique identification number for egos and alters, respectively. Variables *age*, *afemale*, and *arace* contain information about alters' age, gender, and racial category, and differ within egos. For example, ego 1 has three alters who are

TABLE 8.2 *Long form data structure for multilevel modeling in Stata*

egoID	alterID	age	afemale	arace	age	efemale
1	1	28	0	1	25	1
1	2	18	1	2	25	1
1	3	39	0	2	25	1
2	1	22	1	3	28	0
2	2	44	1	1	28	0
3	1	67	0	2	19	0
3	2	36	1	3	19	0
3	3	61	0	3	19	0

TABLE 8.3 *Wide form (or column-wise) data structure*

egoID	age1	afemale1	arace1	age2	afemale2	arace2
1	28	0	1	25	1	2
2	18	1	3	43	0	2
3	59	0	2	37	1	1

aged 28, 18, and 39. Variables *age* and *efemale* contain data about egos, and do not differ across alters nested within a given ego. Ego 1 is twenty-five years old, and all of the alters' cells contain the same value for *age*.

Often, nested data are stored in wide form, and must be transformed, or reshaped, into long form for MLM analysis in Stata. When data are stored in wide form, each ego has only one row of data, and alter attributes are stored in columns in a series of numbered variables (see Table 8.3). This is the format that is referred to as “column-wise” in E-NET (see Chapter 7). There is typically a variable naming convention that must be followed when reshaping data, and this differs across statistical packages.

Step 1: Null random intercept model. The first step in multilevel model building is to compute a null (i.e., empty) random intercept model. This allows (1) testing whether MLM is needed, and (2) determining a baseline level of intraclass correlation. The Stata *xtmixed* syntax and output for the null random intercept model predicting support are as follows:

```
xtmixed support || EG0ID:, mle variance
```

```
Performing EM optimization:
```

```
Performing gradient-based optimization:
```

```
Iteration 0: log likelihood = -24652.639
```

```
Iteration 1: log likelihood = -24652.638
```

```
Computing standard errors:
```

```
Mixed-effects ML regression
```

```
Group variable: EG0ID
```

```
Number of obs      = 19,417
```

```
Number of groups   = 1,050
```

```
Obs per group:
```

```
min      = 2
```

```
avg      = 18.5
```

```
max      = 67
```

```
Wald chi2(0)      = .
```

```
Prob > chi2       = .
```

```
Log likelihood = -24652.638
```

```
-----+-----
support |      Coef.   Std. Err.      z    P>|z|   [95% Conf. Interval]
-----+-----
   _cons | .8383393   .0090775   92.35  0.000   .8205477   .8561309
-----+-----
```


Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]	
EGOID: Identity					
	var(_cons)	.0433297	.0037613	.0365507	.0513659
	var(Residual)	.7134136	.0074401	.6989794	.728146
LR test vs. linear model: chibar2 (01) = 380.44 Prob >= chibar2 = 0.0000					

Commands after the comma are optional. The *mle* command specifies that maximum likelihood estimation be used, and *variance* requests that the variances (and later covariances) for the random effects be included in the output. The variable after the two vertical lines is the grouping variable – ego identification number if modeling alters nested in egos. It is the unique identifier shared by each Level 1 observation in a Level 2 unit.

This very basic model provides several critical pieces of information. First, the number of observations at Level 1 is 19,417. This is the number of unique alters mentioned by ego respondents. The number of groups, or egos, is 1,050, consistent with the sample size for the NCCS. The minimum number of alters mentioned by any ego is two, while the maximum is sixty-seven, with an average of eighteen and a half. This can be interpreted as the minimum, maximum, and mean network size.

The coefficient next to *_cons* is the weighted average intercept, or the mean number of support resources for which alters are activated. This fixed estimate is 0.84, or slightly less than one type of support resource. At the bottom of the output, the *var(cons)* is the intercept variance at the ego level, or the variance between egos. In contrast, *var(Residual)* provides the intercept variance at the alter level, or the variance within egos. The variance is larger at the alter level than at the ego level, suggesting that the intraclass correlation will be closer to zero than to one.

The output from the *estat* command confirms this, providing an ICC of 0.06. This suggests modest correlation between alters nested in the same ego with respect to number of support resources provided. In other words, there is substantial variation within ego networks. However, despite the small ICC, the results of the likelihood-ratio test provided at the bottom of the model indicate that a random intercept is necessary to adjust for clustering at the ego level ($p < .00001$). Had this p -value been $\geq .05$, we could have safely used a standard OLS regression model.

estat icc
Intraclass correlation

Level	ICC	Std. Err.	[95% Conf. Interval]	
EGOID	.0572581	.0047832	.0485728	.0673863

Typically, a researcher would move to Step II at this point. However, because the NCCS contains a third level of analysis, determining whether a three-level model is necessary to account for clustering at the community level is required. The command for a null three-level MLM in Stata, where *PLACE* is the grouping identification variable at Level 3 is as follows:

```
xtmixed support | PLACE:, | EGOID:, mle variance
```

Performing EM optimization:

Performing gradient-based optimization:

```
Iteration 0:  log likelihood = -24651.429
```

```
Iteration 1: log likelihood = -24651.411
```

```
Iteration 2: log likelihood = -24651.411
```

Computing standard errors:

Mixed-effects ML regression Number of obs = 19,417

Group Variable	No. of Groups	Observations per Group		
		Minimum	Average	Maximum
PLACE	50	63	388.3	705
EGOID	1,050	2	18.5	67

```

Log likelihood = -24651.411
Wald chi2(0) = .
Prob > chi2 = .

```

support	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
cons	.8390371	.0107098	78.34	0.000	.8180462 .860028

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
PLACE: Identity				
var(_cons)	.0015827	.0012343	.0003433	.0072979
EGOID: Identity				
var(_cons)	.0418485	.0037877	.0350459	.0499715
var(Residual)	.7133894	.0074396	.6989561	.7281208

LR test vs. linear model: $\chi^2(2) = 382.89$ Prob > $\chi^2 = 0.0000$

Note: LR test is conservative and provided only for reference.

support	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
egofem	-.0151588	.018436	-0.82	0.411	-.0512927	.0209752
altfem	.0697531	.0124629	5.60	0.000	.0453263	.0941799
_cons	.8114269	.0146299	55.46	0.000	.7827529	.8401009

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]	
EGOID: Identity					
	var(_cons)	.0439183	.0037895	.037085	.0520106
	var(Residual)	.7121235	.0074287	.6977115	.7268333

LR test vs. linear model: $\chi^2(01) = 387.81$ Prob >= $\chi^2 = 0.0000$

The findings suggest that egos who are women do not differ significantly from egos who are men with respect to the number of support resources elicited from alters. However, at the alter level, egos seek 0.07 more support resources from women alters, on average, compared to men alters ($p < .001$). The Wald chi-square test, which tests whether the coefficients for ego and alter gender are simultaneously equal to zero, is rejected at $p < .00001$, suggesting that the model has better fit than a null model. However, the variances for the random intercept and residuals have changed very little, suggesting that the model does not explain much variance.

To address the potential problem of cluster confounding, an aggregated Level 2 version of the Level 1 variable, alter gender, is added. Contextual variables are computed using the `egen` command in Stata, as follows:

```
bysort EGOID: egen netfem=mean(altfem)
lab var netfem "Proportion female in network"
gen netfem10=netfem*10
lab var netfem10 "Percent female in network in tens"
```

In addition, the scale of the contextual variable is changed such that a one-unit change corresponds to a 10 percentage-point increase in percent women in the network. The model that includes this contextual variable provides the following output:

```
xtmixed support egofem altfem netfem10 || EGOID:, mle variance
Performing EM optimization:
Performing gradient-based optimization:
Iteration 0: log likelihood = -24624.612
Iteration 1: log likelihood = -24624.611
```

Computing standard errors:
Mixed-effects ML regression
Group variable: EGOID

Number of obs = 19,409
Number of groups = 1,050
Obs per group:
min = 2
avg = 18.5
max = 67

Wald chi2(3) = 40.11
Prob > chi2 = 0.0000

Log likelihood = -24624.611

support	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
egofem	.0189512	.0216578	0.88	0.382	-.0234973	.0613998
altfem	.0761756	.0126522	6.02	0.000	.0513778	.1009735
netfem10	-.0217038	.0073214	-2.96	0.003	-.0360534	-.0073541
_cons	.8996575	.0332023	27.10	0.000	.8345822	.9647329

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
EGOID: Identity				
var(_cons)	.0431032	.0037524	.0363419	.0511224
var(Residual)	.7121663	.0074291	.6977535	.7268769

LR test vs. linear model: chibar2(01) = 377.53 Prob >= chibar2 = 0.0000

These results illustrate the importance of including contextual effects. Controlling for alter gender, each 10 percentage-point increase in percent women is associated with a 0.02 reduction in the number of support resources elicited from alters ($p<.01$). In other words, the gender composition of the network has the opposite effect of alter gender. Being embedded in a network with many women reduces the number of support resources elicited from any given alter, holding that alter’s gender constant, even though being a woman alter increases the amount of support that egos seek out.

In addition, the coefficient for alter gender has increased slightly, and should now be closer to the true effect. The smaller coefficient in the model without the contextual effect reflected cluster confounding (Hoffman 2015). That is, the apparent effect of alter gender was a combined effect of Level 1 alter gender and Level 2 network gender composition – two measures that now reveal opposite effects on support elicitation.

Step III: Random coefficient model with covariates. The third step in multilevel model building is to determine whether the effects of any

Level 1 variables differ across Level 2 units. If true, this would require the inclusion of an additional random variable for cluster-specific slopes. The random coefficient model is computed in Stata using the following command (note that `///` allows the long `xtmixed` command to be broken and continued on the next line):

```
xtmixed support egofem altfem netfem10 || EG0ID: altfem, mle ///
covariance(unstructured) variance
```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log likelihood = -24628.189

Iteration 1: log likelihood = -24621.43

Iteration 2: log likelihood = -24621.409

Iteration 3: log likelihood = -24621.409

Computing standard errors:

Mixed-effects ML regression

Number of obs = 19,409

Group variable: EG0ID

Number of groups = 1,050

Obs per group:

min = 2

avg = 18.5

max = 67

Wald chi2(3) = 44.13

Prob > chi2 = 0.0000

Log likelihood = -24621.409

support	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+						
egofem	.010089	.0215817	0.47	0.640	-.0322104	.0523884
altfem	.0827772	.0130589	6.34	0.000	.0571822	.1083722
netfem10	-.0215531	.007329	-2.94	0.003	-.0359177	-.0071886
_cons	.9005753	.0330399	27.26	0.000	.8358183	.9653323

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
-----+				
EG0ID: Unstructured				
var(altfem)	.0092829	.00718	.0020385	.0422728
var(_cons)	.0376667	.0051298	.0288426	.0491905
cov(altfem,_cons)	.0030878	.0048649	-.0064472	.0126228
-----+				
var(Residual)	.7098437	.0075845	.695133	.7248657

LR test vs. linear model: chi2(3) = 383.93

Prob > chi2 = 0.0000

Note: LR test is conservative and provided only for reference.

The coefficient for `altfem` is now the weighted average effect of alter gender on number of support resources in each ego network. This

coefficient is larger in the random coefficient model compared to the random intercept model, and the standard errors are smaller. At the bottom of the output, the `var(altfem)` is the random slope variance at the ego level, `var(_cons)` is the random intercept variance at the ego level, and `var(Residual)` is the residual variance at the alter level. The `cov(altfem, _cons)` is the covariance of the random slopes and random intercepts. This value suggests that there is a very small, positive relationship between random intercepts and slopes such that alter gender has a larger effect among egos with higher levels of support at the intercept.

After storing estimates from the random intercept and random coefficient models, a likelihood ratio test is used to formally examine whether adding the random slope improves model fit:

```
quietly xtmixed support egofem altfem netfem10 || EG0ID:, mle variance
est store ri
quietly xtmixed support egofem altfem netfem10 || EG0ID: altfem, mle
variance
est store rc
lrtest rc ri
Likelihood-ratio test                                LR chi2(1) =    6.02
(Assumption: ri nested in rc)                        Prob > chi2 =  0.0142
```

The null hypothesis can be rejected at $p < .05$, suggesting that the random coefficient model provides a better fit for the data than the random intercept model. In other words, the effect of alter gender does vary significantly between egos.

Step IV: Cross-level interaction effects. In the above steps, we determined that egos elicit significantly more support resources from women than men, but that the gender of the ego does not influence the amount of support elicited from alters. We can assess whether there is a gender similarity effect using a cross-level interaction model. In other words, it is possible that the effect of an alter being a woman (Level 1) is stronger when the ego is also a woman (Level 2), perhaps because people prefer seeking support from similar others (i.e., homophily). We add an interaction term for alter gender * ego gender to the previous random coefficient model, in addition to interacting ego gender with the contextual variable (note that `##` is used in Stata to generate interactions):

```
xtmixed support altfem##egofem c.netfem10##egofem || EG0ID: altfem, ///
mle covariance(unstructured) variance
```

Performing EM optimization:

Performing gradient-based optimization:

```
Iteration 0: log likelihood = -24602.469
Iteration 1: log likelihood = -24593.455
Iteration 2: log likelihood = -24593.327
Iteration 3: log likelihood = -24593.322
Iteration 4: log likelihood = -24593.322
```

Computing standard errors:

Mixed-effects ML regression
Group variable: EG0ID

Number of obs = 19,409
Number of groups = 1,050

Obs per group:

min = 2
avg = 18.5
max = 67

Log likelihood = -24593.322

Wald chi2(5) = 103.07
Prob > chi2 = 0.0000

support	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
1.altfem	-.0234568	.018998	-1.23	0.217	-.0606921	.0137786
1.egofem	-.0578108	.0740929	-0.78	0.435	-.2030302	.0874086
altfem#egofem						
1 1	.191793	.0255892	7.50	0.000	.1416391	.2419468
netfem10	-.020383	.0112081	-1.82	0.069	-.0423506	.0015845
egofem#c.netfem10						
1	-.0033288	.0147954	-0.22	0.822	-.0323272	.0256697
_cons	.9356164	.0481098	19.45	0.000	.841323	1.02991

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
EG0ID: Unstructured				
var(altfem)	.0022581	.0066916	6.78e-06	.7518713
var(_cons)	.0358574	.0049609	.0273411	.0470265
cov(altfem,_cons)	.0067692	.0045788	-.002205	.0157434
var(Residual)	.7093482	.007568	.6946691	.7243374

LR test vs. linear model: chi2(3) = 385.69

Prob > chi2 = 0.0000

Note: LR test is conservative and provided only for reference.

The interaction term *altfem#egofem* is positive and significant, indicating the presence of a cross-level interaction. Specifically, the effect of an alter being a woman when the ego is man is close to zero and is nonsignificant ($b=-0.02$; $p=0.22$). In contrast, the effect of an alter being a woman when the ego is also a woman is $b=0.17$ ($-0.02 + 0.19$), and is significant at $p<.001$. In short, women are disproportionately likely to elicit support from other women. However, the interaction between ego gender and the gender composition of the network (*egofem#c.netfem10*) is not significant, suggesting that the effect of the gendered network context does not vary significantly by ego gender. We could safely drop this interaction term.

Given that gender similarity matters at the level of the dyad, we might also test whether men and women behave differently with respect to eliciting support from men versus women in distinct network contexts. For example, women may be especially unlikely to elicit support from men in networks made up of a high percentage of women. This kind of pattern might be due to social norms around caregiving and support, which are often conceptualized as the responsibility of women, and women may be particularly sensitive to these norms. However, with few women around to provide support, women may rely more heavily on men out of necessity. Testing this model requires a three-way cross-level interaction between alter gender (Level 1), ego gender (Level 2), and network gender composition (Level 2). If significant, the three-way interaction term suggests that the effect of alter gender on elicitation of support could depend both on the gender of the ego and on the gender composition of the network. The Stata output for this three-way cross-level interaction model is:

```
xtmixed support altfem##egofem##c.netfem10 || EG0ID: altfem, mle ///
covariance(unstructured) variance
```

```
Performing EM optimization:
```

```
Performing gradient-based optimization:
```

```
Iteration 0: log likelihood = -24595.879
Iteration 1: log likelihood = -24585.913
Iteration 2: log likelihood = -24585.733
Iteration 3: log likelihood = -24585.718
Iteration 4: log likelihood = -24585.718
Iteration 5: log likelihood = -24585.718
```

```
Computing standard errors:
```

```
Mixed-effects ML regression
Group variable: EG0ID
```

```
Number of obs    = 19,409
Number of groups  = 1,050
```

```

                                Obs per group:
                                min =      2
                                avg =    18.5
                                max =     67

                                Wald chi2(7)    =   118.60
                                Prob > chi2      =   0.0000

Log likelihood = -24585.718
-----+-----
              support |      Coef.   Std. Err.      z    P>|z|    [95% Conf. Interval]
-----+-----
              1.altfem |   -.2320519   .0764134   -3.04   0.002   -.3818193   -.0822844
              1.egofem |   -.2469914   .0900621   -2.74   0.006   -.4235099   -.070473
              altfem#egofem |
                1 1 |   .6266911   .1146873    5.46   0.000    .4019081    .8514741
              netfem10 |
                1 |   -.0375262   .0127753   -2.94   0.003   -.0625654   -.012487
              altfem#c.netfem10 |
                1 |   .0482862   .0171529    2.82   0.005    .014667    .0819053
              egofem#c.netfem10 |
                1 |   .0363773   .018064    2.01   0.044    .0009724    .0717822
              altfem#egofem#c.netfem10 |
                1 1 |   -.088062   .0225937   -3.90   0.000   -.1323448   -.0437792
              _cons |   1.003057   .0538028   18.64   0.000    .8976056    1.108509
-----+-----

Random-effects Parameters |   Estimate   Std. Err.   [95% Conf. Interval]
-----+-----
EGOID: Unstructured
      var(altfem) |   .0013376   .0011913   .0002335   .0076639
      var(_cons) |   .0360479   .0045872   .0280906   .0462592
      cov(altfem,_cons) |   .0069439   .0028184    .00142    .0124678
-----+-----
      var(Residual) |   .7089275   .0073965   .6945778   .7235737
-----+-----

LR test vs. linear model: chi2(3) = 386.76          Prob > chi2 = 0.0000

Note: LR test is conservative and provided only for reference.

```

The three-way interaction term *altfem#egofem#c.netfem10* is significant at $p < .001$, supporting the presence of a three-way interaction. These complex interactions are difficult to interpret using coefficients alone, so the *margins* command is used to produce predicted values at different levels of all three variables. Though network composition is held at 25 percent women and 75 percent women as an illustration, any number and combination of values that make substantive sense can be used:

margins, at (netfem10=(2.5 7.5) egofem=(0 1) altfem=(0 1))

Adjusted predictions Number of obs = 19,409

Expression : Linear prediction, fixed portion, predict()

1._at	:	altfem	=	0
		egofem	=	0
		netfem10	=	2.5
2._at	:	altfem	=	0
		egofem	=	0
		netfem10	=	7.5
3._at	:	altfem	=	0
		egofem	=	1
		netfem10	=	2.5
4._at	:	altfem	=	0
		egofem	=	1
		netfem10	=	7.5
5._at	:	altfem	=	1
		egofem	=	0
		netfem10	=	2.5
6._at	:	altfem	=	1
		egofem	=	0
		netfem10	=	7.5
7._at	:	altfem	=	1
		egofem	=	1
		netfem10	=	2.5
8._at	:	altfem	=	1
		egofem	=	1
		netfem10	=	7.5

	Delta-method					
	Margin	Std. Err.	z	P> z	[95% Conf. Interval]	

_at						
1	.9092415	.0248349	36.61	0.000	.860566	.9579171
2	.7216106	.0467031	15.45	0.000	.6300741	.813147
3	.7531933	.041555	18.13	0.000	.671747	.8346396
4	.7474487	.0293871	25.43	0.000	.689851	.8050465
5	.797905	.0354082	22.53	0.000	.7285062	.8673039
6	.8517048	.0514178	16.56	0.000	.7509278	.9524819
7	1.048393	.0428644	24.46	0.000	.9643802	1.132406
8	.8437692	.0227667	37.06	0.000	.7991473	.8883911

Our findings indicate that the predicted number of support resources is highest (1.05) when women are seeking support from women in networks where women are in the minority (combination 7, where *altfem* = 1, *egofem* = 1, and *netfem10* = 2.5). In contrast, the lowest predicted number of support resources is 0.72, occurring when men elicit support

from men in networks where women are in the majority (combination 2, where $altfem = 0$, $egofem = 0$, and $netfem10 = 7.5$). These findings are consistent with theories about gender norms surrounding caregiving and support seeking.

A graph of predicted values provides a visualization of how ego and alter gender effects vary across different gendered network contexts. Figure 8.13 is generated using the *marginsplot* command after *margins*. Concentrating on the right side of the figure, where network composition is majority women, both men and women egos show a slight preference for eliciting support resources from alters who are women. However, on the left, where network composition is majority men, effects of ego and alter gender emerge. Specifically, egos that are men are more likely to seek support from men alters rather than women alters in this network context. In contrast, women are substantially more likely to elicit support from women than from men in networks where women are in the minority. Put differently, when women supporters are in short supply, men are content to elicit support from other men, while women gravitate strongly toward the few women in their networks.

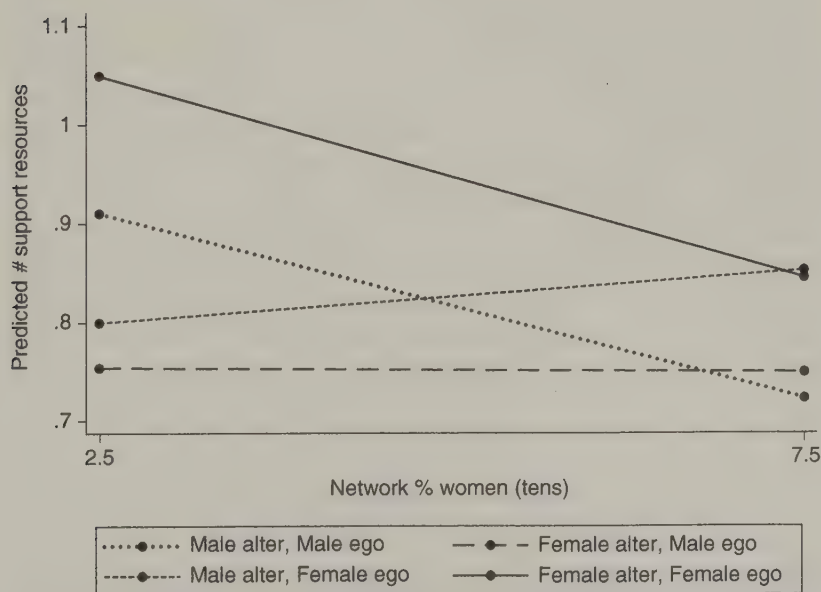


FIGURE 8.13 Predicted values of number of support resources for three-way cross-level interaction, Northern California Community Study

8.4.15 Conclusion

Multivariate regression models are the mainstay of ego network research. In this chapter we discussed how regression can be used to test theory, where ego networks are conceptualized as independent variables, dependent variables, or both. Levels of observation in ego network research were defined, distinguishing these from levels of analysis. While standard OLS regression is appropriate for dependent variables observed at the ego level, multilevel modeling is often required for outcomes observed at the alter level. Moreover, multilevel modeling offers a number of advantages for examining research questions about relationships or alter attributes, including the ability to model cross-level interactions. We introduced multilevel variance-components models for ego network analysis, and illustrated the standard progression of multilevel model-building and interpretation in Stata using an ego network dataset. Chapter 9 will discuss an extension of these models for longitudinal analysis of ego network dynamics.

Modeling Ego Network Dynamics

While most egocentric network research provides only a snapshot of networks at one point in time, recent work has focused on the dynamic and adaptable nature of personal social networks. Applying a rich set of theories to a range of substantive applications, social scientists are interested in examining how and why social networks evolve. Gradual changes common to the rhythm of daily life are punctuated by substantial shifts that correspond to important life events or role transitions like moving to a new location, retirement, and marriage. Understanding network change is necessary for explaining the effects of networks on individual and group behavior and outcomes (Moody et al. 2005), and for identifying the mechanisms driving micro-processes (Pescosolido 1992). Yet, capturing network dynamics, and in particular distinguishing real change from renewed interaction, tie latency, and measurement error, requires a unique set of methodological tools. In addition to providing an overview of what we currently know about egocentric network dynamics, this chapter familiarizes readers with commonly used methods for longitudinal data collection and the analysis of change in social network characteristics.

9.1 EGOCENTRIC NETWORK DYNAMICS: THEORETICAL AND CONCEPTUAL FOUNDATIONS

Although existing theory and research make it clear that personal social networks are dynamic in nature, the overwhelming majority of studies measure the aggregate of past social interactions captured as static ties at one point in time (Moody et al. 2005). A common strategy is to ask

respondents to name all of the individuals they spoke with about important matters, for example, during the past six months. This masks the changing patterns of social interaction that occurred during that time period. In fact, we know that networks exhibit a substantial amount of membership turnover, or tie churn (Sasovova et al. 2010; Halgin & Borgatti 2012; Perry & Pescosolido 2012). In their longitudinal study of adults in Toronto, Ontario, Wellman and colleagues found that only about 27 percent of intimate ties persisted ten years later (1997). In similar studies, 25 percent to 33 percent of caregivers' support networks were present after a decade (Suitor & Keeton 1997), and only about 22 percent of recent widows' ties persisted over the course of a year (Morgan et al. 1997).

However, despite this turnover in the individuals that make up a network, the structure and composition of ego networks (e.g., density, percent kin) tend to remain fairly stable over time (Lubbers et al. 2010; Wellman et al. 1997). These findings are indicative of a pattern of replacement, wherein lost or latent ties are substituted by new or formerly inactive ones (Small, Pamphile, & McMahan 2015). Because it requires insight into the comings and goings of specific alters from personal social networks (i.e., dynamics at the alter level), the replacement process demands a multilevel approach to analyzing network dynamics. Moreover, understanding the typical structural organization of personal social networks and common antecedents of network change is central to effectively developing methodological strategies for observing, explaining, and intervening in dynamics.

9.1.1 The Core/Periphery Structure of Egocentric Networks

Personal social networks tend to be comprised of two basic components: A small, densely knit, and cohesive core and a larger set of weaker and more loosely connected alters that comprises the periphery (Borgatti & Everett 2000; Wellman et al. 1997). Figure 9.1 depicts an ego network with a core/periphery structure, where alters 1, 2, 3, 4, 7 and 8 comprise the core. This structure is strongly related to dynamic patterns in egocentric networks such that the core tends to remain stable over time, while the periphery is characterized by a fair amount of instability and membership turnover (Morgan et al. 1997). The approximately 25 percent of ties that persist over time largely inhabit the core network, a substantial proportion of which are often immediate kin (Wellman et al. 1997). These long-term associations tend to be founded in a shared history and

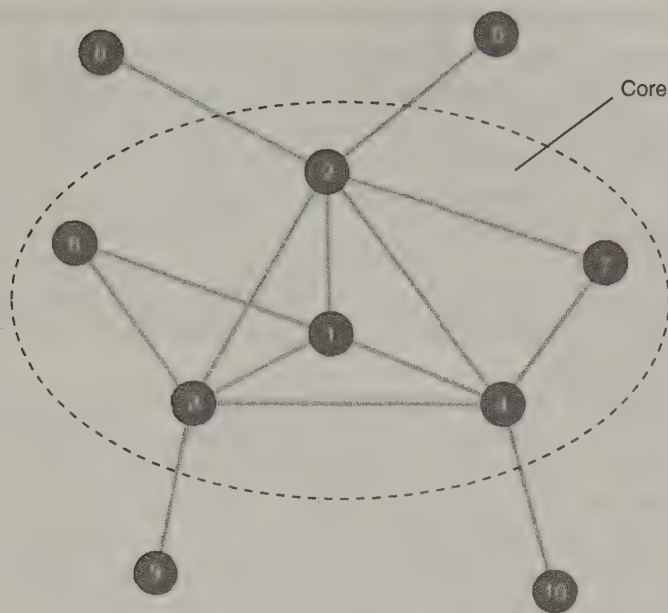


FIGURE 9.1 Egocentric network with a core/periphery structure

mutual feelings of obligation and, usually, affection. The remaining associations – the periphery – are adaptable, transitory, sparsely connected, and functionally specific. This core-periphery structure suggests a need to investigate both stability and instability in social networks over time, and to closely examine specific sectors of networks and their potentially divergent characteristics, interactions and exchanges, and consequences.

Some researchers have found evidence for a core/periphery structure with more subtle gradations. For example, Wellman and Gulia (1999b: 85) identified three loosely defined groups in the personal networks of East Yorkers: “3–6 socially close, intimate ties, 5–15 less strong but still significant ties, and approximately 1,000 acquaintances and latent (but often still mobilizable) relationships.” In their study of personal social networks in France, Degenne and Forsé (2004) observed a similar structure. They identified (1) a core network of fewer than ten very close confidants (the average core size was three); (2) an active periphery of about twenty regular interaction partners; (3) about 200–300 significant people; and (4) an additional several thousand latent ties that had potential to be activated. These structures are often depicted using multiple concentric circles, where inner rings signify closer ties to alters.

The core/periphery structure poses important challenges to the study of cross-sectional networks (Bernard et al. 1987; Killworth et al. 1990). If individuals engage in sporadic and brief periods of meaningful contact, as when a good friend visits only occasionally, the likelihood of these relationships being present in a snapshot of a network is nearly random. Similarly, when peripheral ties are not captured, they are typically assumed to be absent rather than inactive. As Morgan and colleagues (1997: 10) noted, instability in personal networks is not synonymous with real change, and "recognizing the existence of instability means that we expect a certain amount of 'coming and going' in the membership of most personal networks."

Measuring and examining peripheral networks and their instability is essential for understanding patterns of interaction and the structure of connection rather than simply identifying characteristics of support networks (Bernard et al. 1990). Although the structure of two egos' core networks may be identical, their extended networks are likely to be highly unique (Hammer 1983). Individuals may have very dense extended networks (e.g., large families), from which they receive homogeneous ideas, information, support, and feedback. Alternatively, the periphery may provide access to diverse and non-redundant resources from alters that do not interact with one another. Moreover, the characteristics and structure of the periphery can either facilitate or impede recruitment into the core, affecting whether core ties that dissolve are replaced.

Hammer's (1983) discussion of social class differences in social networks and their implications for health illustrates the importance of distinguishing core and peripheral networks. She observed that while empirical research has identified few systematic social class differences in the characteristics of core networks, measures of peripheral networks (e.g., number of non-kin, frequency of contact with weak ties, or the average number of alters with whom ego comes into contact in a given day) diverge markedly and confer advantages for those in higher socioeconomic status groups (e.g., Granovetter 1983; Lin 1999b; Pool & Kochen 1978). Hammer argued that the relatively bounded and closed peripheral networks of individuals in low-socioeconomic status groups may help explain class disparities in health. Specifically, a large and heterogeneous peripheral network serves a number of important social and psychological functions that may protect health, including providing valuable social feedback and opportunities for access to diverse resources, as well as facilitating recruitment to the core network.

9.1.2 Latency and Tie Activation in Personal Networks

Closely linked to the core/periphery structure of networks is a process called the activation of latent ties. Ties reflect relational states, which are nearly always “on” (at least until a tie is dissolved), including kinship, friendship, and colleagues. However, they are also characterized by differing forms and levels of interaction, or relational events. They may be active or inactive at any given point in time. The extended periphery of personal social networks typically contains members that are less salient and infrequently called upon and, as a result, are likely to be inactive at any particular point in time (Bernard et al. 1991; Morgan et al. 1997). Activation occurs when one or more of these latent ties is mobilized by an ego for a particular function, or, alternatively, alters become more interactive or functional of their own volition (Perry and Pescosolido 2015). In short, an alter moves from the periphery into the core, or at least into a more active inner circle, for any of a variety of reasons.

Tie activation often occurs when ego’s needs or circumstances change (e.g., Hurlbert et al. 2000), making activation processes a critical mechanism in the reciprocal relationship between individual biographies and network dynamics. Consequently, existing research on changes in functionality or frequency of contact with alters largely focuses on tie activation for particular kinds of support or resources. For example, an unexpected employment search might prompt a call to a well-connected former colleague or associate, even if that tie had not been accessed for many years. Thus, activation may be strategic and proactive on the part of an ego seeking support from alters perceived to be the most willing and able to help (Perry & Pescosolido 2015). However, activation is not one-sided. Alters may self-activate if they perceive changes in ego’s needs, or if their own situation evolves. For instance, the activation of an extended kinship tie for assistance with childcare may be prompted both by ego’s new parenthood status and by the alter’s recent retirement from the paid labor force. Ties may also be activated coercively, such as when alters push ego to seek medical care (Pescosolido, Gardner, & Lubell 1998). Finally, because weak and infrequently activated associates often have social networks that overlap very little with ego’s ties, latent ties may provide access to information, individuals, and other resources that are unavailable within the more stable and dense set of core ties (Granovetter 1973). The activation of these latent ties, then, represents a crucial coping mechanism for individuals facing unfamiliar challenges and periods of elevated support needs.

9.1.3 Network Disruption: Exogenous Shocks to Personal Network Systems

Often, changes in the nature, activity, or function of dyads are attributed, accurately or not, to endogenous dynamics in a network system (Leik & Chalkley 1997; Perry 2014). Variation in factors like conflict, self-disclosure, interpersonal competence and attraction, reciprocity, and burden can alter relationship dynamics and lead to network losses and gains (Collins & Miller 1994; Pescosolido 1992; Rivera, Soderstrom, & Uzzi 2010; White & Klein 2008). However, there is growing recognition that a substantial amount of network change is caused by exogenous shocks to personal networks, or network disruption (Perry 2006).

Wellman and colleagues (1997) theorized that personal network dynamics operate similarly to biological evolution. That is, networks undergo gradual, random ebbs and flows in membership punctuated by “intense rapid shifts” (p. 47) that correspond to significant events like divorce or parenthood. Disruptive events and transitions can both restrict access to existing network ties and provide opportunities for new associations. In either case, changes often alter network structure, function, and content in meaningful ways. While some research has focused on tragic or unexpected events like natural disaster (Erikson 1976; Varda et al. 2009) or serious illness (Bury 1982; Perry & Pescosolido 2012), most has examined normative life course episodes and transitions (e.g., job loss, widowhood, marriage and divorce, residential mobility; see Wrzus et al. 2013 for a meta-analysis).

Disruptive events and transitions influence social network dynamics because they often alter opportunity structures – including shared environments and activities – that help sustain relationships (Feld & Carter 1998). Disruptions may result in a loss of contact with alters, as when a retiree or unemployed person no longer engages in routine interaction with former coworkers (Jones 1991). Because contact is necessary for exchanging resources and developing and maintaining intimacy, relationships characterized by little contact often dissolve or become latent (Wellman et al. 1997). Alternatively, when one or both members of a dyad no longer engage in the same activities (e.g., attending church together, bowling in a league together), the foundation of mutual interests, goals, problems, and conversation topics may break down over time. In short, people have less in common (or more in common, in the case of entry into new contexts or roles). In addition, events like employment transitions, disability, and parenthood can alter

available resources, including time and income, which may ultimately affect ego's ability or willingness to reciprocate support or actively maintain existing ties.

In the end, the structure of an ego network may influence the extent of network disruption following a crisis or transition. In loosely connected networks, the loss of one bridging tie can isolate an ego from formerly accessible alters. Divorced individuals are often cut off from the friends and family members of their ex-partners (Kalmijn & van Groenou 2005). These social losses represent an independent source of psychological distress for divorcees, widows, laid off employees, and others experiencing significant role change (Gerstel, Riessman, & Rosenfield 1985). Consequently, network disruption is important, not only as an explanation for network dynamics, but as a mechanism in the link between life events and adjustment and other outcomes.

9.2 MEASURING CHANGE IN EGOCENTRIC NETWORKS

Network dynamics are most commonly measured using panel data (i.e., longitudinal or cohort data), where repeated measures are collected from the same sample at multiple points in time. Critical challenges arise in the collection of longitudinal ego network data. A foundational issue is the ability to distinguish real network change from reporting error. Another lies in identifying alter-level dynamics that underlie aggregate changes in structural or other network characteristics. Here, these problems are outlined and methods for managing them are discussed.

9.2.1 Distinguishing Real Change from Error

As described in detail (Chapter 4), the name generator method for eliciting alter names is prone to various kinds of errors, including forgetting. This is problematic in the context of a longitudinal design because the researcher often cannot determine whether an alter is not mentioned in a particular wave of data collection because of a recall error or because the relationship has changed in some meaningful way. In a systematic review, about 20 percent of core alters, on average, were forgotten in test-retest conditions (Brewer 2000), with the rate of forgetting being substantially higher for larger networks with greater range (Bell et al. 2007). Such errors are likely not random because egos are disproportionately likely to recall alters that are consequential or are linked in memory to other alters they have already named.

Another threat to data quality in longitudinal ego network designs is panel conditioning (Warren & Halpern-Manners 2012). Bias may result when respondents learn from prior interviews and respond differently in follow-up waves of a study. The data collection process for egocentric networks relies on burdensome loop options. Respondents are asked to provide multiple events or objects (i.e., names) and subsequently answer a series of questions about each one (i.e., name generators). In these cases, respondents may learn that the number of names elicited is directly related to the amount of time and effort required to complete the interview. This can lead to intentional omission of alters by respondents in follow-up waves of longitudinal network studies. For example, drawing on discussion network data from a large intervention study in Ghana, Valente, Dougherty, and Stammer (2016) identified evidence of respondent learning at the community level. Specifically, respondents' network size decreased significantly within specific communities as the study progressed. Those interviewed on day one provided more names, on average, than those interviewed forty-five days later. Similarly, a three-wave panel study revealed a significant decrease in the mean, median, and standard deviation of network size over a four-year period, producing an average decline of 2.7 alters over the study period (Eagle & Proeschold-Bell 2015). After rejecting alternative explanations (e.g., attrition bias), the authors concluded that their results support the existence of panel conditioning in the network name generator task.

Interviewer learning is another source of potential bias in longitudinal network research (Eagle & Proeschold-Bell 2015; Van der Zouwen & Van Tilburg 2001). Particularly if interviewer compensation occurs per completion, interviewers may not prompt for additional alters or may increase the pace of the name generator task. These behaviors have been shown to influence the number of alters named (Marsden 2003). While interviewer effects can be problematic in cross-sectional network research (Chapter 4), these problems may be exacerbated in longitudinal designs. Interviewer interest and enthusiasm may wane over the course of a lengthy study period.

Finally, respondents and interviewers sometimes commit spelling mistakes, typos, or other errors that make it difficult to match alters across waves in a longitudinal design. In these cases, a stable relation could be recorded as both a tie added and lost. These errors are especially common in self-administered surveys, including web-based designs (Vehovar et al. 2008). They can often be corrected at the data cleaning stage, but this is

a difficult and time-consuming process, and may require a certain amount of probabilistic matching.

9.2.2 Identifying Underlying Alter Dynamics

The relative stability of aggregate characteristics of ego networks belies a substantial amount of underlying turnover in the specific alters that make up a network. Even a significant change in measures like network size or mean closeness of the network over time offers little insight into the mechanisms driving those patterns at the dyadic level. For example, a researcher may observe that an ego's mean network closeness has decreased over two waves of a study. Three possible explanations could underlie this finding. First, ego may have dropped one or more very close alters from the network, reducing the overall mean. Second, ego may have added one or more weak ties, perhaps weak because the relationships are still developing. Third, ego might have become less close (e.g., through conflict or some network disruption) to stable members of the existing network, with no alter losses or additions. Being able to distinguish these patterns is critical because they have very different implications for ego. Adding weak ties while maintaining strong ones could be advantageous for outcomes like status mobility or wellbeing. Losing access to strong ties or experiencing a weakening of stable ties would likely have the opposite effect.

A recent mixed-methods study of first-year graduate students at an American university illustrates the merits of accounting for alter-level dynamics in personal social networks. Drawing on network disruption theory and three waves of data, Small, Pamphile, and McMahan (2015) tested competing models of network change among egos experiencing a major life transition and entering a new social context – graduate school. They found that core discussion networks changed rapidly due to evolving obligations, institutional environments, and shared activities. The authors' findings supported a substitution explanation (as opposed to stasis, expansion, or shedding), wherein network members were replaced by new alters as egos adjusted to their transition. This study highlights the contextual nature of even core support networks, and demonstrates that network dynamics, specifically replacement processes, are often functional and exogenously driven.

9.2.3 Strategies for Measuring and Describing Network Dynamics

Capturing dynamics at the level of alters and distinguishing real change from error requires that information about prior alters is incorporated into subsequent interviews. In the Indianapolis Network Mental Health Study, interviewers brought a list of alters mentioned in response to each name generator to follow-up face-to-face interviews (Pescosolido 1990). Respondents were first prompted to name their current alters. They were then shown the list of alters from the previous interview. Respondents were asked why they did not name omitted alters from the prior wave. The open-ended responses were recorded and later analyzed for content (Perry 2012). Respondents described multiple reasons for change, but the most common was an active decision to cease interactions and discussions that were not helpful or that caused problems.

In addition to producing analyzable open-ended data, this strategy improves the quality of network data. Any mistakes were corrected immediately, including respondent recall errors (i.e., forgetting) and cases where an alter's name was misspelled or a different name was given (Wright & Pescosolido 2002). Moreover, because respondents were given the option to either provide an explanation for why they did not include a particular alter or to add that alter, respondent panel conditioning effects were minimized. In a face-to-face interview, it is more cognitively difficult to provide a false open-ended response (i.e., a lie by commission) than to prevent true information about an alter from coming to light (i.e., a lie by omission; Gombos 2006). Respondents were prohibited through this method from doing the latter. This strategy was later adopted and validated in other longitudinal egocentric network studies (e.g., the National Social Life, Health, and Aging Project; Cornwell et al. 2014).

Once data are collected, a number of measures of network change can be computed. At the alter level, the number or proportion of alters dropped, added, and retained (i.e., stable ties) can be calculated as separate measures. At times, a single measure that captures the overall level of membership turnover or tie churn in a network is preferred. Turnover across two waves of data can be measured by pooling networks from both waves (Equation 9.1)

$$\frac{N_d + N_a}{N_u} \quad (9.1)$$

N_d represents the number of alters dropped between two time points, N_a is the number of alters added during the same period, and N_u is the total

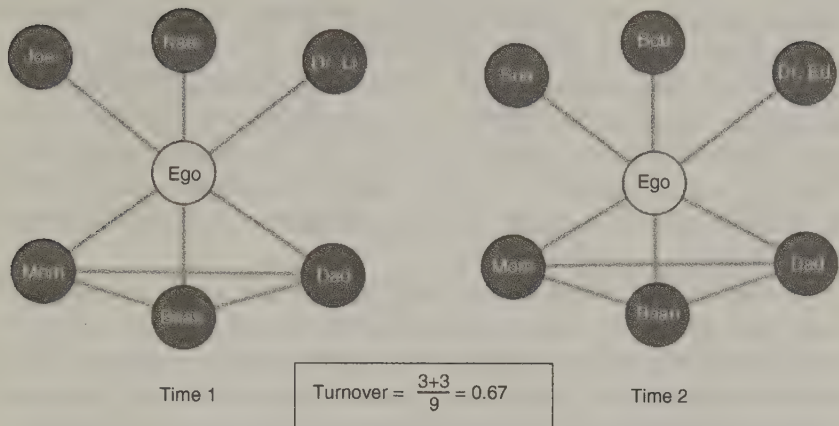


FIGURE 9.2 Calculating membership turnover in a personal social network

number of unique alters pooled across time points (see example in Figure 9.2). Turnover is essentially a proportional measure of instability as a function of total network size, and ranges from 0 to 1. Turnover may be very high even when network size, composition, or other aggregate measures remain stable.

To describe dynamics, aggregate network characteristics can be compared over time, testing for significant change using bivariate statistics. Simply presenting a measure of the difference between two time points may also be useful (e.g., frequency of contact at Wave 2 – Wave 1). To assess alter-level change, descriptive statistics on the average number or proportion of alters dropped, added, or retained, as well as the measure of network turnover, can be provided. Another strategy is to compare the characteristics of alters in each of these categories. For example, Cornwell and coauthors (2014) found that retained alters in older Americans' networks were more likely to be kin, household members, emotionally close, and in frequent contact relative to alters that were dropped or added over a five-year period. These kinds of analyses provide basic insights into the kinds of alter or tie characteristics that motivate network stability or instability at the dyadic level.

9.3 MULTILEVEL MODELING OF NETWORK CHANGE

A researcher might be interested in determining how network density influences a person's self-reported level of stress over a five-year period.

Suppose egos have been interviewed about their networks and their stress experiences annually. The resulting dataset might contain up to five observations (time points) per ego respondent, providing an opportunity to explore causal research questions not possible with cross-sectional data. Examining change over time – not only how stress changes over the study period, but also how variations in network characteristics like density influence the trajectory of change in stress – would capitalize on these longitudinal data.

Though longitudinal data have many advantages, repeated measures over time create dependence that prohibits the use of ordinary least squares (OLS) regression models. In the case of panel data, dependence – or correlation between observations nested in the same cluster – takes the form of autocorrelation, or serial correlation. This means that there is similarity between observations in a series separated by some unit of time. Put differently, people are more similar to themselves across time than they are to other people, on average. One person's typical level of stress is going to be different than another's because some people are just habitually more stressed. There are many reasons for this kind of dependency, including a whole host of variables that often cannot be controlled (e.g., personality traits, upbringing, prior experiences). A given person's "normal" level of stress may be higher because they have a sensitive or high-strung temperament. These typical differences between egos violate the OLS regression assumption that all observations are independent. For panel data, observations are usually not independent. They are correlated within egos. MLM explicitly models this dependence, relaxing the assumption that observations are independent. In a two-level model, we would say that stress observations over time (Level 1) are nested within egos (Level 2). This data structure is depicted in Figure 9.3.

Longitudinal models aim to estimate the effects of time on the dependent variable. Including a time measure (e.g., months in the study, survey wave, year, or age) as a covariate reveals whether the outcome changes in some patterned way over the course of the study. For example, when studying occupational mobility after college, the time coefficient indicates whether occupational prestige significantly increases or decreases, on average, in the years following graduation. This kind of regression – one that includes time as a covariate – is called a growth model, and is an important tool in the effort to understand social dynamics.

Of course, there are instances in ego network research in which growth, or change over time, may not be linear. Many outcomes of interest in ego network research exhibit a *variable rate of change*. Examples include

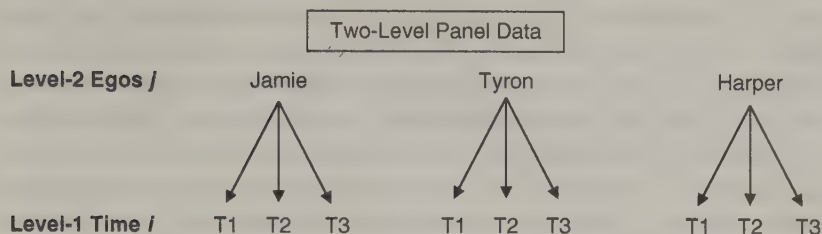


FIGURE 9.3 Panel data structure for egocentric network analysis

health declines (which are slow until later life, and then increase more rapidly) or learning (which tends to increase more quickly at first, and then level off). Network change may also be *nonlinear*. For instance, substantial membership turnover may follow in the months after a disruption like job loss. The network may eventually stabilize, with turnover declining. Nonlinear growth is modeled using a growth curve model, which is a polynomial model of time. Both time and time-squared terms are used to assess nonlinear change in the dependent variable over time.

Longitudinal data provide an opportunity to simultaneously examine both static and dynamic processes. A question in the stress study might be whether egos who have more densely knit networks report higher or lower levels of stress, on average, than egos with more sparse networks. Also of interest is whether having a more densely connected network in one year of the study compared to another year is associated with higher or lower stress levels than usual. Essentially, longitudinal modeling distinguishes the effects of average differences between egos from differences within egos over time. The former, *between-person difference*, reflects how a person is different from others in the data, while the latter, *within-person difference*, describes how a person is different from their own "normal" right now. In the between-person effect, the comparison group is other egos. In the within-person effect, the comparison group is oneself.

Understanding the distinction between within- and between-person difference clarifies why we might want to measure both effects in a model of change over time. The between-person effect targets factors that differentiate the people or groups being studied (e.g., do people with higher levels of educational attainment have larger networks than those with lower levels of education?). Much social science research

falls into this category. Yet, between-person effects do not reveal anything about change over time. The within-person effect addresses this directly by providing a fairly robust test of the effect of some change in one variable on the amount of change in another. In testing the impact of a network intervention (e.g., peer-to-peer support for weight loss), for example, a central aim is to assess whether that intervention is associated with a change in participants' normal eating and exercise behaviors. Conflating this within-person effect with average differences in eating and exercise relative to other participants would present a problem. It is cleaner to compare Sarah now to Sarah before the network intervention rather than comparing Sarah to other people who did not have the intervention. More to the point, there are many reasons (besides the intervention) that could explain individual differences. The within-person strategy controls for these unmeasured, static person-level covariates. A model that combines these effects into one coefficient would also present a problem. In longitudinal MLM, when one effect is separated into distinct between-person and within-person effects, these become two variables in the models. The variation within and between people is decomposed so their effects can be examined separately.

In addition to decomposing the variance into between-person and within-person effects, MLMs also allow within-person effects to vary across egos. In the intervention study, constraining the effect of the network intervention to be the same across all study participants is unreasonable. Some egos will be more susceptible to peer influence than others, resulting in a larger observed change in their diet and exercise behaviors after the intervention compared to before. For others, peer counseling and motivation may not produce any change at all. Allowing the intervention to have different effects for different egos while simultaneously estimating the average effect across all egos is optimal. Multilevel models permit this kind of flexibility.

Two conditions are critical for using longitudinal MLMs. First, the dependent variable must change over time. If a researcher is interested in predicting a static characteristic of ego, longitudinal multilevel modeling is inappropriate because there is no variation over time. For example, all of a person's repeated observations have the same value for race. Second, observations at the ego level must be independent of one another, and ideally drawn from a probability sample. That is, ego respondents should not be sampled from the same school, household, or organization in such a way that their survey responses are likely to be correlated, unless this

extra layer of dependence can be explicitly modeled (using, for example, a 3-level model).

In the following sections, we detail the features of MLM that make this statistical method ideal for modeling social network dynamics, and identify interpretable elements of these models. However, the majority of this discussion focuses on longitudinal models that are not specific to ego networks, *per se*. They can be used to predict change in any kind of outcome. Later, three-level models, where a longitudinal component is added to an analysis of alters nested within egos, are discussed. Reading Chapter 8, which introduces the cross-sectional multilevel model for hierarchical data structures (e.g., alters nested in egos), will facilitate understanding this section.

We conclude with a step-by-step guide to building longitudinal multilevel models using an existing dataset and the statistical software program Stata. The general discussion of MLM follows a similar step-by-step progression, beginning with a simple model of the effects of time, and later adding other covariates.

9.3.1 The Longitudinal Random Coefficient Model

A major feature of longitudinal analysis is the explicit modeling of time as an independent variable. To answer a fundamental question: how does the outcome of interest change over time? Does it increase or decrease, on average? Are these dynamic patterns statistically different from chance or random fluctuations? Modeling and interpreting the effect of time as an independent variable represents the first step in building more complex explanatory models that include covariates. Individuals tend to change over time, or grow and develop, at different rates. Consequently, we often observe heterogeneity in the effects of a time variable across egos, which requires a random coefficient model. A random coefficient allows the slope for time to vary across people in the dataset.

To explicitly model dependence, multilevel models manage the random parts of a regression model differently than OLS regression (Chapter 8). The multilevel model splits the residual (or error) ε_i from an ordinary regression model into separate error components. The original residual term becomes three terms in the random coefficient model. Each person gets a random intercept and random coefficient, which are shared by every repeated observation of the same person, and a new (now smaller) residual term, which is unique for every observation. In short, MLM removes the part of the residual that is correlated across observations within the

same person and models it as separate terms. This residual error ε_{ij} no longer violates the assumption of independence. In the longitudinal multi-level model, i represents observations over time and j represents egos. The new random components provide information about how much dependence exists in the data (i.e., how similar repeated observations for the same ego are to each other).

The longitudinal random coefficient model includes a person-specific random intercept and slope for the time variable, and is often called a growth model. Other random coefficients for covariates that change over time may be included, as well, if the slope is theorized to vary across egos. The formula for a growth model (Equation 9.2) is adapted from Raudenbush and Bryk (2002). It includes a fixed intercept β_o , which is interpreted like the y-intercept in an ordinary regression model, as well as an ego-specific random intercept ζ_{oj} . There is also a fixed effect of time β_1 , interpreted as the change in the dependent variable associated with a one-unit change in $Time_{ij}$, and an ego-specific random coefficient ζ_{1j} . Finally, there is an error term ε_{ij} for each observation.

$$y_{ij} = (\beta_o + \zeta_{oj}) + (\beta_1 + \zeta_{1j})Time_{ij} + \varepsilon_{ij} \quad (9.2)$$

If a nonlinear growth curve model is specified, the time-squared (and time-cubed, if applicable) term will also have its own random coefficient. It would be inappropriate to allow the effect of linear time to vary across egos, and yet constrain the nonlinear effect (or the shape of the curve) to be uniform for each person in the sample. Also, although random components of our multilevel model (i.e., the random intercept, random coefficient, and residual) are generally not interpreted, assumptions are made about their mean, distribution, variance, and covariance (see Chapter 8 for detail).

In Figure 9.4 one ego has two observations – say, a baseline measurement and a six-month follow-up. In this illustration, time (the independent variable) has a larger effect for this ego compared to average. This ego also has an intercept that is greater than average for this sample. This is shown by a slope $(\beta_1 + \zeta_{1j})$ that is steeper than the overall β_1 , and an intercept $(\beta_o + \zeta_{oj})$ that is above the overall intercept β_o .

Determining whether different groups experience distinct growth trajectories, or different effects of the time covariate, is often of interest. This is easily accomplished using a cross-level interaction model, where the effect of time at Level 1 is allowed to differ by a Level 2 group variable (see Chapter 8). For example, do men and women, smokers and nonsmokers,

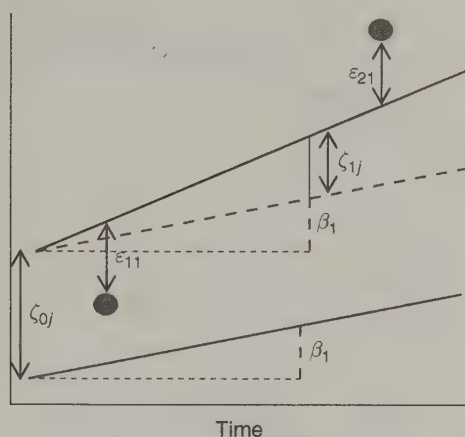


FIGURE 9.4 The longitudinal random coefficient model with between- and within-person differences

or blue-collar and white-collar workers exhibit different dynamic patterns? To test this, a multiplicative interaction term can be included, which is the product of time and a dichotomous group indicator (or set of dichotomous group indicators, if more than two categories). A likelihood-ratio test determines whether we need to allow the effect of time to vary across egos; that is, whether egos do “grow” at different rates. This test compares the random coefficient model to the random intercept model. A rejected null hypothesis suggests no significant difference between the two models. That is, the random intercept model can be used instead of the random coefficient model.

It is possible to quantify and interpret the extent to which repeated observations within egos are correlated. In the longitudinal multilevel model, the intraclass correlation (ICC) is interpreted as a measure of within-person correlation (Raudenbush & Bryk 2002). Higher values indicate that observations over time nested within egos are more similar to one another. The ICC represents the standardized way of expressing the amount of dependency due to egos being more similar to themselves over time than to other egos. The larger the ICC, the more problematic it is to ignore dependence and run ordinary regression models.

Finally, though we have thus far limited our discussion of dynamics to predicting ego’s outcomes or characteristics of ego’s networks over time, it is often useful to examine dynamic attributes of alters. For example,

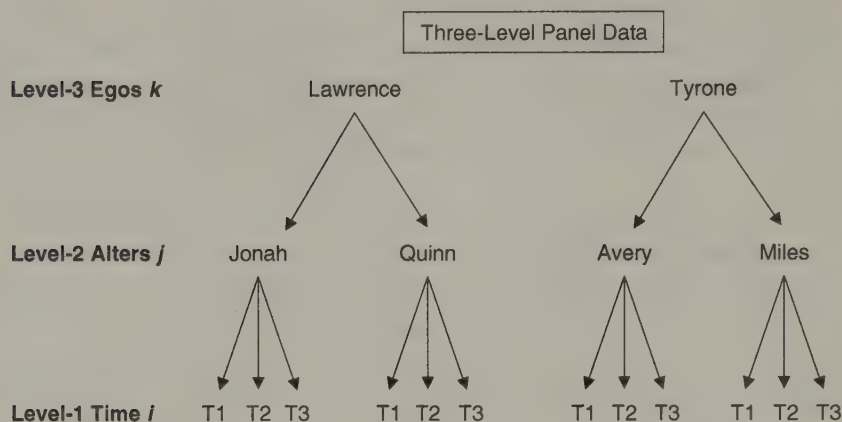


FIGURE 9.5 Longitudinal and hierarchical three-level data structure for ego network analysis

how does the level of support each alter provides to ego change over time as a function of alter or ego characteristics? This kind of analysis would require a hierarchical model, where alters are nested in egos (as discussed in Chapter 8). Modeling alter-level dynamics leverages data at three levels, where observations over time are nested in alters, which are in turn nested in egos (see Figure 9.5). This strategy has a number of advantages, such as allowing estimation of coefficients at the ego (e.g., network density) and alter (e.g., alter frequency of contact) levels simultaneously, making it possible to identify the relative effects of dynamic processes occurring at different levels of observation (Wellman & Frank 2001). Moreover, modeling alter level variables provides opportunities for examining cross-level interactions, such as moderation of the effects of alter attributes by personal network characteristics. This represents an analysis of whether and how the network context changes the direction or magnitude of the influence of properties of alters or relationships. Such data will be unbalanced – meaning that every alter will not be observed at every time point – because of membership turnover in networks. However, the multilevel random coefficient model does not require balanced data. It produces coefficients on the basis of the data available. More information about estimating and interpreting three-level multilevel models, provided in Chapter 8, can be applied in a straightforward way to growth modeling.

9.3.2 An Illustration of the Longitudinal Random Coefficient Model: Network Dynamics following Parenthood

Researchers interested in explaining how network size changes after parenthood could conduct a longitudinal survey using egocentric methodology. A cohort composed of childless adults and age-matched recent parents could be interviewed about their friends and family members at three points in time – immediately following the birth, when the child is five years old, and when the child is nine years old. Clustering is likely within ego respondents such that some egos have larger networks, on average, than others. Some may come from larger families, for example, or live among a sociable group of neighbors.

The repeated observations of two fictional egos – Sue and Sam – are presented in Figure 9.6. In this example, Sue has provided data at all three time points. Her network size is lowest at Time 1, and highest at Time 3. Though Sue's network is larger, on average, than other egos (all of her observations are above the grand mean β_0), substantial within-person variation is observed. Sam has also provided data at three time points. His network is smaller, on average, than other egos. He has less within-person variation than Sue. In these data, variation between persons is

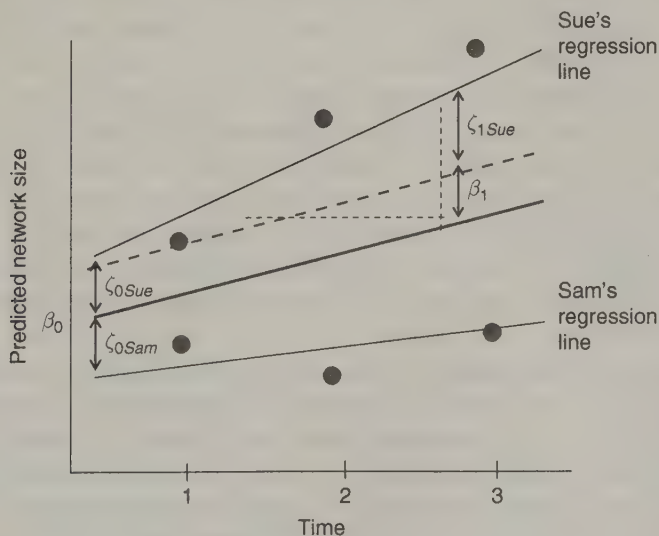


FIGURE 9.6 Scatterplot with fitted random coefficient regression line for the effects of time on network size among new parents

greater than variation within persons. That is, egos' network size tends to be more different from other egos' than from their own networks at different time points. This is evidence of dependence.

The independent variable in this growth model is time. Effects of time are typically modeled alone prior to adding other predictors to establish how the dependent variable changes longitudinally. Subsequently, a time-invariant (i.e., a variable that does not change over time) independent variable could be added representing our two groups, parents and non-parents. The hypothesis that new parents begin the study with smaller networks is supported by intercept differences between the two groups. However, as time passes after becoming new parents, networks might become larger among parents relative to nonparents. Having an infant or toddler can be socially isolating (you can't take them anywhere), but this likely improves as children become more self-sufficient. Also, as children enter school, people tend to develop relationships with other families, perhaps widening their friendship circle. An interaction between the continuous time variable and a binary parenthood variable would test this hypothesis.

In this model, constraining the effect of time to be equal across persons in the study is inappropriate since networks likely change at different rates. In Figure 9.6, the effect of time is larger among Sue's observations compared to the average coefficient. Her network grows at a faster rate than is typical. Consequently, her random slope is positive. The total estimated effect of time for Sue is equal to the overall effect β_1 plus her cluster-specific random coefficient ζ_{1j} . Sam's within-cluster regression line has a smaller slope compared to the average effect, so his random coefficient is negative. In addition, the pattern observed for these egos suggests a positive correlation between random intercepts and slopes. That is, the larger a person's network, on average, when time is equal to zero, the faster their network grows over time. People with larger networks at the beginning of the study experience faster network growth over time.

Figure 9.7 displays the estimated effects for all the egos in our sample simultaneously. Both the intercept and the coefficient are random, or stochastic. That is, each ego has their own intercept for network size and their own sloped line based on their rate of network growth. The overall intercept β_0 and slope β_1 (depicted with the heavy, solid line), useful for hypothesis testing, are weighted averages of each ego's intercept and slope. If an interaction term had been included to test whether parents' growth is larger in magnitude compared to those

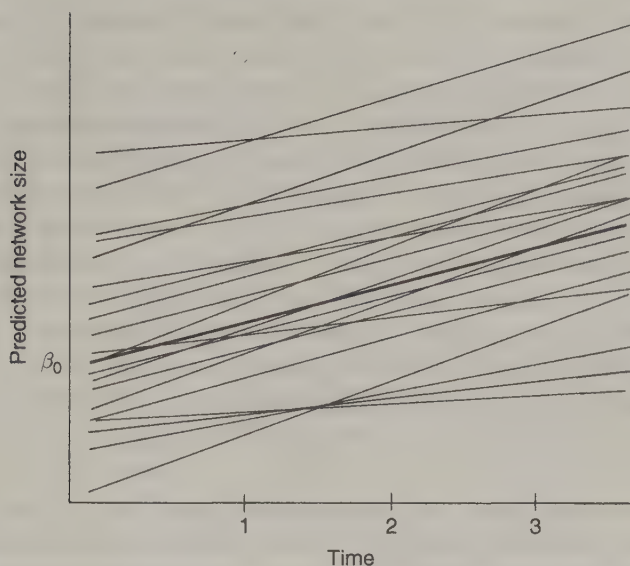


FIGURE 9.7 Individual regression lines from a random coefficient regression model predicting network size over time

without children, there would be two distinct overall intercepts and slopes (and associated random intercepts and slopes). One pair would represent parents and another pair would represent nonparents.

9.3.3 Variance Decomposition: Between-Person and Within-Person Effects

As noted previously, longitudinal data provide an opportunity to examine how change in one variable is associated with change in another. In order to isolate the effects of change within persons from typical differences between people, one variable is split into separate within-person and between-person variables. Within-person effects describe how and why a person deviates from their own “normal” conditions right now, while between-person effects describe how and why a person is different from someone else, on average. This is critical because all time-variant variables (i.e., measures that change over time) contain information about both within-person and between-person effects. Not splitting the variable into these two kinds of effects would prohibit discerning why a person’s outcome is better (or worse) at a particular time point. Is it (1) because that person typically has better outcomes than others, or (2) because life

conditions have changed to make that person's outcomes better than usual, or (3) both?

This strategy, variance decomposition, is related to the issue of cluster confounding described in Chapter 8. In cross-sectional MLM, where alters are nested in egos, the problem of combined coefficients is addressed using a contextual effect. This strategy permits estimation of the "true" effect of covariates that vary across alters, and assesses the influence of network context (e.g., alter closeness versus mean network closeness). In longitudinal MLM, the problem is solved by explicitly splitting the two effects that are confounded when only one variable is used (Hoffman 2015). Failing to address cluster confounding in longitudinal models not only causes the kinds of interpretation problems described above, but can lead to biased estimates and standard errors for variables that change over time.

Variance decomposition is accomplished by splitting all variables at Level 1 (i.e., those that change over time) into purely between-person and purely within-person effects. The between-person effect is equal to the person mean of all repeated observations (Equation 9.3).

$$BPx_j = \bar{X}_j \quad (9.3)$$

For example, a person might be surveyed in three waves, each time indicating how many smokers are in the network. The network contains five smokers at Time 1, eight smokers at Time 2, and six smokers at Time 3. The person mean (or between-person variable) for number of smokers would be $(5 + 8 + 6)/3$ or 6.33 alters. It does not vary over time within a respondent. A person gets the same value in each panel, or wave, of the study.

The within-person effect captures change over time. It is equal to a person's deviation from their own person mean at a given time point, and is often called person mean centering (Equation 9.4).

$$WPx_{ij} = x_{ij} - \bar{X}_j \quad (9.4)$$

Returning to the example above, there would be three distinct within-person observations. At Time 1, the within-person number of smokers is $5 - 6.33$, or -1.33 . At Time 1, this ego has fewer smokers than normal. At Time 2, within-person number of smokers is $8 - 6.33$, or $+1.67$. At Time 2, the network contains more smokers than usual. At Time 3, the within-person number of smokers is $6 - 6.33$, or -0.33 . These are the within-person deviations from the typical number of smokers.

The within-person variable could be used to determine whether deviations from one's average have consequences for ego's outcomes at a given time point (e.g., attitudes toward smoking, or likelihood of starting or quitting smoking). When the network contains more smokers than usual, an ego might be less likely to quit smoking. By removing variation related to the person mean (the between-person effect), the confounding effects of time-invariant characteristics are controlled. For example, it is possible that a positive relationship between number of smokers in the network (the original variable x_{ij} , or the combined effect of within- and between-person network influence) and propensity to quit smoking is actually due to a third variable – educational attainment – which is correlated with both the independent variable of interest and our dependent variable. People with lower levels of educational attainment are more likely to smoke, and less likely to quit smoking. Because of homophily and shared foci of activity (e.g., being in similar occupations), individuals with lower levels of education are more likely to form relationships with others in their own social class who are also more likely to be smokers. Thus, any correlation between number of smokers in the network and likelihood of quitting smoking could be due to the influence of socioeconomic status on both variables, not due to social influence. Omitted confounding variables create a condition referred to as *endogeneity*.

However, decomposing the variance into within-person and between-person effects of number of smokers in the network removes the threat to causality associated with confounding variables that do not change over the course of our study. This includes any unobserved social conditions or processes that influenced what the network looked like when the ego entered the study, and are also related to outcomes of interest. Any confounding time-invariant characteristics will be correlated with the between-person effect (i.e., an ego's typical number of alters who smoke) rather than the within-person effect (i.e., how ego's number of alters who smoke differs from usual), which purely reflects change over time. However, confounding effects of time-varying characteristics (e.g., employment status) can still create endogeneity and threaten causal inference in longitudinal models, making it important to control for these to the extent possible.

9.3.4 An Illustration of Between-Person and Within-Person Effects: Network Dynamics following Parenthood

In the example of changes in network size following the transition to parenthood, some feature of life that changes across the parenthood

career might be proposed as the mechanism underlying network dynamics. In other words, in addition to modeling growth trajectories using the time variable (i.e., how networks change), we might target the identification of characteristics that explain network change (i.e., why networks change). To this end, we might measure community engagement at each point in time using an index of activities like volunteering, participation in clubs, teams, or organizations, and attendance at community events. Engagement provides opportunities for forming new ties and sustaining existing ones on the basis of shared foci of activity (Feld 1981), perhaps contributing to larger networks.

In this analysis, the effects of a person's usual level of community engagement (i.e., some people are habitually more involved than others) must be disaggregated from changes in community engagement (perhaps as a function of parenthood). To accomplish this, the variance in community engagement could be decomposed into within-person and between-person variables. Both are included in the growth model. The person mean, or between-person effect, indicates whether egos who are more engaged in their communities, on average, tend to have larger personal networks than other egos who are less engaged. The deviation from the person mean, or within-person effect, indicates whether being more engaged right now compared to usual is associated with reporting larger networks than ego's own average.

Moreover, the effect of within-person variation in community engagement should be allowed to differ across egos. Being involved in one's community may not have the same effect for every person. In this case, another random coefficient for within-person change in community engagement would be added to the model, resulting in two random coefficients (one for time and one for within-person engagement) and one random intercept (for mean differences in network size when all independent variables are jointly equal to zero). Of course, neither a random coefficient for the binary parenthood variable nor for between-person community engagement would be included because these do not vary over time or across repeated observations. However, testing whether the effect of within-person engagement does actually differ across egos (i.e., each ego needs their own slope for community engagement) is an important step. This is accomplished using a likelihood-ratio test comparing the growth model with one random coefficient for time and the growth model with two random coefficients.

The correlation between random slopes and random coefficients, or $\rho_{2,1}$, can be interpreted. A negative value would suggest that the influence of community engagement is smaller in magnitude among egos who have larger networks, on average. This pattern might indicate that there is some maximum number of sustainable personal relationships. Exposure to potential new friends has diminishing returns among those who have larger networks at the outset.

9.3.5 Longitudinal MLM for Categorical Outcomes

Often the outcomes of interest to network researchers studying dynamics are not continuous. They might be binary (e.g., whether or not ego joins a social movement), categorical (e.g., whether ego votes republican, democrat, or independent), ordinal (e.g., perceived network density on a Likert scale), or counts (e.g., number of sexual partners). Growth models can be fitted for these outcomes, often called “categorical” as a family. There are multilevel versions of logistic regression for binary dependent variables, ordinal regression for ordinal variables, and multinomial regression for categorical variables. In addition, both poisson and negative binomial regression can be fitted for longitudinal count outcomes (see Rabe-Hesketh & Skrondal 2008 for an excellent guide to multilevel modeling for categorical outcomes in Stata).

9.4 STEP-BY-STEP GUIDE TO LONGITUDINAL MULTILEVEL MODEL-BUILDING

This section illustrates the standard progression of multilevel model-building for longitudinal data using a modified dataset from a panel study of adolescents in the United States. Data were collected every six months beginning in 2007 to understand students’ experiences in the American public education system. In total, 2,655 participants provided data on up to seven different occasions.

These data allow an examination of social network predictors of intergroup bias; that is, the tendency to systematically evaluate your own ethnic group more favorably than other ethnic groups. Intergroup bias is assessed using a self-report measure of affective in-group and out-group evaluations. Higher scores indicate more positive in-group affect and more negative out-group affect (i.e., more intergroup bias). This dependent variable is labeled *ingroup*, and has a value for each person

at each point in time. In addition, the variable *time* reflects six-month intervals and ranges from one to seven. The key independent variable of interest is the effective size of the friendship network. Effective size is the number of alters minus the mean number of ties that each alter has to other alters in the network. It taps into the degree of nonredundancy in the network, or the extent to which ego has access to a larger number of alters that could bring novel ideas, attitudes, or information to the relationship. Put differently, someone with a higher effective size has a network that contains more structural holes, which are associated with greater opportunities for exposure to diverse ideas and information (e.g., about racial and ethnic outgroups). A control for ego gender (*female*, where 1=female and 0=male) is added. Finally, *id* contains unique ego respondent identification numbers. Theoretically, this captures the essence of Allport's (1954) contact hypothesis, which suggests that greater exposure to diverse others decreases prejudice.

9.4.1 Step I: Null Random Intercept Model

The first step in longitudinal multilevel model building is to compute a null (i.e., empty) random intercept model. This tests whether MLM is needed. A multilevel model is almost always necessary for panel data because of the strong tendency for people to be more similar to themselves over time than they are to other people. The Stata *xtmixed* syntax and output for the null random intercept model predicting *ingroup* are as follows:

```
xtmixed ingroup || id:, mle variance
```

```
Performing EM optimization:
```

```
Performing gradient-based optimization:
```

```
Iteration 0: log likelihood = -29610.812
```

```
Iteration 1: log likelihood = -29610.812
```

```
Computing standard errors:
```

```
Mixed-effects ML regression
```

```
Group variable: id
```

```
Number of obs      =      16,388
```

```
Number of groups   =       2,655
```

```
Obs per group:
```

```
min =          6
```

```
avg =         6.2
```

```
max =          7
```

```
Wald chi2(0)      =          .
```

```
Prob > chi2       =          .
```

```
Log likelihood = -29610.812
```

```
-----+-----
```

ingroup	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]
__cons	21.16808	.0352159	601.10	0.000	21.09906 21.2371

```
-----+-----
```

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]	
id: Identity					
	var(_cons)	3.063383	.0904135	2.891204	3.245816
	var(Residual)	1.410444	.0170212	1.377474	1.444202
LR test vs. linear model: chibar2(01) = 11834.56 Prob >= chibar2 = 0.0000					

The commands after the comma are optional. The *mle* command specifies that maximum likelihood estimation be used, and *variance* requests that the variances (and later covariances) for the random effects be included in the output. The variable after the two vertical lines is the grouping variable, or ego identification number. It is the unique identifier shared by each repeated observation nested in a particular ego.

This very basic model provides some important information. First, the number of observations at Level 1 is 16,388. The number of groups, or egos, is 2,655, consistent with the number of adolescents in the study. In addition, the minimum number of observations over time provided by any ego is six, while the maximum is seven, with an average of 6.2 per ego. The coefficient next to *_cons* is the weighted average intercept, or the mean score on the intergroup bias measure. This fixed estimate is 21.17, reflecting a moderate degree of bias since the scale has a potential range of ten to thirty. At the bottom of the output, the *var(_cons)* is the intercept variance at the ego level, or the variance between egos. In contrast, *var(Residual)* provides the intercept variance at the observation level, or variance within egos. The variance is larger at the ego level than at the observation level, suggesting that our intraclass correlation will be closer to one than to zero. The output from the *estat* command confirms this, providing an ICC of 0.68:

```
estat icc
```

```
Intraclass correlation
```

Level	ICC	Std. Err.	[95% Conf. Interval]	
id	.6847344	.0069568	.670943	.6982078

This result suggests strong correlation between observations nested in the same ego with respect to the measure of interclass bias. In other words, most of the variation is between egos rather than within egos. As expected, the results of the likelihood-ratio test provided at the bottom of the model indicate that a random intercept is necessary to adjust for dependence at

the ego level ($p < .00001$). Had this p -value been $\geq .05$, a standard OLS regression model would be appropriate.

9.4.2 Step II: Random Coefficient Growth Model

The second step in longitudinal multilevel model building is to run a growth model by adding the effect of time. A random coefficient for time allows the trajectory of change in intergroup bias to vary across egos. We add a random coefficient in Stata by including the variable representing time after the ego identifier and the colon. Each ego gets their own slope bump or deficit, depending how their rate of change differs from the weighted average. Some adolescents will become more biased attitudinally over time, while others will become less biased toward other groups:

```
. xtmixed ingroup time || id: time, mle covariance(unstructured) variance
Performing EM optimization:
Performing gradient-based optimization:
Iteration 0:  log likelihood = -21977.524
Iteration 1:  log likelihood = -21977.524
Computing standard errors:
Mixed-effects ML regression              Number of obs      =      16,388
Group variable: id                      Number of groups   =       2,655
                                         Obs per group:
                                         min =              6
                                         avg =             6.2
                                         max =              7
                                         Wald chi2(1)      =    10330.69
                                         Prob > chi2       =      0.0000
-----+-----
      ingroup |      Coef. | Std. Err.      z    P>|z|     [95% Conf. Interval]
-----+-----
      time |   .484861 |   .0047704   101.64  0.000    .4755112   .4942107
      _cons |  19.42742 |   .0425943   456.10  0.000   19.34394   19.51091
-----+-----
Random-effects Parameters | Estimate   Std. Err.   [95% Conf. Interval]
-----+-----
id: Unstructured
      var(time) |   .0403047 |   .0016851 |   .0371336   .0437465
      var(_cons) |   4.498923 |   .1322974 |   4.246955   4.765839
      cov(time,_cons) |  -.2541307 |   .0122849 |  -.2782087  -.2300527
-----+-----
      var(Residual) |   .3800625 |   .0051096 |   .3701787   .3902101
-----+-----
LR test vs. linear model: chi2(3) = 23980.85          Prob > chi2 = 0.0000
Note: LR test is conservative and provided only for reference.
```


The coefficient for *time* is the weighted average effect of time on intergroup bias. The coefficient for *time* is positive and significant, suggesting that each 6-month increase in time is associated with a 0.48-unit increase in intergroup bias. In other words, adolescents become slightly more biased as they age. At the bottom of the output, *var(time)* is the slope variance at the ego level, *var(_cons)* is the intercept variance at the ego level, and *var(Residual)* is the residual variance at the observation level. The *cov(time,_cons)* is the covariance of the random slopes and random intercepts. A value of -0.25 indicates that there is a negative correlation between random intercepts and random slopes such that *time* has a smaller effect (i.e., a less steep positive slope) among egos with higher initial levels of intergroup bias (i.e., when *time*=0). In other words, if a person starts out as more biased, their rate of increase in bias is less pronounced over time.

After storing estimates from the random intercept and random coefficient models, a likelihood ratio test formally examines whether adding the random coefficient for *time* improves model fit:

```
quietly xtmixed ingroup time || id: , mle variance
est store ri
quietly xtmixed ingroup time || id: time, mle covariance(unstructured) variance
est store rc
lrtest rc ri
Likelihood-ratio test                                LR chi2(2) = 1772.37
(Assumption: ri nested in rc)                        Prob > chi2 = 0.0000
```

We can reject the null hypothesis at $p < .00001$, suggesting that the random coefficient model provides a better fit for the data than the random intercept model. In other words, the effect of *time* does vary significantly between egos.

9.4.3 Step III: Random Coefficient Growth Curve Model

After running a linear growth model, the next step is to assess the linearity of *time* using a polynomial model. This is tested by generating and adding a squared (and potentially cubed) term for *time*. A random slope is added for the squared term to allow both the linear and nonlinear components of the effect of *time* to vary across egos:

```
gen timesq=time*time
xtmixed ingroup time timesq || id: time timesq, mle cov(unstructured) variance
Performing EM optimization:
```

Performing gradient-based optimization:

Iteration 0: log likelihood = -21701.758
Iteration 1: log likelihood = -21628.157
Iteration 2: log likelihood = -21611.583
Iteration 3: log likelihood = -21611.501
Iteration 4: log likelihood = -21611.501

Computing standard errors:

Mixed-effects ML regression
Group variable: id

Number of obs = 16,388
Number of groups = 2,655
Obs per group:
min = 6
avg = 6.2
max = 7
Wald chi2(2) = 10354.03
Prob > chi2 = 0.0000

Log likelihood = -21611.501

ingroup	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
time	.7762489	.0149563	51.90	0.000	.7469351	.8055627
timesq	-.0402395	.0018892	-21.30	0.000	-.0439422	-.0365368
_cons	19.02596	.0466154	408.15	0.000	18.93459	19.11732

Random-effects Parameters		Estimate	Std. Err.	[95% Conf. Interval]	
id: Unstructured					
var(time)		.1971488	.0166764	.1670294	.2326995
var(timesq)		.0021189	.0002623	.0016624	.0027007
var(_cons)		4.780778	.158855	4.47935	5.10249
cov(time,timesq)		-.0183112	.0020371	-.0223038	-.0143186
cov(time,_cons)		-.4590467	.0416632	-.540705	-.3773883
cov(timesq,_cons)		.0231357	.0049341	.0134651	.0328063
var(Residual)		.3351888	.0050523	.3254314	.3452388

LR test vs. linear model: chi2(6) = 24681.07
Prob > chi2 = 0.0000

Note: LR test is conservative and provided only for reference.

The coefficient for *timesq* is statistically significant, indicating that a nonlinear specification of time is appropriate. A positive coefficient for *time* and a small negative coefficient for *timesq* indicates that inter-group bias increases more rapidly at first and then tapers off. At the bottom of the output, *var(time)* and *var(timesq)* are the slope variances for the two time variables at the ego level, *var(_cons)* is the intercept variance at the ego level, and *var(Residual)* is the residual variance at the observation level. With two random coefficients in the model, we now have three covariance terms to interpret. The *cov(time,_timesq)* is the covariance of the two random slopes, and suggests that egos with a steeper slope for time have a more negative

effect of time squared, meaning that their slope tapers off more quickly compared those with a less pronounced rate of growth. The $cov(time, _cons)$ is the covariance of the random slopes for linear time and random intercepts. It is now -0.46 in the nonlinear model, indicating a more negative correlation compared to the linear model. Linear time has a smaller effect among egos with higher initial levels of intergroup bias. Moreover, $cov(timesq, _cons)$ – the covariance of the random slopes for nonlinear time and random intercepts – is slightly positive, meaning that there is less tapering off of bias over time among egos with higher starting values of intergroup bias, on average. If a person starts out as more biased, their rate of increase in bias over time is less pronounced and more linear. If a person starts out as less biased, their rate of growth in bias is steeper, but tapers off quickly.

The nonlinear effect of time can be plotted using the *margins* and *marginsplot* commands in Stata. These can produce and graph marginal effects and predicted values.

```
margins, over(time)
```

Predictive margins	Number of obs				=	16,388
Expression	: Linear prediction, fixed portion, predict()					
over	: time					

	Delta-method					
	Margin	Std. Err.	z	P> z	[95 percent Conf. Interval]	
time						
1	19.76197	.0404329	488.76	0.000	19.68272	19.84121
2	20.4175	.0376314	542.57	0.000	20.34374	20.49125
3	20.99255	.036387	576.92	0.000	20.92123	21.06387
4	21.48712	.035373	607.44	0.000	21.41779	21.55645
5	21.90121	.0342964	638.59	0.000	21.83399	21.96843
6	22.23483	.0341037	651.98	0.000	22.16799	22.30167
7	22.48796	.0368975	609.47	0.000	22.41565	22.56028

This command outputs the fixed (i.e., ignoring random coefficients and intercepts) predicted values of intergroup bias, including confidence intervals, at each value of time 1–7. We can then plot these predicted values to see the nonlinear relationship between time and intergroup bias using the *marginsplot* command. Figure 9.8 confirms our crude interpretation of the shape of the growth curve that we deduced from the coefficients for time and time-squared, but provides a better indication of the magnitude of change. Though the effect is nonlinear, the shape of the curve is not particularly pronounced.

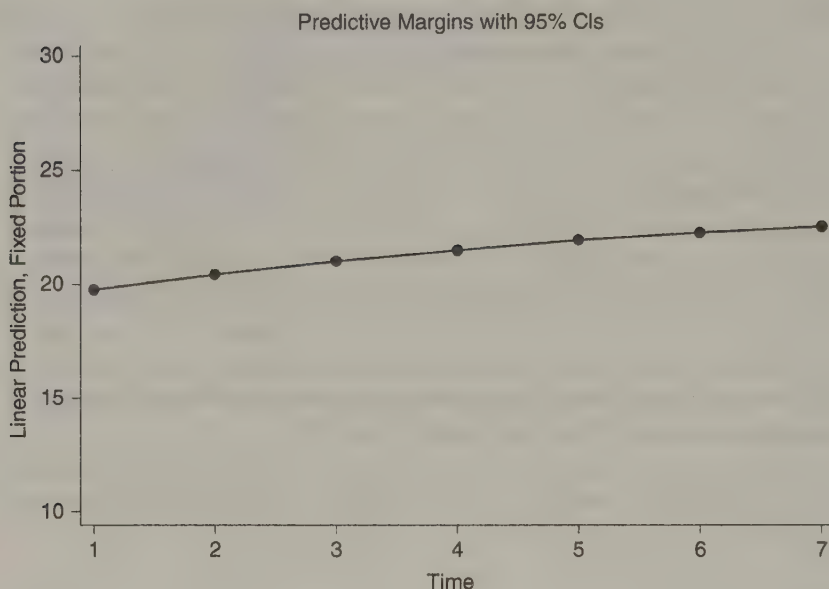


FIGURE 9.8 Predicted values of the nonlinear effect of time on intergroup bias

9.4.4 Step IV: Random Coefficient Growth Curve Model with Covariates

After determining the appropriate model of change over time, other covariates are added. To examine how effective size is related to intergroup bias, controlling for ego gender, the model is first run without decomposing the variance of effective size into within-person and between person effects. There will be one coefficient for effective size that contains information about egos' typical effective size, and how effective size changes over time. This coefficient will be an uninterpretable blend of the within-person and between-person effects:

```
xtmixed ingroup time timesq female effsize || id: time timesq, ///
mle cov(unstructured) variance
```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log likelihood = -20696.676

Iteration 1: log likelihood = -20684.795

Iteration 2: log likelihood = -20634.095

Iteration 3: log likelihood = -20632.501
 Iteration 4: log likelihood = -20631.526
 Iteration 5: log likelihood = -20631.522
 Iteration 6: log likelihood = -20631.522

Computing standard errors:

Mixed-effects ML regression
 Group variable: id

Number of obs = 16,388

Number of groups = 2,655

Obs per group:

min = 6

avg = 6.2

max = 7

Wald chi2(4) = 15286.90

Prob > chi2 = 0.0000

Log likelihood = -20631.522

ingroup	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
time	.6174285	.0147822	41.77	0.000	.588456	.646401
timesq	-.0306107	.0018637	-16.42	0.000	-.0342636	-.0269578
female	-.2547956	.0526805	-4.84	0.000	-.3580474	-.1515438
effsize	-1.195567	.0241707	-49.46	0.000	-1.24294	-1.148193
_cons	21.49627	.0673236	319.30	0.000	21.36432	21.62822

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
id: Unstructured				
var(time)	.1585951	.0156033	.130781	.1923246
var(timesq)	.0018233	.0002515	.0013913	.0023895
var(_cons)	2.983241	.1122258	2.771195	3.211511
cov(time,timesq)	-.0153852	.0019371	-.0191818	-.0115886
cov(time,_cons)	-.3290074	.0344544	-.3965368	-.2614779
cov(timesq,_cons)	.016707	.0041395	.0085938	.0248202
var(Residual)	.3362576	.0051088	.3263922	.3464213

LR test vs. linear model: chi2(6) = 17373.13

Prob > chi2 = 0.0000

Note: LR test is conservative and provided only for reference.

The findings suggest that egos who are women report 0.25 fewer units of intergroup bias than men, on average ($p < .001$). Also, each one-unit increase in effective size is associated with an estimated 1.20 fewer units of intergroup bias. People with more structural holes exhibit less bias, perhaps because they are exposed to less redundant, or more heterogeneous, attitudes and ideas through interpersonal contact. However, the coefficient for effective size could be biased because it combines within-person and between-person variation. Further, the interpretation is problematic because the comparison group is both other egos and other time points.

9.4.5 Step V: Random Coefficient Growth Curve Model
with Decomposed Variance

To address the issue of cluster confounding, or combined within-person and between-person effects, the effective size variable is split into two distinct variables. The between-person variable will be equal to the person mean (or mean across observations) for each ego. The within-person variable will be equal to the value at a given time point minus the person mean, reflecting ego’s deviation from their own typical effective size over time. A random coefficient for within-person change in effective size is added, since network dynamics are unlikely to have uniform effects for all egos in the sample. The commands for creating these variables in Stata, and the resulting model are:

```
bysort id: egen bpeffsize=mean(effsize)
lab var bpeffsize "Between-person effective size"

gen wpeffsize=effsize-bpeffsize
lab var wpeffsize "Within-person effective size"

xtmixed ingroup time timesq female bpeffsize wpeffsize || id: time ///.
timesq wpeffsize, mle cov(unstructured) variance

Performing EM optimization:
Performing gradient-based optimization:
Iteration 0: log likelihood = -20042.749
Iteration 1: log likelihood = -20011.666
Iteration 2: log likelihood = -20003.04
Iteration 3: log likelihood = -20002.989
Iteration 4: log likelihood = -20002.989

Computing standard errors:
Mixed-effects ML regression
Group variable: id
Number of obs = 16,388
Number of groups = 2,655
Obs per group:
    min = 6
    avg = 6.2
    max = 7
Wald chi2(5) = 17977.19
Prob > chi2 = 0.0000
Log likelihood = -20002.989
```

ingroup	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
time	.6274892	.0143286	43.79	0.000	.5994057	.6555727
timesq	-.0318582	.0017738	-17.96	0.000	-.0353347	-.0283818
female	-.3914295	.0486846	-8.04	0.000	-.4868497	-.2960094
bpeffsize	-2.078811	.0396746	-52.40	0.000	-2.156572	-2.001051
wpeffsize	-1.146953	.0425629	-26.95	0.000	-1.230375	-1.063532
_cons	22.97311	.0805438	285.22	0.000	22.81524	23.13097

Random-effects Parameters	Estimate	Std. Err.	[95% Conf. Interval]	
id: Unstructured				
var(time)	.1057585	.0148859	.0802611	.1393559
var(timesq)	.0011378	.0002291	.0007667	.0016885
var(wpeffs~e)	1.2221	.0972933	1.045542	1.428474
var(_cons)	2.398278	.1034916	2.203779	2.609943
cov(time,timesq)	-.0094228	.0017984	-.0129476	-.005898
cov(time,wpeffs~e)	.0974626	.0284721	.0416583	.1532668
cov(time,_cons)	-.2181323	.0329024	-.2826198	-.1536448
cov(timesq,wpeffs~e)	-.0025268	.0035706	-.009525	.0044715
cov(timesq,_cons)	.0066692	.0038362	-.0008496	.0141879
cov(wpeffs~e,_cons)	-.1282915	.0749478	-.2751865	.0186036
var(Residual)	.3061565	.0048295	.2968357	.3157699
LR test vs. linear model: chi2(10) = 18320.36 Prob > chi2 = 0.0000				

Note: LR test is conservative and provided only for reference.

Both the between-person and within-person effects of effective size are significant, suggesting that variance decomposition is necessary to produce unbiased estimates. Comparing the estimates from the two models (with and without variance decomposition) reveals that the coefficient for effective size was -1.20 in the previous model, and is now -2.08 for the between-person coefficient and -1.15 for the within-person coefficient. The between-person effect is substantially different from (and in this case larger than) the within-person effect, demonstrating the hazards of ignoring cluster confounding.

This model also permits straightforward interpretations of the effects of network effective size. A one-unit increase in network effective size at a given time point, compared to ego's average, is associated with an estimated 1.15-unit reduction in intergroup bias ($p < .001$). That is, during times when egos have higher effective size than normal, their intergroup bias is lower than usual. In addition, each 1-unit increase in average effective size compared to other egos is associated with an estimated 2.08-unit decrease in intergroup bias ($p < .001$). In other words, egos whose networks are typically characterized by higher effective size, that is, which contain fewer redundant ideas and attitudes, tend to be less biased relative to other egos. Lagging the within-person independent variable should be considered only if there is reason to believe that there is a delay in the causal influence of effective size on intergroup bias (see Rabe-Hesketh & Skrondal 2008).

The additional variances and covariances at the bottom of the output also provide information. The *var(wpeffs~e)* is the slope variance for

network effective size at the ego level. The `cov(time, wpeffs~e)` output provides the covariance of the random slopes for time and effective size, while `cov(timesq, wpeffs~e)` is the covariance for time-squared and effective size. These indicate that among egos with a more positive coefficient for network effective size, the rate of growth in intergroup bias over time is steeper and has a more pronounced taper toward the end of the study period. Because the coefficient for network effective size is negative, this indicates that adolescents who exhibit a weaker relationship between effective size and intergroup bias also tend to experience greater increases in bias over time, though the curve is more tapered. It is possible that there are other factors driving increasing intergroup bias among adolescents whose friendship networks have a less pronounced effect, such as biased family members. Alternatively, there may be something about the content of those networks – such as friends' levels of bias – that explains this pattern, but is not measured here.

Finally, to be certain that the random coefficient for within-person intergroup bias is necessary, a likelihood ratio test is conducted after storing estimates from both models. This test determines whether the additional random slope improves model fit.

```
quietly xtmixed ingroup time timesq female bpeffsize wpeffsize || id: ///
time timesq, mle cov(unstructured) variance
est store rc2

quietly xtmixed ingroup time timesq female bpeffsize wpeffsize || id: ///
time timesq wpeffsize, mle cov(unstructured) variance
est store rc3

lrtest rc3 rc2

Likelihood-ratio test                                LR chi2(4) = 603.34
(Assumption: rc2 nested in rc3)                     Prob > chi2 = 0.0000
```

The results suggest that the null hypothesis can be rejected at $p < .05$. The model with a random coefficient for within-person effective size provides a better fit for the data than the model which constrains these slopes to be equal. In other words, the effect of network effective size over time does vary significantly between egos.

9.5 CONCLUSION

This chapter introduced key concepts that guide research on social network dynamics, such as the core-periphery structure and tie activation. In addition to providing a conceptual foundation, this chapter familiarized

readers with commonly used methods for longitudinal data collection and the analysis of change in social network characteristics. Strategies for distinguishing real change from tie inactivity (i.e., latency) and methodological artifact were provided. In addition, measures of change in the specific individuals who make up a network (i.e., membership turnover or tie churn), unique to egocentric network analysis, were also described. Finally, building on Chapter 8, we introduced the basic growth curve model for analyzing longitudinal data and methods for assessing within-person change over time, and provided a step-by-step guide to building such models.

Relating Egocentric and Sociocentric Network Analysis

As discussed throughout this book, an ego network consists of the constellation of persons and relationships surrounding an ego. What hasn't been made explicit, to this point, is that ego networks can be obtained either through an egocentric research design or a sociocentric design. In an egocentric design, the ego is the respondent, and all of the information about the alters and their ties to other alters is obtained from ego. In a whole network or sociocentric design, every node is a respondent, and information about the alter-alter ties is obtained from the alters themselves. In the analysis, each node in turn can be designated as ego and the pattern of ties among ego's alters can be determined. In short, ego networks are present and can be analyzed in both egocentric and sociocentric research designs. This suggests that we can apply the distinction between egocentric and sociocentric to both data and analysis, creating a two-by-two classification of network research.

In Table 10.1, quadrants A and D refer to egocentric and sociocentric research in pure form. The former is the explicit subject of this book, and the latter is discussed in a variety of other works, including Wasserman and Faust (1994) and Borgatti, Everett, and Johnson (2013). In this chapter, we focus on the two off-diagonal quadrants, which mix elements of egocentric work and sociocentric work. Quadrant B refers to studies where a sociocentric approach to egocentric data is employed, including (1) converting egocentric data to sociocentric data, and (2) analyzing egocentric data with sociocentric software and concepts. Quadrant C refers to studies that collect whole network data, but in the analysis make use of measures that utilize only ego's immediate network neighborhood, ignoring the larger structure of the network in which they are

TABLE 10.1 *Typology of studies based on ego network vs sociocentric data or analysis*

		Analysis	
		Ego	Whole
Data	Ego	A. Egocentric analysis of egocentric data	B. Whole network analysis of egocentric data
	Whole	C. Egocentric analysis of whole network data	D. Whole network analysis of whole network data

embedded. The chapter ends with a discussion of cognitive social structure (CSS) research designs, which are neither egocentric nor sociocentric, but have elements of both (Krackhardt 1987).

10.1 QUADRANT B: WHOLE NETWORK ANALYSIS OF EGOCENTRIC DATA

Quadrant B refers to the application of whole network analysis to ego network data. Two different methods accomplish this. The first merges ego network data into a whole network, analyzing the result as if it were a sociocentric dataset. The second treats each ego network as a whole network in itself and applies standard sociocentric analytical procedures.

10.1.1 Merging Ego Networks to Form a Sociocentric Network

One way to conduct whole network analysis of ego network data is to combine the data from distinct ego networks into one large sociocentric network. This is often referred to as the “connect the dots” approach (Weeks et al. 2002). With ego network data collection, knowing the identities of either the egos or alters is not essential. Indeed, one of the advantages of the ego network design lies in the anonymity offered to egos and the people they name in consent agreements. But in certain situations, the identities of the alters can be determined *post hoc*; for example, when the respondents are chosen from the same community and no effort is made to disguise the alters.

For instance, a respondent might say she does drugs with Joe, whom she identifies as a white male in his twenties who hangs out on 4th street. If there is a respondent named Joseph Smith whose attributes are consistent with the description, a tentative match can be made. In fact, a study of drug injectors on the streets of Hartford, Connecticut by Weeks et al.

(2002) did just that. The authors' organization – the Institute for Community Research – worked with the community over many years, using trained fieldworkers who came to know the streets and drug community well. The fieldworkers could often tell researchers with some degree of certainty that alter “Joe” in Survey 1 was probably the person most people knew as “Jose,” and who had participated in an earlier study. In this kind of setting, where intensive long-term fieldwork is being conducted, reconstructing large portions of the whole network from the egocentric data is possible, albeit with some error.

There is software that facilitates the process of identifying egos and alters across ego networks. Using entity resolution techniques from computer science, a software package called SPIDER (Young & Hopkins 2015) helps identify individuals mentioned in a network survey. It can make use of a host of alter attributes, such as gender, age, location, and activities, and it reduces the impact of spelling errors on data quality by matching names phonetically. This type of software can also be used to connect alters from different waves of a longitudinal ego network study when alter pseudonyms – which are prone to error – have been used.

The difficulty with connecting the dots, of course, is that there will be error. In many cases, the researcher will have failed to spot that an alter mentioned by one ego is the same person mentioned by a different ego. This kind of error – a false negative – represents a failure to detect that two nodes are the same. In this case, there are many nodes in the data that are actually the same person. Each of these “selves” will be incomplete, leading to miscalculation of network measures for many nodes, not just the misidentified ones. In other cases, the researcher will have mistakenly equated two or more nodes as the same person – a false positive – creating a conglomerate ego that has too many ties, and, in all likelihood, too much diversity of contacts. These composite nodes will show up as more central than either of their constituent selves should have been.

To address these kinds of errors, the best option is to apply a bootstrapping procedure to generate confidence intervals around the measures (Snijders & Borgatti 1999). For example, a researcher might measure the centrality of each node on the reconstructed network. Knowing there is error, the network is treated as a sample from the true network. The next step is to introduce the same kinds of errors into the data that are likely there, randomly merging some nodes and randomly splitting apart others. This generates another sample, and another set of centrality measures. Doing this thousands of times establishes a mean

centrality for each node, along with a confidence interval around it. Alternatively, the number of samples in which node A scored higher than node B can be counted and, using scaling techniques, the relative positions of each node on a centrality continuum can be inferred (Torgerson 1958).

Another set of challenges associated with this kind of data must be addressed. Since the only respondents are the original egos, any alters mentioned by them that were not also egos will not have been surveyed. The latter will not have any outgoing ties. If the network is defined in terms of the whole set of all individuals surveyed or named, care must be taken in selecting appropriate analyses. For example, if we were to ignore the direction of ties and calculate degree centrality (i.e., the number of ties each node has), nodes that were original ego respondents would tend to be more central. They not only had the opportunity to name others, but others could name them. Conversely, in calculating indegree (i.e., the number of *incoming* nominations a node has), the scores for all nodes are comparable because every node had an equal opportunity to have been named. Alternatively, the network can be restricted only to nodes that were respondents, so that each node is on equal footing. However, this does come at the cost of losing data and reducing the representativeness of results.

10.1.2 Treating Ego Networks as Whole Networks

Another way to apply a whole network approach to ego network data is to treat each ego network as a separate sociocentric network. In a sense, this is similar to methods presented in Chapter 7, where whole network concepts such as density, number of components, and fragmentation, among others, were applied to ego networks. There, the objective was to develop ego-level variables that could be correlated with other ego characteristics. Here, instead of focusing on ego, the focus is on the networks themselves, treating them no differently than networks of classroom friendships or coauthorship networks within scientific disciplines. In a sociocentric study, the boundaries of the network can be chosen using a variety of criteria, such as drug injectors in a certain neighborhood. To treat egocentric networks as sociocentric means that ties to ego are treated as just another criterion for setting the boundaries of a network. It is similar to recording daily interactions among the customers at a coffee shop. Each day produces a new and different network, just as each ego provides a distinct network. When a sociometric approach to analyzing

ego network data is employed, the method of generating alters (and thus, networks) is ignored and analysis proceeds as with any other whole network, perhaps using meta-analytic techniques to summarize across networks.

As an example, Faust and Skvoretz (2002) analyzed a heterogeneous collection of forty-two whole networks. The networks included a red deer dominance network, a cow-to-cow nose-licking network, a monkey grooming network, a senate bill co-sponsorship network, and an organizational advice network. They used an early form of exponential random graph models (ERGM; Frank & Strauss 1986; Wasserman & Pattison 1996). ERGMs, or p^* models, are essentially logit models predicting the log-odds of a tie between two nodes, conditional on a particular network structure and set of covariates. The observed set of ties is treated as one possible pattern of ties that might emerge among a given set of nodes, and the goal is to identify factors associated with higher log-odds of a tie existing (e.g., shared node attributes). Using ERGMs, Faust and Skvoretz (2002) estimated the tendency of each network to develop certain micro-structures, such as the ones shown in Figure 10.1. By fitting the

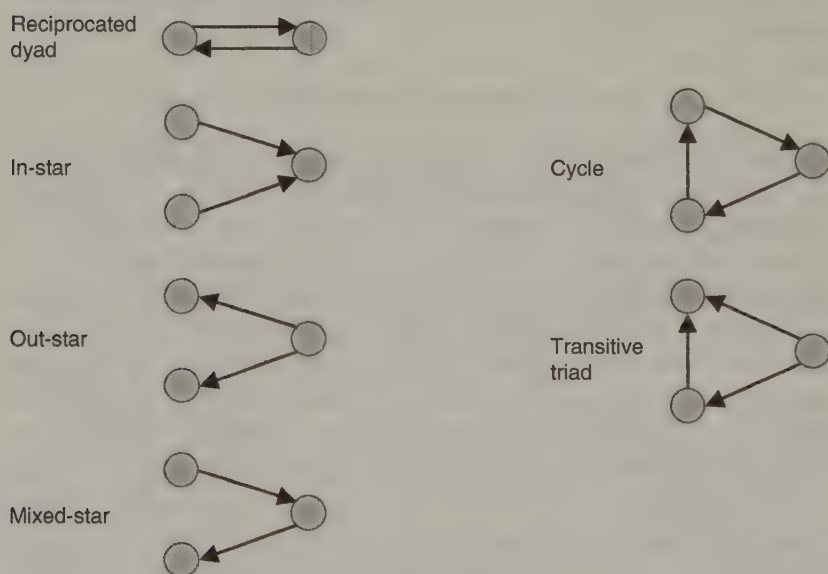


FIGURE 10.1 Micro-configurations in an ERGM analysis of forty-two whole networks

Source: Faust and Skvoretz (2002)

same model to each network, they were able to compare the sizes of the parameters across networks. For instance, the red deer dominance and cow nose-licking networks were high in transitivity.

Having obtained ERGM estimates for each network, one approach would be to conduct a cluster analysis to identify sets of networks with similar estimates. However, the strategy employed by Faust and Skvoretz (2002) is more interesting. Since exponential random graph models predict the presence or absence of ties, the parameter estimates generated in one dataset may be used to predict ties in another dataset. This is similar to estimating a regression model on one sample and using the estimated model to make predictions on a different dataset. Using ERGM models for each network to predict ties in all of the other networks, they determined which models did a good job of predicting the ties present in a network that was not used to create the model. For example, the model for silvereye birds was able to accurately predict ties in the red deer network. This result indicates a high level of similarity in underlying structure between the red deer dominance network and the silvereye fighting network. Examining similarities among all pairs of networks, they determined that, in general, type of relation (e.g., positive vs agonistic) was more important in structuring networks than the kinds of nodes involved (e.g., primate or bird).

Faust and Skvoretz's (2002) research reflects a widespread interest in the network science community in identifying universal laws that govern all networks, or at least certain classes of them. Similar methods could be used to analyze ego networks as a series of whole networks. The goal would be to determine whether certain kinds of ego network structures (e.g., a core-periphery structure) are more common than others. Alternatively, this method could be used to identify commonalities in network structures among particular types of egos (e.g., those of high versus low socioeconomic status).

10.1.3 Applying Whole Network Software to Egocentric Data

Quadrant B of Table 10.1 refers to the application of sociocentric approaches to egocentric data. So far, we have examined linking ego networks to form a whole network, and analyzing ego networks as if they were not properties of an ego. In this section, we take a more practical turn, discussing how to apply sociocentric software to egocentric data. Two different strategies are presented: automated analysis of separate ego networks and extraction of ego network properties from one large network.

TABLE 10.2 *Example batch file automation scripts in UCINET*

A. Batch file automation script for generating ego network measures using hard coding:	
->egonet1 n = removenode(egonet1 1) //removes first node and calls egonet1 n	
->egonet1ncoh = cohesion(egonet1 n) //runs battery of whole network measures	
->egonet2 n = removenode(egonet2 1) //removes first node and calls egonet2 n	
->egonet2ncoh = cohesion(egonet2 n) //runs battery of whole network measures	
//repeat for each egonet	
B. Batch file automation script for generating ego network measures using forfiles:	
->\$ \$n = removenode(\$ \$ 1) //removes first node	
->\$ \$ncoh = cohesion(\$ \$N) //runs battery of whole network measures	
C. Batch file automation script for drawing ego networks using forfiles:	
->draw \$ \$ //draws network with ego included	
->draw \$ \$n //draws network with ego excluded	

In the first approach, a separate network (i.e., a separate dataset) is constructed for each ego network. These can then be analyzed one after another to generate network measures for each ego. In programs like UCINET, this can be automated in a batch file. Example A of Table 10.2 shows a UCINET script in which a series of ego networks are analyzed, key results are compiled into a single dataset, and these variables are then joined with other data collected about ego. In the script, ego is assumed to be included in the data. For the purposes of the example, ego is always the first node. The symbol // indicates the beginning of a comment. After running the script, an output file of the form "egonet#ncoh" will have been created for each ego network. Then, in the UCINET main menu, the Join Columns command is executed to merge all of the egonet#ncoh files into a single file. This file is exported to Excel and subsequently into a statistical software package.

If a consistent file naming convention is used, the process can be further simplified by having the script file take a parameter instead of hard coding a filename (Example B in Table 10.2). The script is repeated for all files that have a similar name, such as any file in a given folder that begins with "egonet". After creating this batch file script, a forfiles command is

TABLE 10.3 A block-diagonal matrix containing four ego networks

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
2	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	1	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
4	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	1	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	1	1	0	1	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	1	0	1	1	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0	0
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	1
17	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	1
19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	1

Note: Ego is the first node in each ego network

executed in the command line interface using: `->forfiles egonet* run doit.txt $$.` The `forfiles` command looks through the current folder and finds every file of the form `egonet1`, `egonet2`, etc., and executes the `doit.txt` script once for each file, substituting every instance of `$$` with the actual filename `egonet1`, `egonet2`, etc. The same approach is used to command UCINET to draw each ego network (Example C in Table 10.2).

The second method for applying whole network analysis software to egocentric data combines all of the ego networks into a single data matrix as shown in Table 10.3. In the matrix, the first ego network occupies rows and columns 1 through 7. The second ego network occupies rows and columns 8 through 14, etc. There are no ties between nodes of one ego network and nodes of another ego network (i.e., block diagonal matrices). UCINET has a menu option (graph union) that constructs the joint matrix from all the separate matrices. The entire matrix can be analyzed at once (e.g., computing structural holes measures), producing scores for all nodes, including alters (which are ignored). The scores are then extracted

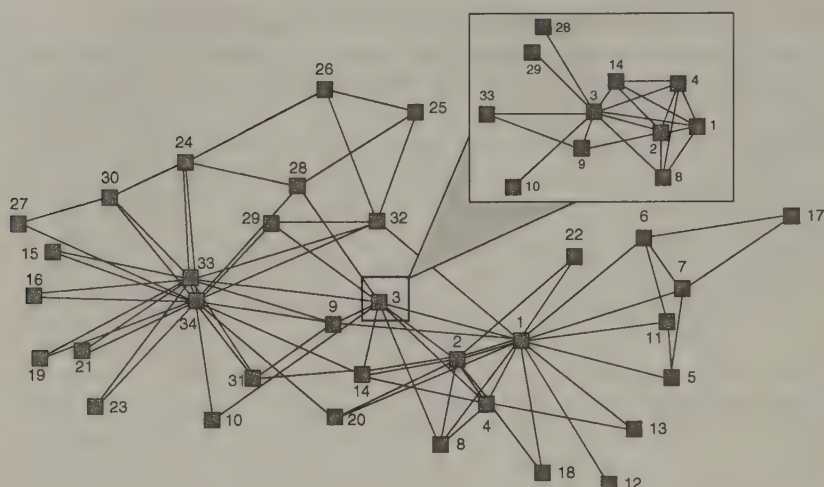


FIGURE 10.2 Ego network (inset) corresponding to Node 3 in a whole network

for all of the nodes that are egos and merged with the dataset containing ego-level variables.

10.2 QUADRANT C: EGOCENTRIC ANALYSIS OF WHOLE NETWORK DATA

Since every whole network dataset contains the ego network associated with each node (Figure 10.2), analyses that can be conducted using ego network data can also be conducted with whole network data.¹ All of the measures described in Chapter 7, which characterize a node's immediate social environment, can be calculated. However, because ego networks generated from sociocentric research designs are subtly different from the ego networks generated from egocentric designs, such measures must be interpreted differently. Moreover, all of the limitations of sociocentric designs discussed in Chapter 2 apply here.

10.2.1 Other-Perceived versus Self-Reported Alter Data

In egocentric research designs, information about the alters – including attributes and ties to other alters – are obtained from ego. They are

¹ In practice, data collected in egocentric studies tends to be both richer and shallower than that collected in sociocentric studies. Typically, egocentric studies collect more kinds of ties between ego and alter, but fewer kinds of ties between alters and alters.

perceived data. In sociocentric designs, alter information is self-reported by the alters. In general, alter-reported data are considered to be more accurate. For example, alters generally know better than egos how old they are, what degrees they have earned, and how much alcohol they drink.² Alters are also expected to be experts on their own ties. However, accuracy is not necessarily the goal (Chapter 2). Most explanations for diffusion or adoption of innovation hinge on egos imitating or adopting what they perceive others to be doing. That ego's alters all binge drink at parties does not affect ego's drinking behavior unless ego is aware of this. Similarly, if egos *believe* that all their friends are binge drinking, they may follow suit, even if, in reality, none of them do. Though not a network study, this is exactly what Perkins (2002) found. "Dramatic misrepresentations" of peer norms supporting binge drinking on college campuses had the significant effect of increasing drinking. The impact was greatest among socially connected students. Thus, when studying these kinds of influence processes, an egocentric design is preferable. Ego's perceptions are the proximal cause. However, in cases where actual alter provision of needed resources is critical (for example, for ego's ability to weather a financial crisis), calculating measures using sociocentric data is preferable. The resources must truly be accessible and must be accessed, not simply perceived to exist.

Similar considerations apply to ties. One theorized consequence of densely connected ego networks is a loss of autonomy. Egos are constrained by alters communicating with each other (Burt 1992). For example, if ego's friends are friends, ego cannot be unkind to one friend without the others finding out about it. Being embedded in a dense network safeguards nodes against malfeasance (Jones, Hesterly, & Borgatti 1997). But ties among alters do not have to exist in reality for this mechanism of constraint to work. It is enough that ego believes the ties to be there. Conversely, the benefits of greater nonredundant information attributed to open networks only accrue if, in reality, there are few ties among the alters. Ego's perceptions have little impact.

In short, though measures developed for sociocentric data can be computed for ego network data, the ability of those measures to predict ego outcomes will vary. It depends whether the mechanisms generating

² An exception to this general expectation would be items that have a strong social desirability component. For example, ego might give us a better estimate of how good a mother a given alter is than the alter would.

these outcomes require ego's awareness, ego's perceptions, or the actual existence of resources and processes.

10.2.2 Incoming versus Outgoing Ties

In ego network analysis, there are no incoming ties because the alters are not interviewed. While a sociocentric design might reveal that ego has a secret admirer, egocentric data collection cannot. Technically, certain ego network measures – such as reciprocity and indegree (the number of incoming ties ego has) – can only be calculated using sociocentric data. In this sense, ego network analysis techniques are not quite the same as those applied to sociocentric data.

However, making a change to egocentric data collection methods can permit observation of directed ties. Specifically, in addition to asking egos who they seek advice from on health-related matters, egos could be asked to report who seeks out their advice. At the cost of additional burden on respondents, this permits measurement of *perceived* reciprocity. Because it is based on ego's perceptions, it is likely to be much higher than when calculated using sociocentric data. One classic reason for this is the presence of a cultural schema that leads people to assume reciprocity (deSoto 1960). Another lies in sociocentric data itself. The tie from A to B is measured by a different instrument (i.e., a different respondent) than the tie from B to A. If A and B have different understandings of the word “advice,” they may give different answers about each other, even if they perceive the same relationship. An advantage of egocentric data is that a consistent definition of advice is being used.

The name generator workaround also permits measurement of a kind of ersatz indegree, which can be used as a measure of felt support and contribution. If egos perceive that many people look to them for help and advice, they are likely to feel that they are making a difference in people's lives. With sociocentric data, indegree would not be interpreted this way because it would require that egos be aware of incoming nominations, which might not be the case. Rather, indegree is usually associated with status, influence, prestige, and, in the case of affective ties, popularity (Borgatti et al. 2013).

10.2.3 Non-Choices

Ego networks, unlike whole networks, do not, by definition, contain non-choices. Ego's nominations for any given kind of relation are observable, but we cannot know who they do not nominate. This

has implications for the measurement of homophily. To illustrate, suppose that in an egocentric study we measured gender homophily by counting the proportion of a person's ties that are the same gender as ego. The problem is, properly speaking, homophily refers to preference for one's own kind, but what is measured by the proportion homophilous measure is the result of a process that has elements of both preference and availability. If 80 percent of ego's coworkers are white, observing that 80 percent of ego's friends at work are white is not necessarily indicative of racial homophily, but of availability. Given the context, the result is consistent with ego choosing friends independently of race.

Methods for adjusting for availability or distribution of particular characteristics in the population (Chapter 7) require an estimate of the proportion of whites, for example, in each ego's social environment. Homophily is then measured in terms of deviations from the ambient proportions. The difficulty, of course, is deciding what constitutes a person's social environment. Who could they have chosen as a friend, but didn't? Because egos operate in multiple social worlds simultaneously (e.g., families, work places, neighborhoods), it is nearly impossible to determine any ego's true potential for forming ties with various types of alters. A person could live in a segregated neighborhood, but work in an integrated environment. Quantifying the potential for developing friendships with people of other races in this case is quite difficult.

Whole network research simplifies this problem by assuming that the respondent could have chosen anyone in the study (e.g., from the roster). This makes it possible to compare the percentage of whites among friends with the percentage of whites among nonfriends. If the percentage of whites among both friends and nonfriends is 80 percent, then the person is not exhibiting a preference for whites.

The difference between egocentric and sociocentric measures of homophily is shown in the crosstab shown in Table 10.4. The table cross-classifies an ego's relations with all other nodes in the network with respect to whether they have a tie, and whether they are in the same group. Cell A gives the number of alters to whom ego has a tie and are of the same race as ego. Cell B holds the number of alters to whom ego does not have a tie, but are of the same race as ego. Cell C contains the number of alters to whom ego has a tie and are of a different race. Finally, cell D gives the number of nodes to whom ego does not have a tie and are of a different race.

TABLE 10.4 *Cross-tabulation of dyads involving ego*

		Has tie		
		1	0	
Same group	1	A	B	A+B
	0	C	D	C+D
		A+C	B+D	N

Note: For each potential alter, the researcher determines whether ego has a tie to that alter (column variable) and whether ego and alter belong to the same group (row variable).

Measures of homophily for egocentric data, such as the proportion homophilous and the E-I index, only make use of the data in the first column (where "has a tie" is 1). The proportion homophilous is $A/(A+C)$ and the E-I index is $(C-A)/(C+A)$. Information about who ego is *not* tied to is unavailable. In sociocentric data, who ego chose not to befriend is known, making it possible to evaluate to what extent $A/(A+C)$ is larger than $B/(B+D)$. One of the most common measures is the correlation coefficient, which can be calculated from the contingency table shown in Table 10.4 using the following formula:

$$r = \frac{ad - bc}{\sqrt{(a+b)(a+c)(c+d)(b+d)}} \quad (1)$$

Less common, but perhaps more desirable, is Yule's Q, defined as:

$$q = \frac{ad - bc}{ad + bc} \quad (2)$$

The advantage of Yule's Q is that it is unaffected by either the number of ties or the number of same-group nodes. For example, if the number of people in one group doubled, Yule's Q would not change. In Table 10.4, the values of A and B would be multiplied by two. Looking at Equation 2, multiplying A and B by two multiplies both the terms in the numerator and the denominator by two, so Yule's Q is unaffected.

In sum, the correlation coefficient and Yule's Q make use of information about ties that are not present. This provides a more accurate assessment of homophily that is only possible when sociocentric data are subjected to ego network analysis.

10.3 COGNITIVE SOCIAL STRUCTURES

In 1984, David Krackhardt introduced the concept of cognitive social structures (CSS), an approach that has expanded rapidly in the past several decades (see Brands 2013 for a review). CSS represents a design alternative to both the whole network and the egocentric approach. Like the sociocentric design, CSS works from a set roster of potential alters that is the same for all egos. But like the ego network design, a CSS study proceeds by asking egos for their perceptions of (1) attributes of each alter, and (2) who is connected to whom among the alters. This generates an alter-by-alter adjacency matrix of perceived ties for each ego, as in ego network research. Measuring perceived ties opens up the possibility of answering a wealth of new research questions. For example, a well-known finding is that being perceived to have a tie with a high status alter has positive outcomes, contributing to a person's reputation as a high performer (Kilduff & Krackhardt 1994). Moreover, the effect holds even after controlling for the actual existence of the tie.

The CSS literature provides important insights about biases that are typically encountered in ego network data. Multiple studies have shown that in reporting friendship ties among their friends, egos perceive much greater levels of transitivity than actually exist (de Soto 1960; Kilduff et al. 2008). In other words, an ego that perceives that alters A and B are friends, and that B and C are friends, is extremely likely to report that A and C are friends as well, even if A and C do not see it that way. This phenomenon results from cognitive/cultural schema associated with friendship that leads egos to expect transitivity (Kilduff et al. 2008).

Not surprisingly, a major theme of CSS research is the accuracy of social perceptions. People vary widely in their ability to perceive relationships among those around them (Krackhardt 1987). Moreover, it is beneficial to egos if their perceptions of the relations among their alters are accurate. For example, controlling for rank, managers with more accurate perceptions of alter-alter ties tend to be seen as more powerful in their organizations (Krackhardt 1990). Likewise, Simpson and collaborators (2011) found that individuals with more accurate social perceptions received higher payoffs in an experimental bargaining game. Johnson and Orbach (2002) showed that congressional lobbyists – whose jobs depend on understanding relations among legislators – made excellent informants about social ties among legislators, and were, in fact, more accurate on average than the legislators themselves when reporting on relationships between legislators.

In Krackhardt's original design, the set of alters in a CSS are the same as the set of egos, resulting in a three-dimensional dataset that is $N \times N \times N$. However, as the Johnson and Orbach (2002) study demonstrated, this doesn't have to be the case. The observers of a network don't have to be part of the network. This is useful for studying network of people who would be unwilling to be interviewed, such as US Senators, terrorists, and Hollywood celebrities. Instead of interviewing the nodes themselves, we can use key informants to tell us about the network. These would be persons close enough to the target individuals to report relatively accurately on their ties, but who are also willing to cooperate with the researcher.

10.4 CONCLUSION

Because this book is specifically about ego network research, it emphasizes the separation between the worlds of egocentric and sociocentric research. However, there are many points of commonality between them beyond their shared theoretical base. This chapter has illustrated that the egocentric/sociocentric distinction applies both to data and to analysis. This creates opportunities to mix and match, for example, applying a sociocentric sensibility to ego network data. This chapter also considered the cognitive social structure approach, which combines elements of both sociocentric and ego network approaches.

The Promise and Challenge of Ego Networks in Network Science

This book has addressed theoretical foundations, fundamental research designs, and core methodological issues in ego network research. We have emphasized that a focus on networks reflects a specific relational view of the operation of the social world, as well as the natural one. In contrast, the scientific enterprise has long been characterized by the unquestioned prominence of the individual unit – the single particle, the rights of each citizen, and the sovereignty of each society. That logic has dominated much of philosophy and science since ancient times. Even in social science, the individual unit was reified after World War II, in part because of a methodological swing from single communities to the nation-state, and from ethnography to survey methods (Coleman 1986). In this context, it became commonplace to believe, for example, that the health of individuals depended, in large part, on their own health behaviors and decisions, underscoring personal responsibility to eat well, exercise regularly, and avoid stress (see Lalonde 1974). In short, the individual, divorced from larger social structures and communities, was seen as the causal agent and the appropriate target of intervention.

The rise of social networks has represented a sea change in the way that scientists across disciplines theorize and analyze both the social and natural worlds. For example, Christakis and Fowler's (2007) findings on diffusion of obesity, smoking, and happiness through social networks in the Framingham Heart Study upended the standard individually oriented approach to medical practice under every premise, from the original germ theory of disease to the contemporary focus on personalized medicine. As described in Chapter 1, ego network research was an early catalyst of this scientific revolution, and the egocentric approach continues to

provide unique insights and innovations. However, there are areas of opportunity for advancement in ego network research, and perhaps places where this approach has stalled. As a set of final considerations, we turn briefly to promising directions, some new and others representing a return to classic ways of thinking.

11.1 A RETURN TO EGO NETWORK TYPOLOGIES

At the turn of the century, Wellman and Potter (1999) raised the issue of theoretical network typologies, asking whether they would provide unique insights beyond an approach relying on variables. Sociology, along with the other social sciences, had developed by noting a seismic shift in Western societies and conceptualizing that shift typologically. The labels varied – *gemeinschaft* to *gesellschaft*, mechanic to organic, folk to urban, agrarian to industrial, sacred to secular – but the common goal was to capture the nature and meaning of change. Nearly a century later, debates centered on the need (or not) for a new type to describe another large-scale shift (e.g., postindustrial, late industrial, postmodern; Freeman & Winch 1957; Pescosolido & Rubin 2000). However, the notion of typology was not popular when Wellman and Potter raised it in 1999. Early social science methods using typological approaches had been replaced with multivariate analyses. In network science specifically, typologies such as parochial and cosmopolitan forms (Suchman 1966) had been eschewed in favor of component elements like size, density, centrality, etc. in personal networks.

Wellman and Potter (1999) argued that a focus on endless single measures of an ego's network does not really reflect the totality of the way personal networks surround and influence individuals. Using their East Yorkers data, they found four characteristics to be central to the description of network types – predominance of immediate kin/friends, frequency of contact, range of contacts (size and heterogeneity), and intimacy (number/proportion of close ties). They proposed that these four dimensions might be the central building blocks of network types, noting that they do not represent a continuum. These ideas challenge whether a multivariate specification of a number of network descriptors can possibly capture the holistic nature of networks. In yet another example, Moolenaar et al. (2012) found seven distinct types of networks among elementary school teachers that mapped onto the intersection of two dimensions – instrumental/expressive networks and interdependence/independence. Their multidimensional scaling analysis begins to move

past structural elements to ones that focus on sentiment, or writ large, culture.

The early focus in the historical typological arguments was on a set of characteristics that created a unique societal type, defining and structuring social lives differentially. Each network form created a *weltanschung* (Ward 1907) that oriented behavior and shaped lifestyle and life chances. In the network tradition, Simmel (1955) laid out a basic theory of the structure of modern and premodern networks, and discussed their effects on identity, prejudice, and life chances. More recently, these have been reconceptualized as modal types that vary across individuals both within and across locations and historical periods (Pescosolido & Rubin 2000). In this view, social networks “bind” time and space for individuals and societies (Giddens 1990: 14). Social network structures constitute social spaces among identities and provide a structure that links social interaction, culture, and society (Stryker 1980; White 1992: 70). Different structures offer alternatives for community identity as well as personal identity, setting the characteristic nature of opportunities and limitations for both the group and the individual. They are not the only social forms in each era, but circumscribe individuals’ “moral personalities” in different ways (Simmel 1955).

The first type of social structure, according to Simmel, is depicted in Figure 11.1. Characterized by social networks that form concentric circles, each larger group envelopes the smaller ones, placing a heavy burden of loyalty, commitment, and prescribed behaviors on individuals. Giddens (1990: 18) and Blau (1993: 40) described individuals “enmeshed in capsulated, polymorphous communities” with “single-faceted, generalist roles” and “a common cultural orientation,” the latter of which was suspicious of anyone outside the limited geographical boundary of place. The accident of birth determines not only family and community, but occupation, religion, and political statuses. This lack of individual freedom is balanced by a sense of security in a well-reinforced safety net. While Simmel saw this form as dominant in premodern society, it is also found in contemporary society in areas where extreme geographical or religious segregation are present (e.g., Wilson 1998).

The second form in Simmel’s typology is characterized by intersecting circles, with an individual’s “moral personality” formed by the point of intersection of many different groups (Simmel 1955: 141; Figure 11.2). Networks can be chosen (e.g., work) or inherited (e.g., family). Networks may be overlapping and of long duration, yet there is

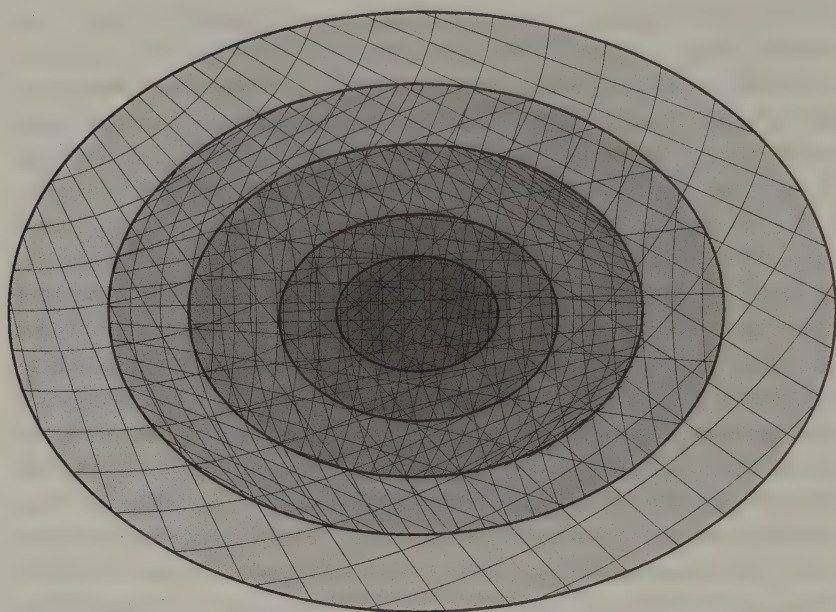


FIGURE 11.1 Social network formation in the premodern era (grayscale)

Source: Simmel (1955)

a greater element of choice. The individual as a social unit takes on greater prominence, with diverse network ties producing stronger individual personalities and greater potential for change across the life course. With more autonomy and individuality comes a greater burden of constructing interests, higher levels of tolerance, and trust in abstract systems such as banking and transportation. Yet, new freedoms also bring contextual uncertainty, weaker support networks, potential for cross-network conflicts and, as Simmel (1955: 141) stated, “psychological tensions.” Simmel described intersecting circles as the archetypal network structure of modern society.

Figure 11.3 depicts the third network type as a “spoke” – the model in contemporary society. Drawing from a wealth of critique and concern with how the organizational and interpersonal bases of social life changed over the last decades of the twentieth century, this new model typological form connects distinct social circles, some of which are only loosely bound. As Simmel (1955: 146) pointed out, overlapping social circles cannot occur if the social groups involved are far apart in purpose and in the demands they place on individuals. Four characteristic features are

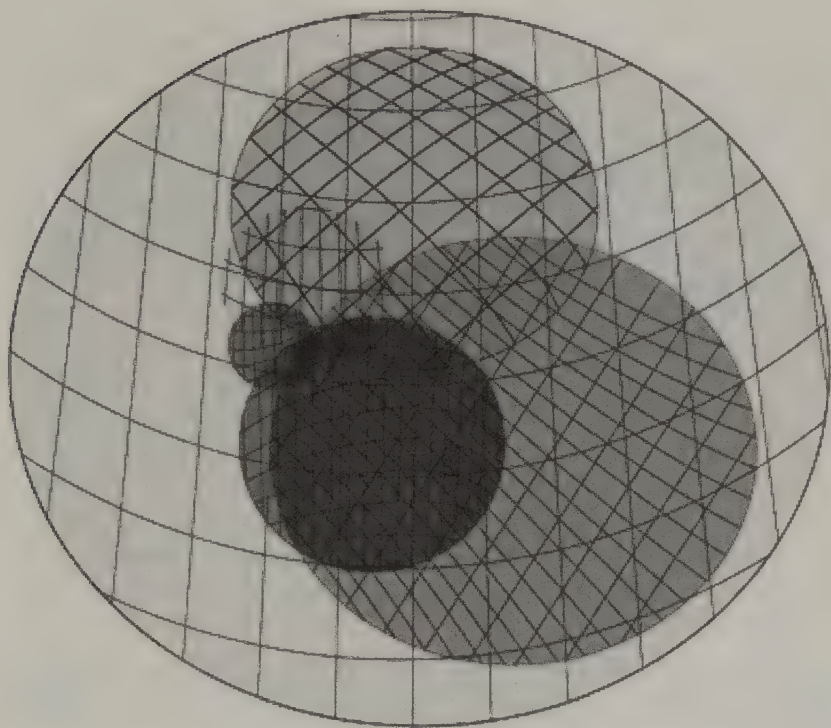


FIGURE 11.2 Social network formation in the contemporary era (grayscale)

Source: Simmel (1955)

(1) individuals not in social networks, but standing outside of networks, (2) ties to institutions that are multiple and often temporary, not single and lifelong (e.g., multiple marriages, multiple jobs over the career), (3) interactions more often through electronic, social media, or other new forms of communication rather than face-to-face, and (4) a dramatic increase in the speed and scale of communication, media, and transportation. The basis for this network structure lies in the recognition that social life is based on serial, ephemeral, short-term, contingent relationships with comparably limited social contracts. With more weak and integrative ties come increased opportunities, but also a new tenuousness of social life that exceeds even Simmel's modern form. On the one hand, with the individual in a critical and central role as the integrator of social relationships, freedom and choice exist on an unprecedented scale. On the other hand, this structure also requires constant negotiation between network ties because social relations can neither be taken for granted nor can they

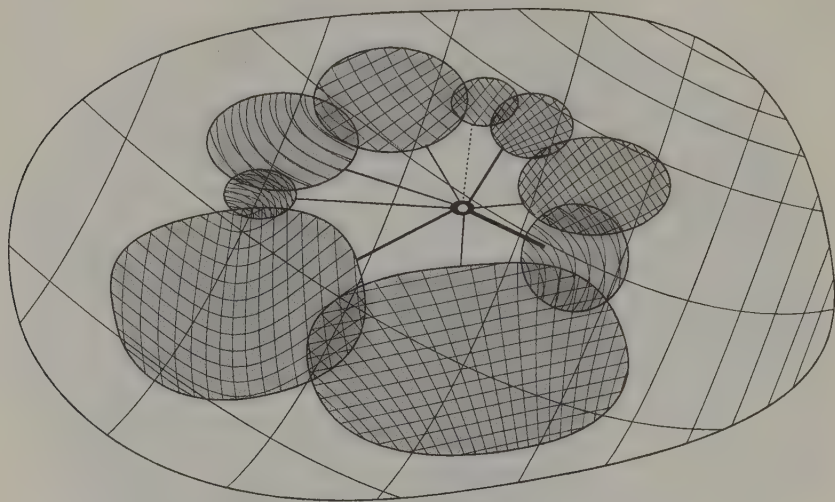


FIGURE 11.3 Social network formation in the modern era (grayscale)

Source: Simmel (1955)

survive for long without attention. There is a fragility to social networks without the webs of relations that crisscross social life (Pescosolido & Rubin 2000).

In essence, following Wellman and Potter (1999), if we think of these not as time periods but as basic structural forms in any society (even if some are more likely than others in certain historical periods), we begin to develop a theoretical typology that allows the development and testing of hypotheses. All network structures advantage some people while disadvantaging others. For example, those who profit from the spoke structure are likely to be those in power (as in all social forms), those who are resource rich, and those on the cutting edge of societal developments (e.g., technology). However, the temporary and contingent nature of work (without benefits) for many working-class and middle-class individuals represents not newly found freedom, but new economic insecurity. Further, any conditions (e.g., serious illness) or events (e.g., divorce) that hinder an individual's ability to maintain ties are likely to have devastating consequences. In the end, there have always been individuals whose social relations extended across place, even if it was the "world" as it was constructed across tribes or ancient societies. Similarly, there are still tight-fisted social networks where individuals' lives are "secure" but heavily constrained (e.g., North Korea, cults).

11.2 CULTURE AND EGO NETWORKS

Typologies come closer to thinking about what is held inside networks of different forms. However, at the end of the last century Burt wrote: “Network content is rarely a variable in the studies – analysts agree that informal coordination through interpersonal networks is important as a form of social capital, but their eyes go shifty like a cornered ferret if you push past the network metaphor for details about how specific kinds of relations matter” (Burt 1997: 357). While there have been improvements since then, for the most part, network researchers have continued to note a culture gap (Emirbayer & Goodwin 1994) but few have incorporated it into their research.

DiMaggio (2011), among others, has taken on the issue of culture and social networks. Along with Embayer and Goodwin (1994), he makes the bold claim that “relationality is central to understanding how cultural systems operate” (2011: 286) and that “network analysis is the natural methodological framework for empirically developing insights from leading theoretical approaches to cultural analysis.” DiMaggio argues that in social systems where “groups” tightly define systems (e.g., in Simmel’s premodern, concentric circle system described above), culture is easily ascribed and shared. However, in contemporary society, “fluid, open-ended, functionally differentiated networks” (2011: 290) define social life. The processes that shape meaning, social identity, and belonging have a fundamentally different, and increasingly more complex nature. As a result, “thought communities” reflect the duality that belief systems are constituted in social networks and simultaneously influence the individuals in them. For ego networks, it follows that it is not just their structure that can be measured and identified. Rather, in a parallel manner, the cultural content of ego networks is inextricably tied to them, and represents an essential part of how they influence social life. The central question that remains is how to move past simply identifying these cultures, as in the many examples through which DiMaggio illustrates the duality of culture and networks.

For many network scientists, a critical goal is to understand how network structure and culture jointly influence social behavior and social processes. Thus, the measurement of “culture” becomes an important, but far from obvious, part of data collection. One way to do this is through the name interpreters, where respondents are asked substantive (not structural or relational) questions about alters. That is, rather than only asking about role relationship (i.e., kin, coworker) or tie dimensions (e.g.,

closeness), trying to get at culture requires asking questions about beliefs, attitudes, and norms. This is by no means easy or standard. Asking them as part of name interpreters has at least three major limitations. First, cultural constructs such as religious beliefs, attitudes toward doctors, or racial prejudice are best measured using scales that have established reliability and validity, but contain anywhere from four to twenty plus items. This is not feasible in a name interpreter scheme where each item would be asked for each tie, with the number of questions increasing geometrically as ties mentioned increase. Second, even if the resources or respondents with a tolerance for such a lengthy approach were available, choosing one cultural scale would have a heavy weight of importance. Third, even if feasible theoretically and methodologically, egos' reports of what they think their ties believe may be inaccurate. This is the least of the potential problems because research has shown that such perceptions can have as powerful an impact on individuals' attitudes and behaviors as "reality." In fact, the classic research on college drinking finds that students' perceptions of the drinking behavior of other students is more influential on their own drinking than others' actual behavior (Martin et al. 2010).

Few methodological solutions to the measurement of culture in ego networks have been proposed. One possibility is using RDS methods to interview ego's alters (see Chapter 3). This may be the single best existing approach, but the one least likely to be implemented due to cost and logistical issues. Moreover, this approach does not measure ego's perceptions of alters' attitudes, values, and beliefs. Other less optimal strategies are to use a single cultural item in a name interpreter module, or to employ proxy measures adapted from psychometrically validated scales that ask about the network as a whole. All of these options have important weaknesses, making this area of research ripe for methodological innovation.

In the end, the introduction of culture into networks remains a relatively new frontier. A social network perspective has guided studies of the production of culture and cultural markets, as well as analyses of networks of cultural actors or identities. Moreover, as we have shown in earlier chapters, for example in Vilna Bashi's study of immigrants, researchers who have taken a more ethnographic approach have done the most to see networks as structure holding cultural value systems. The question becomes how can these insights and approaches be merged more successfully into more quantitative studies of ego networks or into mixed method studies of networks?

11.3 MERGING THE STRENGTHS OF EGOCENTRIC AND SOCIOCENTRIC APPROACHES

The final issue that stands at the cutting edge of network research is how to take advantage of multiple kinds of network data, drawing on the two major types – egocentric and sociocentric (see Chapters 2 and 10). The two most significant benefits of an ego network design are (1) the ability to use probability sampling, and thus make generalizable inferences about the social world, and (2) the flexibility and unbounded nature of ego networks, which permit simultaneous examination of Simmel's multiple overlapping social circles across different domains (e.g., work, family, neighborhood). However, sociocentric designs hold the major advantage of being able to map the broader network structure in which ego networks develop and change over time. This structure includes measurement of the *absence* of ties – a potentially critical missing piece in the landscape of personal community networks (see Chapter 2).

A hybrid study design – where ego network data are collected in the context of a whole network design – has the potential to combine insights from both approaches. For example, sociocentric data collection methods might be used to study a classroom of children, but an ego network instrument could be added. Each child would be given a roster of the class members and asked about friendships, communication, help-seeking, bullying, etc. Following that, a standard egocentric study could be conducted to assess the child's relationships outside of school, including with family, neighborhood children and other peers, and other adults outside the family. The results of the ego network study could be very useful in explaining patterns identified in the whole network data. Further, ego network data could be directly incorporated into whole network analysis by aggregating ego network characteristics and treating them as attributes of nodes. For example, ego networks might explain a child's peripheral position in the class network, despite good social skills. Knowing that the child is happily embedded in an absorbing network of friends and family outside the classroom could explain the lack of interest in deepening friendships within the classroom.

Another approach is to build up a whole network context using ego networks as the foundation. Recently, Moody (2017) suggested that a return to geographically delimited network studies, most clearly community studies, represents the best avenue to obtaining sociocentric insights from ego networks. He defines a “network hyper-sample,” where the research design extends a number of steps to reach a point

where interconnections within the community become general and saturation is achieved. In a random walk design, for example, the sampling unit is the whole network, not individuals, and the goal is to characterize the social structure of one community. However, because the whole network is built from a galaxy of ego networks, rich information on ties unbounded by a particular context is retained and can be incorporated into analyses. Further, because a subset of each ego's alters is randomly selected into the sample from the nominated ties, an element of probability is introduced that permits the derivation of unbiased indicators and estimates of their precision (Heckathorn 2002). That is, this approach addresses the inference limitations of typical sociocentric designs.

The utility of such hybrid approaches is illustrated by a geographically delimited community study of seventy-five rural villages in Karnataka, India (Banjee et al. 2013). Half of all households in these Indian villages were surveyed using a random sample stratified by religion and geographic sublocations. Employing a name generator approach, alters were elicited across twelve distinct domains, including friendship, visiting, borrowing, lending, advice, instrumental support, family, and church members. Subsequently, a whole network of directed ties was built using egocentric nominations both within and across villages. Jackson, Rodriguez-Barraquer, and Tan (2012) used these rich network data to examine informal encouragement of cooperative behavior through "supported" patterns of exchange of favors. Here, supported ties were defined as relationships between two exchange partners that shared a common friend. The authors reasoned that exchanges of favors are often infrequent and unreciprocated, suggesting that they are probably not reinforced by the promise or potential for direct benefit. However, when exchange relationships are embedded in a "social quilt" of many supported interdependencies (i.e., tree-unions of completely connected subnetworks; see Figure 11.4), individuals may be compelled to cooperate under threat of social exclusion. This is precisely the pattern that Jackson, Rodriguez-Barraquer, and Tan (2012) identified using sociocentric data from rural Indian villages, providing insight into the embedding of egos' decisions in the network structure of friendship and cooperative behavior in a large community. In short, individual behavior within ego networks was reinforced by the broader social structure. The next step requires that similar community studies be conducted in other areas of the world – and particularly in the global North – to test the generalizability of these findings.

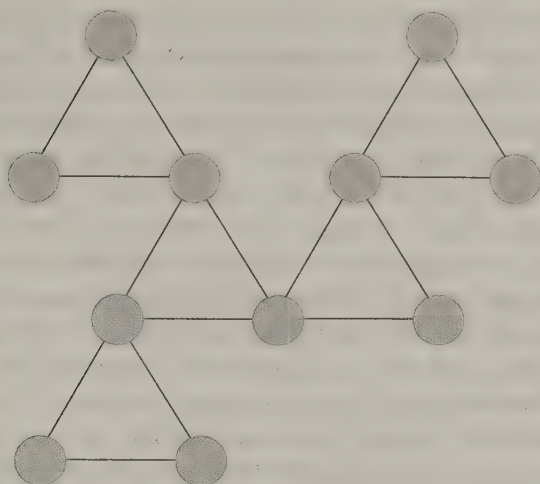


FIGURE 11.4 A cooperation-supporting social quilt of connected subnetworks
Source: Jackson, Rodriguez-Barraquer, and Tan (2012)

11.4 A FINAL NOTE ON THE ROLE OF SOCIAL NETWORKS IN NETWORK SCIENCE

As a final note, we turn back to larger issues confronting the role of ego networks in network science, returning to an observation made in Chapter 1. Too often, network science is seen as a method of analysis – a tool that can be applied to any and every problem, and, more importantly, to every data set. If a basic premise of the network paradigm is that the interconnection among particles, people, and societies constitutes the engine of action, then network science does apply, or potentially can be applied, to every problem. However, while network science is often a correct way of thinking, it is not a universally applicable methodological tool. As method, networks represent a different way of defining the nature of data. In our experience, researchers are often attracted to the idea of connectedness, stimulated in their own work by recent findings from network science. However, that enthusiasm often dampens when the expectation of applying network methods to their existing data is not easily accomplished. This feeling is reinforced when the difficulty of collecting or finding suitable networked data becomes clear. In this sense, network science requires a different kind of methodological thinking.

The criticism that network science overemphasizes methodology is, in some ways, a fair one. In the twentieth century, much of the research on networks was led by methodologists solving the unique analytic problems associated with the violation of the case independence assumption that underlies traditional statistics. This includes the sociometric school's development of the sociogram in the 1930s (Moreno & Jennings 1938); White, Boorman, and Brieger's (1976) solution to blockmodeling using structural equivalence and average path length; and more recently, natural and physical scientists' large-scale, powerful algorithms to analyze data from phone records, proteins, and travel routes (Barabasi & Albert 1999). Each represented clear advancements that spurred theoretical and analytic work. However, not all of these advancements were greeted with enthusiasm, at all or for long. Visualizing even moderate-sized networks was difficult using the sociogram. Blockmodeling also had size limits that frustrated researchers for decades. And, while the entry of natural and physical scientists dramatically reenergized network research, their claims of "novel" assumptions that broke through scientific stalemates provoked reactions that were more shock than awe. As Scott (1991) notes, some of the new breed of network scientists fails to understand both the history of social network research and the earlier collaborations between sociologists and physicists, from Engles to Homans to White. In fact, sociology's founder, August Comte, originally named the discipline of sociology "social physics" to reflect the similarity of metaphorical imagery in social science that speaks of the "forces" that impact human behavioral systems (Scott 1991).

More importantly, the critique of networks as method cautions against methodological dependence without adequate theorizing. At the turn of the last century, Wellman (1999: xiv) reminded us that tools can become worldviews. The rapid development of nonparametric statistics and computer packages such as SAS and SPSS reified the notions of independence and of individuals analytically isolated from one another in the social and health sciences. Likewise, the search for a universal power law structure (i.e., the unequal distribution of influence within networks) across all big data masks the importance of thinking through its meaning and relevance. Social and physical scientists have argued against having too universal a view of scientific phenomena. Similarly, both have argued that human social networks are not like other networks, despite attempts by others to fold them in with proteins, computer grids, and insects (Newman & Park 2003).

In the end, social network scientists have always embraced a broader view of networks – whether the work was theoretical or empirical, whether boundary issues led to egocentric or sociocentric approaches, and whether the data collected were quantitative or qualitative. To close off any possible way of theorizing, collecting data, or analyzing network connections is unwise, if not dangerous. Old Kuhnian notions that one paradigm replaces another has been revealed as more an exercise in academic power (and networks), rather than based in scientific “truth.” Further, paradigm replacement may have described the development of disciplines and theories in the twentieth century. But these are outdated in a scientific century that accepts the complexity that Nobel Prize winner Elinor Ostrom (2009) embraced, or that E.O. Wilson foresaw in 1984. While both were visionary, even their visions were extremely different, with the latter calling for the fusion of the sciences into one and the former seeing the unique and almost essential puzzle pieces that each brings to the table of discovery. Eventually, the optimal result will be a common ground with the strongest elements from each discipline and broad segments of science being brought to bear on complex, contemporary social problems, perhaps with networks as the linking mechanism (Pescosolido 2006; Pescosolido et al. 2017).

Network Science has many faces, each of them unique and contributing to an understanding of how connectedness represents a major mechanism of human, natural, and technological action. It is in our best interests to explore all of them as well as their interconnections. To that end, we hope that this book has provided a platform for understanding egocentric networks as they are conceptualized, measured, and analyzed at present, and also a vision for how this approach can be taken further in the future.

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Egocentric network analysis is used widely across the social sciences, and is increasingly being employed in business, informatics, and the natural sciences. This approach focuses on individuals sampled from a population, and their direct social ties. A fundamental tenet is that each of these individuals is embedded in a personal community – partially of their own creation and nearly unique to them – whose composition and structure have important consequences for wellbeing and life chances. Egocentric network analysis requires a distinct set of methodological tools that overlaps only minimally with sociocentric network analysis, in which there is a common network boundary shared by all actors. However, until now there has been no single comprehensive reference for studying ego networks. This book covers the landscape of egocentric network theory, research design, and analysis, with an emphasis on application to engaging substantive examples and empirical data. *Egocentric Network Analysis: Foundations, Methods, and Models* provides a readable blueprint for the novice, but also serves as a guide to state-of-the art innovation for the experienced researcher.

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